

Critical Research Methods in Psychology

CRITICAL RESEARCH METHODS IN PSYCHOLOGY

STEPHANIE D'COSTA; MIREILLE
UKEYE; MAKENZIE O'NEIL; AND
REBECCA ANGUIANO



Critical Research Methods in Psychology Copyright © by Stephanie D'Costa; Mireille Ukeye; Makenzie O'Neil; and Rebecca Anguiano is licensed under a [Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License](https://creativecommons.org/licenses/by-nc-sa/4.0/), except where otherwise noted.

[Critical Research methods in Psychology](#) by Stephanie D'Costa, Makenzie O'Neil, Mimi Ukeye, & Rebecca Anguiano is licensed under [CC BY-NC-SA 4.0](https://creativecommons.org/licenses/by-nc-sa/4.0/), except where otherwise noted. Some parts of this collection have more restrictive licenses. Please note those licenses when you use this collection.

This book was produced with Pressbooks (<https://pressbooks.com>) and rendered with Prince.

CONTENTS

Land, Labor, & Community	xiii
Acknowledgements	
About the Authors	xvii
Rebecca Anguiano; Mireille Ukeye; Makenzie O'Neil; and Stephanie D'Costa	
Table of Contents	xxiii
License Attributions	xxx

Part I. Frameworks

1. Methods of Knowing	3
Rebecca Anguiano	
2. Reflexivity and Positionality	20
Stephanie D'Costa	
3. Relational Accountability	33
Stephanie D'Costa	

4. Research Paradigms & Theories	39
Stephanie D'Costa; Makenzie O'Neil; and Sujay Sabnis	
5. Critical Theory, Critical Theories, and Research	61
Rebecca Anguiano	
6. QuantCrit	80
Stephanie D'Costa	
7. Research Ethics	100
Mireille Ukeye	

Part II. Research Practices

8. Scientific Method	157
Makenzie O'Neil	
9. Populations & Samples	171
Stephanie D'Costa and Makenzie O'Neil	
10. Variables	187
Rebecca Anguiano	
11. Research Questions	196
Mireille Ukeye	

12. Developing a Hypothesis	211
Stephanie D'Costa	
13. Validity in Research	224
Makenzie O'Neil	

Part III. Psychological Measurement

14. Understanding Psychological Measurement	239
Rebecca Anguiano	
15. Reliability and Validity of Measurement	260
Rebecca Anguiano	
16. Practical Strategies for Psychological Measurement	277
Stephanie D'Costa	

Part IV. Methods

17. Experimental Design	295
Rebecca Anguiano	

18. Single-Subject Research Designs	321
Stephanie D'Costa	
19. Non-Experimental Research	359
Mireille Ukeye	
20. Correlational Research	372
Stephanie D'Costa	
21. Quasi-Experimental Research	392
Stephanie D'Costa	
22. Qualitative Research	416
Rebecca Anguiano	
23. Overview of Survey Research	435
Stephanie D'Costa	

Part V. Statistics

24. Introduction to Statistics: Descriptive & Inferential	459
Rebecca Anguiano	
25. Frequency Distributions	490
Makenzie O'Neil	
26. Central Tendency	511
Mireille Ukeye and Makenzie O'Neil	

27. Variability	519
Rebecca Anguiano	
28. The Standard Normal Distribution and Z-Scores	537
Makenzie O'Neil	
29. Probability and the Normal Distribution	558
Makenzie O'Neil	
30. Sampling Distributions	580
Mireille Ukeye	
31. Introduction to Hypothesis Testing using Z-scores	601
Makenzie O'Neil	
32. T-Tests	646
Stephanie D'Costa	
33. Analysis of Variance	692
Rebecca Anguiano	
34. Correlations	739
Makenzie O'Neil	
35. Linear Regression	802
Makenzie O'Neil	

Part VI. Writing & Presenting Research

36. American Psychological Association (APA) Style Mireille Ukeye	841
37. Dissemination Mireille Ukeye	864
38. Posters & Presentations Stephanie D'Costa	874
Appendix A: Standard Normal Distribution Table (z Table) Stephanie D'Costa; Mireille Ukeye; Makenzie O'Neil; and Rebecca Anguiano	897
Appendix B: t Distribution Table (t Table) Stephanie D'Costa; Mireille Ukeye; Makenzie O'Neil; and Rebecca Anguiano	927
Appendix C: Critical Values for F (F Table) Stephanie D'Costa; Mireille Ukeye; Makenzie O'Neil; and Rebecca Anguiano	934
Appendix D: Critical Values for Pearson's r (Correlation Table) Stephanie D'Costa; Mireille Ukeye; Makenzie O'Neil; and Rebecca Anguiano	941

Appendix E: Chi-Square Table	949
Stephanie D'Costa; Mireille Ukeye; Makenzie O'Neil; and Rebecca Anguiano	

LAND, LABOR, & COMMUNITY ACKNOWLEDGEMENTS

This textbook is an offering to our students, colleagues, and communities to envision a research practice that is grounded in justice. As a project team, we also acknowledge that time and energy dedicated to this book were not in isolation—the land, labor, and community all contributed to the creation of this text.

We worked on this text on the lands of the [Muwekma Ohlone peoples](#), multiple Indigenous nations who were violently forced to give up their land, language, and knowledge systems due to colonization. Their continued existence speaks to the power of resistance. Our home institution, Saint Mary's College of California, created the following acknowledgment that we would like to share.



One or more interactive elements has been excluded from this version of the text. You can view them online here:

<https://smc.pressbooks.pub/critresearchmethodspsyd/?p=18#oembed-1>

Video i.1. Land Acknowledgement by Saint Mary's College of California

As settlers on this land, we hope to be in right relationship with the land and all living things. One way we do that is by contributing Shummi Tax to support the rematriation of land.

We also acknowledge the Labor that allows us to access the privilege of being members of academic institutions. The United States' wealth was founded on the exploitation of enslaved Africans, and our current unequal wealth distribution is connected to a pivotal historical legacy. In the San Francisco Bay Area, where this textbook was created, Black folks came for work opportunities and experienced segregation, discrimination, and exploitation at high rates¹. Nevertheless, they became the leaders of liberation movements (e.g., the Black Panther Party) and political advocacy.

We are here today because of a legacy of resistance. And while we acknowledge the harms that have happened and

1. . Moore, S. A. W. (2000). *To Place Our Deeds: The African American Community in Richmond, California, 1910-1963*. Univ of California Press.

currently happen—we also name the ways which our communities have held us.

OER for Social Justice Grant

The creation of this textbook was funded by the OER for Social Justice Grant. Our deepest appreciation for the entire team for providing funding, training, and structure to the textbook creation process. Special shoutout to Swetta Abeyta (SMC), Karna Younger, & Theresa Huff for all your incredible support in this process.

Advisory Panel

The research team invited psychology practitioners, researchers, and students to serve as advisory panel members for this text. The ask was to have criticality be centered in our work. Twelve people were invited to serve in this capacity, and the final text benefited immensely from their thoughtful questions, reflections, and edits.

Panel members [In alphabetical order]

Wendy Castillo

Dawn Henderson

Lakhvir Kaur

Kimberly Knesting-Lund

Dominique La Barrie

Seanna Leath

Aaron J. Neal

Juan R. Pantoja-Patiño

Mary Phan

Christen Priddie

Arnold Rodriguez Robles

Sujay Sabnis

Acknowledgment of FIPSE Funding

The contents of this book were developed under a grant from the Fund for the Improvement of Postsecondary Education, (FIPSE), U.S. Department of Education. However, those contents do not necessarily represent the policy of the Department of Education, and you should not assume endorsement by the Federal Government. For more information about OER for Social Justice, [visit our project website](#).

ABOUT THE AUTHORS

Rebecca Anguiano; Mireille Ukeye;
Makenzie O'Neil; and Stephanie D'Costa

Stephanie D'Costa, Ph.D, LP



Figure i.1. Dr. Stephanie D'Costa

Dr. Stephanie D'Costa is a South Asian, cisgender woman residing on Ohlone lands (Northern California). She immigrated to North America with her family when she was four years old. She is currently an associate professor of counseling at [Saint Mary's College of California](#). Dr. D'Costa earned her Ph.D from the University of Wisconsin-Madison in Educational Psychology. She is passionate about community-based research that empowers culturally and linguistically minoritized children and families. Her [research](#) focuses on a) critically exploring the impact of oppressive forces on minoritized youth and b) building culturally sustaining practices in academics, behaviors, and mental health for k-12 students. She frequently engages with student researchers and

has a number of publications that include student authors. Dr. D’Costa has [published](#) on the need for QuantCrit in school psychology and is committed to teaching about research that empowers marginalized communities. While educational disparities brought her into academia, the resistance and future liberation of marginalized communities keeps her dreaming. Dr. D’Costa is also a licensed clinical psychologist with a specialty in trauma-informed practices for children. When she is not working, she enjoys having chai with her backyard chickens, hiking with her dog, and connecting with friends. You can follow her at [@dr_dcosta](#) (Instagram)

Mireille Ukeye, Ph.D, NCSP



Figure i.2. Dr. Mireille Ukeye

Mireille Ukeye, Ph.D., NCSP is a bilingual school psychologist and an assistant professor of school psychology in the School of Education at Saint Mary’s College of CA. She is originally from East Africa and resides on Ohlone lands (Northern California). The three main areas that reflect her research are equitable practices within the multi-tiered system of support (MTSS), specifically in Title 1 schools, implementation of response to intervention with a focus on English Language Learners (ELL), and school-

based problem-solving approaches to behavioral management. Her research uses multiple modes of investigation to better understand autism spectrum disorders (ASD) and other behavioral disorders to determine the most effective approaches to treating and supporting students with ASD and other behavioral needs in schools. Additionally, her research looks at different ways that mental health providers can support families of children with disabilities. She has been awarded a grant to introduce a mindfulness program in schools that aims to help high school students cope with stress and anxiety following school closures. She is a grant reviewer for the California Department of Education's Nita M. Lowey's 21st Century Community Learning Centers and After School Safety and Enrichment for Teens Program, and SAMHSA. In addition, she also received assessment grants as a co-recipient, which will be utilized to support various assessment projects focused on improving course designs at both the departmental and program levels.

Makenzie O'Neil, Ph.D



Figure i.3 Dr. Makenzie O'Neil

Dr. Makenzie O'Neil is a White cisgender woman who resides on Ohlone lands (Northern California). She is an Associate Professor of Psychology at [Saint Mary's College of California](#). She earned her Ph.D. at Arizona State University in Social Psychology. Her research focuses on examining how discrete positive emotions

impact intergroup relations, including stereotyping, prejudice, and discrimination. She has recently published research examining the impacts of intergroup helping on attitudes and cooperation between groups. Dr. O'Neil is passionate about fostering excitement for conducting research in undergraduate students from diverse and minoritized backgrounds. In addition to teaching research methods and statistics courses at the undergraduate level for nearly a decade, and spearheading the implementation of a lab component to a research methods course, she has worked closely with many student researchers to develop their ideas, including assisting with student authorship on empirical manuscripts.

Rebecca Anguiano, Ph.D, NCSP, LEP

Smiling person with curly hair,
white-rimmed glasses and gold
hoops, yellow cardigan over a
blue shirt, outdoors.

*Figure i.4. Dr. Rebecca
Anguiano*

Dr. Rebecca Anguiano is
a Professor and Department
Chair in the Counseling
Department at [Saint Mary's
College of California](#). She
earned her PhD in School
Psychology from the

University of California, Berkeley and is a credentialed, bilingual (Spanish-English) school psychologist and licensed educational psychologist. As a Brown Chicana woman, Dr. Anguiano's personal experiences have motivated her to focus her research and clinical work on lifting the voices of historically marginalized communities, especially Black and Brown children and their families, as they interface with pre-K-12 schools. Dr. Anguiano's research has explored the translating practices of immigrant families, community organizing, and popular education as methods for family engagement in schools, and culturally responsive data-based decision making to inform interventions for emergent bilingual students. She has been published in outlets such as the School Community Journal, the Journal of Youth and Adolescence, and Cultural Diversity and Ethnic Minority Psychology. Dr. Anguiano has provided training to community agencies and schools on trauma-informed

practices, strength-based psychoeducational evaluations of bilingual children and youth, socially just family engagement, and anti-racist self-care for school leaders during the COVID-19 global pandemic.

Media Attributions

- MU
- MO

TABLE OF CONTENTS

1. Frameworks

- Knowledge Creation
- Positionality
- Relational Accountability
- Research Paradigms
- Critical Theory and its impact on Research
- QuantCrit
- Research Ethics

2. Research Practices

- Scientific Method
- Populations & Samples
- Variables
- Research Questions
- Hypotheses
- Validity in Research

3. Psychological measurement

- Understanding Psychological Measurement
- Reliability & Validity in Measurement
- Practical Strategies in psychological measurement

4. **Methods**

- Experimental
- Single Case Design
- Non-Experimental
- Qualitative
- Miscellaneous

5. **Statistics**

- Descriptive and Inferential
- Frequency Distribution
- Central Tendency
- Variability
- The Standard Normal Distribution and Z-Scores
- Probability and the Normal Distribution
- Sampling Distribution
- Introduction to Hypothesis Testing using Z-scores
- T-Tests
- ANOVAs
- Correlations

- Linear Regression

6. Writing

- APA Style
- Dissemination
- Posters & Presentations

LICENSE ATTRIBUTIONS

About this OER

Critical Research Methods in Psychology by Stephanie D’Costa, Mirelle Ukeye, Makenzie O’Neil, and Rebecca Anguiano is licensed under [CC BY-NC-SA 4.0](#), except where otherwise noted. Some parts of this collection have more restrictive licenses. Please note those licenses when you use this collection.

While we, the primary authors, created original content for this OER, *Critical Research Methods in Psychology* is possible through the generosity of contributors to the OER as well as creators who openly licensed their works to allow us to remix or otherwise include their work. Included copyrighted materials are utilized under fair use. If you have questions about the use of the works cited or attributed in *Critical Research Methods in Psychology*, please contact us at sd43@stmarys-ca.edu.

The following texts were remixed for inclusion in this text:

- [Research Methods in Psychology, 4th Edition](#) by Rajiv S. Jhangiani; I-Chant A. Chiang; Carrie Cuttler; and Dana C. Leighton

- [Practicing and Presenting Social Research](#) by Oral Robinson and Alexander Wilson
- [Scientific Inquiry in Social Work](#) by Matthew DeCarlo
- [Introduction to Statistics in the Psychological Sciences](#) by Linda R. Cote Ph.D.; Rupa G. Gordon Ph.D.; Chrislyn E. Randell Ph.D.; Judy Schmitt; and Helena Marvin
- [ReCentering Psych Stats](#) by Lynette H. Bikos
- [Statistics Through an Equity Lens \[Revised Edition\]](#) by Yvonne Anthony

PART I

FRAMEWORKS

1.

METHODS OF KNOWING

Rebecca Anguiano

Learning Objectives

By the end of this chapter, you will be able to

1. Describe the 5 methods of acquiring knowledge.
2. Describe the benefits and problems with each method of acquiring knowledge.
3. Describe ontology, epistemology, and

axiology and their connection to our research assumptions.

4. Explain how ontology, epistemology, and axiology inform a researcher's methodology.

Take a minute to ponder some of what you know and how you acquired that knowledge. Perhaps you know that you should make your bed in the morning because your mother or father told you this is what you should do. Perhaps you know that swans are white because all of the swans you have seen are white. Perhaps you know that your friend is lying to you because she is acting strange and won't look you in the eye. But should we trust knowledge from these sources? The methods of acquiring knowledge can be broken down into five categories, each with its own strengths and weaknesses.

Intuition

The first method of knowing is intuition. When we use our intuition, we are relying on our guts, our emotions, and/or our instincts to guide us. Rather than examining facts or using

rational thought, intuition involves believing what feels true. The problem with relying on intuition is that our intuitions can be wrong because they are driven by cognitive and motivational biases rather than logical reasoning or scientific evidence. While the strange behavior of your friend may lead you to think s/he is lying to you, it may just be that s/he is preoccupied with some other issue that is irrelevant to you. However, weighing alternatives and thinking of all the different possibilities can be paralyzing for some people, and sometimes decisions based on intuition are actually superior to those based on analysis (people interested in this idea should read Malcolm Gladwell's book *Blink*)¹. Gladwell argues that intuition is also shaped by experience, expertise, and patterns, not just instinct, and therefore can be a powerful tool for decision making. Conversely, while snap judgments can be highly effective, they are also vulnerable to bias, stereotypes, and hidden assumptions. Overall, it is important to think about how decisions are made, respecting intuition and also recognizing its limits, especially in contexts involving power, inequality, and bias.

1. Gladwell, M. E. (2005). *Blink: The power of thinking without thinking*. (9th ed.). New York: Little, Brown & Co.

Authority

Perhaps one of the most common methods of acquiring knowledge is through authority. This method involves accepting new ideas because some authority figure states that they are true. These authorities include parents, the media, doctors, priests and other religious authorities, the government, and professors. While in an ideal world we should be able to trust authority figures, history has taught us otherwise, and many instances of atrocities against humanity are a consequence of people unquestioningly following authority (e.g., Salem Witch Trials, Nazi War Crimes). On a more benign level, while your parents may have told you that you should make your bed in the morning and children know they need to listen to their parents, making your bed provides the warm, damp environment in which mites thrive. Keeping the sheets open provides a less hospitable environment for mites. These examples illustrate that the problem with using authority to obtain knowledge is that they may be wrong, they may just be using their intuition to arrive at their conclusions, and they may have their own reasons to mislead you. Nevertheless, much of the information we acquire is through authority because we don't have time to question and independently research every piece of knowledge we learn through authority. But we can learn to evaluate the credentials of authority figures, to evaluate the methods they used to

arrive at their conclusions, and evaluate whether they have any reasons to mislead us.

Rationalism

Rationalism involves using logic and reasoning to acquire new knowledge. Using this method, premises are stated and logical rules are followed to arrive at sound conclusions. For instance, if I am given the premise that all swans are white and the premise that this is a swan, then I can come to the rational conclusion that this swan is white without actually seeing the swan. The problem with this method is that if the premises are wrong or there is an error in logic, then the conclusion will not be valid. For instance, the premise that all swans are white is incorrect; there are black swans in Australia. Also, unless formally trained in the rules of logic, it is easy to make an error. For example, let's say a student observes that in their survey, most participants who exercise regularly report good mental health. The student concludes that exercise *causes* good mental health. However, without formal training in logic, the student has made an error by confusing correlation with causation. Other factors—such as social support or income—may influence both exercise and mental health. This example illustrates how, without training in logical reasoning, researchers can draw incorrect conclusions even when using real data. Nevertheless, if the premises are correct and logical

rules are followed appropriately, then this is a sound means of acquiring knowledge.

Empiricism

Empiricism involves acquiring knowledge through observation and experience. Once again, many of you may have believed that all swans are white because you have only ever seen white swans. For centuries, people believed the world was flat because it appeared to be flat. These examples and the many visual illusions that trick our senses illustrate the problems with relying on empiricism alone to derive knowledge. We are limited in what we can experience and observe, and our senses can deceive us. Moreover, our prior experiences can alter the way we perceive events. Nevertheless, empiricism is at the heart of the scientific method. Science relies on observations. But not just any observations, science relies on structured observations, which are known as systematic empiricism.

The Scientific Method

The **scientific method** is a process of systematically collecting and evaluating evidence to test ideas and answer questions. While scientists may use intuition, authority, rationalism, and empiricism to generate new ideas, they don't stop there.

Scientists go a step further by using systematic empiricism to make careful observations under various controlled conditions in order to test their ideas, and they use rationalism to arrive at valid conclusions. Systematic empiricism builds on empiricism by emphasizing that observations must be collected using organized, transparent, and replicable procedures. Rather than relying on causal or informal observation, systematic empiricism uses clearly defined methods, careful measurement, and explicit rules for gathering and analyzing data. While the scientific method is the most likely of all of the methods to produce valid knowledge, like all methods of acquiring knowledge, it also has its drawbacks. One major problem is that it is not always feasible to use the scientific method; this method can require considerable time and resources. Another problem with the scientific method is that it cannot be used to answer all questions. As described in the following section, the scientific method can only be used to address empirical questions. This book and your research methods course are designed to provide you with an in-depth examination of how psychologists use the scientific method to advance our understanding of human behavior and the mind.

Ontology, Epistemology, and Axiology in Research

Before a researcher can embark on a scientific inquiry or identify their research methods, they should first carefully

consider the lenses that are shaping their view of the world around them. This includes a consideration of ontology, epistemology, and axiology in research.



One or more interactive elements has been excluded from this version of the text. You can view them online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=26#oembed-1>

Video 1.1. [Ontology epistemology methodology and methods | ontology and epistemology in research examples](#) by [Dr. Dee](#)

English transcript of Ontology epistemology, methodology, and methods video

[Music]

Ontology epistemology methodology and methods

Do you find the terms ontology and epistemology alien and confusing? You are not alone, but today I am going to simplify these terms for you and give you a great way to remember them. Before we get stuck in, please hit the subscribe button and notification bell so you know when we upload other videos like this.

When we design a research study, we always think of the methods. The methods we choose say a lot about our

assumptions, even if these assumptions are not obvious to us. Let's think of our methods as the tip of an iceberg. Lying just below the iceberg's tip is the methodology, which people often confuse with methods. The 'ology' means that there has been a debate or study, so methodology is the study of methods. Methodology is the set of discussions, debates, and choices you have made to get to your method. It considers your research question, your skill set, resources, time, etc. Lying even lower beneath the surface are your ontological and epistemological beliefs. Let's talk about two types of ontology: realism and relativism. There are other ontologies, but that's the subject for another video.

In a nutshell, realists believe that truth exists, so this belief leads them to certain ways of generating knowledge. What we believe about how to generate knowledge is our epistemological belief. Because realists believe there is a single truth out there, they seem to find this truth using what we have come to know as the scientific method. They try to discover new knowledge using objective approaches where you look from outside, make measurements, and draw conclusions. Relativists, however, believe that there are multiple realities and that we can only learn about these realities by understanding the meaning that are research participants give to the phenomenon we are studying. Let's take an example. Say you want to know if Sierra Leone is a wealthy country. If you believe that wealth is measured in economic terms, then your task would be pretty simple. You only have to identify

economic indicators of wealth such as Gross Domestic Product, and you would have your answer. However, if you believe that wealth could have multiple meanings, then you may embark on a different kind of study, one in which you visit Sierra Leone, spend time with the people and discuss what wealth means to them. So, there you have it – ontology and epistemology explained, and the link to your methodology clarified. [ifD – Qualitative Research Specialists screen]

Ontology

Ontology refers to the assumptions one makes about what is real. Generally, when conducting research in psychology, researchers will either take on an ontology based in realism or relativism.

Realism refers to the idea that there is an objective reality separate from human perception. Think back to when you learned about the gravitational force or the mitochondria of a cell. You were learning about the theories and observations that make up our understanding of the physical world. We assume that gravity is real and that the mitochondria of a cell are real. With a powerful microscope, mitochondria are easy to spot and observe, and we can theorize about their function in a cell. The gravitational force is invisible, but clearly apparent from observable facts, like watching an apple fall. The theories about gravity have changed over the years, and those improvements in theory were made when existing

theories fell short in explaining observations. If we weren't able to perceive mitochondria or gravity, they would still be there, doing their thing, because they exist independent of our observation of them.

By contrast, relativism refers to the belief that social concepts do not have an existence apart from human thought. Said differently, relativism asserts that knowledge and truth are socially and culturally situated, meaning that what is considered true, real, or valid, depends on historical, cultural, and social context rather than universal standards. Thus, the emphasis is on examining how people understand the world around them. Unlike realists who seek a single, universal truth, the relativist sees a collection of truths that are created and shared within a social and cultural environment. For example, in the Diagnostic and Statistical Manual (DSM) classification of mental health disorders, there is a list of [culture-bound syndromes](#) that only appear in certain cultures. For example, *susto* describes a unique cluster of symptoms experienced by people in Latin American cultures after a traumatic event that focus on the body. Indeed, many of these syndromes do not fit within a Western conceptualization of mental health because Western culture differentiates less between the mind and body. To a Western scientist, *susto* may seem less real than PTSD. To someone from Latin America, their symptoms may not fit neatly into the PTSD framework developed within Western society. A realist would seek to identify which is objective "real": *susto* or PTSD. But a relativist would acknowledge that

the truth of these disorders is rooted in the cultural context in which it exists.

Epistemology

Epistemology refers to one's beliefs about how we generate knowledge, or how we come to know what is real or true, including both objective and subjective truths. Objective truths are claims about reality that are considered independent of individual beliefs, feelings, or perspectives. They are expected to be verifiable through shared standards, evidence, or observation, and to hold regardless of who observes them. Subjective truths are truths that are based on personal experiences, interpretations, meanings, or feelings. They depend on individuals' perspectives and social contexts and may vary across people or groups.

Generally, scientists approaching research with an objective ontology and epistemology will use quantitative methods designed to draw unbiased data-driven conclusions to arrive at scientific truth. However, given a relativistic ontology, a subjectivist epistemology is also important to consider. Subjective truths do not exist independent of human observation because they are the product of the human mind. We negotiate what is true in the social world through language, arriving at a consensus and engaging in debate. Epistemologically, a scientist seeking subjective truth assumes that truth lies in what people say—in their words. A scientist

uses qualitative methods to analyze words or other media to understand their meaning and generate knowledge of what is subjectively true.

Axiology

Axiology refers to how a scientist believes their own values impact the ways in which they conduct research. A positivist approach to conducting science argues that research should be value-free and guided solely by objective truth (see Research Paradigms chapter for more information on this). However, this approach is often critiqued for leading to blind spots in one's research, and many have suggested that research can never be truly value-free. Thus, axiology invites researchers to reflect on how their pre-held beliefs may influence every part of the scientific process, from the research questions we may generate, to how we conduct a study, to the conclusions we are able to draw. As Dr. Lynette Pretorius writes, "This reflection is essential for maintaining integrity and transparency in research, as it highlights the ways in which personal and cultural values influence the research process, from the formulation of questions to the dissemination of results."²

2. Pretorius, L. (2024). Demystifying research paradigms: Navigating ontology, epistemology, and axiology in research. *The Qualitative Report*, 29(10), 2698-2715. <https://doi.org/10.46743/2160-3715/2024.7632>

Accounting for Ontology, Epistemology, and Axiology in Research Methodologies

In their seminal text, *Indigenous Statistics*, Indigenous researchers Walter and Anderson (2016)³ use the concepts epistemology, ontology, and axiology to demonstrate how quantitative methodologies and statistics often reflect the cultural framework or dominant ideology of the context (e.g., country) in which they are produced without realizing it. That is to say, researchers and statisticians collect, analyze, and interpret data using specific ontological, epistemological, and axiological assumptions that they either do not explicitly name or are not aware of. Walter and Anderson assert that this results in methodologies that create “documentation of difference, deficit, and dysfunction” (p. 10) in the research generated about marginalized peoples or communities because of the inherent biases present in the underlying assumptions of all research endeavors.

3. Walter, M., & Andersen, C. (2016). *Indigenous statistics: A quantitative research methodology*. Routledge.

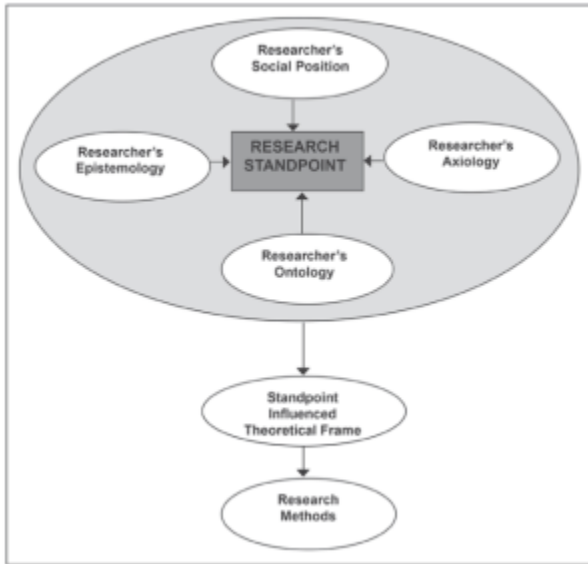


Figure 1.1. Walter & Anderson's (2016) conceptualization of a research methodology from the text [Indigenous Statistics \[PDF\]](#)

In the figure above, Walter and Anderson (2016) summarize how they define methodology, accounting for how a researcher's social position, such as their culture, race, gender, or socioeconomic background, and in turn their ontology, epistemology, and axiology, informs or creates their research standpoint. They go on to define methodology as made up of three parts that are all linked: standpoint, theoretical frame, and methods. Our standpoint consists of one's social position and underlying beliefs about ourselves and what constitutes knowledge, i.e., our ontological, epistemological, and axiological assumptions. By doing this, the underlying

assumptions that inform our research methodology choices are made explicit, and therefore our potential biases are also made visible.

Equity Activity: Social Position and Epistemology, Ontology, and Axiology



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=26#h5p-1>

License & Attribution

“Methods of Knowing” by Rebecca Anguiano and Makenzie O’Neil is adapted from “[Methods of Knowing](#)” by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler, and Dana C. Leighton, which is licensed CC BY-NC-SA 4.0. and “[Paradigms, theories, and how they shape a researcher’s approach](#)” by Matthew DeCarlo and is licensed CC BY-NC-SA 4.0

“Methods of Knowing” is licensed under CC BY-NC-SA 4.0.

“[Ontology epistemology methodology and methods.](#)” Standard YouTube License.

2.

REFLEXIVITY AND POSITIONALITY

Stephanie D'Costa

Learning Objectives

By the end of this chapter, you will be able to

1. Identify inclusive language to use in their positionality statements.
2. Articulate positionalities when writing & presenting their research findings.
3. List actionable steps in their research presentations to address social issues.

Thinking through your motivations for research is an act of

reflection. Reflection on one's motivations and positionality is an essential part of every stage of research.

Reflexivity

The ability to be **reflexive** is vital to the process of picking a research question, conducting research, and analyzing data. To be reflexive is to be able to examine and react to your own emotions, motives, and situation¹. In psychology research, this requires the ability to critically recognize your influencers and your influence on others. Holland (1999)² expounds that reflexivity is the ability to take account of one's self and the effects of personality or presence of the researcher on the investigation (p. 464). Reflexivity means being sensitive to "how relations of power operate in the research process" (Kirby et al., 2017 p. 50) and affect your relationship with, and perspective of, the subject.

As the subject of psychological research is complex, dynamic, and sometimes conducted upon populations for which you are removed or have privilege over, taking stock of your own position (with its institutional supports, privileges

1. Cambridge Dictionary (2021). Reflexivity. Cambridge Dictionary. <https://dictionary.cambridge.org/dictionary/english/reflexivity>

2. Holland, R. (1999). Reflexivity. Human relations, 52(4), 463-484.

and limitations) is essential for both ethical (the application of moral principles and professional code of conduct to research) and epistemic (the philosophy concerning the nature of knowledge) reasons. Recognition of ethics ensures that there is not exploitation taking place in your research, and epistemology ensures that your own biases are accounted for.

Positionality

A related concept to reflexivity is positionality. **Positionality** describes one's worldview and the position one adopts about research and its social and political content (Holmes, 2020, p.1). This involves taking stock of

‘where the researcher is coming from’, [and] concerns ontological assumptions (an individual's beliefs about the nature of social reality and what is knowable about the world), epistemological assumptions (an individual's beliefs about the nature of knowledge) and assumptions about human nature and agency (individual's assumptions about the way we interact with our environment and relate to it) (Holmes, 2020, p.1-2)

Because psychological research is by nature, rarely value-free, researchers must account for their motivations. Beliefs, values, and interests are shaped by our personal identities (e.g. experiences, gender, race, ethnicity, sexuality, (dis)ability

statuses, political allegiances, social class, geographic location, history, etc.) and the systems that privilege or marginalize these identities. These positionalities influence our research interests and topics, the perspectives we adopt in carrying out research, our motivations, how we conduct the research, and the outcomes. Positionality also determines the subject we investigate, the participants we choose and the methods we use to conduct research ³. Hence, if you are uncertain about a research topic or you know the topic but are uncertain about how to narrow it down, it might be worthwhile to think about your positionality. Think about your identity, what you believe about social processes (such as inequality), what you have learned, what your experiences are, and see if that could help you to narrow down your research topic.

Positionality Statements

Intentionally reflecting on your positionality is an important part of the research process. Hence, researchers frequently invest time in developing **positionality statements** and including them in their papers. Reflecting on your positionality is not only important in helping you to decide on

3. Holmes, A. G. D. (2020). Researcher Positionality—A Consideration of Its Influence and Place in Qualitative Research—A New Researcher Guide. *Shanlax International Journal of Education*, 8(4), 1-10.

a topic, it can also help shape your methodology and interpret your findings. Positionality statements are also important because our identities and lived realities create biases in how we interpret and view the world. An awareness of our own biases, as researchers, enhances our understanding of how our lived experiences, shaped by systems of power and privilege, inform the way that we conduct research. Below are two examples of positionality statements.

Examples of Positionality Statements

- I position myself as a *bricoleur*, layering feminist standpoint theory and postcolonial theory, and propose the collaborative data collection and analysis techniques, with particular attention to ethical and cultural sensitivity, using a social constructivist approach to grounded theory...In light of postcolonial critiques of Western researchers and international development, I have often wondered: Am I doing more harm than good? The privilege that accompanies my social

location as a White, upper class, Canadian, academic woman means that, despite good intentions, my efforts to support education in postcolonial contexts risk being patronizing, insulting, threatening, imperialist, and recolonizing⁴ (p. 1-2)

- Canada is not my birthplace and English is not my first language. I was born in Nigeria in the 90s and came to Canada as a very young child who spoke no English at all, but rather who conversed fluently in my native Igbo. As far as citizenship, I hold a Nigerian and Canadian passport. If identity is to be so simply ascribed, one would say that I am a Black, Igbo, Nigerian-Canadian woman...The simplicity of identities is also what hides the complexity of bellowing and the illusion of agency in determining the totality of who it is that we are ⁵ (p.43)

4. Vanner, C. (2015). Positionality at the center: Constructing an epistemological and methodological approach for a western feminist doctoral candidate conducting research in the postcolonial. *International Journal of Qualitative Methods*, 14(4), 1-12.

In both examples, the researchers are forthright about what influenced their research and their interpretations of social reality that are influenced by their positionality. There are several advantages to this openness. Being candid about our positionality increases the credibility of our research and provides contexts for users of our research. Reflexivity and positionality also improve the authority and validity of our knowledge⁶. We encourage you to develop your own positionality statement.

There are also some critiques to the inclusion of positionality statements that should be considered. As people read your positionality they may develop assumptions about you. For example, being candid about your connection to a topic may indicate your limited experience (e.g. being an outsider to the research area and having limited previous experience working with the population). Additionally, some researchers may not believe that positionality statements matter or belong in the research process given the context of their research paradigm (See the section on Positivism in [Chapter 4](#)).

5. Odozor, E. T. (2020). Making peace with movement: dislocation and the Black Diaspora. In G. S. Dei, E. Odozor and A. V. Jimenez (eds.), *Cartographies of Blackness & Black Indigenities* (pp.41-50). Myers Education Press.

6. Smith, L. T. (1999). *Decolonising Methodologies: Research and Indigenous Peoples*. University of Otago Press.

Writing Positionality Statements

A good positionality statement describes one's epistemological position (i.e. how one views the world in terms of their philosophy, personal beliefs, theoretical influence and perspectives which guide the research) as well other potential influences on research such as personal characteristics and identities in terms of gender, age, social class, ethnicity and political beliefs (see Holmes, 2020, p. 4). It should also address any predetermined position that the researcher takes (e.g., participant, as an insider or outsider to the population being studied, theoretical influences, etc.), the research context, and a reflexive opinion about how these might affect the research process. Hence, taking stock of positionality requires understanding how "one's position in the social hierarchy vis-a-vis other groups potentially 'limits or broadens' one's understanding of others" ⁷ (p. 48). This means interrogating what biases you may have of the groups being studied and how your own social location may influence that bias. Consider the following questions: does your disdain for slow customer service perhaps come from your never having to work in the service industry? Or the opposite? Does your idealization of agricultural work perhaps come from your only having done

7. Kirby, S. L., Greaves, L., & Reid, C. (2017). *Experience research social change: Methods beyond the mainstream*. University of Toronto Press.

non-physical city labour? By answering questions about why and how you have come to study your topic, you will be clearer about your presuppositions and forthright with your reader about your relationship to the subject matter.

How to Write a Positionality Statement

Writing a positionality statement helps you to intentionally reflect on your identity, life history, experiences, values, and the things/issues that are important to you. This reflection can help you determine what aspect of your identity is of broader sociological interest, which can be useful in narrowing your research interests. Even if you already know what topic you want to research, a positionality statement can help you to focus your research on issues that are important to who you are or to your political/world views. Here are some things to include in your positionality statement:

- Identity characteristics (e.g., age, gender, sexuality, ethnicity, social class, disability)

status, citizenship, immigration status, religion, marital status etc.), and how these identities are shaped by societal structures and systems to privilege and marginalize various communities.

- Life experiences (previous or current job, volunteering activities, membership in advocacy groups etc.)
- Political, philosophical and theoretical beliefs (lens through which you view and interpret the world)
- Relationship to phenomena of interest (insider and/or outsider status)
- Clearly reflecting on how these characteristics, experiences, and history relate to the phenomena being researched.

Remember, that it is more than just articulating your identities but more importantly about the research at hand, how do your identities relate to “x” topic. For instance, take the first example statement in the [Positionality Statements](#)

- The author clearly states their identity characteristics as a “White, upper-class, Canadian, academic woman”

- She demonstrates professional life experiences in this area of study.
- The author orients her position in theory as “layering feminist standpoint theory and postcolonial theory,...with particular attention to ethical and cultural sensitivity, using a social constructivist approach to grounded theory.”
- Throughout the statement, she recognizes herself as holding an outsider position, which is demonstrated in statements such as “The privilege that accompanies my social location...means that, despite good intentions, my efforts to support education in postcolonial contexts risk...”.
- Finally, there is clear reflexivity taking place throughout the statement, with the author stating how her own identities and position of privilege may be perceived by the very communities she seeks to serve: “...risk being patronizing, insulting, threatening, imperialist, and recolonizing.”

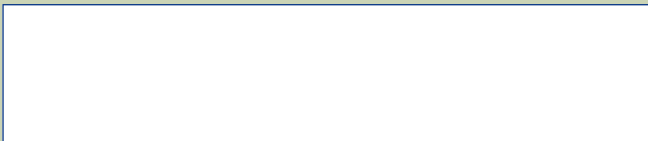
Additional tips:

- There is no limit on the length of your

positionality statement, and often the length will be based on the intended use of the positionality statement. For instance, a book's positionality statement that is integrating multiple concepts may be a fair bit longer than an empirical research article. However, a good guideline to follow is to try to keep it within a paragraph.

- Get a friend or close acquaintance—ideally with knowledge relevant to the topic being researched—to read your draft positionality statement and inform you of any personal details that you might have overlooked that you may have that need more reflexive engagement.

Equity Activity: Developing a Positionality Statement





An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=28#h5p-2>

License & Attribution

“Reflexivity & Positionality” by Stephanie D’Costa is adapted from “[A Note on Reflexivity and Positionality](#)” by Oral Robinson and Alexander Wilson is licensed CC BY-NC 4.0.

“Reflexivity & Positionality” is licensed under CC BY-NC-SA 4.0.

3.

RELATIONAL ACCOUNTABILITY

Stephanie D'Costa

Learning Objectives

By the end of this chapter, you will be able to

1. Describe key aspects of participatory action research.
2. Articulate the benefits of involving the community in your research project.
3. Identify ways to engage in relational accountability in your research.

In addition to social justice and applying research solutions to

participants' lives, research should actively benefit participants and meaningfully include the communities they belong to. Historically, social research has been conducted in isolation from its subject matter and has been patronizing toward vulnerable populations¹. To counteract this trend, we encourage you to design your research to benefit participants. For example, through volunteering with youth, you might decide to design and evaluate a program to help with their education. Participation Action Research (PAR) embodies helping participants to create solutions to their problems. The following video and examples provide more context about the ways PAR works and can support research.



One or more interactive elements has been excluded from this version of the text. You can view them online here:

<https://smc.pressbooks.pub/critresearchmethodspsycho/p=30#oembed-1>

1. Tuck, E. (2009). "Suspending Damage: A Letter to Communities." https://pages.ucsd.edu/~rfrank/class_web/ES-114A/Week%204/TuckHEdR79-3.pdf

Video 3.1 [Participatory Action Research](#) by [Vera Institute of Justice](#)

Combining Research and Respect

When thinking about your research topic, you must also think about how the processes and outcomes show respect and concern for your participants' welfare. Practicing reciprocity, monitoring power dynamics, obtaining ongoing consent, permitting participants to tell their stories, and empowering them as allies in the research process are among the important guiding principles that convey respect. According to Hutchinson and colleagues (1994), when we embed these research principles into our engagement with participants, we create opportunities for healing. For example, qualitative interviews can be cathartic, validating participants' feelings and experiences (self-acknowledgment), offering a sense of purpose, fostering self-awareness (gaining new perspectives through reflection); promoting healing (by providing an outlet for emotional expression) and amplifying voices of the disenfranchised². [2]. This means that when considering issues

2. Hutchinson, S. A., Wilson, M. E., & Wilson, H. S. (1994). Benefits of participating in research interviews. *Image: The Journal of Nursing Scholarship*, 26(2), 161-166.

you are already connected to or wish to connect with, you should also consider the interest, agency and proximity of the groups who may be willing to be a part of your study. This will be partly conditioned by your positionality.

Relational Accountability

Relational accountability refers to the ethical responsibility researchers have to the Indigenous people and communities their work involves and represents. Importantly, it emphasizes the idea that the relationship established by research must be approached with honesty, respect, and reciprocity ³. “As a researcher,” Wilson (2001) asserts, “you are answering to all your relations,” implying that your research ought to: (1) know its relations and (2) gain knowledge “so as to fulfill [your] end of the relationship” (p. 177). With Wilson’s concept in mind, we suggest engaging in relational accounting as you generate your research question – considering your relationship to the voices and variables that will form the datum of your argument – and making yourself accountable to the group that is granting you access to their knowledge and experiences. Not only will you better understand your

3. Wilson, S. (2001). What is an Indigenous research methodology?. Canadian journal of native education, 25(2).

research participants, but so will your purpose for gathering information about them.

Equity Activity: Relational Accounting in Research



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=30#h5p-3>

License & Attribution

“Relational Accountability” by Stephanie D’Costa is adapted from “[Relational Accountability](#)” by

Oral Robinson and Alexander Wilson is licensed
CC BY-NC 4.0.

“Reflexivity & Positionality” is licensed under CC
BY-NC-SA 4.0.

4.

RESEARCH PARADIGMS & THEORIES

Stephanie D'Costa; Makenzie O'Neil; and
Sujay Sabnis

Learning Objectives

By the end of this chapter, you will be able to

1. Define a paradigm and describe its significance.
2. Identify and describe four predominant paradigms found in psychological research.
3. Identify underlying beliefs that support those paradigms.
4. Articulate the difference between quantitative

& qualitative data.

5. Articulate the difference between quantitative & qualitative data's roles in addressing issues of equity.

The terms paradigm and theory are often used interchangeably in social science, although social scientists do not always agree whether these are identical or distinct concepts. In this text, we will make a clear distinction between the two ideas. Regarding them as analytically distinct will provide a useful framework for understanding the connections between research methods and social scientific ways of thinking.

Paradigms in social science

Research paradigm refers to a *shared* set of often-unspoken assumptions in a scientific community about what can be known, and what is the best way to know it (Kuhn, 1962).¹ This shared set of assumptions about the nature of knowledge will guide the research practices of that particular community,

1. See Kuhn's seminal work for more on paradigms: Kuhn, T. (1962). The structure of scientific revolutions. Chicago, IL: University of Chicago Press

and distinguish it from other research communities that follow a different paradigm. We will discuss four such research communities that exist today in Western research landscape – post positivism, constructivism, criticalism, and postmodernism.

Note that the specific order in which we list these research communities is important because each paradigm can be said to have emerged in opposition to the one before it. For example, post positivists form the oldest and the most traditional community of researchers. This paradigm of research is what most people are thinking of when they think of a ‘scientific study’. Some social scientists who were trained in the postpositivist paradigm were dissatisfied with the constraints and limitations of this paradigm, and this discontent led them to rebel, and break away to form a separate research community. From this community, emerged the paradigm that we today know as constructivists. The emergence of critical and postmodern paradigm follows a similar trajectory.

Although this is an oversimplified version of the history of research paradigms, framing it this way can help novice researchers grasp the significance of each paradigm, and better understand why each paradigm attracts passionate defenders and opponents. We believe this approach will help readers avoid the common trap of seeing each research paradigm like an item one must pick from a shopping catalogue when embarking on a new research study. Instead, we wish for the

reader to understand each paradigm as a living and evolving *community* of researchers formed around a particular belief system about the nature of reality and knowledge.

In this chapter, we introduce the key beliefs of each paradigm community. However, remember that no community is a monolith. As with any community, these communities are characterized by internal debates, tensions, and generational rifts. Most of time, these internal tensions among researchers within a



Figure 3.1. The lens through which you view the world is the “paradigm”, which provides the framework through which you make sense human experiences.

paradigm remain under the surface without significantly disrupting the overall community. In rare instances, the tensions can bubble over due to growing discontent among a faction of researchers who feel their traditional mode of research is no longer able to adapt to the newer developments happening around them. After trying in vain to reform the existing paradigm, some of these researchers decide to make a clean break from the existing paradigm, and end up in a breakaway group which goes on to become its own paradigm that is much more responsive to the changing world. After enjoying a period of success, this paradigmatic community may undergo a similar cycle. According to Thomas Kuhn, the

philosopher of science who popularized the term paradigm, this is the story of how science has evolved over thousands of years.

Now let's look at four of the most common social scientific paradigms that might guide you as you begin to think about conducting research. Remember that each paradigm is a living breathing community of researchers, rather than a checklist of words. Thus, when we describe the key beliefs of each paradigm community, we mean that in the broadest sense. We encourage readers to do further reading to familiarize themselves with the internal differences within each community.

Post Positivism

Post-positivism is the first paradigm we will consider, as it represents the most conventional understanding of science as it is practiced today. As the prefix 'post-' indicates, this paradigm is a development over positivism, the original paradigm of science since 1800s, before it received some upgrades in the 1960s to become Positivism 2.0, also known as post positivism. Although post positivists updated some of the philosophical claims of positivism to keep up with the times, the two do not differ significantly in terms of how they 'do' science. For this reason, we do not treat positivism and post positivism as two separate paradigms. Some textbooks and articles geared toward newcomers may even use the two

interchangeably, although post positivism is the more accurate term for describing the reigning paradigm of science today.

Post positivists are guided by the principles of objectivity, “knowability,” and deductive logic. Post positivists argued that reality exists independently of the human mind, and that through a process of elimination, we can get closer to knowing this reality (although we will never reach it). Social scientists who operate from this paradigm retain positivism’s emphasis on sensory data (i.e., data which can be directly seen or heard) over other forms of data such as people’s subjective perceptions and interpretation of their surroundings. Post positivism also calls for a value-free science, one in which researchers minimize their emotional investments or agendas in a quest for the objective truth.

While the intention of post positivism to seek objective truth free from researcher bias seems in keeping with the fundamental goals of science, in practice this paradigm can lead to two possible missteps in how one conducts research. First, a researcher may believe they are conducting a study objectively, and thus their conclusions are also devoid of personal biases. However, they may be limited by their own positionality to see an alternative vantage point on the information. Take, for instance, scientific studies of intelligence where white/European scientists used “intelligence tests” to measure the intelligence of different racial groups. When white participants scored significantly more than non-white participants, they interpreted this to

mean that white people are more intelligent than non-white people. They failed to consider that the score differences might have less to do with any actual differences in intelligence, and more to do with the specific way in which they had conceptualized constructs such as “intelligence” and “smartness”. They were defining these constructs in ways that felt most familiar to them as white upper-class men living in WEIRD societies. WEIRD refers to an acronym for societies that are Western, Educated, Industrial, Rich, Democracies, and are the minority of the population of the world (recent data suggested that WEIRD people represent about 12% of the world’s population but nearly 90% of psychological samples!). Thus, even when a researcher takes a post positivist approach, guided primarily by the pursuit of empirical, objective truth, their own perspective and positionality still impact the conclusions they draw.

A second misstep of post positivist perspective is that its emphasis on information which can be observed through our senses severely restricts what can be studied. For example, social scientists cannot directly observe people’s thoughts, passions, emotions, desires, and intentions because these are not material objects such as atoms or microorganisms. Due to the demands of their paradigm, they have to find some way to make these objects material, by converting them into a paper-pencil surveys or even brain imaging technologies. But these survey responses and MRI scans are extremely poor substitutes for that person’s internal world. And any theory built out of

this data is itself going to be extremely poor version of what's going in people's minds. Thus, while the postpositivist paradigm has contributed to many groundbreaking discoveries, its reliance on empiricism can only get us so far before we are faced with the choice of stopping and turning back, or transitioning to a newer paradigm that allows us to go further into the study of the social world.

Constructionism

Building from the positivist paradigm, another predominant paradigm in social sciences is constructivism (although some may prefer the term interpretivism or naturalism (Tashakkori & Teddlie, 1998)². Peter Berger and Thomas Luckman (1966)³ are credited by many for having developed this perspective in sociology. While postpositivists seek to understand the objective reality of a situation, constructivists argue people create a reality as opposed to a reality existing out there waiting to be discovered. Because people create truth rather than discover it, truth is different based on who you ask, and people change their definitions of truth all the time based on their

2. Tashakkori, A. & Teddlie, C. (1998). *Mixed methodology: Combining qualitative and quantitative approaches*. Sage.

3. Berger, P. L., & Luckman, T. (1966). *The social construction of reality: A treatise in the sociology of knowledge*. New York, NY: Doubleday.

interactions with other people. According to this paradigm, we create reality through our interactions and the interpretations that we have of our interactions.

Researchers operating within this framework take keen interest in how people come to socially agree, or disagree, about what is real and true. To better understand the idea of relativity and constructivism, consider how certain hand gestures can connote different meanings across the globe. When someone raises their middle finger in North America, we can safely assume that neither the person raising the finger nor the person to whom the finger is being directed are very happy. Other gestures, such as the thumbs up, may carry similar eyebrow-raising meanings in other societies and cultures. While the thumbs up gesture has a particular meaning in North American culture, that is not shared across cultures (Wong, 2007).⁴ So where does the meaning of hand gestures reside? Does it reside inside the raised middle finger? If post positivists were to dissect a hand with a raised middle finger or examine it under an X-Ray machine, would they find the source of the obscenity in there? Constructivists argue that the meaning of the gesture does not lie in the physical hand

4. For more about how the meanings of hand gestures vary by region, you might read the following blog entry: Wong, W. (2007). The top 10 hand gestures you'd better get right. Retrieved from <http://www.languagetrainers.co.uk/blog/2007/09/24/top-10-hand-gestures>

itself but in the gesture's intended meaning and the gesture's interpretation. In other words, the meaning of the raised finger lies in the social context in which it occurred. Outside of this context, the middle finger is just another finger. But what if the intended meaning of the initiator and the interpretation of the receiver of the gesture are at odds with one another, then who is "right"? A constructivist would claim that there is no one universal truth, and that both the initiator's and the receiver's version is "true" even though the two truths contradict each other.

It would be a mistake to think of the constructivist perspective as only individualistic. While individuals may construct their own realities, groups as small as married couples and as large as nations also agree on notions of truth. In other words, the meanings that we construct have power beyond the individual people who create them. Therefore, constructivists are equally interested in the initial creation of meanings and the ways that people and groups work to redefine meanings.

However, a person's (or couple's or nation's) constructed reality does not exist in a vacuum separate from the world around them, and constructivism has historically not accounted for how things like power dynamics, privilege, and historical context impact the construction of 'truth'. For instance, if a white male boss at a company comments about a Black female employee's hair being "unprofessional," the boss' reality may be that they are being helpful to their employee

by suggesting how she should present herself in the workplace. But the employee's reality is that she experienced a microaggression, and now feels harmed or unsafe in the workplace. Whose reality is "true"? Was the boss merely being helpful or was he being racist? Constructivism may explore how different parties involved in this conflict interpret the situation, and how cultural and historical factors (e.g., the stigmatization of Black women's hair) influence their respective perspectives. But constructivists do not take sides, or inquire whose version of truth is more likely to be believed in a society. In this case, the white boss' version of truth is more likely to be believed than the Black female employee's version because of a combination of white privilege and male privilege.

Thus one of the discontents with constructivism was its lack of attention to the *role of power in shaping who gets to construct the truth*. Constructivism has significantly advanced our appreciation and understanding of how "truth" varies depending on a person's vantage point; however, it did not highlight the structural and systemic foundations that privilege one vantage point over the others.

Critical

Thus, a third paradigm is the critical paradigm. At its core, the community of researchers associated with the critical paradigm is focused on understanding the role of power in shaping social reality. Although some rather diverse perspectives are included

here, the critical paradigm generally includes ideas developed by early social theorists, such as Max Horkheimer (Calhoun, Gerteis, Moody, Pfaff, & Virk, 2007),⁵ and later works developed by feminist scholars, such as Nancy Fraser (1989).⁶ and critical race theorists like Derrick Bell (Bell, 2005).⁷ All of these theorists were interested in understanding how social institutions like media, academia, church, market, and judiciary are engaged in creating a reality that justifies and normalizes the oppression of different groups. For example, a critical race scholar may study how a city government maintains segregation in the city by building fewer bus routes between the predominantly white and predominantly Black parts of the city. Given that the city officials are bound to deny the accusations of racism in shaping this policy, the researcher would be especially interested in doing a deep dive into the city's bus system, understanding how it works, who runs it, how decisions are made, what rationales or factors influence these decisions, and how the system impacts the Black residents of the city. The researcher may then publish these

5. Calhoun, C., Gerteis, J., Moody, J., Pfaff, S., & Virk, I. (Eds.). (2007). *Classical sociological theory* (2nd ed.). Malden, MA: Blackwell.

6. Fraser, N. (1989). *Unruly practices: Power, discourse, and gender in contemporary social theory*. Minneapolis, MN: University of Minnesota Press.

7. Bell, D. (2005). *Silent Covenants: Brown v. Board of Education and the Unfulfilled Hopes for Racial Reform*. OUP.

findings in local media, and/or use it to build a legal case against the city.

Thus, unlike the positivist paradigm, the critical paradigm posits that social science can never be truly objective or value-free. Further, this paradigm operates from the perspective that scientific investigation should be conducted with the express goal of social change in mind. Researchers in the critical paradigm might begin with the knowledge that systems are biased against others, such as women or marginalized ethnic groups, and use their research projects to foster positive change in the research participants and the systems being studied. In this way, they differ from both post positivists and constructionists who believe a researcher must remain impartial and aloof in order to be able to do good research. In sum, the critical paradigm not only studies power imbalances but also seeks to change them. More on the critical paradigm will be discussed in the following chapters.

Postmodernism

Finally, postmodernism is a paradigm that challenges almost every way of knowing that many social scientists take for granted (Best & Kellner, 1991).⁸ While positivists claim that

8. Best, S., & Kellner, D. (1991). *Postmodern theory: Critical interrogations*. New York, NY: Guilford.

there is an objective, knowable truth that can be grasped by a researcher, postmodernists would say that there is not. While social constructionists may argue that truth is constructed by individuals, postmodernists would claim that the individual is herself a product of the society and therefore the ‘truth’ she constructs is not truly coming from her, but from the society as a whole. Finally, while the critical paradigm may focus on seeking justice and equity for the oppressed, postmodern researchers may encourage them to think more mindfully about how their specific notions of justice or equity are also limited by the dominant ideologies (such as capitalism, incrementalism, etc.) in the society where they grew up, and encourage them to imagine what justice could look like outside of that frame. As you might imagine, the postmodernist paradigm poses quite a challenge for researchers. How do you study something that may or may not be real, and that only exists when a society speaks or talks about it? This fascinating question is worth pondering as you begin to think about conducting your own research. Part of the value of the postmodern paradigm is its emphasis on the limitations of human knowledge. Table 3.1 summarizes each of the paradigms discussed here.

Table 3.1 Social scientific paradigms

Paradigm	Emphasis	Assumption
Post Positivism	Objectivity, knowability, and deductive logic	Society can and should be studied empirically and scientifically.
Constructionism	Truth as varying, socially constructed, and ever-changing	Reality is created collectively. Social context and interaction frame our realities.
Critical	Power, inequality, and social change	Social science can never be truly value-free and should be conducted with the express goal of social change in mind.
Postmodernism	Inherent problems with previous paradigms	Truth is always bound within historical and cultural context. There are no universally true explanations.

Let's work through an example. If we are examining a problem like cannabis addiction, what would a social scientific investigation look like in each paradigm? A postpositivist study may focus on proving that cannabis addiction truly exists, and may precisely measure the prevalence of cannabis abuse and finding out the key causes or contributors to this problem. Forgoing the investment in proving whether or not cannabis addiction exists, a constructivist might focus on how people who are *constructed by the society* as “cannabis addicts”

understand their relationships with cannabis. In addition, the researchers may also look at how their family members or community perceives them and their substance use. In doing so, constructivists seek to illuminate the multiple ways to understand cannabis users. A study in the critical paradigm may investigate the racist history of laws criminalizing cannabis use, and document the disproportionate impact of these laws on minoritized people. A postmodern study may deconstruct the concept of ‘addiction’ and ‘treatment’ to show how the language used by positivists, constructionists, and critical researchers, to talk about this issue is drawing from some of the same discourses that contributed to the criminalization of cannabis in the first place. The postmodern research may also highlight how the uncritical use of this language on “addiction” is preventing them from imagining other more helpful ways to engage with this issue. These examples should illustrate how one topic can be investigated across each paradigm.

Social science theories

Much like paradigms, theories provide a way to look at the world and understand human interaction. Paradigms are grounded in over-arching, general assumptions about the nature of reality and truth, and what it means to ‘know’ something. Theories, on the other hand, describe more specific phenomena. A common definition for theory in social science

is “a systematic set of interrelated statements intended to explain some aspect of social life” (Rubin & Babbie, 2017, p. 615).⁹ At the core, theories can provide explanations for many phenomena. They help us answer the “why” and “how” questions we often have about the patterns we observe in social life. While paradigms may point us in a particular direction with respect to our “why” questions, theories more specifically map out the explanation, or the “how,” behind the “why.”



Just as researchers might examine the same topic from different levels of inquiry or paradigms, they could also investigate the same topic from different theoretical perspectives. In this case, their research questions could be the

9. Rubin, A., and Babbie, E. R. (2017). *Research methods for social work* (9th ed.). Belmont: Wadsworth

same, but the way they conceptualize their observations will be largely shaped by theory.

Within each area of specialization in psychology, there are many theories that aim to explain more specific types of interactions. For example, the study of sexual harassment includes different theories that posit varying explanations of why harassment occurs. One theory, first developed by criminologists, is called routine activities theory. It posits that sexual harassment is most likely to occur when a workplace lacks unified groups and when potentially vulnerable targets and motivated offenders are both present (DeCoster, Estes, & Mueller, 1999).¹⁰ Other theories of sexual harassment, called relational theories, suggest that a person's relationships, such as their marriages or friendships, are the key to understanding why and how workplace sexual harassment occurs and how people will respond to it when it does occur (Morgan, 1999).¹¹ Relational theories focus on the power that different social relationships provide (e.g., married people who have supportive partners at home might be more likely than those who lack support at home to report sexual harassment when it occurs). Finally, feminist theories of sexual harassment take

10. DeCoster, S., Estes, S. B., & Mueller, C. W. (1999). Routine activities and sexual harassment in the workplace. *Work and Occupations*, 26, 21–49.

11. Morgan, P. A. (1999). Risking relationships: Understanding the litigation choices of sexually harassed women. *The Law and Society Review*, 33, 201–226.

a different stance. These theories posit that the organizational structure of our gender system, where those who are the most masculine have the most power, best explains why and how workplace sexual harassment occurs (MacKinnon, 1979).¹² As you might imagine, the theory a researcher applies to examine the topic of sexual harassment will shape the questions the researcher asks. It will also shape the explanations the researcher provides for why harassment occurs.

For students beginning their study of a new topic, it may be intimidating to learn that there are so many theories beyond what you've learned in your theory classes. What's worse is that there is no central database of different theories on your topic. However, as you review the literature in your topic area, you will learn more about the theories that scientists have created to explain how your topic works in the real world. In addition to peer-reviewed journal articles, books are another valuable source to learn theories related to your topic. Books often contain works of theoretical and philosophical importance that are beyond the scope of an academic journal.

12. MacKinnon, C. 1979. *Sexual harassment of working women: A case of sex discrimination*. New Haven, CT: Yale University Press.

Paradigm and Theory in Psychology

In addition to paradigms and theories, the levels of analysis and the order in which one proceeds in the research process influence the questions we ask about the social world, how we ask them, and even what we may find. A micro-level study of gangs will look much different than a macro-level study of gangs. In some cases, you could apply multiple levels of analysis to your investigation, but doing so isn't always practical or feasible. Therefore, it is crucial to understand the different levels of analysis and be aware of which level you are employing. One's theoretical perspective will also shape a study. In particular, the chosen theory will not only shape the way a question is asked but also which topic will be investigated in the first place. Further, commitment to one theory over another may limit the kinds of questions you pose and could result in missing other possible explanations.

Social science is not fundamentally biased due to the limitations of paradigms and theories, but at the same time, it can never be completely value-free. Social constructionists, criticalists, and postmodernists might point out that bias is always a part of research to at least some degree. Our job as researchers is to recognize and address our biases as part of the research process, if an imperfect part. We frame and conduct our work by using our own theories, levels of analysis, temporal processes, and paradigms. Understanding those

frames and approaches is crucial not only for successfully embarking upon and completing any research-based investigation, but also for responsibly reading and understanding others' work.

Equity Activity: Understanding the Types of Paradigms

Based on the study description, determine which paradigm the research falls under.



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=34#h5p-57>

License & Attribution

“Research Paradigms” by Stephanie D’Costa and Makenzie O’Neil is adapted from “[Paradigms, theories, and how they shape a researcher’s approach](#)” by Matthew DeCarlo and is licensed CC BY-NC-SA 4.0; and “[WEIRD Societies](#)” by Shelly Volsche & Annemarie Hasnain is licensed CC BY-NC-SA 4.0.

“Research Paradigms” is licensed under CC BY-NC-SA 4.0.

5.

CRITICAL THEORY, CRITICAL THEORIES, AND RESEARCH

Rebecca Anguiano

Learning Objectives

By the end of this chapter, you will be able to

1. Define Critical Theory and describe its role as a Marxist, structuralist tradition within philosophy, the social sciences, and research.
2. Distinguish Critical Theory from other critical theories, explaining how perspectives such as

Critical Race Theory, intersectionality, and postcolonial or anticolonial theories emerged in response to, or in dialogue with, the limitations of class-centered framework

3. Apply critical theories as tools for uncovering, interrupting, and dismantling systems of oppression and injustice in psychological and educational research and practice.

What is Critical Theory?

Critical Theory refers to a specific intellectual tradition that emerged in the early twentieth century in the Frankfurt School, a group of European social scientists and philosophers working within a Marxist, structuralist framework. Grounded in critiques of class struggles and capitalism, Critical theory seeks to expose how economic structures shape social relations and produce inequality, with the explicit goal of informing political action and societal transformation ¹. A defining

1. Rush, F. (Ed.). (2004). *The Cambridge Companion to Critical Theory*. Cambridge: Cambridge University Press.

feature of Critical Theory is its rejection of the separation between theory and practice: knowledge production is understood as inherently political, and research is meant to support emancipation from systems of domination.

Critical Theory is interdisciplinary in nature, drawing on philosophy, sociology, economics, and political theory to analyze how power operates through social institutions. However, its primary analytic focus has historically centered economic exploitation and class struggle. While this focus has been generative, it has also been the subject of sustained critique for inadequately accounting for forms of oppression not fully reducible to class, such as racism, colonialism, patriarchy, homophobia, transphobia and other forms of oppression ².

As a result of these critiques, a wide range of **critical theories** (lowercase, plural) have emerged. These theories are not subtypes or strands of Critical Theory, but rather distinct perspectives that develop in dialogue with—and often in response to—the limitations of classical Marxist analyses. For example, Critical Race Theory arose from dissatisfaction with Marxism’s inability to adequately theorize race and racism; queer theory emerged in part as a response to the ways both

2. Celikates, Robin and Jeffrey Flynn, “Critical Theory (Frankfurt School)”, The Stanford Encyclopedia of Philosophy (Winter 2023 Edition), Edward N. Zalta & Uri Nodelman (eds.).

Marxism and liberal rights frameworks failed to account for sexuality and gender variance; and decolonial and anticolonial theories challenge Eurocentric assumptions embedded in Western knowledge production itself. While these approaches differ in focus and method, they share a commitment to interrogating power, exposing structural injustice, and linking scholarship to social change.

Critical Perspectives and Research in Psychology and Education

As discussed in earlier sections, one of the main purposes of this textbook is to support students in approaching research methods in psychology and education with critical thinking. This is important because historically, the field of psychology has often used research to perpetuate harm towards marginalized communities. Traditional research paradigms have often claimed objectivity and neutrality while failing to acknowledge that all research is shaped by particular epistemological assumptions, social positions, and power relations (Zuberi & Bonilla-Silva, 2008)³.

3. Zuberi, T., & Bonilla-Silva, E. (Eds.). (2008). *White logic, white methods: Racism and methodology*. Rowman & Littlefield Publishers.

Research methodologies are influenced by researchers' positionality, beliefs about how knowledge is constructed, decisions about what questions or topics are of value, and assumptions about what counts as legitimate evidence (Walter & Anderson, 2016)⁴. When these influences remain unexamined, research can reproduce deficit-based narratives about marginalized communities and reinforce existing systems of oppression.

Applying critical theories to research methods allows scholars to challenge these assumptions and reimagine the research process itself. Rather than treating research as neutral or detached, critical approaches foreground justice, power, and accountability, asking how research can be used to name harm, expose structural inequities, and support transformative change. In this chapter, we introduce several influential **critical theories**—including Critical Race Theory, intersectionality, and decolonial or anticolonial theory—to explore how they inform research in psychology and education and provide tools for uncovering and challenging injustice. Throughout this chapter, we use the term Critical Theory to refer specifically to the Marxist tradition originating in the Frankfurt School. We use critical theories (lowercase, plural) to refer to a broader set of theoretical frameworks—such as

4. Walter, M., & Andersen, C. (2016). *Indigenous statistics: A quantitative research methodology*. Routledge.

Critical Race Theory, intersectionality, and decolonial or anticolonial theory—that emerged in response to, critique of, or dialogue with Critical Theory’s limitations, particularly its narrow focus on class and political economy.

Critical Race Theory

Critical Race Theory (CRT) articulates both a theory and movement aimed at naming and dismantling the relationships among race, racism, and power (Delgado & Stefancic, 2017)⁵. It spans diverse fields including law, education, medicine, and social sciences, describing how racism is baked into systems that ultimately maintain power for white bodies and perpetuate harm upon racialized ones (Proctor & Rivera, 2022)⁶. CRT also emerged as a response to the limitations of Marxist and class-based frameworks, including Critical Theory, which many scholars argued were insufficient for explaining the persistence and centrality of racism. Like many critical theories, CRT includes a political movement or activist component, wherein theory and research are not separated

5. Delgado, R. & Stefancic, J. (2017). *Critical race theory: An introduction*. New York University Press.

6. Proctor, S. L., & Rivera, D. P. (Eds.). (2022). *Critical theories for school psychology and counseling: a foundation for equity and inclusion in school-based practice*. Routledge, Taylor & Francis Group.

from the ultimate goal of transforming unjust, and in this case, racist systems. It is broader than civil rights or anti-racist movements, which focus on incremental progress, placing race and racism as embedded in the fabric of all institutions and society at large.

CRT emerged in the post-Civil Rights era during a time when color-blind legislation was essentially stalling or rolling back advances that had been made during the civil rights movement. Foundational CRT scholar, Derrick Bell (1992)⁷, argued that racism is a permanent aspect of our society that often goes unacknowledged or remains invisible. This makes racism insidious and, therefore, extremely challenging to address. Critical race theory built on two important movements, radical feminism and critical legal studies. Radical feminism exposes the connection between social roles and gendered power relations, as well as the everyday practices, routines, and habits that create patriarchy (Delgado & Stefancic, 2017). CRT also built upon radical activist movements, including the Black Power and Chicano movements, and critical figures like W.E.B. DuBois and Fredrick Douglass.

7. Bell, D. (1992). *Faces at the bottom of the well*. Basic Books.

Core Tenets of CRT

Although CRT shares with other critical theories a commitment to structural analysis and social transformation, its core tenets are distinct and specifically focused on race and racism. The first of these is that racism is an *ordinary* or a normal and permanent feature of United States (U.S.) society (Bell, 1992). As a result, racism is embedded in the systems and structures of everyday life, making it difficult to see and therefore challenging to undo. CRT scholars also agree that white supremacy benefits the dominant group in ways that mean white elites will likely not want to eradicate racism because of the material and psychological privileges it affords them. Said another way, white people will not be interested in supporting anti-racist efforts unless they also benefit from it, a tenet also known as *interest convergence* (Delgado & Stefancic, 2017).

A third core tenet of CRT is the idea that *race is socially constructed*, as opposed to fixed or biologically determined. As a result, racial categories are created, manipulated, and utilized when convenient for those in power. In this way, it is actually racism that creates and perpetuates race, based on changing perceptions of skin color or other physical features of racialized people (Proctor & Rivera, 2022). CRT scholars also note the ways in which different minoritized groups are racialized differently at different times, usually in response to the needs of the economy or labor market, an idea known as

differential racialization (Delgado & Stefancic, 2017). Related to this idea is the concept of ***intersectionality and antiessentialism***. We will discuss intersectionality more in detail in the next section, but this tenet conveys that every individual has a complex, multilayered identity and cannot be reduced or essentialized into just one category. Finally, CRT's last tenet specifically highlights the importance of lifting up the lived experiences and stories of marginalized people as a means to curing silence and naming types of discrimination so that they can be fought against (Delgado & Stefancic, 2017). This process is known as ***counter-storytelling***, a powerful CRT tool used to highlight the humanity of marginalized peoples by calling attention to the contradictions, misconceptions, and harm that racism imposes on people of color.

CRT's Impact on Research in Education and Psychology

CRT has had a significant impact on the fields of psychology and education. In psychology, for example, CRT has supported researchers in understanding the psychological impact of racism on the well-being of people of color, including areas like identity development, trauma, and overall mental health (e.g., Helms, 1990). In education, CRT scholars have used this lens to provide structural and systemic explanations for disparities in achievement or discipline

practices between white students and students of color. For example, Gloria Ladson-Billings, a prominent educational researcher, famously described what she coined to be the “education debt,” explaining that the achievement gap between Black and white students was actually the result of structural inequities and systemic neglect across generations (Ladson-Billings, 2006). In the next chapter, we discuss QuantCrit, a critical methodological framework grounded in CRT that originated from the education field (Crawford et al., 2018). Applying QuantCrit to our understanding of statistics allows us to critically examine taken-for-granted assumptions in research and utilize quantitative methods for social justice purposes. Below is a video outline *What Critical Race Theory Actually Is — and Isn’t* narrated by historian and author Ibram X. Kendi.



One or more interactive elements has been excluded from this version of the text. You

can view them online here:

<https://smc.pressbooks.pub/critresearchmethodspsyd/?p=36#oembed-1>

Video 5.1 *[What Critical Race Theory Actually Is – and Isn’t](#)*
by *[NowThis Impact](#)*

Intersectionality

Intersectionality is a critical theory grounded in Black feminist thought with a focus on the unique experience of women of color who find themselves at the intersection of race, class, and gender oppression (Hancock, 2016)⁸. It developed in dialogue with critical race theory and decolonial thought by highlighting the simultaneous effects of belonging to multiple social categories that are constructed by connected systems of oppression and power. Put succinctly, an intersectional lens allows us to understand how the various identities one carries, such as race, gender, class, and sexuality, all interact to create a unique, multidimensional experience of privilege and marginalization.

Kimberlé Crenshaw (1989)⁹, a critical legal scholar, coined the term intersectionality. She critiqued how antidiscrimination law frequently left out the experiences of Black women whose identities place them at the intersection of both racial and gender discrimination. She argued that Black women did not just experience gender oppression as white

8. Hancock, A. M. (2016). *Intersectionality: An intellectual history*. Oxford University Press.

9. Crenshaw, K. (1989). Demarginalizing the intersection of race and ex: A Black feminist critique of antidiscrimination doctrine, feminist theory, and antiracist politics. *University of Chicago Legal Forum*, 8, 139-167.

women did or racial oppression as Black men did, but rather experienced the unique and cumulative effects of both identities. She asserted that “the intersectional experience is greater than the sum of racism and sexism, any analysis that does not take intersectionality into account cannot sufficiently address the particular manner in which Black women are subordinated” (p. 140). In this way, Crenshaw paved the way for the application of intersectional thought by moving beyond single-axis analysis to reveal interlocking systems of oppression.

Core Tenets of Intersectionality

Patricia Hill Collins is another foundational intersectionality scholar who elaborated on the interconnected nature of power systems. She outlines the core constructs of intersectionality as (1) relationality; (2) power; (3) social inequality; (4) social context; (5) complexity; and (6) social justice (Hill Collins, 2019)¹⁰. **Relationality** refers to the idea that systems of power, like class, race, and gender, are actually connected and mutually constructed. This means that they maintain one another, creating an interlocking web that upholds these hierarchies. Intersectionality theory posits that systems of

10. Hill Collins, P. (2019). Intersectionality as critical social theory. Duke University Press.

power co-create one another and work together to produce *social inequalities*. Social inequality is the systematic and enduring unequal distribution of resources, opportunities, and power across social groups, shaped by social structures and institutions. *Social context* is important in terms of how knowledge is produced among different communities and how this relates to systems of power and what types of knowledge are valued. Intersectional work is also, by nature, *complex* because it accounts for multiple, interacting social categories that create unique individual experiences in response to interlocking systems of oppression. Finally, *social justice* is a core construct in intersectionality theory and research because it demands that theory and practice be connected. Like all critical theories, intersectionality inherently includes an activism component wherein we are obligated to use this critical theory to not only reveal inequities but also change them.

Intersectionality's Impact on Research in Education and Psychology

Intersectionality has had important impacts on psychology and education research and practice. It has helped us understand how intersecting oppressions affect mental health. For example, a Black queer woman will experience unique stressors and challenges as she encounters racist, sexist, and

homophobic systems in society. In education research, intersectional frameworks have helped reveal how systemic barriers intersect to perpetuate academic and discipline disparities for students from marginalized backgrounds. Research indicates that Black girls tend to be disproportionately suspended from school compared to their white female counterparts (Crenshaw, 2016)¹¹. The unique experience of Black girls in school is a result of their intersecting race and gender identities, which must both be understood in order to end punitive disciplinary practices that push these young women out of school. The video below features Dr. Bettina Love, a prominent Education Researcher, discussing why she thinks it is so critical to focus on the needs of Black girls in school.



One or more interactive elements has been excluded from this version of the text. You can view them online here:

<https://smc.pressbooks.pub/critresearchmethodspsycho/p=36#oembed-2>

11. Crenshaw, K. (2016). Black Girls Matter: Pushed Out, Overpoliced and Underprotected. African American Policy Forum.

Video 5.2 [*Bettina Love: On Black Girls, Discipline, and Schools*](#)
by [*Education Week*](#)

Decolonial and Anticolonial Frameworks

Decolonial and Anticolonial frameworks are two additional important critical theories that have had a wide-reaching impact across many disciplines. These two schools of thought critically investigate and seek to dismantle the effects of colonialism on knowledge, land, language, identity, culture, and power (Quijano, 2000)¹². Colonialism, in its most basic sense, is when a country or region is dominated politically, socially, culturally, and economically by another country. Decolonial and anticolonial scholar activists argue that colonialism has lasting effects well beyond formal decolonization. While decolonial scholars focus especially on the residue of colonialism that remain in the ways of knowing and being of colonized peoples, anticolonial scholars also stress the importance of resistance to and freedom from colonialism via the survival and sovereignty of Indigenous peoples, demanding tangible political changes such as land rematriation

12. Quijano, A. (2000). Coloniality of power, Eurocentrism and Latin America. *Nepantla: Views from South*, 1(3), 533–580.

and the reclamation of cultural languages and governing systems (Tuck & Yang, 2012)¹³. Taken together, both these frameworks seek to dismantle colonial systems and structures from the mind, body, spirit, land, and power systems. Furthermore, decolonial and anticolonial theories explicitly critique Western epistemologies, including Marxist Critical Theory, for reproducing Eurocentric assumptions about knowledge, progress, and liberation.

Core Tenets of Decolonial and Anticolonial Frameworks

Both Decolonial and Anticolonial theorists emphasize that colonialism is not a thing of the past but rather a ***persisting power*** present in current systems and structures (Quijano, 2000). This central tenet is critical because it highlights the ongoing effects of colonialism not just in terms of political control but also in its difficult-to-see forms, such as what types of ways of knowing are valued. This leads to a second core tenet of decolonial thought, a concept called ***epistemic decolonization***, which emphasizes centering indigenous ways of thinking and producing knowledge, which are typically devalued in comparison to Eurocentric knowledge (Mignolo,

13. Tuck, E., & Yang, K. W. (2012). Decolonization is not a metaphor. *Decolonization: Indigeneity, Education & Society*, 1(1), 1–40.

2011). Decolonial scholar Walter Mignolo advocated for what he termed *epistemic disobedience*, or actively resisting colonial ways of thinking and lifting up indigenous epistemology, such as oral traditions and land-based knowledge. Building on this, another core tenet in an Anticolonial framework is the *reclamation of Indigenous sovereignty*. Anticolonial scholars assert that decolonization demands action. This includes things like returning land to indigenous peoples, supporting the preservation and resurgence of indigenous language and culture. Relatedly, anticolonial thought fights against the cultural assimilation that colonization forces on colonized peoples. Anticolonial scholar Linda Tuhiwai Smith (2012) discusses decolonizing methodologies and tools like storytelling and art as a means toward recovering indigenous ways of knowing and being.

Decolonial and Anticolonial Frameworks' Impact on Research in Education and Psychology

Decolonial and Anticolonial thought have influenced research, practice, and political movements. These scholars have exposed Eurocentric hierarchies in ways of thinking and behaving that have resulted in trauma and internalized oppression caused by colonialism (Fanon, 1961). Decolonial frameworks also help us to understand and heal these wounds from the inside out. It has also highlighted the effects of

colonization on Indigenous peoples of the U.S. For example, the U.S. government funded Indian boarding schools in the late 1800s that forced assimilation on indigenous communities. Children were removed from their families and subjected to all kinds of abuse and cultural erasure (National Native American Boarding School Healing Coalition, 2021). Decolonial thought makes colonial violence clear and advocates for land rematriation, the affirmation of indigenous traditions, and psychological healing from trauma. In psychology, Decolonial thought also supports community-based research in which research is conducted with communities, not for them, ensuring voice and engagement. In the video below, psychologist and researcher Dr. Thema Bryant Davis discusses why we need to decolonize psychology.



One or more interactive elements has been excluded from this version of the text. You can view them online here:

<https://smc.pressbooks.pub/critresearchmethodspych/?p=36#oembed-3>

Video 5.3 [Why we need to decolonize psychology](#) | [Thema Bryant](#) | [TEDxNashville](#) by [TEDx Talks](#)

Equity Activity: Why we need to decolonize psychology



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=36#h5p-5>

License & Attribution

“Critical Theory and its Impact on Research” is licensed under CC BY-NC-SA 4.0.

“[What Critical Race Theory actually is – and isn’t](#)” and “[Bettina Love: On Black girls, discipline, and schools](#)” and “[Why we need to decolonize psychology](#).” Standard YouTube License.

6.

QUANTCRIT

Stephanie D'Costa

Learning Objectives

By the end of this chapter, you will be able to

1. Analyze the ways that research was used to further oppress individuals from marginalized backgrounds.
2. List ways that their social location impacts the research process.
3. Determine critical research practices for consuming and creating research.

What is QuantCrit?



Figure 6.1. Research Methodology Tree. [\[Image Description\]](#)

We have discussed both how research is shaped by paradigms and ways of knowing. We have also acknowledged that no research is free from bias, and much of our historical research was used to perpetuate oppression and harm. It is also important that we infuse critical theories into the research process, as it allows us to engage in research as a form of *resistance*. QuantCrit is one such critical theory that

uses the principles of critical race theory (for definition and overview, please see the [previous chapter](#)) and applies them to how we engage and conduct quantitative research. In the following sections, each of the main tenets of QuantCrit will be explored along with examples of how it's applied. Additionally, some considerations for researchers will also be included.

Tenet 1: The Centrality of Racism

The first tenet of quantCrit acknowledges the centrality of **racism**. Racism is embedded in our societies and institutional structures, and without intentionality, quantitative methods serve to perpetuate existing racial hierarchies¹.

Example

In understanding college and career aspirations, Marrun and colleagues (2025) used a quantCrit lens to pair racially and ethnically minoritized (REM)²

-
1. Crawford, C. E., Demack, S., Gillborn, D., & Warmington, P. (2018). Quants and crits: Using numbers for social justice (or, how not to be lied to with statistics) In *Understanding critical race research methods and methodologies* (pp. 125–137). Routledge.
 2. The term racially and ethnically minoritized refers to individuals who self-identify as African American/ Black, Hispanic/Latinx, Asian American/Pacific Islander, or Native American. This term intentionally recognizes that we are not describing

students' aspirations with their access to resources that would support their success. Their study was framed by an exploration of the [myth of meritocracy](#) as a cultural narrative that leads to race-evasive policies³ and practices in terms of accessing higher education. Post-secondary students were invited to complete surveys of their experiences and findings indicated that students across racial groups had high aspirations of going to college or a career. However, the researchers found limited exposure from college recruiters, limited conversations with current teachers about their college experience, and continued under-enrollment in advanced high school coursework. By connecting the centrality of racism to college and career success, the researchers

these communities by any biological differences but instead recognizing the way systemic racism impacts their existence. We acknowledge the limitation of this term as this aggregation can obfuscate how racism impacts these communities differently given the historical ways oppression has shown up within the U.S.

3. policies that ignore the consideration of race

demonstrated the structural gaps between aspirations and access⁴.

Research Considerations

1. Include a positionality statement (Please refer back to chapter 2) in your research and consider how your positionality impacts your research topic and process. As you review research, consider who the authors are and how they arrived at this research study. This could be found in the methods section, if authors identify their reflexivity. It could also be in the author biographies (if included).
2. Include asset-based language in your research questions. Make sure your questions consider the ways that power structures impact the outcomes for marginalized

Tenet 2: Numbers Are Not

4. Marrun, N. A., Mireles, D., Beck, J. S., & Clark, C. (2025). The (non) numeric truths behind college and career aspirations: a QuantCrit analysis of the precarious pathways for secondary students of color. *Race Ethnicity and Education*, 1-23.

Neutral

With our historical post positivist lens towards research (See chapter 4), there was an assumption that quantitative numbers obtained through statistical analyses were neutral and to be accepted at face value. QuantCrit challenges this underlying belief and argues that research has historically been used to uphold oppressive systems and to perpetuate dominant, elite values⁵. Many of the ways this happens in research are insidious, such as not considering race in a study or using deficit or blaming language.

Example

An example of numbers that are not neutral comes to light when creating outcome measures to understand individuals' perspectives. Marroquín-

5. Zuberi, T., & Bonilla-Silva, E. (Eds.). (2008). *White logic, white methods: Racism and methodology*. Rowman & Littlefield Publishers.

Flores and colleagues (2024) wanted to better understand college students from minoritized backgrounds' experience in entering the STEM field. While traditional studies that explore this topic focus on deficit language towards minoritized students, this study sought to center an asset-based approach called [Community Cultural Wealth](#) to demonstrate the ways individuals leverage their strengths in challenging circumstances. They also intentionally used a QuantCrit approach to guide the process for developing a 100-item survey. In reviewing their own experiences and interviews with college students, they found that data needed to be captured in the intersections between discrete cultural wealth areas. For example, they describe how a queer college student described challenges with familial strengths given her family's religious background but was able to identify the ways her family supported her dreams and aspirations for college. The researchers decided to use a factor model with cross-loading to allow for space to interpret data from an intersectional lens. By centering the researchers' lived experiences and minoritized college students' perspectives, they were able to choose a statistical model to interpret

the data in ways that aligned with the individual's lived experiences.⁶ By centering the researchers' lived experiences and minoritized college students' perspectives, they were able to choose a statistical model to interpret the data in ways that aligned with individuals' lived experiences.

Research Considerations

1. Consider the type of statistical analysis you choose to answer the questions you have. Remember that variables don't exist in a vacuum, and you should provide context for how you include them in your study. For example, “controlling” (or holding constant) for things like socioeconomic status (SES) doesn't take into account the constellation of things that are connected to that variable (e.g., higher likelihood of under-resourced schools with higher populations of students from low SES

6. Marroquín-Flores, R. A., Tijerina, R. M., Tedeschi, M., Banjara, S., Warmesley, R., McFather, L., ... & Limeri, L. B. (2024). Using a QuantCrit Approach to Develop and Collect Evidence of Validity for a Measure of Community Cultural Wealth. *CBE—Life Sciences Education*, 23(4), ar55.

- backgrounds) and may impact outcome measures.
2. Consider who is in the sample of the study and whether the findings are truly generalizable. What limitations arise when cultural expectations are not accounted for?

Tenet 3: Categories Are Not Natural or Given

This tenet is explicit about the fact that race is a *social* construct. While research has consistently demonstrated that there are no biological differences between racial groups (in fact, we see greater levels of variability within groups than between groups), individuals' experience of *racism* impacts their physical, psychological, and economic well-being. Many research studies include data about race without an analysis of what that category truly implies. This tenet also applies to categories like disability in which QuantCrit acknowledges that these categories are not stable across time and are connected to sociopolitical and historical contexts⁷.

7. Cruz, R. A., Kulkarni, S. S., & Firestone, A. R. (2021). A QuantCrit analysis of context, discipline, special education, and disproportionality. *AERA Open*, 7, 23328584211041354.

Example

Research has indicated that there has been disproportionate **disproportionate** representation of children of color receiving special education services. In order to qualify for these services, students need to be assessed and qualify for a disability that warrants additional educational services to support their learning. However, there is often variability in how disproportionality is calculated in the research⁸. For example, Sullivan and colleagues (2020) describe how Asian students are often missing from conversations about special education disproportionality, given that they are treated as a monolith. Asian students are often seen as being underrepresented in special education,

8. D’Costa, S., Grant, S., Kulkarni, T., Crossing, A., Zahn, M., & Tanaka, M. L. (2024). A call for QuantCrit methodologies: Unpacking the need for a critical lens in school psychology research. *School psychology international*, 45(3), 254-279.

which leaves them off the radar for needing additional educational supports. However, when data is disaggregated, the findings indicate that Pacific Islander students (who are often lumped into the Asian category) are qualifying for special education at higher rates⁹. By disaggregating data, we can identify which students need additional support and consider reasons for why schools are not meeting their needs. Statistical analyses have also been weaponized to *resegrete* Black & Latine students in special education¹⁰. QuantCrit can be used to elucidate these patterns and invite researchers to consider anti-racist special education practices.

-
9. Sullivan, A. L., Kulkarni, T., & Chhuon, V. (2020). Making visible the invisible: Multistudy investigation of disproportionate special education identification of US Asian American and Pacific Islander students. *Exceptional children*, 86(4), 449-467.
10. Bell, N. S., Collier, Z., Vélez, V. N., & Ford, D. Y. (2025). CritSEM: Advancing QuantCrit to examine racialized resegregation in special education. *Journal of Research on Educational Effectiveness*, 18(2), 390-422.

Research Considerations

1. Consider disaggregating your data. Remember that racial categories are not a monolith, and by disaggregating data, you can see the ways that outcomes look different for subgroups.
2. Create opportunities for participants to give input into the categories they identify with. Provide context for why these categories are used in the first place.

Tenet 4: Data Can't Speak For Itself

This tenet speaks to the fact the research is shaped by the researcher and who the researcher is influences a) the types of questions they ask, b) the types of data they collect, c) the analysis procedure they choose, and d) the way they interpret the findings¹¹. Especially in research that has implications for marginalized groups, QuantCrit argues that the interpretation needs to center those who are intimately impacted. This

11. Castillo, W., & Gillborn, D. (2022). How to “QuantCrit:” Practices and questions for education data researchers and users. (EdWorkingPaper: 22-546). Retrieved from Annenberg Institute at Brown University. <https://doi.org/10.26300/v5kh-dd65>

includes engaging marginalized voices throughout the research process.

Example

One example of the different ways researchers approach data can be demonstrated through the conceptualization of discipline data in schools. Time and time again, the research has indicated that Black youth experience higher rates of discipline and more egregious punishments than other groups of students. If researchers are not careful, this data can lead to further stereotyping of Black youth *behavior* without considerations of how school personnel are influenced by white supremacy ideologies and anti-Blackness¹². Additional conversations are needed to identify the ways that schools replicate **carceral** in their desire to control Black bodies instead of

12. Castillo, W., & Strunk, K. K. (2025). *How to QuantCrit: Applying critical race theory to quantitative data in education* (p. 134). Taylor & Francis.

educate them¹³. While data collected may be similar across studies, using critical frameworks for interpretation is vital in not replicating harm.

Research Considerations

1. Be clear about how traditionally marginalized groups have been a part of the research process. This is through the use of critical theories, participant input, research team membership, and the methods you choose for analysis.
2. Provide context for people to understand why your results are meaningful. Look for articles that help the reader demystify the results in ways that are applicable to them.

Tenet 5: Social Justice Orientation

This final tenet of QuantCrit is a strong reminder of the *why*

13. Shedd, C. (2015). Unequal city: Race, schools, and perceptions of injustice. Russell Sage Foundation.

we engage in the research process. Research is a powerful tool and medium to move towards justice or replicate oppression for marginalized groups. From a critical lens, there is no neutral middle ground when it comes to research. This tenet also reminds us that social justice is both a process and an end goal—therefore how we engage in research from the conceptualization to the findings, matters¹⁴. Engaging in quantitative research with intentionality can allow researchers and communities to use data for liberation.

Example

Vargas and colleagues (2024) used a QuantCrit approach when looking at the Equitable Housing Project (EHP). The EHP is a collaboration with an academic institution and a local non-profit that supports houseless community members in the Los

14. D’Costa, S., Grant, S., Kulkarni, T., Crossing, A., Zahn, M., & Tanaka, M. L. (2024). A call for QuantCrit methodologies: Unpacking the need for a critical lens in school psychology research. *School psychology international*, 45(3), 254-279.

Angeles area. The research team included a formerly houseless individual who is pivotal to the partnership with the non-profit and a member of the research design process. In conversations with the non-profit, a social justice question was raised around bias in the assessment process that supported prioritization of resource allocation. The research team reviewed the sociohistorical context, which indicates that racism and sexism play a big role in systemic access to affordable housing. The researchers intentionally used intersectionality theory to frame their research questions. In an educational process, the research team describes five different statistical procedures that they could have used to analyze the demographic data of service utilization for the EHP. Models that use intersectional labels for race and sex found that white men tend to benefit the most from service allocation, and women of color (specifically Black or Latine) are provided less access to resources. The researchers stress that a critical interpretation of the results is needed to promote social justice. When systems hold [race-neutral](#) approaches, they don't consider the unique ways that people of color in

general and women of color in particular experience houselessness¹⁵.

Research Considerations

1. Research implications matter. Don't simply report your findings, but identify ways that your data contributes to the larger narrative. Integrate historical socio-political contexts into your discussion of findings.
2. Consider how your research and data find their way back to support the communities from whom they were collected. What is the practical significance of your work, and how can oppressed groups leverage the data for justice?
3. Make sure the measures you choose are salient to the population you are working with. If they are not, consider what information might be missed from this limitation.

15. Vargas, J. H., & Peet, J. Z. (2024). The first primer for the QuantCrit-curious critical race theorist or psychologist: On intersectionality theory, interaction effects, and AN (C) OVA/regression models. *Journal of Social Issues*, 80(1), 168-217.

Equity Activity: Consuming research from a justice lens

Consider the tenets of QuantCrit and how you could apply them in reading research articles related to your specific topic of interest. Together, let's draft some guiding statements on how you will consider each tenet. If you get stuck at any point in this activity, feel free to review [How to "QuantCrit:" Practices and questions for education data researchers and users](#) as a starting framework.



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=39#h5p-6>

License & Attribution

Research consideration examples were expanded on from Castillo, W., & Gillborn, D. (2022). [*How to “QuantCrit:” Practices and questions for education data researchers and users.*](#) (EdWorkingPaper: 22-546). Retrieved from Annenberg Institute at Brown University. <https://doi.org/10.26300/v5kh-dd65>

“QuantCrit” by Stephanie D’Costa is licensed under CC BY-NC-SA 4.0.

Image Descriptions

Figure 6.1. A poster titled “Research Methodology Tree” depicts a leafy tree with arrows to four parts: at the top-left branch “Methods (details of exactly how we collect data)”; top-right branch “Methodology (how we should best collect data)”; on the trunk “Epistemology (how we should investigate the world)”; and on the roots “Ontology (how we view the world).” The design visualizes how ontology and

epistemology underpin methodology and methods. [\[Return to Figure 6.1\]](#)

Media Attributions

- research tree

7.

RESEARCH ETHICS

Mireille Ukeye

Learning Objectives

By the end of this chapter, you will be able to

1. Describe a simple framework for thinking about ethical issues in psychological research.
2. Give examples of several ethical issues that arise in psychological research, including ones that affect research participants, the scientific community, and society more generally.

Ethics is the branch of philosophy that is concerned with morality, what it means to behave morally, and how people can achieve that goal. It can also



refer to a set of principles and practices that provide moral guidance in a particular field. There is an ethics of business, medicine, teaching, and, of course, scientific research. As the opening example illustrates, many kinds of ethical issues can arise in scientific research, especially when it involves human participants. For this reason, it is useful to begin with a general framework for thinking through these issues.

A Framework for Thinking About Research Ethics

Table 7.1 presents a framework for thinking through the ethical issues involved in psychological research. The rows of Table 7.1 represent four general moral principles that apply to scientific research: weighing risks against benefits, acting responsibly and with integrity, seeking justice, and respecting people's rights and dignity. (These principles are adapted from those in the American Psychological Association [APA] Ethics Code.) The columns of Table 7.1 represent three groups of people that are affected by scientific research: the research participants, the scientific community, and society more

generally. The idea is that a thorough consideration of the ethics of any research project must take into account how each of the four moral principles applies to each of the three groups of people.

Table 7.1 A Framework for Thinking About Ethical Issues in s

Moral principle	Who is affected	
	Research participants	Scientific community
Weighing risks against benefits		
Acting responsibly and with integrity		
Seeking justice		
Respecting people’s rights and dignity		

Moral Principles

Let us look more closely at each of the moral principles and how they can be applied to each of the three groups.

Weighing Risks Against Benefits

Scientific research in psychology can be ethical only if its risks are outweighed by its benefits. Among the risks to research participants are that a treatment might fail to help or even be harmful, a procedure might result in physical or psychological

harm, and their right to privacy might be violated. Among the potential benefits are receiving a helpful treatment, learning about psychology, experiencing the satisfaction of contributing to scientific knowledge, and receiving compensation (e.g., money, course credit, vouchers, tangible goods) for participating. Scientific research can have risks and benefits to the scientific community and to society, too (Rosenthal, 1994).¹ A risk to science is that if a research question is relevant, not relevant to the needs of the science community and/or society, or a study is poorly designed, then the time, money, and effort spent on that research could have been spent on more impactful research. A risk to society is that research results could be misunderstood or misapplied with harmful consequences. The research that mistakenly linked the measles, mumps, and rubella (MMR) vaccine to autism resulted in both of these kinds of harm. Of course, the benefits of scientific research to science and society are that it advances scientific knowledge and can contribute to the welfare of society.

1. Rosenthal, R. M. (1994). Science and ethics in conducting, analyzing, and reporting psychological research. *Psychological Science*, 5, 127–133.



It is not necessarily easy to weigh the risks of research against its benefits because the risks and benefits may not be directly comparable. For example, it is common for the risks of a study to be

primarily to the research participants, but the benefits are primarily for science or society. Consider, for example, Stanley Milgram's original study on obedience to authority (Milgram, 1963).² The participants were told that they were taking part in a study on the effects of punishment on learning and were instructed to give electric shocks to another participant each time that participant responded incorrectly on a learning task. With each incorrect response, the shock became stronger, eventually causing the other participant (who was in the next room) to protest, complain about his heart, scream in pain, and finally fall silent and stop responding. If the first participant hesitated or expressed concern, the researcher said that he must continue. In reality, the other participant was a confederate of the researcher—a helper who pretended to be a real participant—and the protests, complaints, and screams that the real participant heard were an audio recording that

2. Milgram, S. (1963). Behavioral study of obedience. *Journal of Abnormal and Social Psychology*, 67, 371–378.

was activated when he flipped the switch to administer the “shocks.” The surprising result of this study was that most of the real participants continued to administer the shocks right through the confederate’s protests, complaints, and screams. Although this is considered one of the most important results in psychology with implications for understanding events like the Holocaust or the mistreatment of prisoners by US soldiers at Abu Ghraib, it came at the cost of producing severe psychological stress in the research participants.

Was It Worth It?

Much of the debate over the ethics of Milgram’s obedience study concerns the question of whether the resulting scientific knowledge was worth the harm caused to the research participants. To get a better sense of the harm, consider Milgram’s (1963) own description of it.

In a large number of cases, the degree of tension reached extremes that are rarely seen in sociopsychological laboratory studies. Subjects were observed to sweat, tremble, stutter, bite their lips, groan, and dig their fingernails into their flesh....Fourteen of the 40 subjects showed definite signs of nervous

laughter and smiling. The laughter seemed entirely out of place, even bizarre. Full-blown uncontrollable seizures [of laughter] were observed for three subjects. On one occasion we observed a seizure so violently convulsive that it was necessary to call a halt to the experiment (p. 375).

Milgram also noted that another observer reported that within 20 minutes one participant “was reduced to a twitching, stuttering wreck, who was rapidly approaching the point of nervous collapse” (p. 377)

To Milgram’s credit, he went to great lengths to debrief his participants—including returning their mental states to normal—and to show that most of them thought the research was valuable and they were glad to have participated.

Acting Responsibly and With Integrity

Researchers must act responsibly and with integrity. This means carrying out their research in a thorough and culturally competent manner, meeting



their professional obligations, and being truthful. Acting with integrity is important because it promotes trust, which is an essential element of all effective human relationships. Participants must be able to trust that researchers are being honest with them (e.g., about what the study involves), will keep their promises (e.g., to maintain confidentiality), and will carry out their research in ways that maximize benefits and minimize risk. An important issue here is the use of deception. Some research questions (such as Milgram's) are difficult or impossible to answer without deceiving research participants. Thus, acting with integrity can conflict with doing research that advances scientific knowledge and benefits society. We will consider how psychologists generally deal with this conflict shortly.

The scientific community and society must also be able to trust that researchers have conducted their research thoroughly and competently and that they have reported on it honestly. Again, the example at the beginning of the chapter illustrates what can happen when this trust is violated. In this

case, other researchers wasted resources on unnecessary follow-up research, and people avoided the MMR vaccine, putting their children at increased risk of measles, mumps, and rubella. Indeed, many people, including children, have died as a result of parents' misinformed decisions not to vaccinate their children.

Seeking Justice

Researchers must conduct their research in a just manner. They should treat their participants fairly, for example, by giving them adequate compensation for their participation and making sure that benefits and risks are distributed across all participants. For instance, in a study of a new and potentially beneficial psychotherapy, some participants might receive the psychotherapy while others serve as a control group that receives no treatment. If the psychotherapy turns out to be effective, it would be fair to offer it to participants in the control group when the study ends.



At a broader societal level, members of some groups have historically faced more than their fair share of the risks of scientific research, including people who are institutionalized, are disabled, or belong to minoritized racial or ethnic groups. A particularly tragic example of scientific

racism is the Tuskegee syphilis study conducted by the US Public Health Service from 1932 to 1972 (Reverby, 2009).³ The participants in this study were poor African American men in the vicinity of Tuskegee, Alabama, who were told that they were being treated for “bad blood.” Although they were given some free medical care, they were not treated for their syphilis. Instead, they were observed to see how the disease developed in untreated patients. Even after the use of penicillin became the standard treatment for syphilis in the 1940s, these men continued to be denied treatment without being given an opportunity to leave the study. The study was eventually discontinued only after details were made known to the general public by journalists and activists. It is now widely recognized that researchers need to consider issues of justice and fairness at the societal level.

“They Were Betrayed”

In 1997—65 years after the Tuskegee Syphilis Study began and 25 years after it

3. Reverby, S. M. (2009). *Examining Tuskegee: The infamous syphilis study and its legacy*. Chapel Hill, NC: University of North Carolina Press.

ended—President Bill Clinton formally apologized on behalf of the US government to those who were affected. Here is an excerpt from the apology:

So today America does remember the hundreds of men used in research without their knowledge and consent. We remember them and their family members. Men who were poor and African American, without resources and with few alternatives, they believed they had found hope when they were offered free medical care by the United States Public Health Service. They were betrayed.

Read the [full text of the apology](#) at the Clinton White House Archives.

Citational Justice

Justice in psychological research extends beyond the ethical treatment of participants to include how knowledge is created, represented, and credited. Citational justice refers to the ethical responsibility of researchers to cite prior scholarship in ways that are accurate, fair, and inclusive. Because citation practices influence which ideas are

legitimized and which scholars are recognized, they play a meaningful role in shaping the direction and values of the scientific field.

Why Citational Justice Matters

Citation is a primary mechanism through which scientific contributions are acknowledged and rewarded. Decisions about what to cite directly affect scholars' visibility, professional advancement, and access to resources such as funding and collaboration opportunities. Systematic patterns of undercitation, particularly affecting scholars from minoritized racial and ethnic groups, women, scholars from the Global South, and those working in interdisciplinary or community-engaged traditions, can reinforce existing inequities within the scientific community, even when exclusion is unintentional.

Citational justice is also essential for the integrity and quality of scientific knowledge. When reference lists rely heavily on a narrow set of highly cited sources, important perspectives, methods, or findings may be overlooked. This can result in an incomplete or distorted understanding of psychological phenomena. Ethical citation practices strengthen science by ensuring that conclusions are grounded in the full scope of relevant scholarship rather than a limited subset of voices. Beyond academia, citational justice has broader societal implications. Psychological research informs educational

policy, clinical practice, and public decision-making. When scholarship that centers marginalized populations or culturally responsive approaches is excluded from citation, the resulting knowledge base may fail to address the needs of diverse communities. Inclusive citation practices, therefore, contribute to a more equitable and socially responsible application of psychological science.

Practicing Citational Justice

Practicing citational justice involves conducting thorough, thoughtful literature reviews, prioritizing original and foundational sources, and reflecting critically on patterns in one's citations. Although no single study can include all relevant scholarship, ethical researchers remain attentive to whose work is represented, whose is absent, and how their citational choices contribute to the scientific record. In this sense, citation is not merely a technical requirement of academic writing but an ethical practice that reflects the discipline's values.

Respecting People's Rights and Dignity

Researchers must respect people's rights and dignity as human beings. One element of this is respecting their **autonomy**—their right to make their own



choices and take their own actions free from coercion. Of fundamental importance here is the concept of **informed consent**. This means that researchers obtain and document people's agreement to participate in a study before they actually complete the study. Informed consent entails having participants be informed of everything that might reasonably be expected to affect their decision. Consider the participants in the Tuskegee study. Although they agreed to participate in the study, they were not told that they had syphilis but would be denied treatment for it. Had they been told this basic fact about the study, it seems likely that they would not have agreed to participate. Likewise, had participants in Milgram's study been told that they might be "reduced to a twitching, stuttering wreck," it seems likely that many of them would not have agreed to participate. In neither of these studies did participants give true informed consent.

Another element of respecting people's rights and dignity is respecting their **privacy**—their right to decide what

information about them is shared with others. This means that researchers must maintain **confidentiality**, which is essentially an agreement not to disclose participants' personal information without their consent or some appropriate legal authorization. Even more ideally participants can maintain **anonymity**, which is when their name and other personally identifiable information is not collected at all. Anonymity can also include collecting personally identifiable information, however masking or de-identifying such information so participants' information remains anonymous. Take, for instance, utilizing pseudonyms instead of participants' actual names.

Unavoidable Ethical Conflict

It may already be clear that ethical conflict in psychological research is unavoidable. Because there is little, if any, psychological research that is completely risk-free, there will almost always be a conflict between risks and benefits. Research that is beneficial to one group (e.g., the scientific community) can be harmful to another (e.g., the research participants), creating especially difficult tradeoffs. We have also seen that being completely truthful with research participants can make it difficult or impossible to conduct scientifically valid studies on important questions.

Of course, many ethical conflicts are fairly easy to resolve. Nearly everyone would agree that deceiving research

participants and then subjecting them to physical harm would not be justified by filling a small gap in the research literature. But many ethical conflicts are not easy to resolve, and competent and well-meaning researchers can disagree about how to resolve them. Consider, for example, an actual study on “personal space” conducted in a public men’s room (Middlemist, Knowles, & Matter, 1976).⁴ The researchers secretly observed their participants to see whether it took them longer to begin urinating when there was another man (a confederate of the researchers) at a nearby urinal. While some critics found this to be an unjustified assault on human dignity (Koocher, 1977),⁵ the researchers had carefully considered the ethical conflicts, resolved them as best they could, and concluded that the benefits of the research outweighed the risks (Middlemist, Knowles, & Matter, 1977).⁶ For example, they had interviewed some preliminary participants and found

4. Middlemist, R. D., Knowles, E. S., & Matter, C. F. (1976). Personal space invasions in the lavatory: Suggestive evidence for arousal. *Journal of Personality and Social Psychology*, 33, 541–546.

5. Koocher, G. P. (1977). Bathroom behavior and human dignity. *Journal of Personality and Social Psychology*, 35, 120–121.

6. Middlemist, R. D., Knowles, E. S., & Matter, C. F. (1977). What to do and what to report: A reply to Koocher. *Journal of Personality and Social Psychology*, 35, 122–125.

that none of them was bothered by the fact that they had been observed.

The point here is that although it may not be possible to eliminate ethical conflict completely, it is possible to deal with it in responsible and constructive ways. In general, this means thoroughly and carefully thinking through the ethical issues that are raised, minimizing the risks, and weighing the risks against the benefits. It also means being able to explain one's ethical decisions to others, seeking feedback on them, and ultimately taking responsibility for them.

Putting Ethics into Practice

In this section, we look at some practical advice for conducting ethical research in psychology. Again, it is important to remember that ethical issues arise well before you begin to collect data and continue to arise through publication and beyond.

Know and Accept Your Ethical Responsibilities



As the American Psychological Association (APA) Ethics Code notes in its introduction, “Lack of awareness or misunderstanding of an

ethical standard is not itself a defense to a charge of unethical conduct.” This is why the very first thing that you must do as a new researcher is to know and accept your ethical responsibilities. At a minimum, this means reading and understanding the relevant sections of the APA Ethics Code, distinguishing minimal risk from at-risk research, and knowing the specific policies and procedures of your institution—including how to prepare and submit a research protocol for institutional review board (IRB) review. If you are conducting research as a course requirement, there may be specific course standards, policies, and procedures. If any standard, policy, or procedure is unclear—or you are unsure what to do about an ethical issue that arises—you must seek clarification. You can do this by reviewing the relevant ethics codes, reading about how similar issues have been resolved by others, or consulting with more experienced researchers, your IRB, or your course instructor. Ultimately, you as the

researcher must take responsibility for the ethics of the research you conduct.

Identify and Minimize Risks

As you design your study, you must identify and minimize risks to participants. Start by listing all the risks, including risks of physical and psychological harm and violations of confidentiality. Remember that it is easy for researchers to see risks as less serious than participants do or even to overlook them completely. For example, one student researcher wanted to test people's sensitivity to violent images by showing them gruesome photographs of crime and accident scenes. Because she was an emergency medical technician, however, she greatly underestimated how disturbing these images were to most people. Remember too that some risks might apply only to some participants. For example, while most people would have no problem completing a survey about their fear of various crimes, those who have been a victim of one of those crimes might become upset. This is why you should seek input from a variety of people, including your research collaborators, more experienced researchers, and even from nonresearchers who might be better able to take the perspective of a participant.

Once you have identified the risks, you can often reduce or eliminate many of them. One way is to modify the research design. For example, you might be able



to shorten or simplify the procedure to prevent boredom and frustration. You might be able to replace upsetting or offensive stimulus materials (e.g., graphic accident scene photos) with less upsetting or offensive ones (e.g., milder photos of the sort people are likely to see in the newspaper). A good example of modifying a research design is a 2009 replication of Milgram's study conducted by Jerry Burger. Instead of allowing his participants to continue administering shocks up to the 450-V maximum, the researcher always stopped the procedure when they were about to administer the 150-V shock (Burger, 2009).⁷ This made sense because in Milgram's study (a) participants' severe negative reactions occurred after this point and (b) most participants who administered the 150-V shock continued all the way to the 450-V maximum. Thus the researcher was able to compare his results directly with Milgram's at every point up to the 150-V shock and also was able to estimate how many of his participants would have

7. Burger, J. M. (2009). Replicating Milgram: Would people still obey today? *American Psychologist*, 64, 1–11.

continued to the maximum—but without subjecting them to the severe stress that Milgram did. (The results, by the way, were that these contemporary participants were just as obedient as Milgram’s were.)

A second way to minimize risks is to use a **pre-screening** procedure to identify and eliminate participants who are at high risk. You can do this in part through the informed consent process. For example, you can warn participants that a survey includes questions about their fear of crime and remind them that they are free to withdraw if they think this might upset them. Prescreening can also involve collecting data to identify and eliminate participants. For example, Burger used an extensive pre-screening procedure involving multiple questionnaires and an interview with a clinical psychologist to identify and eliminate participants with physical or psychological problems that put them at high risk.

A third way to minimize risks is to take active steps to maintain confidentiality. You should keep signed consent forms separately from any data that you collect and in such a way that no individual’s name can be linked to their data. In addition, beyond people’s sex and age, you should only collect personal information that you actually need to answer your research question(s). If people’s sexual orientation or ethnicity is not clearly relevant to your research question(s), for example, then do not ask them about it. Be aware also that certain data collection procedures can lead to unintentional violations of confidentiality. When participants respond to an oral survey

in a shopping mall or complete a questionnaire in a classroom setting, it is possible that their responses will be overheard or seen by others. If the responses are personal, it is better to administer the survey or questionnaire individually in private or to use other techniques to prevent the unintentional sharing of personal information.

Identify and Minimize Deception

Remember that deception can take a variety of forms, not all of which involve actively misleading participants. It is also deceptive to allow participants to make incorrect assumptions (e.g., about what will be on a “memory test”) or simply withhold information about the full design or purpose of the study. It is best to identify and minimize all forms of deception.

Remember that, according to the APA Ethics Code, deception is ethically acceptable only if there is no way to answer your research question without it. Therefore, if your research design includes any form of active deception, you should consider whether it is truly necessary. Imagine, for example, that you want to know whether the age of college professors affects students’ expectations about their teaching ability. You could do this by telling participants that you will show them photos of college professors and ask them to rate each one’s teaching ability. But if the photos are not really of college professors but of your own family members and

friends, then this would be deception. This deception could easily be eliminated, however, by telling participants instead to imagine that the photos are of college professors and to rate them as if they were.

In general, it is considered acceptable to wait until debriefing before you reveal your research question as long as you describe the procedure, risks, and benefits during the informed consent process. For example, you would not have to tell participants that you wanted to know whether the age of college professors affects people's expectations about them until the study was over. Not only is this information unlikely to affect people's decision about whether or not to participate in the study, but it has the potential to invalidate the results. Participants who know that age is the independent variable might rate the older and younger "professors" differently because they think you want them to. Alternatively, they might be careful to rate them the same so that they do not appear prejudiced. But even this extremely mild form of deception can be minimized by informing participants—orally, in writing, or both—that although you have accurately described the procedure, risks, and benefits, you will wait to reveal the research question until afterward. In essence, participants give their consent to be deceived or to have information withheld from them until later.

Weigh the Risks Against the Benefits



Once the risks of the research have been identified and minimized, you need to weigh them against the benefits. This requires identifying all the benefits.

Remember to consider benefits to the research participants, to science, and to society. If you are a student researcher, remember that one of the benefits is the knowledge you will gain about how to conduct scientific research in psychology—knowledge you can then use to complete your studies and succeed in graduate school or in your career.

If the research poses minimal risk—no more than in people’s daily lives or routine physical or psychological examinations—then even a small benefit to participants, science, or society is generally considered enough to justify it. If it poses more than minimal risk, then there should be more benefits. If the research has the potential to upset some participants, for example, then it becomes more important that the study is well designed and can answer a scientifically interesting research question or have clear practical implications. It would be unethical to subject people to pain, fear, or embarrassment for no better reason than to satisfy one’s personal curiosity. In general, psychological research that has the potential to cause harm that is more than minor or lasts

for more than a short time is rarely considered justified by its benefits.

Create Informed Consent and Debriefing Procedures

Once you have settled on a research design, you need to create your informed consent and debriefing procedures. Start by deciding whether informed consent is necessary according to APA Standard 8.05. If informed consent is necessary, there are several things you should do. First, when you recruit participants—whether it is through word of mouth, posted advertisements, or a participant pool—provide them with as much information about the study as you can with language that is developmentally and culturally appropriate to the participants’ background. For instance, conveying information to individuals whose English is their second language is likely to be communicated differently than to those who are native speakers. Additionally, all information ought to be succinct as well as easy to understand. This will allow those who might find the study objectionable to avoid it. Second, prepare a script or set of “talking points” to help you explain the study to your participants in simple everyday language. This should include a description of the procedure, the risks and benefits, and their right to withdraw at any time. Third, create an informed consent form that covers all the points in Standard 8.02a that participants can read and sign – assuming

participants do not have sensory and dexterity impairments – after you have described the study to them. Your university, department, or course instructor may have a sample consent form that you can adapt for your own study. If not, an Internet search will turn up several samples. Remember that if appropriate, both the oral and written parts of the informed consent process should include the fact that you are keeping some information about the design or purpose of the study from them but that you will reveal it during debriefing.

Debriefing is similar to informed consent in that you cannot necessarily expect participants to read and understand written debriefing forms. So again it is best to write a script or set of talking points with the goal of being able to explain the study in simple, everyday language. During the debriefing, you should reveal the research question and full design of the study. For example, if participants are tested under only one condition, then you should explain what happened in the other conditions. If you deceived your participants, you should reveal this as soon as possible, apologize for the deception, explain why it was necessary, and correct any misconceptions that participants might have as a result. Debriefing is also a good time to provide additional benefits to research participants by giving them relevant practical information or referrals to other sources of help. For example, in a study of attitudes toward domestic abuse, you could provide pamphlets about domestic abuse and referral information to counseling services for those who might want it.

Remember to schedule plenty of time for the informed consent and debriefing processes. They cannot be effective if you have to rush through them.

Get Approval

The next step is to get institutional approval for your research based on the specific policies and procedures at your institution or for your



course. This will generally require writing a protocol that describes the purpose of the study, the research design and procedure, the risks and benefits, the steps taken to minimize risks, and the informed consent and debriefing procedures. Do not think of the institutional approval process as merely an obstacle to overcome but as an opportunity to think through the ethics of your research and to consult with others who are likely to have more experience or different perspectives than you. If the IRB has questions or concerns about your research, address them promptly and in good faith. This might even mean making further modifications to your research design and procedure before resubmitting your protocol.

Follow Through

Your concern with ethics should not end when your study receives institutional approval. It now becomes important to stick to the protocol you submitted or to seek additional approval for anything other than a minor change. During the research, you should monitor your participants for unanticipated reactions and seek feedback from them during debriefing. One criticism of Milgram's study is that although he did not know ahead of time that his participants would have such severe negative reactions, he certainly knew after he had tested the first several participants and should have made adjustments at that point (Baumrind, 1985).⁸ Be alert also for potential violations of confidentiality. Keep the consent forms and the data safe and separate from each other and make sure that no one, intentionally or unintentionally, has access to any participant's personal information.

Finally, you must maintain your integrity through the publication process and beyond. Address publication credit—who will be authors on the research and the order of authors—with your collaborators early and avoid plagiarism in your writing. Remember that your scientific goal is to learn about the way the world actually is and that your scientific

8. Baumrind, D. (1985). Research using intentional deception: Ethical issues revisited. *American Psychologist*, 40, 165–174.

duty is to report on your results honestly and accurately. So do not be tempted to fabricate data or alter your results in any way. Besides, unexpected results are often as interesting, or more so, than expected ones.

From Moral Principles to Ethics Codes

The general moral principles of weighing risks against benefits, acting with integrity, seeking justice, and respecting people's rights and dignity provide a useful starting point for thinking about the ethics of psychological research because essentially everyone agrees on them. As we have seen, however, even people who agree on these general principles can disagree about specific ethical issues that arise in the course of conducting research. This is why there also exist more detailed and enforceable ethics codes that provide guidance on important issues that arise frequently. In this section, we begin with a brief historical overview of such ethics codes and then look closely at the one that is most relevant to psychological research—that of the American Psychological Association (APA).

Historical Overview

One of the earliest ethics codes was the **Nuremberg Code**—a

set of 10 principles written in 1947 in conjunction with the trials of Nazi physicians accused of shockingly cruel research on concentration camp prisoners during World War II. It provided a standard against which to compare the behavior of the men on trial—many of whom were eventually convicted and either imprisoned or sentenced to death. The Nuremberg Code was particularly clear about the importance of carefully weighing risks against benefits and the need for informed consent. The **Declaration of Helsinki** is a similar ethics code that was created by the World Medical Council in 1964. Among the standards that it added to the Nuremberg Code was that research with human participants should be based on a written **protocol**—a detailed description of the research—that is reviewed by an independent committee. The Declaration of Helsinki has been revised several times, most recently in 2004.

In the United States, concerns about the Tuskegee study and others led to the publication in 1978 of a set of federal guidelines called the **Belmont Report**. The Belmont Report explicitly recognized the principle of seeking **justice**, including the importance of conducting research in a way that distributes risks and benefits fairly across different groups at the societal level. It also recognized the importance of **respect for persons**, which acknowledges individuals' autonomy and protection for those with diminished autonomy (e.g., prisoners, children), and translates to the need for informed consent. Finally, it recognized the principle of **beneficence**,

which underscores the importance of maximizing the benefits of research while minimizing harms to participants and society. The Belmont Report became the basis of a set of laws—the **Federal Policy for the Protection of Human Subjects**—that apply to research conducted, supported, or regulated by the federal government. An extremely important part of these regulations is that universities, hospitals, and other institutions that receive support from the federal government must establish an **institutional review board (IRB)**—a committee that is responsible for reviewing research protocols for potential ethical problems. An IRB must consist of at least five people with varying backgrounds, including members of different professions, scientists, and nonscientists, and at least one person not otherwise affiliated with the institution. The IRB helps to make sure that the risks of the proposed research are minimized, the benefits outweigh the risks, the research is carried out in a fair manner, and the informed consent procedure is adequate.

The federal regulations also distinguish research that poses three levels of risk. **Exempt research** is the lowest level of risk and includes research on the effectiveness of normal educational activities, the use of standard psychological measures and surveys of a nonsensitive nature that are administered in a way that maintains confidentiality, and research using existing data from public sources. It is called exempt because once approved, it is exempt from regular, continuous review. **Expedited research** poses a somewhat

higher risk than exempt, but still exposes participants to risks that are no greater than minimal risk (those encountered by healthy people in daily life or during routine physical or psychological examinations). Expedited review is done by one member of the IRB or by a separate committee under the authority of the IRB that can only approve minimal risk research (many departments of psychology have such separate committees). Finally, research that does not qualify for exempt or expedited review is **greater than minimal risk research** must be reviewed by the full board of IRB members.

Ethics Codes

Offered by the Office of Human Subjects Research at the National Institutes of Health, the [Ethical Codes and Research Standards](#) allows you to read the ethics codes discussed in this section in their entirety. They are all highly recommended and, with the exception of the Federal Policy, short and easy to read.

- The Nuremberg Code
- The Declaration of Helsinki
- The Belmont Report
- Federal Policy for the Protection of Human Subjects

APA Ethics Code

The APA's *Ethical Principles of Psychologists and Code of*

Conduct (also known as the **APA Ethics Code**) was first published in 1953 and has been revised several times since then, most recently in 2010. It includes about 150 specific ethical standards that psychologists and their students are expected to follow. Much of the APA Ethics Code concerns the clinical practice of psychology—advertising one’s services, setting and collecting fees, having personal relationships with clients, and so on. For our purposes, the most relevant part is *Standard 8: Research and Publication*. Although *Standard 8* is reproduced here in its entirety, we will consider some of its most important aspects—informed consent, deception, debriefing, the use of nonhuman animal subjects, and scholarly integrity—in more detail.

APA Ethics Code Standard 8: Research and Publication

8.01 Institutional Approval

When institutional approval is required, psychologists provide accurate information about their research proposals and obtain approval prior to conducting the research. They conduct the research in accordance with the approved research protocol.

8.02 Informed Consent to Research

- When obtaining informed consent as required in Standard 3.10, Informed Consent, psychologists inform participants about (1) the purpose of the research, expected duration, and procedures; (2) their right to decline to participate and to withdraw from the research once participation has begun; (3) the foreseeable consequences of declining or withdrawing; (4) reasonably foreseeable factors that may be expected to influence their willingness to participate such as potential risks, discomfort, or adverse effects; (5) any prospective research benefits; (6) limits of confidentiality; (7) incentives for participation; and (8) whom to contact for questions about the research and research participants' rights. They provide opportunity for the prospective participants to ask questions and receive answers. (See also Standards 8.03, Informed Consent for Recording

Voices and Images in Research; 8.05, Dispensing With Informed Consent for Research; and 8.07, Deception in Research.)

- Psychologists conducting intervention research involving the use of experimental treatments clarify to participants at the outset of the research (1) the experimental nature of the treatment; (2) the services that will or will not be available to the control group(s) if appropriate; (3) the means by which assignment to treatment and control groups will be made; (4) available treatment alternatives if an individual does not wish to participate in the research or wishes to withdraw once a study has begun; and (5) compensation for or monetary costs of participating including, if appropriate, whether reimbursement from the participant or a third-party payor will be sought. (See also Standard 8.02a, Informed Consent to Research.)

8.03 Informed Consent for Recording Voices and Images in Research

Psychologists obtain informed consent from research participants prior to recording their voices or images for data collection unless (1) the research consists solely of naturalistic observations in public places, and it is not anticipated that the recording will be used in a manner that could cause personal identification or harm, or (2) the research design includes deception, and consent for the use of the recording is obtained during debriefing. (See also Standard 8.07, Deception in Research.)

8.04 Client/Patient, Student, and Subordinate Research Participants

1. When psychologists conduct research with clients/patients, students, or subordinates as participants, psychologists take steps to protect the prospective participants from adverse consequences of declining or withdrawing from participation.

2. When research participation is a course requirement or an opportunity for extra credit, the prospective participant is given the choice of equitable alternative activities.

8.05 Dispensing With Informed Consent for Research

Psychologists may dispense with informed consent only (1) where research would not reasonably be assumed to create distress or harm and involves (a) the study of normal educational practices, curricula, or classroom management methods conducted in educational settings; (b) only anonymous questionnaires, naturalistic observations, or archival research for which disclosure of responses would not place participants at risk of criminal or civil liability or damage their financial standing, employability, or reputation, and confidentiality is protected; or (c) the study of factors related to job or organization effectiveness conducted in organizational settings for which there is no risk to participants' employability, and confidentiality is protected or

(2) where otherwise permitted by law or federal or institutional regulations.

8.06 Offering Inducements for Research Participation

1. Psychologists make reasonable efforts to avoid offering excessive or inappropriate financial or other inducements for research participation when such inducements are likely to coerce participation.
2. When offering professional services as an inducement for research participation, psychologists clarify the nature of the services, as well as the risks, obligations, and limitations. (See also Standard 6.05, Barter With Clients/Patients.)

8.07 Deception in Research

1. Psychologists do not conduct a study involving deception unless they have determined that the use of deceptive techniques is justified by the study's significant prospective scientific, educational, or applied value and that

effective nondeceptive alternative procedures are not feasible.

2. Psychologists do not deceive prospective participants about research that is reasonably expected to cause physical pain or severe emotional distress.
3. Psychologists explain any deception that is an integral feature of the design and conduct of an experiment to participants as early as is feasible, preferably at the conclusion of their participation, but no later than at the conclusion of the data collection, and permit participants to withdraw their data. (See also Standard 8.08, Debriefing.)

8.08 Debriefing

1. Psychologists provide a prompt opportunity for participants to obtain appropriate information about the nature, results, and conclusions of the research, and they take reasonable steps to correct any misconceptions that participants may have of which the psychologists are

aware.

2. If scientific or humane values justify delaying or withholding this information, psychologists take reasonable measures to reduce the risk of harm.
3. When psychologists become aware that research procedures have harmed a participant, they take reasonable steps to minimize the harm.

8.09 Humane Care and Use of Animals in Research

1. Psychologists acquire, care for, use, and dispose of animals in compliance with current federal, state, and local laws and regulations, and with professional standards.
2. Psychologists trained in research methods and experienced in the care of laboratory animals supervise all procedures involving animals and are responsible for ensuring appropriate consideration of their comfort, health, and humane treatment.
3. Psychologists ensure that all individuals

under their supervision who are using animals have received instruction in research methods and in the care, maintenance, and handling of the species being used, to the extent appropriate to their role. (See also Standard 2.05, Delegation of Work to Others.)

4. Psychologists make reasonable efforts to minimize the discomfort, infection, illness, and pain of animal subjects.
5. Psychologists use a procedure subjecting animals to pain, stress, or privation only when an alternative procedure is unavailable and the goal is justified by its prospective scientific, educational, or applied value.
6. Psychologists perform surgical procedures under appropriate anesthesia and follow techniques to avoid infection and minimize pain during and after surgery.
7. When it is appropriate that an animal's life be terminated, psychologists proceed rapidly, with an effort to minimize pain and in accordance with accepted

procedures.

8.10 Reporting Research Results

- Psychologists do not fabricate data. (See also Standard 5.01a, Avoidance of False or Deceptive Statements.)
- If psychologists discover significant errors in their published data, they take reasonable steps to correct such errors in a correction, retraction, erratum, or other appropriate publication means.

8.11 Plagiarism

Psychologists do not present portions of another's work or data as their own, even if the other work or data source is cited occasionally.

8.12 Publication Credit

- Psychologists take responsibility and credit, including authorship credit, only for work they have actually performed or to which they have substantially contributed. (See also Standard 8.12b, Publication

Credit.)

- Principal authorship and other publication credits accurately reflect the relative scientific or professional contributions of the individuals involved, regardless of their relative status. Mere possession of an institutional position, such as department chair, does not justify authorship credit. Minor contributions to the research or to the writing for publications are acknowledged appropriately, such as in footnotes or in an introductory statement.
- Except under exceptional circumstances, a student is listed as principal author on any multiple-authored article that is substantially based on the student's doctoral dissertation. Faculty advisors discuss publication credit with students as early as feasible and throughout the research and publication process as appropriate. (See also Standard 8.12b, Publication Credit.)

8.13 Duplicate Publication of Data

Psychologists do not publish, as original data, data that have been previously published. This does not preclude republishing data when they are accompanied by proper acknowledgment.

8.14 Sharing Research Data for Verification

1. After research results are published, psychologists do not withhold the data on which their conclusions are based from other competent professionals who seek to verify the substantive claims through reanalysis and who intend to use such data only for that purpose, provided that the confidentiality of the participants can be protected and unless legal rights concerning proprietary data preclude their release. This does not preclude psychologists from requiring that such individuals or groups be responsible for costs associated with the provision of such information.

2. Psychologists who request data from other psychologists to verify the substantive claims through reanalysis may use shared data only for the declared purpose. Requesting psychologists obtain prior written agreement for all other uses of the data.

8.15 Reviewers

Psychologists who review material submitted for presentation, publication, grant, or research proposal review respect the confidentiality of and the proprietary rights in such information of those who submitted it.

Source: You can read the [full APA Ethics Code](#) on APA's website.

Informed Consent

Standards 8.02 to 8.05 are about informed consent. Again, informed consent means obtaining and documenting people's agreement to participate in a study, having informed them of everything that might reasonably be expected to affect their

decision. This includes details of the procedure, the risks and benefits of the research, the fact that they have the right to decline to participate or to withdraw from the study, the consequences of doing so, and any legal limits to confidentiality. For example, some U.S. states require researchers who learn of child abuse or other crimes to report this information to authorities.

Although the process of obtaining informed consent often involves having participants read and sign a **consent form**, it is important to understand



that this is not all it is. Although having participants read and sign a consent form might be enough when they are competent adults with the necessary ability and motivation, many participants do not actually read consent forms or read them but do not understand them. For example, participants often mistake consent forms for legal documents and mistakenly believe that by signing them they give up their right to sue the researcher (Mann, 1994).⁹ Even with competent adults, therefore, it is good practice to tell participants about

9. Mann, T. (1994). Informed consent for psychological research: Do subjects comprehend consent forms and understand their legal rights? *Psychological Science*, 5, 140–143.

the risks and benefits, demonstrate the procedure, ask them if they have questions, and remind them of their right to withdraw at any time—in addition to having them read and sign a consent form.

Note also that there are situations in which informed consent is not necessary. These include situations in which the research is not expected to cause any harm and the procedure is straightforward or the study is conducted in the context of people's ordinary activities. For example, if you wanted to sit outside a public building and observe whether people hold the door open for people behind them, you would not need to obtain their informed consent. Similarly, if a college instructor wanted to compare two legitimate teaching methods across two sections of his research methods course, he would not need to obtain informed consent from his students.

Deception

Deception of participants in psychological research can take a variety of forms: misinforming participants about the purpose of a study, using confederates, using phony equipment like Milgram's shock generator, and presenting participants with false feedback about their performance (e.g., telling them they did poorly on a test when they actually did well). Deception also includes not informing participants of the full design or true purpose of the research even if they are not actively

misinformed (Sieber, Iannuzzo, & Rodriguez, 1995).¹⁰ For example, a study on incidental learning—learning without conscious effort—might involve having participants read through a list of words in preparation for a “memory test” later. Although participants are likely to assume that the memory test will require them to recall the words, it might instead require them to recall the contents of the room or the appearance of the research assistant.

Some researchers have argued that deception of research participants is rarely if ever ethically justified. Among their arguments are that it prevents participants from giving truly informed consent, fails to respect their dignity as human beings, has the potential to upset them, makes them distrustful and therefore less honest in their responding, and damages the reputation of researchers in the field (Baumrind, 1985).¹¹

Note, however, that the APA Ethics Code takes a more moderate approach—allowing deception when the benefits of the study outweigh the risks, participants cannot reasonably be expected to be harmed, the research question cannot be answered without the use of deception, and participants are informed about the deception as soon as possible. This

10. Sieber, J. E., Iannuzzo, R., & Rodriguez, B. (1995). Deception methods in psychology: Have they changed in 23 years? *Ethics & Behavior*, 5, 67–85.

11. Baumrind, D. (1985). Research using intentional deception: Ethical issues revisited. *American Psychologist*, 40, 165–174.

approach acknowledges that not all forms of deception are equally bad. Compare, for example, Milgram's study in which he deceived his participants in several significant ways that resulted in their experiencing severe psychological stress with an incidental learning study in which a "memory test" turns out to be slightly different from what participants were expecting. It also acknowledges that some scientifically and socially important research questions can be difficult or impossible to answer without deceiving participants. Knowing that a study concerns the extent to which they obey authority, act aggressively toward a peer, or help a stranger is likely to change the way people behave so that the results no longer generalize to the real world.

Debriefing

Standard 8.08 is about **debriefing**. This is the process of informing research participants as soon as possible of the purpose of the study, revealing any deception, and correcting any other misconceptions they might have as a result of participating. Debriefing also involves minimizing harm that might have occurred. For example, an experiment on the effects of being in a sad mood on memory might involve inducing a sad mood in participants by having them think sad thoughts, watch a sad video, and/or listen to sad music. Debriefing would be the time to return participants' moods to

normal by having them think happy thoughts, watch a happy video, or listen to happy music.

Nonhuman Animal Subjects

Standard 8.09 is about the humane treatment and care of nonhuman animal subjects. Although most contemporary research in psychology does not involve nonhuman animal subjects, a significant minority of it does—especially in the study of learning and conditioning, behavioral neuroscience, and the development of drug and surgical therapies for psychological disorders.

The use of nonhuman animal subjects in psychological research is similar to the use of deception in that there are those who argue that it is rarely, if ever, ethically acceptable (Bowd & Shapiro, 1993).¹² Clearly, nonhuman animals are incapable of giving informed consent. Yet they can be subjected to numerous procedures that are likely to cause them suffering. They can be confined, deprived of food and water, subjected to pain, operated on, and ultimately euthanized. (Of course, they can also be observed benignly in natural or zoo-like settings.) Others point out that psychological research on nonhuman animals has resulted in many important benefits to humans,

12. Bowd, A. D., & Shapiro, K. J. (1993). The case against animal laboratory research in psychology. *Journal of Social Issues*, 49, 133–142.

including the development of behavioral therapies for many disorders, more effective pain control methods, and antipsychotic drugs (Miller, 1985).¹³ It has also resulted in benefits to nonhuman animals, including alternatives to shooting and poisoning as means of controlling them.

As with deception, the APA acknowledges that the benefits of research on nonhuman animals can outweigh the costs, in which case it is ethically acceptable. However, researchers must use alternative methods when they can. When they cannot, they must acquire and care for their subjects humanely and minimize the harm to them. For more information on the APA's position on nonhuman animal subjects, see the website of the [APA's Committee on Animal Research and Ethics](#).

Scholarly Integrity



Standards 8.10 to 8.15 are about scholarly integrity. These include the obvious points that researchers must not fabricate data or plagiarize. Plagiarism means using others' words or ideas

13. Miller, N. E. (1985). The value of behavioral research on animals. *American Psychologist*, 40, 423–440.

without proper acknowledgment. Proper acknowledgment generally means indicating direct quotations with quotation marks and providing a citation to the source of any quotation or idea used. Self-plagiarism is also considered unethical and refers to publishing the same material more than once. In other words, researchers should not borrow prior phrasing from their other published works, just as students should not submit the same work to more than one class.

The remaining standards make some less obvious but equally important points. Researchers should not publish the same data a second time as though it were new; they should share their data with other researchers, and as peer reviewers, they should keep the unpublished research they review confidential. Note that the authors' names on published research—and the order in which those names appear—should reflect the importance of each person's contribution to the research. It would be unethical, for example, for a faculty member to make themselves the first author on research that was largely conducted by a student.

Equity Activity: Ethical considerations in researching homeless populations

A researcher is conducting a study on mental health challenges faced by individuals experiencing

homelessness. To recruit participants, the researcher collaborates with local shelters and outreach organizations. However, ethical concerns arise:

- **Informed Consent:** Many participants have limited literacy or access to legal knowledge, so the researcher uses visual aids, simplified language, and verbal consent options to ensure understanding.
- **Confidentiality:** Protecting participants' identities is critical, especially when sharing findings with policymakers or the public. Data is anonymized, and any identifying details (e.g., shelter locations) are excluded from publications.



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=41#h5p-7>

References

Holmes, A. G. D. (2020). Researcher Positionality—A Consideration of Its Influence and Place in Qualitative Research—A New Researcher Guide. *Shanlax International Journal of Education*, 8(4), 1-10.

Mott, C., & Cockayne, D. (2017). Citation matters: mobilizing the politics of citation toward a practice of 'conscientious engagement'. *Gender, Place & Culture*, 24(7), 954-973. <https://doi.org/10.1080/0966369X.2017.1339022>

Resource

Gammons, R. (2023, May 22). *Diversity, Equity, and Inclusion in Research: Citation Justice*. University of Maryland Libraries: Research Guides. <https://lib.guides.umd.edu/ResearchEquity/CitationJustice>

License & Attribution

“Research Ethics” by Mireille Ukeye is adapted from “[Research Ethics](#)” by Rajiv S. Jhangiani; I-Chant A. Chiang; Carrie Cuttler; and Dana C. Leighton is licensed CC BY 4.0.

“Research Ethics” is licensed under CC BY-NC-SA 4.0.

PART II

RESEARCH PRACTICES

8.

SCIENTIFIC METHOD

Makenzie O'Neil

Learning Objectives

By the end of this chapter, you will be able to

1. Define science and describe the three fundamental features of science.
2. Explain why psychology is a science.
3. Describe a general model of scientific research in psychology and give specific examples that fit the model.

Many people believe that women tend to talk more than men—with some even suggesting that this difference has a

biological basis. One widely cited estimate is that women speak 20,000 words per day on average and men speak only 7,000. This claim seems plausible, but is it *true*? A group of psychologists led by Matthias Mehl decided to find out. They checked to see if anyone had actually tried to count the daily number of words spoken by women and men. No one had. So these researchers conducted a study in which female and male college students (369 in all) wore audio recorders while they went about their lives. The result? The women spoke an average of 16,215 words per day and the men spoke an average of 15,669—an extremely small difference that could easily be explained by chance. In an article in the journal *Science*, these researchers summed up their findings as follows: “We therefore conclude, on the basis of available empirical evidence, that the widespread and highly publicized stereotype about female talkativeness is unfounded” (Mehl, Vazire, Ramirez-Esparza, Slater, & Pennebaker, 2007, p. 82)¹. Psychology is usually defined as the scientific study of human behavior and mental processes, and this example illustrates the features that make it scientific. mpri

1. ²

2. [1]

What Is Science?

Some people are surprised to learn that psychology is a **science**. They generally agree that astronomy, biology, and chemistry are sciences but wonder what psychology has in common with these other fields. Before answering this question, however, it is worth reflecting on what astronomy, biology, and chemistry have in common with *each other*. It is clearly not their subject matter. Astronomers study celestial bodies, biologists study living organisms, and chemists study matter and its properties. It is also not the equipment and techniques that they use. Few biologists would know what to do with a radio telescope, for example, and few chemists would know how to track a moose population in the wild. For these and other reasons, philosophers and scientists who have thought deeply about this question have concluded that what the sciences have in common is a general approach to understanding the natural world. Psychology is a science because it takes this same general approach to understanding one aspect of the natural world: human behavior.

Features of Science

The general scientific approach has three fundamental features

(Stanovich, 2010)³. The first is **systematic empiricism**. Empiricism refers to learning based on observation, and scientists learn about the natural world systematically, by carefully planning, making, recording, and analyzing observations of it. As we will see, logical reasoning and even creativity play important roles in science, too, but scientists are unique in their insistence on checking their ideas about the way the world is against their systematic observations. Notice, for example, that Mehl and his colleagues did not trust other people's stereotypes or even their own informal observations. Instead, they systematically recorded, counted, and compared the number of words spoken by a large sample of women and men. Furthermore, when their systematic observations turned out to conflict with people's stereotypes, they trusted their systematic observations.

The second feature of the scientific approach—which follows in a straightforward way from the first—is that it is concerned with **empirical questions**. These are questions about the way the world actually is and, therefore, can be answered by systematically observing it. The question of whether women talk more than men is empirical in this way. Either women really do talk more than men, or they do not, and this can be determined by systematically observing how

3. Stanovich, K. E. (2010). *How to think straight about psychology* (9th ed.). Boston, MA: Allyn & Bacon.

much women and men actually talk. Having said this, there are many interesting and important questions that are not empirically testable and that science is not in a position to answer. Among these are questions about values—whether things are good or bad, just or unjust, or beautiful or ugly, and how the world *ought* to be. So although the question of whether a stereotype is accurate or inaccurate is an empirically testable one that science can answer, the question—or, rather, the value judgment—of whether it is wrong for people to hold inaccurate stereotypes is not. Similarly, the question of whether criminal behavior has a genetic basis is an empirical question, but the question of what actions ought to be considered illegal is not. It is especially important for researchers in psychology to be mindful of this distinction.

The third feature of science is that it creates **public knowledge**. After asking their empirical questions, making their systematic observations, and drawing their conclusions, scientists publish their work. This usually means writing an article for publication in a professional journal, in which they put their research question in the context of previous research, describe in detail the methods they used to answer their question, and clearly present their results and conclusions. Increasingly, scientists are opting to publish their work in open-access journals, in which the articles are freely available to all—scientists and nonscientists alike. This important choice allows publicly funded research to create knowledge that is truly public.

Publication is an essential feature of science for two reasons. One is that science is a social process—a large-scale collaboration among many researchers distributed across both time and space. Our current scientific knowledge of most topics is based on many different studies conducted by many different researchers who have shared their work publicly over many years. The second is that publication allows science to be self-correcting. Individual scientists understand that, despite their best efforts, their methods can be flawed and their conclusions incorrect. Publication allows others in the scientific community to detect and correct these errors so that, over time, scientific knowledge increasingly reflects the way the world actually is.

A good example of the self-correcting nature of science is the [“Many Labs Replication Project”](#) [\[New Tab\]](#) – a large and coordinated effort by prominent psychological scientists around the world to attempt to replicate findings from 13 classic and contemporary studies (Klein et al., 2013)⁴. One of the findings selected by these researchers for replication was the fascinating effect, first reported by Simone Schnall and her colleagues at the University of Plymouth, that washing one’s

4. Klein, R. A., Ratliff, K. A., Vianello, M., Adams, R. B., Bahník, S., Bernstein, M. J., . . . Nosek, B. A. (2013). Investigating variation in replicability: A “many labs” replication project. *Social Psychology*, 45(3), 142-152. doi: 10.1027/1864-9335/a000178

hands leads people to view moral transgressions—ranging from keeping money inside a found wallet to using a kitten for sexual arousal—as less wrong (Schnall, Benton, & Harvey, 2008)⁵. If reliable, this effect might help explain why so many religious traditions associate physical cleanliness with moral purity. However, despite using the same materials and nearly identical procedures with a much larger sample, the “Many Labs” researchers were unable to replicate the original finding (Johnson, Cheung, & Donnellan, 2013)⁶, suggesting that the original finding may have stemmed from the relatively small sample size (which can lead to unreliable results) used in the original study. To be clear, at this stage we are still unable to definitively conclude that the handwashing effect does not exist; however, the effort that has gone into testing its reliability certainly demonstrates the collaborative and cautious nature of scientific progress.

For more on this, see the Noba Project’s chapter on [The Replication Crisis in Psychology](#).

5. Schnall, S., Benton, J., & Harvey, S. (2008). With a clean conscience: Cleanliness reduces the severity of moral judgments. *Psychological Science*, 19(12), 1219-1222. doi: 10.1111/j.1467-9280.2008.02227.x

6. Johnson, D. J., Cheung, F., & Donnellan, M. B. (2013). Does cleanliness influence moral judgments? A direct replication of Schnall, Benton, and Harvey (2008). *Social Psychology*, 45(3), 209-215. doi: 10.1027/1864-9335/a000186

Overview of the Scientific Method

Here is the abstract of a 2014 article in the journal *Psychological Science*.

Taking notes on laptops rather than in longhand is increasingly common. Many researchers have suggested that laptop note taking is less effective than longhand note taking for learning. Prior studies have primarily focused on students' capacity for multitasking and distraction when using laptops. The present research suggests that even when laptops are used solely to take notes, they may still be impairing learning because their use results in shallower processing. In three studies, we found that students who took notes on laptops performed worse on conceptual questions than students who took notes longhand. We show that whereas taking more notes can be beneficial, laptop note takers' tendency to transcribe lectures verbatim rather than processing information and reframing it in their own words is detrimental to learning. (Mueler & Oppenheimer, 2014, p. 1159)⁷

In this abstract, the researcher has identified a research question—about the effect of taking notes on a laptop on learning—and identified why it is worthy of

7.⁸

8. [1]

investigation—because the practice is ubiquitous and may be harmful to learning. In this chapter, we give you a broad overview of the various stages of the research process, which will then be expanded upon throughout the rest of this textbook. These stages include finding a topic of investigation, reviewing the literature, refining your research question and generating a hypothesis, designing and conducting an empirical study, analyzing the data, coming to conclusions, and reporting the results.

A Model of Scientific Research in Psychology

Figure 8.1 presents a simple model of scientific research in psychology, which will be expanded on further in [Chapter 12](#). The researchers formulate a research question, conduct an empirical study designed to answer the question, analyze the resulting data, draw conclusions about the answer to the question, and publish the results so that they become part of the research literature (i.e., all the published research in that field). Because the research literature is one of the primary sources of new research questions, this process can be thought of as a cycle. New research leads to new questions, which lead to new research, and so on. Figure 8.1 also indicates that research questions can originate outside of this cycle, either with informal observations or with practical problems that need to be solved. But even in these cases, the researcher would

start by checking the research literature to see if the question had already been answered and to refine it based on what previous research had already found.



Figure 8.1 In this simple model of scientific research in Psychology, informal observations lead to a research question, followed by an empirical study, data analysis, and conclusions that feed the research literature and refine future questions.

The research by Mehl and his colleagues is described nicely by this model. Their research question—whether women are more talkative than men—was suggested to them both by people’s stereotypes and by claims published in the research literature about the relative talkativeness of women and men.

When they checked the research literature, however, they found that this question had not been adequately addressed in scientific studies. They then conducted a careful empirical study, analyzed the results (finding very little difference between women and men), formed their conclusions, and published their work so that it became part of the research literature. The publication of their article is not the end of the story, however, because their work suggests many new questions (about the reliability of the result, about potential cultural differences, etc.) that will likely be taken up by them and by other researchers inspired by their work.



One or more interactive elements has been excluded from this version of the text. You can view them online here:

<https://smc.pressbooks.pub/critresearchmethodspych/?p=45#oembed-1>

Video 8.1 [Studying Brains As We Age](#) by [American Psychological Association](#)

As another example, consider a population health question around disparities in dementia onset based on racial differences. Many psychologists decided to tackle this question scientifically and have documented the presence of these differences across time and geographical location (Shiekh et al.,

2021). Dr. Manly and her research team wanted to understand why this phenomenon was occurring and used the scientific process to collect data and interpret it. She was also intentional about recognizing that Race is really a proxy for Racism, and her findings revealed that many of her African American participants had experienced limited or interrupted schooling due to segregation policies. When she compared dementia rates with other individuals with similar interrupted learning experience there ended up being no racial differences (Manly & Espino, 2004⁹). Each new study was published and became part of the growing research literature on this topic. For instance, housing segregation and income inequality were also later found to influence the onset of dementia (Pohl et al., 2021).

Equity Activity: Implementing the Scientific Method

Based on the video above where Dr. Manly's discusses her research, walk through how she

9. Manly, J. J., & Espino, D. V. (2004). Cultural influences on dementia recognition and management. *Clinics in geriatric medicine*, 20(1), 93-119.

implemented the particular steps of the scientific method.



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=45#h5p-8>

License & Attribution

“Scientific Method” by Makenzie O’Neil is adapted from “[Understanding Science](#)” and “[A Model of Scientific Research in Psychology](#)” by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler, and Dana C. Leighton, which is licensed CC BY-NC-SA 4.0.

“[Understanding driver distraction](#)” and “[Studying](#)”

[brains as we age](#)” by [American Psychological Association](#). Standard YouTube License.

“Scientific Method” is licensed under CC BY-NC-SA 4.0.

9.

POPULATIONS & SAMPLES

Stephanie D'Costa and Makenzie O'Neil

Learning Objectives

By the end of this chapter, you will be able to

1. Describe the difference between a sample and a population.
2. Identify basic sampling methods.
3. Describe how intentionality in sampling can lead to more equitable research.

We are usually interested in understanding a specific group of people. This group is known as the population of interest, or

simply the population. The population is the collection of all people who have some characteristic in common; it can be as broad as “all people” if we have a very general research question about human psychology, or it can be extremely narrow, such as “all freshmen psychology majors at Midwestern public universities” if we have a specific group in mind.

Samples

In statistics, we often rely on a sample—that is, a small subset of a larger set of data—to draw inferences about the larger set. The larger set is known as the population from which the sample is drawn.

Example #1

You have been hired by the National Election Commission to examine how the American people feel about the fairness of the voting procedures in the U.S.

Whom will you ask?

It is not practical to ask every single American how he or she feels about the fairness of the voting procedures. Instead, we query a relatively small number of Americans and draw inferences about the entire country from their responses. The Americans actually queried constitute our sample of the larger population of all Americans.

A sample is typically a small subset of the population. In the case of voting attitudes, we would sample a few thousand Americans drawn from the hundreds of millions that make up the country. In choosing a sample, it is therefore crucial that it not over-represent one kind of citizen at the expense of others. For example, something would be wrong with our sample if it happened to be made up entirely of Florida residents. If the sample held only Floridians, it could not be used to infer the attitudes of other Americans. The same problem would arise if the sample were comprised only of Republicans. Inferences from statistics are based on the assumption that sampling is representative of the population. If the sample is not representative, then the possibility of sampling bias occurs. Sampling bias means that our conclusions apply only to our sample and are not generalizable to the full population.

Sampling bias is an important consideration for researchers who seek to engage in social justice aligned research practices. Historically, many research samples comprised of white men but the findings from these studies were applied to individuals across racial groups. One key example was the creation of the body mass index (BMI) and its connection to health access.

The original research to develop BMI was done in Europe with white men and women. Further research was done in the US in the 1970s but the sample size was still not racial diverse.¹ Nevertheless, the findings were implemented across many racial and ethnic groups, globally. The implications of BMI have led to difficulties with health and insurance access for many racially marginalized communities. It highlights the importance of considering our sampling procedures to ensure that the generalizability of the findings are accurate.

Example #2

We are interested in examining how many math classes have been taken, on average, by current graduating seniors at American colleges and universities during their four years in school.

Whereas our population in example one included all U.S.

1. Pray, R., Riskin, S., & Riskin, S. I. (2023). The history and faults of the body mass index and where to look next: a literature review. *Cureus*, 15(11).

citizens, now it involves just the graduating seniors throughout the country. This is still a large set since there are thousands of colleges and universities, each enrolling many students. (New York University, for example, enrolls 48,000 students.) It would be prohibitively costly to examine the transcript of every college senior. We therefore take a sample of college seniors and then make inferences to the entire population based on what we find. To make the sample, we might first choose some public and private colleges and universities across the United States. Then we might sample 50 students from each of these institutions. Suppose that the average number of math classes taken by the people in our sample was 3.2. We might speculate that 3.2 approximates the number we would find if we had the resources to examine every senior in the entire population. But we must be careful about the possibility that our sample is non-representative of the population. Perhaps we chose an overabundance of math majors, or chose too many technical institutions that have heavy math requirements. Such bad sampling makes our sample unrepresentative of the population of all seniors.

To solidify your understanding of sampling bias, consider the following examples. Try to identify the population and the sample, and then reflect on whether the sample is likely to yield the information desired.

Example #3

A substitute teacher wants to know how students in the class did on their last test. The teacher asks the ten students sitting in the front row to state their latest test scores. He concludes from their report that the class did extremely well.

What is the sample? What is the population? Can you identify any problems with choosing the sample in the way that the teacher did?

In Example #3, the population consists of all students in the class. The sample is made up of just the ten students sitting in the front row. The sample is not likely to be representative of the population. Those who sit in the front row tend to be more interested in the class and tend to perform higher on tests. Hence, the sample may perform at a higher level than the population.

Example #4

A coach is interested in how many cartwheels the average college freshman at his university can do. Eight volunteers from the freshman class step forward. After observing their performance, the coach concludes that college freshmen can do an average of 16 cartwheels in a row without stopping.

In Example #4, the population is the class of all freshmen at the coach's university. The sample is composed of 8 volunteers. The sample is poorly chosen because volunteers are more likely to be able to do cartwheels than the average freshman; people who can't do cartwheels probably did not volunteer! In the example, we are also not told of the gender of the volunteers. Were they all women, for example? That might affect the outcome, contributing to the non-representative nature of the sample (if the school is co-ed).

Sampling Techniques

Next, we need to ask how the sample was selected. The first

question to ask about a sample is whether it was chosen at random. Moore, Notz & Fligner (2013)[4] provide two reasons to choose random sampling. The first reason is to eliminate bias in selecting samples from the list of available individuals. The second reason to use random sampling is that the laws of probability allow trustworthy inference about the population.

If every population member is given an equal chance of sample selection, a random sampling method is used. This is known as probability sampling. Otherwise, a nonrandom or sampling with non-probability is employed.

Probability Sampling

Probability Sampling occurs when the researcher can specify the probability that each member of the population will be selected for the sample. It gives the likelihood of the sample being representative of the population, and is the preferred sampling technique for conducting research, although it can be more difficult to do. There are several types of probability sampling techniques

Simple Random Sampling

Simple Random Sampling (SRS): a reliable method of obtaining information where every single member of a population is chosen randomly, merely by chance. Each person

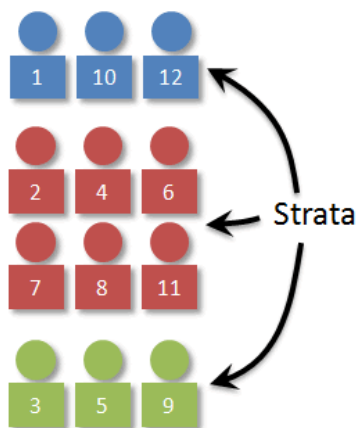
has the same probability of being chosen to be a part of a sample. An SRS of size n individuals from the population is chosen in such a way that every set of n individuals has an equal chance to be the sample actually selected. A specific advantage of simple random sampling is that it is the most straightforward method of probability sampling. A disadvantage is that you may not find enough individuals with your characteristic of interest, especially if that characteristic is uncommon.

Sometimes it is not feasible to build a sample using simple random sampling. To see the problem, consider the fact that both Dallas and Houston competed to be hosts of the 2012 Olympics. Imagine that you had been hired to assess whether most Texans preferred Houston to Dallas as the host, or the reverse. Given the impracticality of obtaining the opinion of every single Texan, you had to construct a sample of the Texas population. But notice how difficult it would have been to proceed by simple random sampling. For example, how would you have contacted those individuals who didn't vote and didn't have a phone? Even among people you found in the telephone book, how could you have identified those who had just relocated to another state (and had no reason to inform you of their move)? What would you have done about the fact that, since the beginning of the study, an additional 4,212 people took up residence in the state of Texas? As you can see, it is sometimes very difficult to develop a truly random

procedure. For this reason, other kinds of sampling techniques have been devised. We now discuss two of them.

Stratified Sampling

Since simple random sampling often does not ensure a representative sample, a sampling method called stratified random sampling is sometimes used to make the sample more representative of the population. This method can be used if the population has a number of distinct “strata” or groups. In stratified sampling, you first identify members of your sample who belong to each group. Then you randomly sample from each of those subgroups in such a way that the sizes of the subgroups in the sample are proportional to their sizes in the population.



Let's take an example: Suppose you were interested in views on capital punishment at an urban university. You have the time and resources to interview 200 students. The student body is diverse with respect to age; many older people work during the day and enroll in night courses

(average age is 39), while younger students generally enroll in day classes (average age of 19). It is possible that night students

have different views about capital punishment than day students. If 70% of the students were day students, it makes sense to ensure that 70% of the sample consisted of day students. Thus, your sample of 200 students would consist of 140 day students and 60 night students. The proportion of day students in the sample and in the population (the entire university) would be the same. Inferences to the entire population of students at the university would therefore be more secure.

Cluster Sampling

Cluster Sampling: a method where statisticians divide the entire population into clusters or sections representing a population. Demographic characteristics, such as race/ethnicity, gender, age, and zip code can be used to identify a cluster. Cluster sampling can be more efficient than simple random sampling, especially where a study takes place over a wide geographic region.

An extended version of cluster sampling is **multi-stage sampling**, where, in the first stage, the population is divided into clusters, and clusters are selected. At each subsequent stage, the selected clusters are further divided into smaller clusters. The process is completed until you get to the last step, where some members of each cluster are selected for the sample. Multi-stage sampling involves a combination of cluster and stratified sampling.

The U.S. Census Bureau uses multistage sampling by first

taking a simple random sample of counties in each state, then taking another simple random sample of households in each county, and collecting data on those households.

Sample Size Matters

Recall that the definition of a random sample is a sample in which every member of the population has an equal chance of being selected. This means that the sampling procedure, rather than the results of the procedure, defines what it means for a sample to be random. Random samples, especially if the sample size is small, are not necessarily representative of the entire population. For example, if a random sample of 20 subjects were taken from a population with an equal number of males and females, there would be a nontrivial probability (.06) that 70% or more of the sample would be female. Such a sample would not be representative, although it would be drawn randomly. Only a large sample size makes it likely that our sample is close to representative of the population. For this reason, inferential statistics take into account the sample size when generalizing results from samples to populations. In later chapters, you'll see what kinds of mathematical techniques ensure this sensitivity to sample size.

Non-Probability Sampling

Alternatively, **Non-probability sampling** occurs when the

researcher cannot specify the probability that each member of the population will be selected for the sample. Non-probability sampling cannot assure the representativeness of a sample as well as probability sampling. However, it tends to be simpler and cheaper for researchers to conduct, and thus psychological research more often involves non-probability sampling techniques.

Some examples of common forms non-probability sampling include: **Convenience sampling** (studying individuals who happen to be nearby and willing to participate); **Snowball sampling** (in which existing research participants help recruit additional participants for the study), and **self-selection sampling** (in which individuals choose to take part in the research on their own accord, without being approached by the researcher directly).

Equity Activity: Applying an equity lens to sampling

The National Survey of American Life (NSAL) is the most comprehensive and detailed study of mental disorders and the mental health of Americans of African descent. The study was conducted by the Program for Research on Black Americans (PRBA) within the Institute for Social Research at the

University of Michigan.² According to Jackson et al. (2004), the study includes “a large, nationally representative sample of African Americans, permitting an examination of the heterogeneity of experience across groups within this segment of the Black American population. Most prior research on Black Americans mental health has lacked adequate sample sizes to systematically address this within-race variation.

Special emphasis of the study is given to the nature of race and ethnicity within the Black population by selecting and interviewing national samples of African American (N=3,570) and Afro-Caribbean (N=1,623) immigrants, and second and older generation populations. National multi-stage probability methods were used in generating the samples. This means that the researcher divided the Americans of African descent into groups (clusters)

2. Jackson JS, Torres M, Caldwell CH, Neighbors HW, Nesse RM, Taylor RJ, Trierweiler SJ, & Williams DR (2004). The National Survey of American Life: A Study of Racial, Ethnic and Cultural Influences on Mental Disorders and Mental Health. *International Journal of Methods in Psychiatric Research*, Volume 13, Number 4

and then randomly sampled within those clusters. For generalizability, probability sampling methods were used in the study for stronger statistical inferences.



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=47#h5p-9>

License & Attribution

“Populations & Samples” by Stephanie D’Costa and Makenzie O’Neil is adapted from “[Inferential Statistics](#)” by Mikki Hebl and David Lane; “[Inferential Statistics: Sampling Methods](#)” by Yvonne Anthony under a [CC BY-NC-SA 4.0 license](#); and Conducting Surveys by Rajiv

Jhangiani, I-Chant A. Chiang, Carrie Cuttler, & Dana C. Leighton under a [CC BY-NC-SA 4.0 license](#).

“Populations & Samples” is licensed under [CC BY-NC-SA 4.0](#).

10.

VARIABLES

Rebecca Anguiano

Learning Objectives

By the end of this chapter, you will be able to

1. Define the concept of a variable, distinguish quantitative from categorical variables, and give examples of variables that might be of interest to psychologists.
2. Explain the differences between independent and dependent variables.
3. Understand the importance of thoughtfully operationalizing variables.

Identifying and Defining the Variables and Population

Variables and Operational Definitions



Figure 10.1 Overview of Variables

A **variable** is a quantity or quality that varies across people or situations. For example, the height of the students enrolled in a university course is a variable because it varies

from student to student. The chosen major of the students is also a variable, as long as not everyone in the class has declared the same major. Almost everything in our world varies, and as such, thinking of examples of constants (things that don't vary) is far more difficult. A rare example of a constant is the speed of light. Variables can be either quantitative or categorical. A **quantitative variable** is a quantity, such as height, that is typically measured by assigning a number to each individual. Other examples of quantitative variables include people's level of talkativeness, how depressed they are, and the number of siblings they have. A **categorical variable** is a quality, such as race or class, and is typically measured by assigning a category label to each individual (e.g., Indigenous, Black, White etc.). Other examples include people's

nationality, their occupation, and whether they are receiving psychotherapy.

After the researcher generates their hypothesis and selects the variables they want to manipulate and measure, the researcher needs to find ways to actually measure the variables of interest. This requires an **operational definition**—a definition of the variable in terms of precisely how it is to be measured. Most variables that researchers are interested in studying cannot be directly observed or measured, and this poses a problem because empiricism (observation) is at the heart of the scientific method. Operationally defining a variable involves taking an abstract construct like depression that cannot be directly observed and transforming it into something that can be directly observed and measured. Most variables can be operationally defined in many different ways. For example, depression can be operationally defined as people's scores on a paper-and-pencil depression scale, such as the Beck Depression Inventory, the number of depressive symptoms they are experiencing, or whether they have been diagnosed with major depressive disorder. Researchers are wise to choose an operational definition that has been used extensively in the research literature.

At the end of the chapter readers will have an opportunity to practice operationalizing equity in the context of academic achievement. Equity is hard to define, let alone operationalize, because it often depends on context, values, and power relations. One way to define equity is that it focuses on

fairness, not sameness, and aims to reduce systemic inequalities while accounting for historical and structural disadvantages. Appropriately operationalizing equity is critical if research is to be connected to social change.

Types of Variables

When conducting research, experimenters often manipulate variables. For example, an experimenter might compare the effectiveness of four types of antidepressants. In this case, the variable is “type of antidepressant.” When a variable is manipulated by an experimenter, it is called an **independent variable**. The experiment seeks to determine the effect of the independent variable on relief from depression. In this example, relief from depression is called a **dependent variable**. In general, the independent variable is manipulated by the experimenter, and its effects on the dependent variable are measured.

Example #1

Can blueberries slow aging? A study indicates that antioxidants found in blueberries may slow down

the process of aging. In this study, 19-month-old rats (equivalent to 60-year-old humans) were fed either their standard diet or a diet supplemented by either blueberry, strawberry, or spinach powder. After eight weeks, the rats were given memory and motor skills tests. Although all supplemented rats showed improvement, those supplemented with blueberry powder showed the most notable improvement.

1. What is the independent variable? (dietary supplement: none, blueberry, strawberry, and spinach)
2. What are the dependent variables? (memory test and motor skills test)

Example #2

Does beta-carotene protect against cancer? Beta-carotene supplements have been thought to protect against cancer. However, a study published in the Journal of the National Cancer Institute suggests this is false. The study was conducted with 39,000

women aged 45 and over. These women were randomly assigned to receive a beta-carotene supplement or a placebo, and their health was studied over their lifetime. Cancer rates for women taking the beta-carotene supplement did not differ systematically from the cancer rates of those women taking the placebo.

1. What is the independent variable? (supplements: beta-carotene or placebo)
2. What is the dependent variable? (occurrence of cancer)

Example #3

How bright is right? An automobile manufacturer wants to know how bright brake lights should be in order to minimize the time required for the driver of a following car to realize that the car in front is stopping and to hit the brakes.

1. What is the independent variable? (brightness of brake

- lights)
2. What is the dependent variable? (time to hit brakes)

Levels of an Independent Variable

If an experiment compares an experimental treatment with a control treatment, then the independent variable (type of treatment) has two levels: **experimental** and **control**. If an experiment were comparing five types of diets, then the independent variable (type of diet) would have 5 levels. In general, the number of levels of an independent variable is the number of experimental conditions.

Discrete and Continuous Variables

Variables such as the number of children in a household are called discrete variables since the possible scores are discrete points on the scale. For example, a household could have three children or six children, but not 4.53 children. Other variables, such as time to respond to a question, are **continuous variables** since the scale is continuous and not made up of discrete steps. The response time could be 1.64 seconds, or it could be 1.64237123922121 seconds. Of course, the practicalities of measurement preclude most measured variables from being truly continuous.

Equity Activity: Operationally defining equity

A researcher wants to understand how issues related systemic inequities impact reading achievement for elementary school children. Before doing so they need operationally define equity in the context of academics. So far, the research has identified the following factors to appropriately assess equity:

- Access to resources
- Student background
- Access to quality curriculum



An interactive HSP element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=49#h5p-10>

License & Attribution

“Variables” by Rebecca Anguiano is adapted from [“Designing a Research Study”](#) by Rajiv S.

Jhangiani, I-Chant A. Chiang, Carrie Cuttler, and Dana C. Leighton, which is licensed CC BY-NC-SA 4.0 and [“Introduction”](#) by Linda R. Cote Ph.D.; Rupa G. Gordon Ph.D., Chrislyn E. Randell Ph.D., Judy Schmitt, and Helena Marvin, which is licensed CC BY-NC-SA 4.0.

“Variables” is licensed under CC BY-NC-SA 4.0.

Media Attributions

- OER_Variables

11.

RESEARCH QUESTIONS

Mireille Ukeye

Learning Objectives

By the end of this chapter, you will be able to

1. Describe some techniques for turning research ideas into empirical research questions and use those techniques to generate questions.
2. Explain what makes a research question interesting and evaluate research questions in terms of their interestingness.
3. Differentiate between quantitative and qualitative research questions and identify

which type of question is most appropriate for a given research purpose.

Research Questions

Psychological research can be approached using quantitative methods, qualitative methods, or a combination of both. The goal of this chapter is not to prioritize one approach over another but to help you generate empirically grounded research questions that match your methodological intentions. Identifying whether a research question is quantitative, qualitative, or mixed-methods is crucial, because the type of question you ask directly shapes your study design, data collection, and analysis. In this section, we provide strategies for developing both quantitative and qualitative research questions and highlight how they differ in structure, purpose, and focus.

Generating Quantitative Research Questions

Quantitative research questions are framed in terms of variables that can be measured numerically and analyzed statistically. These questions often focus on the frequency,

intensity, or prevalence of a behavior or psychological characteristic, or on the relationships among variables. One effective strategy for generating quantitative questions is to review the discussion section of recent empirical articles, where researchers frequently identify unanswered questions or suggest directions for future studies.

You can also generate quantitative questions by thinking about a behavior or psychological construct as a variable and asking: How often does it occur? How strongly is it expressed? How is it related to other variables? If a question has already been studied, it can often be refined by exploring different ways of measuring the variables, examining new populations, or testing contextual influences. In this way, quantitative research questions extend prior work while contributing new insights.

Generating Qualitative Research Questions

Qualitative research questions are designed to explore meaning, experience, process, and context rather than numerical relationships. These questions are particularly useful when a phenomenon is not well understood, when researchers want to center participants' perspectives, or when the goal is to understand how individuals interpret their experiences. Unlike quantitative questions, qualitative

questions are open-ended and flexible, allowing patterns, themes, and insights to emerge from the data.

Qualitative research questions often begin with How, What, or In what ways, focusing on participants' lived experiences, perceptions, or interpretations. For example, rather than asking whether two variables are related, a qualitative question might explore how individuals experience a phenomenon or how they perceive changes over time. These questions are especially valuable in applied areas of psychology, where understanding context and nuance can inform practice, policy, and interventions.

Importantly, qualitative research questions are no less rigorous than quantitative ones. Their rigor depends on conceptual clarity, alignment with the research purpose, and the researcher's ability to collect rich, credible data that illuminate the phenomenon of interest.

Distinguishing Quantitative and Qualitative Research Questions

Table 1 illustrates key differences between quantitative and qualitative research questions. Comparing the two helps clarify how each type reflects different assumptions, goals, and analytic approaches.

Table 1: Examples of Quantitative and Qualitative Research Questions

Focus of Inquiry	Quantitative Research Question	Qualitative Research Question
Frequency or prevalence	How many words do people speak per day on average?	How do individuals describe their daily communication patterns?
Group differences	Is there a difference in talkativeness between same-sex and mixed-sex groups?	How do individuals experience communication in same-sex versus mixed-sex group settings?
Relationships between variables	Is family size related to adult talkativeness?	How do adults perceive the influence of family dynamics on their communication styles?
Applied psychological phenomena	Does taking handwritten notes improve exam performance compared to typing?	How do students describe their experiences and preferences when taking notes by hand versus typing?

By clearly distinguishing between quantitative and qualitative research questions, you can align your questions with the most appropriate methods and analysis strategies. Some studies may even integrate both approaches in a mixed-methods design, using quantitative questions to identify patterns and qualitative questions to provide depth and explanation. Regardless of the approach, well-formulated research questions are the foundation of meaningful and methodologically sound psychological research.

Generating Empirically Testable

Research Questions

Once you have a research idea, you need to use it to generate one or more empirically testable research questions, that is, questions expressed in terms of a single variable or a relationship between variables. One way to do this is to look closely at the discussion section in a recent research article on the topic. This is the last major section of the article, in which the researchers summarize their results, interpret them in the context of past research, and suggest directions for future research. These suggestions often take the form of specific research questions, which you can then try to answer with additional research. This can be a good strategy because it is likely that the suggested questions have already been identified as interesting and important by experienced researchers.

But you may also want to generate your own research questions. How can you do this? First, if you have a particular behavior or psychological characteristic



in mind, you can simply conceptualize it as a variable and ask how frequent or intense it is. How many words on average do people speak per day? How accurate are our memories of traumatic events? What percentage of people have sought professional help for depression? If the question has never been studied scientifically—which is something that you will

learn when you conduct your literature review—then it might be interesting and worth pursuing.

If scientific research has already answered the question of how frequent or intense the behavior or characteristic is, then you should consider turning it into a question about a relationship between that behavior or characteristic and some other variable. One way to do this is to ask yourself the following series of more general questions and write down all the answers you can think of.

- What are some possible causes of the behavior or characteristic?
- What are some possible effects of the behavior or characteristic?
- What types of people might exhibit more or less of the behavior or characteristic?
- What types of situations might elicit more or less of the behavior or characteristic?

In general, each answer you write down can be conceptualized as a second variable, suggesting a question about a relationship. If you were interested in talkativeness, for example, it might occur to you that a possible cause of this psychological characteristic is family size. Is there a relationship between family size and talkativeness? Or it might occur to you that people seem to be more talkative in same-sex groups than mixed-sex groups. Is there a difference in the average level of

talkativeness of people in same-sex groups and people in mixed-sex groups? This approach should allow you to generate many different empirically testable questions about almost any behavior or psychological characteristic.

If, through this process, you generate a question that has never been studied scientifically—which again is something that you will learn in your literature review—then it might be interesting and worth pursuing. But what if you find that it has been studied scientifically? Although novice researchers often want to give up and move on to a new question at this point, this is not necessarily a good strategy. For one thing, the fact that the question has been studied scientifically and the research published suggests that it is of interest to the scientific community. For another, the question can almost certainly be refined so that its answer will still contribute something new to the research literature. Again, asking yourself a series of more general questions about the relationship is a good strategy.

- Are there other ways to define and measure the variables?
- Are there types of people for whom the relationship might be stronger or weaker?
- Are there situations in which the relationship might be stronger or weaker, including situations with practical importance?

For example, research has shown that women and men speak

about the same number of words per day, but this was when talkativeness was measured in terms of the number of words spoken per day among university students in the United States and Mexico. We can still ask whether other ways of measuring talkativeness, perhaps the number of different people spoken to each day, produce the same result. Or we can ask whether studying elderly people or people from other cultures produces the same result. Again, this approach should help you generate many different research questions about almost any relationship.

Evaluating Research Questions

Researchers usually generate many more research questions than they ever attempt to answer. This means they must have some way of evaluating the research questions they generate so that they can choose which ones to pursue. In this section, we consider two criteria for evaluating research questions: the interestingness of the question and the feasibility of answering it.

Interestingness

How often do people tie their shoes? Do people feel pain when you punch them in the jaw? Are women more likely to wear makeup than men? Do people prefer vanilla or chocolate ice cream? Although it would be a fairly simple matter to design a

study and collect data to answer these questions, you probably would not want to because they are not interesting. We are not talking here about whether a research question is interesting to us personally but whether it is interesting to people more generally and, especially, to the scientific community. But what makes a research question interesting in this sense? Here we look at three factors that affect the **interestingness** of a research question: the answer is in doubt, the answer fills a gap in the research literature, and the answer has important practical implications.



First, a research question is interesting to the extent that its answer is in doubt. Obviously, questions that have been answered by

scientific research are no longer interesting as the subject of new empirical research. But the fact that a question has not been answered by scientific research does not necessarily make it interesting. There has to be some reasonable chance that the answer to the question will be something that we did not already know. But how can you assess this before actually collecting data? One approach is to try to think of reasons to expect different answers to the question—especially ones that seem to conflict with common sense. If you can think of reasons to expect at least two different answers, then the question might be interesting. If you can think of reasons to expect only one answer, then it probably is not. The question

of whether women are more talkative than men is interesting because there are reasons to expect both answers. The existence of the stereotype itself suggests the answer could be yes, but the fact that women's and men's verbal abilities are fairly similar suggests the answer could be no. The question of whether people feel pain when you punch them in the jaw is not interesting because there is absolutely no reason to think that the answer could be anything other than a resounding yes.

A second important factor to consider when deciding if a research question is interesting is whether answering it will fill a gap in the research literature. Again, this means in part that the question has not already been answered by scientific research. But it also means that the question is in some sense a natural one for people who are familiar with the research literature. For example, the question of whether taking lecture notes by hand can help improve students' exam performance would be likely to occur to anyone who was familiar with research on note taking and the ineffectiveness of shallow processing on learning.

A final factor to consider when deciding whether a research question is interesting is whether its answer has important practical implications. Again, the question of whether taking notes by hand improves learning has important implications for education, including classroom policies concerning technology use. The question of whether cell phone use impairs driving is interesting because it is relevant to the

personal safety of everyone who travels by car and to the debate over whether cell phone use should be restricted by law.

Feasibility

A second important criterion for evaluating research questions is the **feasibility** of successfully answering them. There are many factors that affect feasibility, including time, money, equipment and materials, technical knowledge and skill, and access to research participants. Clearly, researchers need to take these factors into account so that they do not waste time and effort pursuing research that they cannot complete successfully.

Looking through a sample of professional journals in psychology will reveal many studies that are complicated and difficult to carry out. These include longitudinal designs in which participants are tracked over many years, neuroimaging studies in which participants' brain activity is measured while they carry out various mental tasks, and complex non-experimental studies involving several variables and complicated statistical analyses. Keep in mind, though, that such research tends to be carried out by teams of highly trained researchers whose work is often supported in part by government and private grants. Also, keep in mind that research does not have to be complicated or difficult to produce interesting and important results. Looking through a sample of professional journals will also reveal studies that

are relatively simple and easy to carry out—perhaps involving a convenience sample of university students and a paper-and-pencil task.

A final point here is that it is generally good practice to use methods that have already been used successfully by other researchers. For example, if you want to manipulate people’s moods to make some of them happy, it would be a good idea to use one of the many approaches that have been used successfully by other researchers (e.g., paying them a compliment). This is good not only for the sake of feasibility—the approach is “tried and true”—but also because it provides greater continuity with previous research. This makes it easier to compare your results with those of other researchers and to understand the implications of their research for yours, and vice versa.

Equity Activity: Exploring educational disparities

A researcher investigates whether racial and economic disparities influence students’ access to Advanced Placement (AP) courses.

- Initial Question: “Why do some students take more AP courses than others?”

- Refined Question: “How do school funding levels and racial demographics correlate with access to Advanced Placement courses in urban high schools?”

Refining the question narrows the focus, making it measurable and directly connected to equity issues.

Classroom Activity: The instructor will provide students with broad questions and guide them in refining them using DEIA considerations (e.g., intersectionality).



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=51#h5p-11>

License & Attribution

“Research Questions” by Mireille Ukeye is adapted from “[Generating Good Research Questions](#)” by Rajiv S. Jhangiani; I-Chant A. Chiang; Carrie Cuttler; and Dana C. Leighton is licensed CC BY 4.0.

“Generating Good Research Questions” is licensed under CC BY-NC-SA 4.0.

12.

DEVELOPING A HYPOTHESIS

Stephanie D'Costa

Learning Objectives

By the end of this chapter, you will be able to

1. Distinguish between a theory and a hypothesis.
2. Discover how theories are used to generate hypotheses and how the results of studies can be used to further inform theories.
3. Describe the characteristics of a good hypothesis.
4. Create hypotheses that reflect the complexity

and diversity of human experiences.

Theories and Hypotheses

Before describing how to develop a hypothesis, it is important to distinguish between a theory and a hypothesis. A theory is a coherent explanation or interpretation of one or more phenomena. Although theories can take a variety of forms, one thing they have in common is that they go beyond the phenomena they explain by including variables, structures, processes, functions, or organizing principles that have not been observed directly. Consider, for example, Zajonc's theory of social facilitation and social inhibition (1965)¹. He proposed that being watched by others while performing a task creates a general state of physiological arousal, which increases the likelihood of the dominant (most likely) response. So, for highly practiced tasks, being watched increases the tendency to make correct responses, but for relatively unpracticed tasks, being watched increases the tendency to make incorrect responses. Notice that this theory—which has come to be

1. Zajonc, R. B. (1965). Social facilitation. *Science*, 149, 269–274

called drive theory—provides an explanation of both social facilitation and social inhibition that goes beyond the phenomena themselves by including concepts such as “arousal” and “dominant response,” along with processes such as the effect of arousal on the dominant response.

Outside of science, referring to an idea as a theory often implies that it is untested—perhaps no more than a wild guess. In science, however, the term theory has no such implication. A theory is simply an explanation or interpretation of a set of phenomena. It can be untested, but it can also be extensively tested, well supported, and accepted as an accurate description of the world by the scientific community. The theory of evolution by natural selection, for example, is a theory because it is an explanation of the diversity of life on earth—not because it is untested or unsupported by scientific research. On the contrary, the evidence for this theory is overwhelmingly positive, and nearly all scientists accept its basic assumptions as accurate. Similarly, the “germ theory” of disease is a theory because it is an explanation of the origin of various diseases, not because there is any doubt that many diseases are caused by microorganisms that infect the body.

As we think about theories in the scientific literature, it will be important for us to ask some critical questions. For example, who gets to create theories? Historically, only white males were allowed to have the education and power to promote their theories. This doesn’t mean that marginalized peoples didn’t create theories, but oftentimes, their theories

were ignored, downplayed, or stolen.² The theories we have come to know are not neutral; they are based on the worldview of the person creating them. It is important to realize that psychological theories have real-world implications and can be used to promote oppression.³ One major example of a theory with oppressive outcomes is the concept of eugenics. Eugenics was predicated on the belief that there are inherent differences between groups of people with a particular focus on intellectual differences. This theory then led to the practices of sterilization, segregation, and extermination of individuals deemed intellectually inferior.⁴

A hypothesis, on the other hand, is a specific prediction about a new phenomenon that should be observed if a particular theory is accurate. It is an explanation that relies on just a few key concepts. Hypotheses are often specific predictions about what will happen in a particular study. They are developed by considering existing evidence and using reasoning to infer what will happen in the specific context of interest. Hypotheses are often, but not always, derived from

-
2. Smith, L. T. (2021). *Decolonizing Methodologies: Research and Indigenous Peoples*. Bloomsbury Publishing.
 3. Long, W. (2020). Psychology and oppression. In *Oxford Research Encyclopedia of Psychology*.
 4. Turiel, E. (2020). Eugenics, prejudice, and psychological research. *Human Development*, 64(3), 103-107.

theories. So a hypothesis is often a prediction based on a theory, but some hypotheses are a-theoretical, and only after a set of observations have been made, is a theory developed. This is because theories are broad in nature and they explain larger bodies of data. So if our research question is really original, then we may need to collect some data and make some observations before we can develop a broader theory.

Theories and hypotheses always have this *if-then* relationship. “*If* drive theory is correct, *then* cockroaches should run through a straight runway faster, and a branching runway more slowly, when other cockroaches are present.” Although hypotheses are usually expressed as statements, they can always be rephrased as questions. “Do cockroaches run through a straight runway faster when other cockroaches are present?” Thus, deriving hypotheses from theories is an excellent way of generating interesting research questions.

But how do researchers derive hypotheses from theories? One way is to generate a research question using the techniques discussed in this chapter and then ask whether any theory implies an answer to that question. For example, you might wonder whether expressive writing about positive experiences improves health as much as expressive writing about traumatic experiences. Although this question is an interesting one on its own, you might then ask whether the habituation theory—the idea that expressive writing causes people to habituate to negative thoughts and feelings—implies an answer. In this case, it seems clear that if the habituation

theory is correct, then expressive writing about positive experiences should not be effective because it would not cause people to habituate to negative thoughts and feelings. A second way to derive hypotheses from theories is to focus on some component of the theory that has not yet been directly observed. For example, a researcher could focus on the process of habituation—perhaps hypothesizing that people should show fewer signs of emotional distress with each new writing session.

Among the very best hypotheses are those that distinguish between competing theories. For example, Norbert Schwarz and his colleagues considered two theories of how people make judgments about themselves, such as how assertive they are (Schwarz et al., 1991)⁵. Both theories held that such judgments are based on relevant examples that people bring to mind. However, one theory was that people base their judgments on the *number* of examples they bring to mind, and the other was that people base their judgments on how *easily* they bring those examples to mind. To test these theories, the researchers asked people to recall either six times when they were assertive (which is easy for most people) or 12 times (which is difficult for most people). Then they asked them to judge their own

5. Schwarz, N., Bless, H., Strack, F., Klumpp, G., Rittenauer-Schatka, H., & Simons, A. (1991). Ease of retrieval as information: Another look at the availability heuristic. *Journal of Personality and Social Psychology*, 61, 195–202.

assertiveness. Note that the number-of-examples theory implies that people who recalled 12 examples should judge themselves to be more assertive because they recalled more examples, but the ease-of-examples theory implies that participants who recalled six examples should judge themselves as more assertive because recalling the examples was easier. Thus, the two theories made opposite predictions so that only one of the predictions could be confirmed. The surprising result was that participants who recalled fewer examples judged themselves to be more assertive—providing particularly convincing evidence in favor of the ease-of-retrieval theory over the number-of-examples theory.

Theory Testing

The primary way that scientific researchers use theories is sometimes called the **hypothetico-deductive method** (although this term is much more likely to be used by philosophers of science than by scientists themselves). Researchers begin with a set of phenomena and either construct a theory to explain or interpret them or choose an existing theory to work with. They then make a prediction about some new phenomenon that should be observed if the theory is correct. Again, this prediction is called a hypothesis. The researchers then conduct an empirical study to test the hypothesis. Finally, they reevaluate the theory in light of the new results and revise it if necessary. This process is usually

conceptualized as a cycle because the researchers can then derive a new hypothesis from the revised theory, conduct a new empirical study to test the hypothesis, and so on. As Figure 12.1 shows, this approach meshes nicely with the model of scientific research in psychology presented earlier in the textbook—creating a more detailed model of “theoretically motivated” or “theory-driven” research.

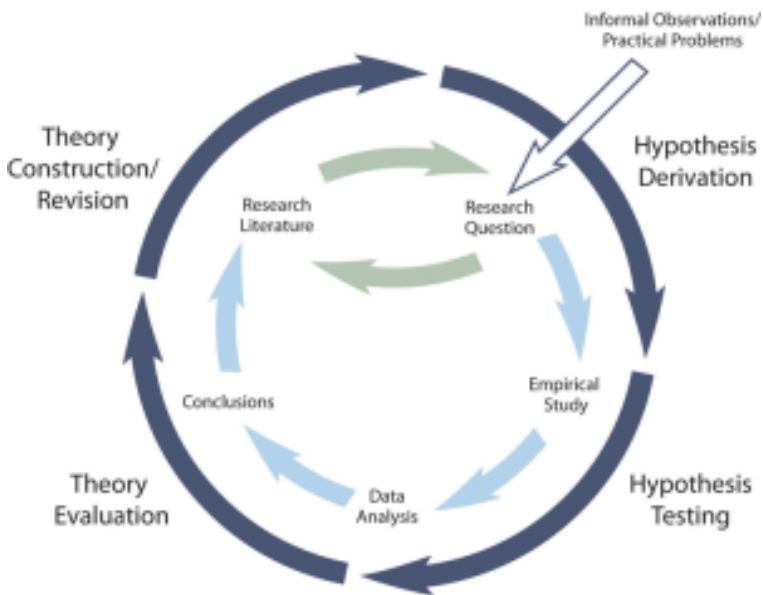


Figure 12.1. Hypothetico-Deductive Method Combined with the General Model of Scientific Research in Psychology. Together they form a model of theoretically motivated research.
[\[Image Description\]](#)

Incorporating Theory into Your Research

When you write your research report or plan your presentation, be aware that there are two basic ways that researchers usually include theory. The first is to raise a research question, answer that question by conducting a new study, and then offer one or more theories (usually more) to explain or interpret the results. This format works well for applied research questions and for research questions that existing theories do not address. The second way is to describe one or more existing theories, derive a hypothesis from one of those theories, test the hypothesis in a new study, and finally reevaluate the theory. This format works well when there is an existing theory that addresses the research question—especially if the resulting hypothesis is surprising or conflicts with a hypothesis derived from a different theory.

To use theories in your research will not only give you guidance in coming up with experiment ideas and possible projects, but it lends legitimacy to your work. Psychologists have been interested in a variety of human behaviors and have developed many theories along the way. Using established theories will help you break new ground as a researcher, not limit you from developing your own ideas.

Characteristics of a Good Hypothesis

There are three general characteristics of a good hypothesis. First, a good hypothesis must be **testable and falsifiable**. We must be able to test the hypothesis using the methods of science, and according to Popper's falsifiability criterion, it must be possible to gather evidence that will disconfirm the hypothesis if it is indeed false. Second, a good hypothesis must be **logical**. As described above, hypotheses are more than just a random guess. Hypotheses should be informed by previous theories or observations and logical reasoning. Typically, we begin with a broad and general theory and use *deductive reasoning* to generate a more specific hypothesis to test based on that theory. Occasionally, however, when there is no theory to inform our hypothesis, we use *inductive reasoning*, which involves using specific observations or research findings to form a more general hypothesis. Finally, the hypothesis should be **positive**. That is, the hypothesis should make a positive statement about the existence of a relationship or effect, rather than a statement that a relationship or effect does not exist. As scientists, we don't set out to show that relationships do not exist or that effects do not occur, so our hypotheses should not be worded in a way to suggest that an effect or relationship does not exist. The nature of science is to assume that something does not exist and then seek to find evidence to prove this wrong, to show that it really does exist. That may

seem backward to you, but that is the nature of the scientific method. The underlying reason for this is beyond the scope of this chapter, but it has to do with statistical theory.

Equity Activity: Applying Intergroup Contact Theory

Allport (1954) theorized that members of different groups (ex. people from different racial backgrounds) could increase positive relationships and attitudes by increasing the contact between groups. This contact need to be mitigated by certain factors, including equal status of participants and the inclusion of shared goals or an activity that required mutual cooperation. This was a particularly important theory in the period of American history that was debating the pros and cons of segregation.

You are working on designing three studies using the principles of intergroup contact theory. Your research question is to determine if there is an increase in positive attitudes between members of different groups. Please draft a hypothesis for each of these studies:



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodspych/?p=54#h5p-12>

License & Attribution

“Developing a Hypothesis” by Stephanie D’Costa is adapted from “[Developing a Hypothesis](#)” by Rajiv S. Jhangiani; I-Chant A. Chiang; Carrie Cuttler; and Dana C. Leighton is licensed CC BY-NC-SA 3.0.

“Practical Strategies for Psychological Measurement” is licensed under CC BY-NC-SA 4.0.

Image Descriptions

Figure 12.1. A circular flowchart shows three linked cycles. The outer dark-blue loop is the scientific research cycle labeled, in clockwise order: Hypothesis Derivation, Hypothesis Testing, Theory Evaluation, and Theory Construction/Revision. A label reading “Informal Observations/Practical Problems” feeds into Hypothesis Derivation. Inside this loop is a light-blue research cycle that lists Empirical Study, then Data Analysis, then Conclusions. Above them is a short green cycle that lists Research Literature, then Research Question. The cycles correspond step for step: the Research Literature and Research Question stages inform Hypothesis Derivation in the outer loop; Empirical Study together with Data Analysis carry out Hypothesis Testing; Conclusions support Theory Evaluation; and Theory Construction/Revision updates the Research Literature and leads to the next Research Question, restarting the process. [[Return to Figure 12.1](#)]

Media Attributions

- 2.3_4.4

13.

VALIDITY IN RESEARCH

Makenzie O'Neil

Learning Objectives

By the end of this chapter, you will be able to

1. Explain what internal validity is and why different methods are considered to be higher or lower in internal validity.
2. Explain what external validity is and evaluate studies in terms of their external validity.
3. Explain the concepts of construct and statistical validity.

Validity in Research

In addition to assessing the validity of individual measures, it is also important when conducting research to evaluate if the study overall is valid. However, answering “is this study valid?” is not as straightforward as it seems because, in psychology, there are many different kinds of validities. Researchers have focused on four validities to help assess whether a study is sound (Judd & Kenny, 1981; Morling, 2014)¹²: internal validity, external validity, construct validity, and statistical validity. We will explore each validity in depth.

Internal Validity

Two variables being statistically related does not necessarily mean that one causes the other. In your psychology education, you have probably heard the phrase, “Correlation does not imply causation.” For example, if it were the case that people who exercise regularly are happier than people who do not exercise regularly, this implication would not necessarily mean

-
1. Judd, C.M. & Kenny, D.A. (1981). *Estimating the effects of social interventions*. Cambridge, MA: Cambridge University Press.
 2. Morling, B. (2014, April). Teach your students to be better consumers. *APS Observer*. Retrieved from <http://www.psychologicalscience.org/index.php/publications/observer/201>

that exercising increases people's happiness. It could mean instead that greater happiness causes people to exercise, or that something like better physical health causes people to exercise *and* be happier.

The purpose of an experiment, however, is to show that two variables are statistically related and to do so in a way that supports the conclusion that the independent variable caused any observed differences in the dependent variable. The logic is based on this assumption: If the researcher creates two or more highly similar conditions and then manipulates the independent variable to produce just one difference between them, then any later difference between the conditions must have been caused by the independent variable. For example, because the only difference between Darley and Latané's conditions was the number of students that participants believed to be involved in the discussion, this difference in belief must have been responsible for differences in helping between the conditions.

An empirical study is said to be high in **internal validity** if the way it was conducted supports the conclusion that the independent variable caused any observed differences in the dependent variable. Thus, experiments are high in internal validity because the way they are conducted—with the manipulation of the independent variable and the control of extraneous variables (such as through the use of random assignment to minimize confounds)—provides strong support for causal conclusions. In contrast, non-experimental research

designs (e.g., correlational designs), in which variables are measured but are not manipulated by a researcher, are low in internal validity.

External Validity

At the same time, the way that research is conducted sometimes leads to a different kind of criticism. In many psychology experiments, the participants are all undergraduate students and come to a classroom or laboratory to fill out a series of paper-and-pencil questionnaires or to perform a carefully designed computerized task. Consider, for example, an experiment in which researcher Barbara Fredrickson and her colleagues had undergraduate students come to a laboratory on campus and complete a math test while wearing a swimsuit (Fredrickson, Roberts, Noll, Quinn, & Twenge, 1998)³. At first, this manipulation might seem silly. When will undergraduate students ever have to complete math tests in their swimsuits outside of this experiment?

The issue we are confronting is that of **external validity**. An empirical study is high in external validity if the way it was

3. Fredrickson, B. L., Roberts, T.-A., Noll, S. M., Quinn, D. M., & Twenge, J. M. (1998). The swimsuit becomes you: Sex differences in self-objectification, restrained eating, and math performance. *Journal of Personality and Social Psychology*, 75, 269–284.

conducted supports generalizing the results to people and situations beyond those actually studied. As a general rule, studies are higher in external validity when the participants and the situation studied are similar to those that the researchers want to generalize to and participants encounter every day, often described as **mundane realism**. Imagine, for example, that a group of researchers is interested in how shoppers in large grocery stores are affected by whether breakfast cereal is packaged in yellow or purple boxes. Their study would be high in external validity and have high mundane realism if they studied the decisions of ordinary people doing their weekly shopping in a real grocery store. If the shoppers bought much more cereal in purple boxes, the researchers would be fairly confident that this increase would be true for other shoppers in other stores. Their study would be relatively low in external validity, however, if they studied a sample of undergraduate students in a laboratory at a selective university who merely judged the appeal of various colors presented on a computer screen; however, this study would have high **psychological realism** where the same mental process is used in both the laboratory and in the real world. If the students judged purple to be more appealing than yellow, the researchers would not be very confident that this preference is relevant to grocery shoppers' cereal-buying decisions because of low external validity, but they could be confident that the visual processing of colors has high psychological realism.

In contrast to internal validity, research designs that involve variables that are measured—particularly in a naturalistic setting—but not manipulated by a researcher, tend to be high in external validity. For experiments, however, the need to manipulate the independent variable and control extraneous variables means that experiments are often conducted under conditions that seem artificial and thus lower in external validity (Bauman, McGraw, Bartels, & Warren, 2014)⁴.

We should be careful, however, not to draw the blanket conclusion that experiments are always low in external validity. Consider field experiments that are conducted entirely outside the laboratory, and may not seem artificial to participants at all. In one such experiment, Robert Cialdini and his colleagues studied whether hotel guests choose to reuse their towels for a second day as opposed to having them washed as a way of conserving water and energy (Cialdini, 2005)⁵. These researchers manipulated the message on a card left in a large sample of hotel rooms. One version of the message emphasized

4. Bauman, C.W., McGraw, A.P., Bartels, D.M., & Warren, C. (2014). Revisiting external validity: Concerns about trolley problems and other sacrificial dilemmas in moral psychology. *Social and Personality Psychology Compass*, 8/9, 536-554.

5. Cialdini, R. (2005, April). Don't throw in the towel: Use social influence research. *APS Observer*. Retrieved from <http://www.psychologicalscience.org/index.php/publications/observer/2005/april-05/dont-throw-in-the-towel-use-social-influence-research.html>

showing respect for the environment, another emphasized that the hotel would donate a portion of its savings to an environmental cause, and a third emphasized that most hotel guests choose to reuse their towels. The result was that guests who received the message that most hotel guests choose to reuse their towels reused their own towels substantially more often than guests receiving either of the other two messages. Given the way they conducted their study, it seems very likely that their result would hold true for other guests in other hotels.

A second reason not to draw the blanket conclusion that experiments are low in external validity is that they are often conducted to learn about psychological processes that are likely to operate in a variety of people and situations. Let us return to the experiment by Fredrickson and colleagues. They found that the women in their study, but not the men, performed worse on the math test when they were wearing swimsuits. They argued that this gender difference was due to women's greater tendency to objectify themselves—to think about themselves from the perspective of an outside observer—which diverts their attention away from other tasks. They argued, furthermore, that this process of self-objectification and its effect on attention is likely to operate in a variety of women and situations—even if none of them ever finds herself taking a math test in her swimsuit. Thus, even though the behavior of participants in the experiment was unlikely to occur in their daily lives (i.e., wearing a

swimsuit to take a test), the underlying psychological processes are generalizable to a variety of women and situations, giving this study more external validity.

Construct Validity

In addition to the generalizability of the results of the study, another element to scrutinize in a study is the **construct validity** or the quality of the study's manipulations or measurements. The research question that Darley and Latané started with is "Does helping behavior become diffused?" They hypothesized that participants in a lab would be less likely to help when they believed there were more potential helpers besides themselves. This conversion from research question to experiment design is called **operationalization**. Darley and Latané operationalized the independent variable of diffusion of responsibility by increasing the number of potential helpers. In evaluating this design, we would say that the construct validity was very high because the experiment's manipulations very clearly speak to the research question; there was a crisis, a way for the participant to help, and by increasing the number of other students involved in the discussion, they provided a way to test diffusion.

What if the number of conditions in Darley and Latané's study changed? Consider if there were only two conditions: one student involved in the discussion or two. Even though we may see a decrease in helping by adding another person, it may

not be a clear demonstration of diffusion of responsibility, just merely the presence of others. We might think it was a form of Bandura's concept of social inhibition. The construct validity would be lower. However, had there been five conditions, perhaps we would see the decrease continue with more people in the discussion, or perhaps it would plateau after a certain number of people. In that situation, we may develop a more nuanced understanding of the phenomenon. But by adding still more conditions, the construct validity may not get higher. When designing your own research, consider how well the research question is operationalized in your study.

Statistical Validity

Statistical validity concerns the proper statistical treatment of data and the soundness of the researchers' statistical conclusions. There are many different types of inferential statistics tests (e.g., *t*-tests, ANOVA, regression, correlation), and statistical validity concerns the use of the proper type of test to analyze the data. When considering the proper type of test, researchers must consider the scale of measure their dependent variable was measured on and the design of their study. Further, many inferential statistics tests carry certain assumptions (e.g., the data are normally distributed), and statistical validity is threatened when these assumptions are not met but the statistics are used nonetheless.

The proper statistical analysis should be conducted on the

data to determine whether the difference or relationship that was predicted was indeed found. Interestingly, the likelihood of detecting an effect of the independent variable on the dependent variable depends not just on whether a relationship really exists between these variables, but also on the number of conditions and the size of the sample. This is why it is important to conduct a power analysis when designing a study, which is a calculation that informs you of the number of participants you need to recruit to detect an effect of a specific size.

Prioritizing Validities

These four big validities—internal, external, construct, and statistical—are useful to keep in mind when both reading about other research and designing your own. However, researchers must prioritize, and often it is not possible to have high validity in all four areas. In nearly every decision that is made when designing a study, there are trade-offs with validity. Making a decision that increases one type of validity can often also decrease other types of validity. Nonetheless, these are choices that researchers need to make and then communicate when disseminating their findings. In Cialdini's study on towel usage in hotels, the external validity was high, but the statistical validity was more modest. This discrepancy does not invalidate the study, but it shows where there may be room for improvement for future follow-up studies (Goldstein,

Cialdini, & Griskevicius, 2008)⁶ Morling (2014) points out that many psychology studies have high internal and construct validity but sometimes sacrifice external validity.

Equity Activity: Exploring educational disparities

A study examines the effectiveness of a **culturally responsive classroom intervention** designed to improve engagement and academic performance among bilingual students. Researchers implement the intervention in a predominantly Spanish-speaking elementary school, where teachers incorporate students' home languages, diverse cultural references, and inclusive teaching strategies into daily lessons. Results show that students in the intervention group demonstrate increased participation, higher test scores, and improved self-esteem compared to a control group.

6. Goldstein, N. J., Cialdini, R. B., & Griskevicius, V. (2008). A room with a viewpoint: Using social norms to motivate environmental conservation in hotels. *Journal of Consumer Research*, 35, 472–482.



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=56#h5p-13>

License & Attribution

“Validity in Research” by Makenzie O’Neil is adapted from “[Experimentation and Validity](#)” by Rajiv S. Jhangiani; I-Chant A. Chiang; Carrie Cuttler; and Dana C. Leighton, which is licensed CC BY-NC-SA 4.0

“Validity in Research” is licensed under CC BY-NC-SA 4.0.

PART III

PSYCHOLOGICAL MEASUREMENT

14.

UNDERSTANDING PSYCHOLOGICAL MEASUREMENT

Rebecca Anguiano

Learning Objectives

By the end of this chapter, you will be able to

1. Define measurement and give several examples of measurement in psychology.
2. Explain what a psychological construct is and give several examples.
3. Distinguish conceptual from operational definitions, give examples of each, and create

- simple operational definitions.
4. Explain scales of measurement.

What Is Measurement?

Measurement is the assignment of scores to individuals so that the scores represent some characteristic of the individuals. This very general definition is consistent with the kinds of measurement that everyone is familiar with—for example, weighing oneself by stepping onto a bathroom scale, or checking the internal temperature of a roasting turkey using a meat thermometer. It is also consistent with measurement in the other sciences. In physics, for example, one might measure the potential energy of an object in Earth's gravitational field by finding its mass and height (which of course requires measuring *those* variables) and then multiplying them together along with the gravitational acceleration of Earth (9.8 m/s^2). The result of this procedure is a score that represents the object's potential energy.

This general definition of measurement is consistent with measurement in psychology, too. (Psychological measurement is often referred to as **psychometrics**.) Imagine, for example, that a cognitive psychologist wants to measure a person's working memory capacity—their ability to hold in mind and

think about several pieces of information all at the same time. To do this, she might use a backward digit span task, in which she reads a list of two digits to the person and asks them to repeat them in reverse order. She then repeats this several times, increasing the length of the list by one digit each time, until the person makes an error. The length of the longest list for which the person responds correctly is the score and represents their working memory capacity. Or imagine a clinical psychologist who is interested in how depressed a person is. He administers the Beck Depression Inventory, which is a 21-item self-report questionnaire in which the person rates the extent to which they have felt sad, lost energy, and experienced other symptoms of depression over the past 2 weeks. The sum of these 21 ratings is the score and represents the person's current level of depression.

The important point here is that measurement does not require any particular instruments or procedures. What it *does* require is *some* systematic procedure for assigning scores to individuals or objects so that those scores represent the characteristic of interest.

Psychological Constructs

Many variables studied by psychologists are straightforward and simple to measure. These include age, height, weight, and birth order. You can ask people how old they are and be reasonably sure that they know and will tell you. Although

people might not know or want to tell you how much they weigh, you can have them step onto a bathroom scale. Other variables studied by psychologists—perhaps the majority—are not so straightforward or simple to measure. We cannot accurately assess people’s level of intelligence by looking at them, and we certainly cannot put their self-esteem on a bathroom scale. These kinds of variables are called **constructs** (pronounced *CON-structs*) and include personality traits (e.g., extraversion), emotional states (e.g., fear), attitudes (e.g., toward taxes), and abilities (e.g., athleticism).

Psychological constructs cannot be observed directly. One reason is that they often represent *tendencies* to think, feel, or act in certain ways. For example, to say that a particular university student is highly extraverted does not necessarily mean that she is behaving in an extraverted way right now. In fact, she might be sitting quietly by herself, reading a book. Instead, it means that she has a general tendency to behave in extraverted ways (e.g., being outgoing, enjoying social interactions) across a variety of situations. Another reason psychological constructs cannot be observed directly is that they often involve internal processes. Fear, for example, involves the activation of certain central and peripheral nervous system structures, along with certain kinds of thoughts, feelings, and behaviors—none of which is necessarily obvious to an outside observer. Notice also that neither extraversion nor fear “reduces to” any particular

thought, feeling, act, or physiological structure or process. Instead, each is a kind of summary of a complex set of behaviors and internal processes.

The Big Five

The Big Five is a set of five broad dimensions that capture much of the variation in human personality. Each of the Big Five can even be defined in terms of six more specific constructs called “facets” (Costa & McCrae, 1992)¹.

1. Costa, P. T., Jr., & McCrae, R. R. (1992). Normal personality assessment in clinical practice: The NEO Personality Inventory. *Psychological Assessment*, 4, 5–13.

Table 14.1 The Big Five Personality Dimensions

Openness to experience	Fantasy	Aesthetics	Feelings	Actions	Ideas	Values
Conscientiousness						
Extroversion						
Agreeableness						
Neuroticism						

Table 14.2 The Big Five Personality Dimensions Expanded

Openness to experience	Fantasy	Aesthetics	Feelings	Actions	Ideas	Values
Conscientiousness	Competence	Order	Dutifulness	Achievement Striving	Self-discipline	Deliberation
Extroversion	Warmth	Gregariousness	Assertiveness	Activity	Excitement seeking	Positive emotions
Agreeableness	Trust	Straight-forwardness	Altruism	Compliance	Modesty	Tender mindedness
Neuroticism	Worry	Anger	Discouragement	Self-consciousness	Impulsivity	Vulnerability

The **conceptual definition** of a psychological construct describes the behaviors and internal processes that make up that construct, along with how it relates to other variables. For example, a conceptual definition of neuroticism (another one of the Big Five) would be that it is people's tendency to experience negative emotions such as anxiety, anger, and sadness across a variety of situations. This definition might also include that it has a strong genetic component, remains fairly stable over time, and is positively correlated with the tendency to experience pain and other physical symptoms.

Students sometimes wonder why, when researchers want to understand a construct like self-esteem or neuroticism, they do not simply look it up in the dictionary. One reason is that many scientific constructs do not have counterparts in everyday language (e.g., working memory capacity). More importantly, researchers are in the business of developing definitions that are more detailed and precise—and that more accurately describe the way the world is—than the informal definitions in the dictionary. As we will see, they do this by proposing conceptual definitions, testing them empirically, and revising them as necessary. Sometimes they throw them out altogether. This is why the research literature often includes different conceptual definitions of the same construct. In some cases, an older conceptual definition has

been replaced by a newer one that fits and works better. In others, researchers are still in the process of deciding which of the various conceptual definitions is the best.

Operational Definitions

An **operational definition** is a definition of a variable in terms of precisely how it is to be measured. These measures generally fall into one of three broad categories. **Self-report measures** are those in which participants report on their own thoughts, feelings, and actions, as with the Rosenberg Self-Esteem Scale (Rosenberg, 1965)². **Behavioral measures** are those in which some other aspect of participants' behavior is observed and recorded. This is an extremely broad category that includes the observation of people's behavior both in highly structured laboratory tasks and in more natural settings. A good example of the former would be measuring working memory capacity using the backward digit span task. A good example of the latter is a famous operational definition of physical aggression from researcher Albert Bandura and his colleagues (Bandura, Ross, & Ross, 1961)³. They let each of

2. Rosenberg, M. (1965). *Society and the adolescent self-image*. Princeton, NJ: Princeton University Press

3. Bandura, A., Ross, D., & Ross, S. A. (1961). Transmission of aggression through

several children play for 20 minutes in a room that contained a clown-shaped punching bag called a Bobo doll. They filmed each child and counted the number of acts of physical aggression the child committed. These included hitting the doll with a mallet, punching it, and kicking it. Their operational definition, then, was the number of these specifically defined acts that the child committed during the 20-minute period. Finally, **physiological measures** are those that involve recording any of a wide variety of physiological processes, including heart rate and blood pressure, galvanic skin response, hormone levels, and electrical activity and blood flow in the brain.

For any given variable or construct, there will be multiple operational definitions. Stress is a good example. A rough conceptual definition is that stress is an adaptive response to a perceived danger or threat that involves physiological, cognitive, affective, and behavioral components. But researchers have operationally defined it in several ways. The Social Readjustment Rating Scale (Holmes & Rahe, 1967)⁴ is a self-report questionnaire on which people identify stressful events that they have experienced in the past year and assigns

imitation of aggressive models. *Journal of Abnormal and Social Psychology*, 63, 575–582.

4. Holmes, T. H., & Rahe, R. H. (1967). The Social Readjustment Rating Scale. *Journal of Psychosomatic Research*, 11(2), 213-218.

points for each one depending on its severity. For example, a man who has been divorced (73 points), changed jobs (36 points), and had a change in sleeping habits (16 points) in the past year would have a total score of 125. The Hassles and Uplifts Scale (DeLongis, Coyne, Dakof, Folkman & Lazarus, 1982) ⁵ is similar but focuses on everyday stressors like misplacing things and being concerned about one's weight. The Perceived Stress Scale (Cohen, Kamarck, & Mermelstein, 1983) ⁶ is another self-report measure that focuses on people's feelings of stress (e.g., "How often have you felt nervous and stressed?"). Researchers have also operationally defined stress in terms of several physiological variables, including blood pressure and levels of the stress hormone cortisol.

When psychologists use multiple operational definitions of the same construct—either within a study or across studies—they are using **converging operations**. The idea is that the various operational definitions are “converging” or coming together on the same construct. When scores based on several different operational definitions are closely related to each other and produce similar patterns of results, this

5. DeLongis, A., Coyne, J. C., Dakof, G., Folkman, S., & Lazarus, R. S. (1982). Relationships of daily hassles, uplifts, and major life events to health status. *Health Psychology, 1*(2), 119-136.

6. Cohen, S., Kamarck, T., & Mermelstein, R. (1983). A global measure of perceived stress. *Journal of Health and Social Behavior, 24*, 386-396.

constitutes good evidence that the construct is being measured effectively and that it is useful. The various measures of stress, for example, are all correlated with each other and have all been shown to be correlated with other variables such as immune system functioning (also measured in a variety of ways) (Segerstrom & Miller, 2004)⁷. This is what allows researchers eventually to draw useful general conclusions, such as “stress is negatively correlated with immune system functioning,” as opposed to more specific and less useful ones, such as “people’s scores on the Perceived Stress Scale are negatively correlated with their white blood counts.”

Levels of Measurement

The psychologist S. S. Stevens suggested that scores can be assigned to individuals in a way that communicates more or less quantitative information about the variable of interest⁸ (Stevens, 1946). For example, the officials at a 100-m race could simply rank order the runners as they crossed the finish line (first, second, etc.), or they could time each runner to the

7. Segerstrom, S. E., & Miller, G. E. (2004). Psychological stress and the human immune system: A meta-analytic study of 30 years of inquiry. *Psychological Bulletin*, 130, 601–630.

8. Stevens, S. S. (1946). On the theory of scales of measurement. *Science*, 103, 677–680

nearest tenth of a second using a stopwatch (11.5 s, 12.1 s, etc.). In either case, they would be measuring the runners' times by systematically assigning scores to represent those times. But while the rank ordering procedure communicates the fact that the second-place runner took longer to finish than the first-place finisher, the stopwatch procedure also communicates *how much* longer the second-place finisher took. Stevens actually suggested four different **levels of measurement** (which he called "scales of measurement") that correspond to four types of information that can be communicated by a set of scores, and the statistical procedures that can be used with the information.

The **nominal level** of measurement is used for categorical variables and involves assigning scores that are category labels. Category labels communicate whether any two individuals are the same or different in terms of the variable being measured. For example, if you ask your participants about their marital status, you are engaged in nominal-level measurement. Or if you ask your participants to indicate which of several ethnicities they identify themselves with, you are again engaged in nominal-level measurement. The essential point about nominal scales is that they do not imply any ordering among the responses. For example, when classifying people according to their favorite color, there is no sense in which green is placed "ahead of" blue. Responses are merely

categorized. Nominal scales thus embody the lowest level of measurement⁹.

The remaining three levels of measurement are used for quantitative variables. The **ordinal level** of measurement involves assigning scores so that they represent the rank order of the individuals. Ranks communicate not only whether any two individuals are the same or different in terms of the variable being measured but also whether one individual is higher or lower on that variable. For example, a researcher wishing to measure consumers' satisfaction with their microwave ovens might ask them to specify their feelings as either "very dissatisfied," "somewhat dissatisfied," "somewhat satisfied," or "very satisfied." The items in this scale are ordered, ranging from least to most satisfied. This is what distinguishes ordinal from nominal scales. Unlike nominal scales, ordinal scales allow comparisons of the degree to which two individuals rate the variable. For example, our satisfaction ordering makes it meaningful to assert that one person is more satisfied than another with their microwave ovens. Such an assertion reflects the first person's use of a verbal label that comes later in the list than the label chosen by the second person.

9. Enter your footnote content here.9. Levels of Measurement. Retrieved from http://wikieducator.org/Introduction_to_Research_Methods_In_Psychology/Theories_and_Measurement/Levels_of_Measurement

On the other hand, ordinal scales fail to capture important information that will be present in the other levels of measurement we examine. In particular, the difference between two levels of an ordinal scale cannot be assumed to be the same as the difference between two other levels (just like you cannot assume that the gap between the runners in first and second place is equal to the gap between the runners in second and third place). In our satisfaction scale, for example, the difference between the responses “very dissatisfied” and “somewhat dissatisfied” is probably not equivalent to the difference between “somewhat dissatisfied” and “somewhat satisfied.” Nothing in our measurement procedure allows us to determine whether the two differences reflect the same difference in psychological satisfaction. Statisticians express this point by saying that the differences between adjacent scale values do not necessarily represent equal intervals on the underlying scale giving rise to the measurements. (In our case, the underlying scale is the true feeling of satisfaction, which we are trying to measure.)

The **interval level** of measurement involves assigning scores using numerical scales in which intervals have the same interpretation throughout. As an example, consider either the Fahrenheit or Celsius temperature scales. The difference between 30 degrees and 40 degrees represents the same temperature difference as the difference between 80 degrees and 90 degrees. This is because each 10-degree interval has

the same physical meaning (in terms of the kinetic energy of molecules).

Interval scales are not perfect, however. In particular, they do not have a true zero point even if one of the scaled values happens to carry the name “zero.” The Fahrenheit scale illustrates the issue. Zero degrees Fahrenheit does not represent the complete absence of temperature (the absence of any molecular kinetic energy). In reality, the label “zero” is applied to its temperature for quite accidental reasons connected to the history of temperature measurement. Since an interval scale has no true zero point, it does not make sense to compute ratios of temperatures. For example, there is no sense in which the ratio of 40 to 20 degrees Fahrenheit is the same as the ratio of 100 to 50 degrees; no interesting physical property is preserved across the two ratios. After all, if the “zero” label were applied at the temperature that Fahrenheit happens to label as 10 degrees, the two ratios would instead be 30 to 10 and 90 to 40, no longer the same! For this reason, it does not make sense to say that 80 degrees is “twice as hot” as 40 degrees. Such a claim would depend on an arbitrary decision about where to “start” the temperature scale, namely, what temperature to call zero (whereas the claim is intended to make a more fundamental assertion about the underlying physical reality).

In psychology, the intelligence quotient (IQ) is often considered to be measured at the interval level. While it is technically possible to receive a score of 0 on an IQ test, such

a score would not indicate the complete absence of IQ. Moreover, a person with an IQ score of 140 does not have twice the IQ of a person with a score of 70. However, the difference between IQ scores of 80 and 100 is the same as the difference between IQ scores of 120 and 140.

Finally, the **ratio level** of measurement involves assigning scores in such a way that there is a true zero point that represents the complete absence of the quantity. Height measured in meters and weight measured in kilograms are good examples. So are counts of discrete objects or events, such as the number of siblings one has or the number of questions a student answers correctly on an exam. You can think of a ratio scale as the three earlier scales rolled up in one. Like a nominal scale, it provides a name or category for each object (the numbers serve as labels). Like an ordinal scale, the objects are ordered (in terms of the ordering of the numbers). Like an interval scale, the same difference at two places on the scale has the same meaning. However, in addition, the same ratio at two places on the scale also carries the same meaning (see Table 14.1).

The Fahrenheit scale for temperature has an arbitrary zero point and is therefore not a ratio scale. However, zero on the Kelvin scale is absolute zero. This makes the Kelvin scale a ratio scale. For example, if one temperature is twice as high as another, as measured on the Kelvin scale, then it has twice the kinetic energy of the other temperature.

Another example of a ratio scale is the amount of money

you have in your pocket right now (25 cents, 50 cents, etc.). Money is measured on a ratio scale because, in addition to having the properties of an interval scale, it has a true zero point: if you have zero money, this actually implies the absence of money. Since money has a true zero point, it makes sense to say that someone with 50 cents has twice as much money as someone with 25 cents.

Stevens's levels of measurement are important for at least two reasons. First, they emphasize the generality of the concept of measurement. Although people do not normally think of categorizing or ranking individuals as measurement, in fact, they are as long as they are done so that they represent some characteristic of the individuals. Second, the levels of measurement can serve as a rough guide to the statistical procedures that can be used with the data and the conclusions that can be drawn from them. With nominal-level measurement, for example, the only available measure of central tendency is the mode. With ordinal-level measurement, the median or mode can be used as indicators of central tendency. Interval and ratio-level measurement are typically considered the most desirable because they permit for any indicators of central tendency to be computed (i.e., mean, median, or mode). Also, ratio-level measurement is the only level that allows meaningful statements about ratios of scores. Once again, one cannot say that someone with an IQ of 140 is twice as intelligent as someone with an IQ of 70 because IQ is measured at the interval level, but one can say that someone

with six siblings has twice as many as someone with three because the number of siblings is measured at the ratio level.

Table 14.3. Summary of Levels of Measurement

Level of Measurement	Category labels	Rank order	Equal intervals	True zero
NOMINAL	X			
ORDINAL	X	X		
INTERVAL	X	X	X	
RATIO	X	X	X	X

Equity Activity: Measuring Sense of Belonging



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=59#h5p-14>

Practice Problems

Test Your Knowledge



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=59#h5p-15>

License & Attribution

“Understanding Psychological Measurement” by Rebecca Anguiano is adapted from [“Understanding Psychological Measurement”](#) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler, and Dana C. Leighton, which is licensed CC BY-NC-SA 4.0.

“Understanding Psychological Measurement” is licensed under CC BY-NC-SA 4.0.

15.

RELIABILITY AND VALIDITY OF MEASUREMENT

Rebecca Anguiano

Learning Objectives

By the end of this chapter, you will be able to

1. Define reliability, including the different types and how they are assessed.
2. Define validity, including the different types and how they are assessed.
3. Describe the kinds of evidence that would be relevant to assessing the reliability and

validity of a particular measure.

Again, measurement involves assigning scores to individuals so that they represent some characteristic of the individuals. But how do researchers know that the scores actually represent the characteristic, especially when it is a construct like intelligence, self-esteem, depression, or working memory capacity? The answer is that they conduct research using the measure to confirm that the scores make sense based on their understanding of the construct being measured. This is an extremely important point. Psychologists do not simply *assume* that their measures work. Instead, they collect data to *demonstrate* that they work. If their research does not demonstrate that a measure works, they should stop using it.

As an informal example, imagine that you have been dieting for a month. Your clothes seem to be fitting more loosely, and several friends have asked if you have lost weight. If at this point your bathroom scale indicated that you had lost 10 pounds, this would make sense, and you would continue to use the scale. But if it indicated that you had gained 10 pounds, you would rightly conclude that it was broken and either fix it or get rid of it.

From a critical perspective, is important to understand that validity is not only a technical issue, but a political one. More

specifically, when marginalized communities are excluded from defining and validating measures that are to be used with them, research may actually silence or misconstrue their experiences. For example, if a school creates a survey to measure “student participation” by measuring how often students speak in class, they may inadvertently leave out students from marginalized communities, such as those who speak another language at home and may participate in other meaningful ways. A valid measure must capture how all students participate. In evaluating a measurement method, psychologists consider two general dimensions: reliability and validity.

Reliability

Reliability refers to the consistency of a measure. Psychologists consider three types of consistency: over time (test-retest reliability), across items (internal consistency), and across different researchers (inter-rater reliability).

Test-Retest Reliability

When researchers measure a construct that they assume to be consistent across time, then the scores they obtain should also be consistent across time. **Test-retest reliability** is the extent to which this is actually the case. For example, intelligence is generally thought to be consistent across time. A person

who is highly intelligent today will be highly intelligent next week. This means that any good measure of intelligence should produce roughly the same scores for this individual next week as it does today. Clearly, a measure that produces highly inconsistent scores over time cannot be a very good measure of a construct that is supposed to be consistent.

Assessing test-retest reliability requires using the measure on a group of people at one time, using it again on the *same* group of people at a later time, and then looking at the test-retest correlation between the two sets of scores. This is typically done by graphing the data in a scatterplot and computing the correlation coefficient. Figure 15.1 shows the correlation between two sets of scores of several university students on the Rosenberg Self-Esteem Scale, administered two times, a week apart. The correlation coefficient for these data is $+.95$. In general, a test-retest correlation of $+.80$ or greater is considered to indicate good reliability.

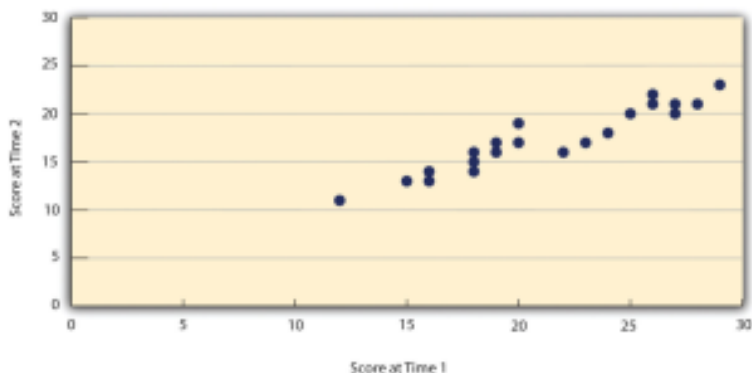


Figure 15.1. Test-Retest Correlation Between Two Sets of Scores of Several College Students on the Rosenberg Self-Esteem Scale, Given Two Times a Week Apart [\[Image Description\]](#)

Again, high test-retest correlations make sense when the construct being measured is assumed to be consistent over time, which is the case for intelligence, self-esteem, and the Big Five personality dimensions. But other constructs are not assumed to be stable over time. The very nature of mood, for example, is that it changes. So, a measure of mood that produced a low test-retest correlation over a period of a month would not be a cause for concern.

Internal Consistency

Another kind of reliability is **internal consistency**, which is the consistency of people's responses across the items on a multiple-item measure. In general, all the items on such

measures are supposed to reflect the same underlying construct, so people's scores on those items should be correlated with each other. On the Rosenberg Self-Esteem Scale, people who agree that they are a person of worth should tend to agree that they have a number of good qualities. If people's responses to the different items are not correlated with each other, then it would no longer make sense to claim that they are all measuring the same underlying construct. This is as true for behavioral and physiological measures as for self-report measures. For example, people might make a series of bets in a simulated game of roulette as a measure of their level of risk seeking. This measure would be internally consistent to the extent that individual participants' bets were consistently high or low across trials.

Like test-retest reliability, internal consistency can only be assessed by collecting and analyzing data. One approach is to look at a **split-half correlation**. This involves splitting the items into two sets, such as the first and second halves of the items or the even- and odd-numbered items. Then a score is computed for each set of items, and the relationship between the two sets of scores is examined. For example, Figure 15.2 shows the split-half correlation between several university students' scores on the even-numbered items and their scores on the odd-numbered items of the Rosenberg Self-Esteem Scale. The correlation coefficient for these data is $+ .88$. A split-half correlation of $+ .80$ or greater is generally considered good internal consistency.

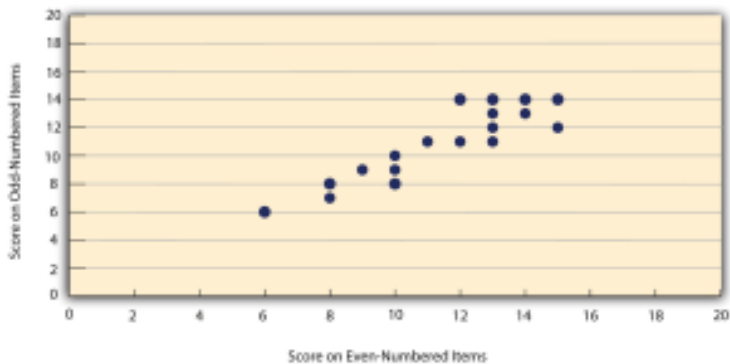


Figure 15.2. Split-Half Correlation Between Several College Students' Scores on the Even-Numbered Items and Their Scores on the Odd-Numbered Items of the Rosenberg Self-Esteem Scale. [Image Description]

Perhaps the most common measure of internal consistency used by researchers in psychology is a statistic called **Cronbach's α** (the Greek letter alpha). Conceptually, α is the mean of all possible split-half correlations for a set of items. For example, there are 252 ways to split a set of 10 items into two sets of five. Cronbach's α would be the mean of the 252 split-half correlations. Note that this is not how α is actually computed, but it is a correct way of interpreting the meaning of this statistic. Again, a value of $+ .80$ or greater is generally taken to indicate good internal consistency.

Interrater Reliability

Many behavioral measures involve significant judgment on the part of an observer or a rater. **Inter-rater reliability** is the extent to which different observers are consistent in their judgments. For example, if you were interested in measuring university students' social skills, you could make video recordings of them as they interacted with another student whom they are meeting for the first time. Then you could have two or more observers watch the videos and rate each student's level of social skills. To the extent that each participant does, in fact, have some level of social skills that can be detected by an attentive observer, different observers' ratings should be highly correlated with each other. Inter-rater reliability would also have been measured in Bandura's Bobo doll study. In this case, the observers' ratings of how many acts of aggression a particular child committed while playing with the Bobo doll should have been highly positively correlated. Inter-rater reliability is often assessed using Cronbach's α when the judgments are quantitative or an analogous statistic called Cohen's κ (the Greek letter kappa) when they are categorical.

Validity

Validity is the extent to which the scores from a measure represent the variable they are intended to. But how do

researchers make this judgment? We have already considered one factor that they take into account—reliability. When a measure has good test-retest reliability and internal consistency, researchers should be more confident that the scores represent what they are supposed to. There has to be more to it, however, because a measure can be extremely reliable but have no validity whatsoever. As an absurd example, imagine someone who believes that people's index finger length reflects their self-esteem and therefore tries to measure self-esteem by holding a ruler up to people's index fingers. Although this measure would have extremely good test-retest reliability, it would have absolutely no validity. The fact that one person's index finger is a centimeter longer than another's would indicate nothing about which one has higher self-esteem.

Discussions of validity usually divide it into several distinct “types.” But a good way to interpret these types is that they are other kinds of evidence—in addition to reliability—that should be taken into account when judging the validity of a measure. Here we consider three basic kinds: face validity, content validity, and criterion validity.

Face Validity

Face validity is the extent to which a measurement method appears “on its face” to measure the construct of interest. Most people would expect a self-esteem questionnaire to include

items about whether they see themselves as a person of worth and whether they think they have good qualities. So a questionnaire that included these kinds of items would have good face validity. The finger-length method of measuring self-esteem, on the other hand, seems to have nothing to do with self-esteem and therefore has poor face validity. Although face validity can be assessed quantitatively—for example, by having a large sample of people rate a measure in terms of whether it appears to measure what it is intended to—it is usually assessed informally.

Face validity is at best a very weak kind of evidence that a measurement method is measuring what it is supposed to. One reason is that it is based on people's intuitions about human behavior, which are frequently wrong. In addition, face validity can also be called into question when power relations or structural inequalities are not explicitly addressed. For example, let's say a sociology research team develops a survey to measure "community cohesion" in urban neighborhoods. The survey appears to have face validity because it includes items like "people in my community help each other" or "I feel my neighbors are friendly." To researchers or professors this seems to accurately measure cohesion. However, when examined critically, this survey actually lacks true face validity for marginalized communities in the neighborhood by leaving out questions about systemic issues, like community members' experiences with policing or discrimination, which shapes social relationships. Therefore,

although at first glance the survey looks valid, it actually fails to measure community cohesion as experienced by those most impacted by structural inequality, demonstrating the limitations of apparent face validity.

Content Validity

Content validity is the extent to which a measure “covers” the construct of interest. For example, if a researcher conceptually defines test anxiety as involving both sympathetic nervous system activation (leading to nervous feelings) and negative thoughts, then his measure of test anxiety should include items about both nervous feelings and negative thoughts. Or consider that attitudes are usually defined as involving thoughts, feelings, and actions toward something. By this conceptual definition, a person has a positive attitude toward exercise to the extent that they think positive thoughts about exercising, feel good about exercising, and actually exercise. So to have good content validity, a measure of people’s attitudes toward exercise would have to reflect all three of these aspects. Like face validity, content validity is not usually assessed quantitatively. Instead, it is assessed by carefully checking the measurement method against the conceptual definition of the construct.

Criterion Validity

Criterion validity is the extent to which people's scores on a measure are correlated with other variables (known as criteria) that one would expect them to be correlated with. For example, people's scores on a new measure of test anxiety should be negatively correlated with their performance on an important school exam. If it were found that people's scores were in fact negatively correlated with their exam performance, then this would be a piece of evidence that these scores really represent people's test anxiety. But if it were found that people scored equally well on the exam regardless of their test anxiety scores, then this would cast doubt on the validity of the measure.

A **criterion** can be any variable that one has reason to think should be correlated with the construct being measured, and there will usually be many of them. For example, one would expect test anxiety scores to be negatively correlated with exam performance and course grades and positively correlated with general anxiety and with blood pressure during an exam. Or imagine that a researcher develops a new measure of physical risk-taking. People's scores on this measure should be correlated with their participation in "extreme" activities such as snowboarding and rock climbing, the number of speeding tickets they have received, and even the number of broken bones they have had over the years. When the criterion is measured at the same time as the construct, criterion validity

is referred to as **concurrent validity**; however, when the criterion is measured at some point in the future (after the construct has been measured), it is referred to as **predictive validity** (because scores on the measure have “predicted” a future outcome).

Criteria can also include other measures of the same construct. For example, one would expect new measures of test anxiety or physical risk taking to be positively correlated with existing established measures of the same constructs. This is known as **convergent validity**.

Assessing convergent validity requires collecting data using the measure. Researchers John Cacioppo and Richard Petty did this when they created their self-report Need for Cognition Scale to measure how much people value and engage in thinking (Cacioppo & Petty, 1982)¹. In a series of studies, they showed that people’s scores were positively correlated with their scores on a standardized academic achievement test, and that their scores were negatively correlated with their scores on a measure of dogmatism (which represents a tendency toward obedience). In the years since it was created, the Need for Cognition Scale has been used in literally hundreds of studies and has been shown to be correlated with a wide variety of other variables, including the effectiveness of an advertisement,

1. Cacioppo, J. T., & Petty, R. E. (1982). The need for cognition. *Journal of Personality and Social Psychology*, 42, 116–131.

interest in politics, and juror decisions (Petty, Briñol, Loersch, & McCaslin, 2009)².

Discriminant Validity

Discriminant validity, on the other hand, is the extent to which scores on a measure are *not* correlated with measures of variables that are conceptually distinct. For example, self-esteem is a general attitude toward the self that is fairly stable over time. It is not the same as mood, which is how good or bad one happens to be feeling right now. So people's scores on a new measure of self-esteem should not be very highly correlated with their moods. If the new measure of self-esteem were highly correlated with a measure of mood, it could be argued that the new measure is not really measuring self-esteem; it is measuring mood instead.

When they created the Need for Cognition Scale, Cacioppo and Petty also provided evidence of discriminant validity by showing that people's scores were not correlated with certain other variables. For example, they found only a weak correlation between people's need for cognition and a measure of their cognitive style—the extent to which they tend to think

2. Petty, R. E., Briñol, P., Loersch, C., & McCaslin, M. J. (2009). The need for cognition. In M. R. Leary & R. H. Hoyle (Eds.), *Handbook of individual differences in social behavior* (pp. 318–329). New York, NY: Guilford Press.

analytically by breaking ideas into smaller parts or holistically in terms of “the big picture.” They also found no correlation between people’s need for cognition and measures of their test anxiety and their tendency to respond in socially desirable ways. All these low correlations provide evidence that the measure is reflecting a conceptually distinct construct.

Equity Activity: Are standardized tests valid measures for university admissions?

Some scholars argue that universities should no longer utilize Scholastic Assessment Tests (SATs) or Graduate Record Examinations (GREs) for admissions into undergraduate or graduate programs. These researchers cite (a) the limited predictive validity of these tests for accurately predicting success in school and (b) bias and inequality in the standardized tests themselves, with certain racial and socioeconomic groups scoring, on average, lower than others.



An interactive H5P element has been excluded from this version of the text. You

can view it online here:

[https://smc.pressbooks.pub/
critresearchmethodpsych/?p=63#h5p-16](https://smc.pressbooks.pub/critresearchmethodpsych/?p=63#h5p-16)

License & Attribution

“Reliability and Validity of Measurement” by Rebecca Anguiano is adapted from “[Reliability and Validity of Measurement](#)” by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler, and Dana C. Leighton, which is licensed CC BY-NC-SA 4.0.

“Reliability and Validity of Measurement” is licensed under CC BY-NC-SA 4.0.

Image Descriptions

Figure 15.1. A rectangular scatterplot has Score at Time 1 on the x-axis (0–30) and Score at Time 2 on the y-axis (0–30). About twenty dark-blue dots run diagonally from the lower-left (around 12–14 on both axes) to the upper-right (around 27–29 on both axes). The upward clustering indicates high test–retest reliability: individuals who scored higher at Time 1 tended to score higher at Time 2. [\[Return to Figure 15.1\]](#)

Figure 15.2. Rectangular scatterplot with tan background. The x-axis is Score on Even-Numbered Items (0–20); the y-axis is Score on Odd-Numbered Items (0–20). About 16 dark-blue points form an upward trend from roughly (6,4) to around (15,12–14), clustering between $x=10-15$ and $y=8-14$. The diagonal pattern indicates good internal consistency: participants who scored high on even items also tended to score high on odd items. [\[Return to Figure 15.2\]](#)

Media Attributions

- Figure 4.2 test-retest
- 4.3

16.

PRACTICAL STRATEGIES FOR PSYCHOLOGICAL MEASUREMENT

Stephanie D'Costa

Learning Objectives

By the end of this chapter, you will be able to

1. Specify the four broad steps in the measurement process.
2. Explain how you would decide whether to use an existing measure or create your own.
3. Describe multiple strategies to identify and locate existing measures of psychological

constructs.

4. Describe several general principles for creating new measures and for implementing existing and new measures.
5. Create a simple plan for assessing the reliability and validity of an existing or new measure.

So far in this chapter, we have considered several basic ideas about the nature of psychological constructs and their measurement. But now imagine that you are in the position of actually having to measure a psychological construct for a research project. How should you proceed? Broadly speaking, there are four steps in the measurement process: (a) conceptually defining the construct, (b) operationally defining the construct, (c) implementing the measure, and (d) evaluating the measure. In this section, we will look at each of these steps in turn. It is important to note that creation of a psychological measure can be a long process and in some cases can even serve as its own research project when validating it on a national level.

Conceptually Defining the Construct

Having a clear and complete conceptual definition of a construct is a prerequisite for good measurement. For one thing, it allows you to make sound decisions about exactly how to measure the construct. If you had only a vague idea that you wanted to measure people's "memory," for example, you would have no way to choose whether you should have them remember a list of vocabulary words, a set of photographs, a newly learned skill, an experience from long ago, or have them remember to perform a task at a later time. Because psychologists now conceptualize memory as a set of semi-independent systems, you would have to be more precise about what you mean by "memory." If you are interested in long-term episodic memory (memory for previous experiences), then having participants remember a list of words that they learned last week would make sense, but having them try to remember to execute a task in the future would not. In general, there is no substitute for reading the research literature on a construct and paying close attention to how others have defined it.

Operationally Defining the Construct

Once you have a conceptual definition of the construct you are interested in studying, it is time to operationally define the construct. Recall an operational definition is a definition of the variable in terms of precisely how it is to be measured. Since most variables are relatively abstract concepts that cannot be directly observed (e.g., stress), and observation is at the heart of the scientific method, conceptual definitions must be transformed into something that can be directly observed and measured. Most variables can be operationally defined in many different ways. For example, stress can be operationally defined as people's scores on a stress scale such as the Perceived Stress Scale (Cohen, Kamarck, & Mermelstein, 1983)¹, cortisol concentrations in their saliva, or the number of stressful life events they have recently experienced. As described below, operationally defining your variable(s) of interest may involve using an existing measure or creating your own measure.

1. Cohen, S., Kamarck, T., & Mermelstein, R. (1983). A global measure of perceived stress. *Journal of Health and Social Behavior*, 24, 386-396.

Using an Existing Measure

It is usually a good idea to use an existing measure that has been used successfully in previous research. Among the advantages are that (a) you save the time and trouble of creating your own, (b) there is already some evidence that the measure is valid (if it has been used successfully), and (c) your results can more easily be compared with and combined with previous results. In fact, if there already exists a reliable and valid measure of a construct, other researchers might expect you to use it unless you have a good and clearly stated reason for not doing so.

If you choose to use an existing measure, you may still have to choose among several alternatives. You might choose the most common one, the one with the best evidence of reliability and validity, the one that best measures a particular aspect of a construct that you are interested in (e.g., a physiological measure of stress if you are most interested in its underlying physiology), or even the one that would be easiest to use. For example, the Ten-Item Personality Inventory (TIPI) is a self-report questionnaire that measures all the Big Five personality dimensions with just 10 items (Gosling, Rentfrow, & Swann, 2003)². It is not as reliable or valid as longer and more

2. Gosling, S. D., Rentfrow, P. J., & Swann, W. B., Jr. (2003). A very brief measure of the Big Five personality domains. *Journal of Research in Personality*, 37, 504–528.

comprehensive measures, but a researcher might choose to use it when testing time is severely limited.

When an existing measure was created primarily for use in scientific research, it is usually described in detail in a published research article and is free to use in your own research—with a proper citation. You might find that later researchers who use the same measure describe it only briefly but provide a reference to the original article, in which case, you would have to get the details from the original article. The American Psychological Association also publishes the *Directory of Unpublished Experimental Measures* and [*PsycTESTS*](#), which are extensive catalogs/collections of measures that have been used in previous research. Many existing measures—especially those that have applications in clinical psychology—are proprietary. This means that a publisher owns the rights to them and that you would have to purchase them. These include many standard intelligence tests, the Beck Depression Inventory, and the Minnesota Multiphasic Personality Inventory (MMPI). Details about many of these measures and how to obtain them can be found in other reference books, including *Tests in Print* and the *Mental Measurements Yearbook*. There is a good chance you can find these reference books in your university library.

Creating Your Own Measure

Instead of using an existing measure, you might want to create

your own. Perhaps there is no existing measure of the construct you are interested in, or existing ones are too difficult or time-consuming to use. Or perhaps you want to use a new measure specifically to see whether it works in the same way as existing measures—that is, to evaluate convergent validity. In this section, we consider some general issues in creating new measures that apply equally to self-report, behavioral, and physiological measures.

First, be aware that most new measures in psychology are really variations of existing measures, so you should still look to the research literature for ideas. Perhaps you can modify an existing questionnaire, create a paper-and-pencil version of a measure that is normally computerized (or vice versa), or adapt a measure that has traditionally been used for another purpose. For example, the famous Stroop task (Stroop, 1935)³—in which people quickly name the colors that various color words are printed in—has been adapted for the study of social anxiety. People high in social anxiety are slower at color naming when the words have negative social connotations such as “stupid” (Amir, Freshman, & Foa, 2002)⁴.

When you create a new measure, you should strive for

3. Stroop, J. R. (1935). Studies of interference in serial verbal reactions. *Journal of Experimental Psychology*, 18, 643–662.

4. Amir, N., Freshman, M., & Foa, E. (2002). Enhanced Stroop interference for threat in social phobia. *Journal of Anxiety Disorders*, 16, 1–9.

simplicity. Remember that your participants are not as interested in your research as you are and that they will vary widely in their ability to understand and carry out whatever task you give them. You should create a set of clear instructions using simple language that you can present in writing or read aloud (or both). It is also a good idea to include one or more practice items so that participants can become familiar with the task and to build in an opportunity for them to ask questions before continuing. It is also best to keep the measure brief to avoid boring or frustrating your participants to the point that their responses start to become less reliable and valid.

The need for brevity, however, needs to be weighed against the fact that it is nearly always better for a measure to include multiple items rather than a single item. There are two reasons for this. One is a matter of content validity. Multiple items are often required to cover a construct adequately. The other is a matter of reliability. People's responses to single items can be influenced by all sorts of irrelevant factors—misunderstanding the particular item, a momentary distraction, or a simple error such as checking the wrong response option. But when several responses are summed or averaged, the effects of these irrelevant factors tend to cancel each other out to produce more reliable scores. Remember, however, that multiple items must be structured in a way that allows them to be combined into a single overall score by summing or averaging. To measure “financial responsibility,” a student might ask people

about their annual income, obtain their credit score, and have them rate how “thrifty” they are—but there is no obvious way to combine these responses into an overall score. To create a true multiple-item measure, the student might instead ask people to rate the degree to which 10 statements about financial responsibility describe them on the same five-point scale.

Finally, the very best way to assure yourself that your measure has clear instructions, includes sufficient practice, and is an appropriate length is to test several people. Observe them as they complete the task, time them, and ask them afterward to comment on how easy or difficult it was, whether the instructions were clear, and anything else you might be wondering about. Obviously, it is better to discover problems with a measure before beginning any large-scale data collection.

Implementing the Measure

You will want to implement any measure in a way that maximizes its reliability and validity. In most cases, it is best to test everyone under similar conditions that, ideally, are quiet and free of distractions. Participants are often tested in groups because it is efficient, but be aware that it can create distractions that reduce the reliability and validity of the measure. As always, it is good to use previous research as a

guide. If others have successfully tested people in groups using a particular measure, then you should consider doing it too.

Be aware also that people can react in a variety of ways to being measured, which reduces the reliability and validity of the scores. Although some disagreeable participants might intentionally respond in ways meant to disrupt a study, participant reactivity is more likely to take the opposite form. Agreeable participants might respond in ways they believe they are expected to. Some participants might engage in **socially desirable responding**, doing or saying things because they think it is the socially appropriate thing. For example, people with low self-esteem agree that they feel they are a person of worth, not because they really feel this way, but because they believe this is the socially appropriate response and do not want to look bad in the eyes of the researcher. Additionally, research studies can have built-in **demand characteristics**: subtle cues that reveal how the researcher expects participants to behave. For example, a participant whose attitude toward exercise is measured immediately after she is asked to read a passage about the dangers of heart disease might reasonably conclude that the passage was meant to improve her attitude. As a result, she might respond more favorably because she believes she is expected to by the researcher. Finally, your own expectations can bias participants' behaviors in unintended ways.

There are several precautions you can take to minimize these kinds of reactivity. One is to make the procedure as clear and

brief as possible so that participants are not tempted to vent their frustrations on your results. Another is to guarantee participants' anonymity and make clear to them that you are doing so. If you are testing them in groups, be sure that they are seated far enough apart that they cannot see each other's responses. Give them all the same type of writing implement so that they cannot be identified by, for example, the pink glitter pen that they used. You can even allow them to seal completed questionnaires into individual envelopes or put them into a drop box where they immediately become mixed with others' questionnaires. Although informed consent requires telling participants what they will be doing, it does not require revealing your hypothesis or other information that might suggest to participants how you expect them to respond. A questionnaire designed to measure financial responsibility need not be titled "Are You Financially Responsible?" It could be titled "Money Questionnaire" or have no title at all. Finally, the effects of your expectations can be minimized by arranging to have the measure administered by a helper who is "blind" or unaware of its intent or of any hypothesis being tested. Regardless of whether this is possible, you should standardize all interactions between researchers and participants—for example, by always reading the same set of instructions word for word.

Evaluating the Measure

Once you have used your measure on a sample of people and have a set of scores, you are in a position to evaluate it more thoroughly in terms of reliability and validity. Even if the measure has been used extensively by other researchers and has already shown evidence of reliability and validity, you should not assume that it worked as expected for your particular sample and under your particular testing conditions. Regardless, you now have additional evidence bearing on the reliability and validity of the measure, and it would make sense to add that evidence to the research literature.

In most research designs, it is not possible to assess test-retest reliability because participants are tested at only one time. For a new measure, you might design a study specifically to assess its test-retest reliability by testing the same set of participants at two separate times. In other cases, a study designed to answer a different question still allows for the assessment of test-retest reliability. For example, a psychology instructor might measure his students' attitude toward critical thinking using the same measure at the beginning and end of the semester to see if there is any change. Even if there is no change, he could still look at the correlation between students' scores at the two times to assess the measure's test-retest reliability. It is also customary to assess internal consistency for any multiple-item measure—usually by looking at a split-half correlation or Cronbach's α .

Criterion validity can be assessed in various ways. For example, if your study included more than one measure of the same construct or measures of conceptually distinct constructs, then you should look at the correlations among these measures to be sure that they fit your expectations. Note also that a successful experimental manipulation also provides evidence of criterion validity. Recall that MacDonald and Martineau manipulated participants' moods by having them think either positive or negative thoughts, and after the manipulation, their mood measure showed a distinct difference between the two groups. This simultaneously provided evidence that their mood manipulation worked *and* that their mood measure was valid.

But what if your newly collected data casts doubt on the reliability or validity of your measure? The short answer is that you have to ask why. It could be that there is something wrong with your measure or how you administered it. It could be that there is something wrong with your conceptual definition. It could be that your experimental manipulation failed. For example, if a mood measure showed no difference between people who you instructed to think positive versus negative thoughts, maybe it is because the participants did not actually think the thoughts they were supposed to, or that the thoughts did not actually affect their moods. In short, it is “back to the drawing board” to revise the measure, revise the conceptual definition, or try a new manipulation.

Equity Activity: Supporting queer youth's mental health outcomes

You are working with your research team to create a project that determines the efficacy of an [Attachment Based Family Therapy \(ABFT\) for LGBTQ youth](#) to reduce anxiety and depression in your local community mental health clinic. To start the project, you need to clarify the variables you are going to use and how you will measure them.



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=65#h5p-17>

License & Attribution

“Practical Strategies for Psychological Measurement” by Stephanie D’Costa is adapted from “[Practical Strategies for Psychological Measurement](#)” by Rajiv S. Jhangiani; I-Chant A. Chiang; Carrie Cuttler; and Dana C. Leighton is licensed CC BY 4.0.

“Practical Strategies for Psychological Measurement” is licensed under CC-BY 4.0.

PART IV

METHODS

17.

EXPERIMENTAL DESIGN

Rebecca Anguiano

Learning Objectives

By the end of this chapter, you will be able to

1. Explain what an experiment is and recognize examples of studies that are experiments and studies that are not experiments.
2. Distinguish between the manipulation of the independent variable and control of extraneous variables and explain the importance of each.
3. Explain the difference between between-subjects and within-subjects experiments, list

some of the pros and cons of each approach, and decide which approach to use to answer a particular research question.

4. Define random assignment, distinguish it from random sampling, explain its purpose in experimental research, and use some simple strategies to implement it
5. Define several types of carryover effect, give examples of each, and explain how counterbalancing helps to deal with them.

What Is an Experiment?

As we saw earlier in the book, an **experiment** is a type of study designed specifically to answer the question of whether there is a causal relationship between two variables. In other words, whether changes in one variable (referred to as an **independent variable**) cause a change in another variable (referred to as a **dependent variable**). Experiments have two fundamental features. The first is that the researchers manipulate, or systematically vary, the level of the independent variable. The different levels of the independent variable are called **conditions**. For example, in Darley and Latané's experiment, the independent variable was the number of

witnesses that participants believed to be present. The researchers manipulated this independent variable by telling participants that there were either one, two, or five other students involved in the discussion, thereby creating three conditions. For a new researcher, it is easy to confuse these terms by believing there are three independent variables in this situation: one, two, or five students involved in the discussion, but there is actually only one independent variable (number of witnesses) with three different levels or conditions (one, two, or five students). The second fundamental feature of an experiment is that the researcher exerts **control** over, or minimizes the variability in, variables other than the independent and dependent variables. These other variables are called **extraneous variables**. Darley and Latané tested all their participants in the same room, exposed them to the same emergency situation, and so on. They also randomly assigned their participants to conditions so that the three groups would be similar to each other to begin with. Notice that although the words manipulation and control have similar meanings in everyday language, researchers make a clear distinction between them. They manipulate the independent variable by systematically changing its levels and control other variables by holding them constant.

This chapter introduces **experimental research methods**, focusing on how researchers design studies to test causal relationships between variables. You will learn what defines an **experiment**, including the roles of **independent variables**,

dependent variables, and **conditions**, and why **manipulation** and **control** are essential for drawing valid conclusions. The chapter also examines **extraneous variables**, how they create noise in data, and the strategies researchers use to manage them, such as **random assignment**, **matched groups**, and **counterbalancing**. Finally, you will compare **between-subjects** and **within-subjects** designs, explore **order and carryover effects**, and learn how researchers choose the most appropriate design to answer different types of research questions.

Manipulation of the Independent Variable

Again, to **manipulate** an independent variable means to change its level systematically so that different groups of participants are exposed to different levels of that variable, or the same group of participants is exposed to different levels at different times. For example, to see whether expressive writing affects people's health, a researcher might instruct some participants to write about traumatic experiences and others to write about neutral experiences. The different levels of the independent variable are referred to as conditions, and researchers often give the conditions short descriptive names to make it easy to talk and write about them. In this case, the conditions might be called the "traumatic condition" and the "neutral condition."

Notice that the manipulation of an independent variable must involve the active intervention of the researcher. Comparing groups of people who differ on the independent variable before the study begins is not the same as manipulating that variable. For example, a researcher who compares the health of people who already keep a journal with the health of people who do not keep a journal has not manipulated this variable and therefore has not conducted an experiment. This distinction is important because groups that already differ in one way at the beginning of a study are likely to differ in other ways too. For example, people who choose to keep journals might also be more conscientious, more introverted, or less stressed than people who do not. Therefore, any observed difference between the two groups in terms of their health might have been caused by whether or not they keep a journal, or it might have been caused by any of the other differences between people who do and do not keep journals. When something varies in a study that is not the independent or dependent variables, it is called an **extraneous variable**. Thus, the active manipulation of the independent variable is crucial for eliminating extraneous variables and therefore potential alternative explanations for the results.

Of course, there are many situations in which the independent variable cannot be manipulated for practical or ethical reasons and therefore an experiment is not possible. For example, whether or not people have a traumatic experience in early childhood cannot be manipulated, making it impossible

to conduct an experiment on the effect of childhood traumatic experiences on the development of depression or anxiety in adulthood. This caveat does not mean it is impossible to study the relationship between childhood traumatic experiences and mental health in adulthood—only that it must be done using non-experimental approaches. We will discuss this type of methodology in detail later in the book.

Independent variables can be manipulated to create two conditions, and experiments involving a single independent variable with two conditions are often referred to as a **single-factor two-level design**. However, sometimes greater insights can be gained by adding more conditions to an experiment. When an experiment has one independent variable that is manipulated to produce more than two conditions, it is referred to as a **single-factor multi-level design**. So rather than comparing a condition in which there was one witness to a condition in which there were five witnesses (which would represent a single-factor two-level design), Darley and Latané's experiment used a single factor multi-level design, by manipulating the independent variable to produce three conditions (a one witness, a two witnesses, and a five witnesses condition).

Control of Extraneous Variables

As we have seen previously in the chapter, an extraneous variable is anything that varies in the context of

a study other than the independent and dependent variables. In an experiment on the effect of expressive writing on health, for example, extraneous variables would include participant variables (individual differences) such as their writing ability, their diet, and their gender. In this study, **writing ability**, **diet**, and **gender** are extraneous variables because they vary across participants and can influence health outcomes independently of the expressive writing intervention. Writing ability may affect how effectively participants engage in expressive writing, diet directly influences physical and mental health, and gender is associated with systematic differences in health, stress, and emotional expression. Because these factors can affect the dependent variable without being part of the independent variable, they introduce alternative explanations for any observed health effects. Extraneous variables pose a problem because many of them are likely to have some effect on the dependent variable. These influencing factors can make it difficult to separate the effect of the independent variable from the effects of the extraneous variables (i.e. the effect of expressive writing on health versus the effects of writing ability, diet, and gender on health), which is why it is important to control extraneous variables by holding them constant.

Extraneous Variables as “Noise”

Extraneous variables make it difficult to detect the effect of the independent variable in two ways. One is by adding variability

or “noise” to the data. Imagine a simple experiment on the effect of mood (happy vs. sad) on the number of happy childhood events people are able to recall. Participants are put into a negative or positive mood (by showing them a happy or sad video clip) and then asked to recall as many happy childhood events as they can. The two leftmost columns of Table 17.1 show what the data might look like if there were no extraneous variables and the number of happy childhood events participants recalled was affected only by their moods. Every participant in the happy mood condition recalled exactly four happy childhood events, and every participant in the sad mood condition recalled exactly three. The effect of mood here is quite obvious. The effect of mood is obvious because there is **no variability within each mood condition**, so the consistent one-event difference between the happy and sad groups can be clearly attributed to mood rather than to any other influences. In reality, however, the data would probably look more like those in the two rightmost columns of Table 17.1. Even in the happy mood condition, some participants would recall fewer happy memories because they have fewer to draw on, use less effective recall strategies, or are less motivated. And even in the sad mood condition, some participants would recall more happy childhood memories because they have more happy memories to draw on, they use more effective recall strategies, or they are more motivated. Although the mean difference between the two groups is the same as in the idealized data, this difference is much less obvious in the

context of the greater variability in the data. Thus, one reason researchers try to control extraneous variables is so their data look more like the idealized data in Table 17.1, which makes the effect of the independent variable easier to detect (although real data never look quite *that* good).

Table 17.1. Hypothetical Noiseless Data and Realistic “noisy” Data

Idealized “noiseless” data	Idealized “noiseless” data	Realistic “noisy” data
Happy mood	Sad mood	Happy mood
4	3	3
4	3	6
4	3	2
4	3	4
4	3	5
4	3	2
4	3	3
4	3	1
4	3	6
4	3	8
$M = 4$	$M = 3$	$M = 4$

One way to control extraneous variables is to hold them constant. This technique can mean holding situation or task variables constant by testing all participants in the same

location, giving them identical instructions, treating them in the same way, and so on. It can also mean holding participant variables constant. For example, many studies of language limit participants to right-handed people, who generally have their language areas isolated in their left cerebral hemispheres¹. Left-handed people are more likely to have their language areas isolated in their right cerebral hemispheres or distributed across both hemispheres, which can change the way they process language and thereby add noise to the data.

In principle, researchers can control extraneous variables by limiting participants to one very specific category of person, such as 20-year-old, heterosexual, female, right-handed psychology majors. The obvious downside to this approach is that it would lower the **external validity** of the study—in particular, the extent to which the results can be generalized beyond the people actually studied. For example, it might be unclear whether results obtained with a sample of younger lesbian women would apply to older gay men. In many situations, the advantages of a diverse sample (increased external validity) outweigh the reduction in noise achieved by a homogeneous one.

1. ²

2. [1]

Between-Subjects Experiments

In a **between-subjects experiment**, each participant is tested in only one condition. For example, in Darley and Latané's experiment each participant was assigned to one of three conditions: the one witness, two witnesses, or five witnesses condition. Another example might include a researcher with a sample of 100 university students who assigns half of them to write about a traumatic event and the other half to write about a neutral event. It is essential in a between-subjects experiment that the researcher assigns participants to conditions so that the different groups are, on average, highly similar to each other. Those in a trauma condition and a neutral condition, for example, should include a similar proportion of men and women, and they should have similar average IQs, similar average levels of motivation, similar average numbers of health problems, and so on. This matching is a matter of controlling these extraneous participant variables across conditions so that they do not become confounding variables.

Random Assignment

The primary way that researchers accomplish this kind of control of extraneous variables across conditions is called **random assignment**, which means using a random process to decide which participants are tested in which conditions. Do not confuse random assignment with random

sampling, which is discussed in a previous chapter on [Population and Samples](#). Random assignment is a method for assigning participants in a sample to the different conditions, and it is an important element of all experimental research in psychology and other fields, too.

In its strictest sense, random assignment should meet two criteria. One is that each participant has an equal chance of being assigned to each condition (e.g., a 50% chance of being assigned to each of two conditions). The second is that each participant is assigned to a condition independently of other participants. Thus, one way to assign participants to two conditions would be to flip a coin for each one. If the coin lands heads, the participant is assigned to Condition A, and if it lands tails, the participant is assigned to Condition B. For three conditions, one could use a computer to generate a random integer from 1 to 3 for each participant. If the integer is 1, the participant is assigned to Condition A; if it is 2, the participant is assigned to Condition B; and if it is 3, the participant is assigned to Condition C. In practice, a full sequence of conditions—one for each participant expected to be in the experiment—is usually created ahead of time, and each new participant is assigned to the next condition in the sequence as they are tested. When the procedure is computerized, the computer program often handles the random assignment.

One problem with coin flipping and other strict procedures for random assignment is that they are likely to result in

unequal sample sizes in the different conditions. Unequal sample sizes are generally not a serious problem, and you should never throw away data you have already collected to achieve equal sample sizes. However, for a fixed number of participants, it is statistically most efficient to divide them into equal-sized groups. It is standard practice, therefore, to use a kind of modified random assignment that keeps the number of participants in each group as similar as possible. One approach is **block randomization**. In block randomization, all the conditions occur once in the sequence before any of them is repeated. Then they all occur again before any of them is repeated. Within each of these “blocks,” the conditions occur in a random order. Again, the sequence of conditions is usually generated before any participants are tested, and each new participant is assigned to the next condition in the sequence. Table 17.2 shows such a sequence for assigning nine participants to three conditions. The Research [Randomizer website](#) will generate block randomization sequences for any number of participants and conditions. Again, when the procedure is computerized, the computer program often handles the block randomization.

Table 17.2. Block Randomization Sequence for Assigning Nine Participants to Three Conditions

Participant	Condition
1	A
2	C
3	B
4	B
5	C
6	A
7	C
8	B
9	A

Random assignment is not guaranteed to control all extraneous variables across conditions. The process is random, so it is always possible that just by chance, the participants in one condition might turn out to be substantially older, less tired, more motivated, or less depressed on average than the participants in another condition. However, there are some reasons that this possibility is not a major concern. One is that random assignment works better than one might expect, especially for large samples. Another is that the inferential statistics that researchers use to decide whether a difference between groups reflects a difference in the population takes the “fallibility” of random assignment into account. Yet another reason is that even if random assignment does result in a

confounding variable and therefore produces misleading results, this confound is likely to be detected when the experiment is replicated. The upshot is that random assignment to conditions—although not infallible in terms of controlling extraneous variables—is always considered a strength of a research design.

Matched Groups

An alternative to simple random assignment of participants to conditions is the use of a **matched-groups design**. Using this design, participants in the various conditions are matched on the dependent variable or on some extraneous variable(s) prior to the manipulation of the independent variable. This guarantees that these variables will not be confounded across the experimental conditions. For instance, if we want to determine whether expressive writing affects people's health, then we could start by measuring various health-related variables in our prospective research participants. We could then use that information to rank-order participants according to how healthy or unhealthy they are. Next, the two healthiest participants would be randomly assigned to complete different conditions (one would be randomly assigned to the traumatic experiences writing condition and the other to the neutral writing condition). The next two healthiest participants would then be randomly assigned to complete different conditions, and so on, until the two least healthy participants. This

method would ensure that participants in the traumatic experiences writing condition are matched to participants in the neutral writing condition with respect to health at the beginning of the study. If at the end of the experiment, a difference in health was detected across the two conditions, then we would know that it is due to the writing manipulation and not to pre-existing differences in health.

Within-Subjects Experiments

In a **within-subjects experiment**, each participant is tested under all conditions. Consider an experiment on the effect of a defendant's physical attractiveness on judgments of his guilt. Again, in a between-subjects experiment, one group of participants would be shown an attractive defendant and asked to judge his guilt, and another group of participants would be shown an unattractive defendant and asked to judge his guilt. In a within-subjects experiment, however, the same group of participants would judge the guilt of both an attractive and an unattractive defendant.

The primary advantage of this approach is that it provides maximum control of extraneous participant variables. Participants in all conditions have the same mean IQ, same socioeconomic status, same number of siblings, and so on—because they are the very same people. Within-subjects experiments also make it possible to use statistical procedures that remove the effect of these extraneous participant variables

on the dependent variable and therefore make the data less “noisy” and the effect of the independent variable easier to detect. We will look more closely at this idea later in the book.

However, not all experiments can use a within-subjects design, nor would it be desirable to do so.

Carryover Effects and Counterbalancing

The primary disadvantage of within-subjects designs is that they can result in order effects. An **order effect** occurs when participants’ responses in the various conditions are affected by the order of conditions to which they were exposed. One type of order effect is a carryover effect. A **carryover effect** is an effect of being tested in one condition on participants’ behavior in later conditions. One type of carryover effect is a **practice effect**, where participants perform a task better in later conditions because they have had a chance to practice it. Another type is a **fatigue effect**, where participants perform a task worse in later conditions because they become tired or bored. Being tested in one condition can also change how participants perceive stimuli or interpret their task in later conditions. This type of effect is called a **context effect (or contrast effect)**. For example, an average-looking defendant might be judged more harshly when participants have just judged an attractive defendant than when they have just judged an unattractive defendant. Within-subjects

experiments also make it easier for participants to guess the hypothesis. For example, a participant who is asked to judge the guilt of an attractive defendant and then is asked to judge the guilt of an unattractive defendant is likely to guess that the hypothesis is that defendant attractiveness affects judgments of guilt. This knowledge could lead the participant to judge the unattractive defendant more harshly because he thinks this is what he is expected to do. Or it could make participants judge the two defendants similarly in an effort to be “fair.”

Carryover effects can be interesting in their own right. (Does the attractiveness of one person depend on the attractiveness of other people that we have seen recently?) But when they are not the focus of the research, carryover effects can be problematic. Imagine, for example, that participants judge the guilt of an attractive defendant and then judge the guilt of an unattractive defendant. If they judge the unattractive defendant more harshly, this might be because of his unattractiveness. But it could be instead that they judge him more harshly because they are becoming bored or tired. In other words, the order of the conditions is a confounding variable. The attractive condition is always the first condition, and the unattractive condition the second. Thus, any difference between the conditions in terms of the dependent variable could be caused by the order of the conditions and not the independent variable itself.

There is a solution to the problem of order effects, however, that can be used in many situations. It is **counterbalancing**,

which means testing different participants in different orders. The best method of counterbalancing is **complete counterbalancing** in which an equal number of participants complete each possible order of conditions. For example, half of the participants would be tested in the attractive defendant condition, followed by the unattractive defendant condition, and the other half would be tested in the unattractive condition, followed by the attractive condition. With three conditions, there would be six different orders (ABC, ACB, BAC, BCA, CAB, and CBA), so some participants would be tested in each of the six orders. With four conditions, there would be 24 different orders; with five conditions, there would be 120 possible orders. With counterbalancing, participants are assigned to orders randomly, using the techniques we have already discussed. Thus, random assignment plays an important role in within-subjects designs just as in between-subjects designs. Here, instead of randomly assigning to conditions, they are randomly assigned to different orders of conditions. In fact, it can safely be said that if a study does not involve random assignment in one form or another, it is not an experiment.

A more efficient way of counterbalancing is through a Latin square design, which randomizes through having equal rows and columns. For example, if you have four treatments, you must have four versions. Like a Sudoku puzzle, no treatment can repeat in a row or column. For four versions of four treatments, the Latin square design would look like:

Table 17.3.
Latin Square
Design
Example

A	B	C	D
B	C	D	A
C	D	A	B
D	A	B	C

You can see in the diagram above that the square has been constructed to ensure that each condition appears at each ordinal position (A appears first once, second once, third once, and fourth once) and each condition precedes and follows each other condition one time. A Latin square for an experiment with 6 conditions would be 6 x 6 in dimension, one for an experiment with 8 conditions would be 8 x 8 in dimension, and so on. So, while complete counterbalancing of 6 conditions would require 720 orders, a Latin square would only require 6 orders.

Finally, when the number of conditions is large, experiments can use **random counterbalancing** in which the order of the conditions is randomly determined for each participant. Using this technique, every possible order of conditions is determined, and then one of these orders is randomly selected for each participant. This is not as powerful a technique as complete counterbalancing or partial counterbalancing using a Latin squares design. Use of random

counterbalancing will result in more random error, but if order effects are likely to be small and the number of conditions is large, this is an option available to researchers.

There are two ways to think about what counterbalancing accomplishes. One is that it controls the order of conditions so that it is no longer a confounding variable. Instead of the attractive condition always being first and the unattractive condition always being second, the attractive condition comes first for some participants and second for others. Likewise, the unattractive condition comes first for some participants and second for others. Thus, any overall difference in the dependent variable between the two conditions cannot have been caused by the order of conditions. A second way to think about what counterbalancing accomplishes is that if there are carryover effects, it makes it possible to detect them. One can analyze the data separately for each order to see whether it had an effect.

When 9 Is “Larger” Than 221

Researcher Michael Birnbaum has argued that the lack of context provided by between-subjects designs is often a bigger problem than the context effects created by within-subjects designs. To demonstrate this problem, he asked participants to

rate two numbers on how large they were on a scale of 1-to-10 where 1 was “very very small” and 10 was “very very large”. One group of participants was asked to rate the number 9, and another group was asked to rate the number 221 (Birnbaum, 1999)³. Participants in this between-subjects design gave the number 9 a mean rating of 5.13 and the number 221 a mean rating of 3.10. In other words, they rated 9 as larger than 221! According to Birnbaum, this difference is because participants spontaneously compared 9 with other one-digit numbers (in which case it is *relatively* large) and compared 221 with other three-digit numbers (in which case it is *relatively* small).

Simultaneous Within-Subjects Designs

So far, we have discussed an approach to within-subjects designs in which participants are tested in one condition at a time. There is another approach, however, that is often used

3. Birnbaum, M.H. (1999). How to show that $9 > 221$: Collect judgments in a between-subjects design. *Psychological Methods*, 4(3), 243-249.

when participants make multiple responses in each condition. Imagine, for example, that participants judge the guilt of 10 attractive defendants and 10 unattractive defendants. Instead of having people make judgments about all 10 defendants of one type followed by all 10 defendants of the other type, the researcher could present all 20 defendants in a sequence that mixed the two types. The researcher could then compute each participant's mean rating for each type of defendant. Or imagine an experiment designed to see whether people with social anxiety disorder remember negative adjectives (e.g., "stupid," "incompetent") better than positive ones (e.g., "happy," "productive"). The researcher could have participants study a single list that includes both kinds of words and then have them try to recall as many words as possible. The researcher could then count the number of each type of word that was recalled.

Between-Subjects or Within-Subjects?

Almost every experiment can be conducted using either a between-subjects design or a within-subjects design. This possibility means that researchers must choose between the two approaches based on their relative merits for the particular situation.

Between-subjects experiments have the advantage of being conceptually simpler and requiring less testing time per

participant. They also avoid carryover effects without the need for counterbalancing. Within-subjects experiments have the advantage of controlling extraneous participant variables, which generally reduces noise in the data and makes it easier to detect any effect of the independent variable upon the dependent variable. Within-subjects experiments also require fewer participants than between-subjects experiments to detect an effect of the same size.

A good rule of thumb, then, is that if it is possible to conduct a within-subjects experiment (with proper counterbalancing) in the time that is available per participant—and you have no serious concerns about carryover effects—this design is probably the best option. If a within-subjects design would be difficult or impossible to carry out, then you should consider a between-subjects design instead. For example, if you were testing participants in a doctor's waiting room or shoppers in line at a grocery store, you might not have enough time to test each participant in all conditions and therefore would opt for a between-subjects design. Or imagine you were trying to reduce people's level of prejudice by having them interact with someone of another race. A within-subjects design with counterbalancing would require testing some participants in the treatment condition first and then in a control condition. But if the treatment works and reduces people's level of prejudice, then they would no longer be suitable for testing in the control condition. This difficulty is true for many designs that involve a treatment

meant to produce long-term change in participants' behavior (e.g., studies testing the effectiveness of psychotherapy). Clearly, a between-subjects design would be necessary here.

Remember also that using one type of design does not preclude using the other type in a different study. There is no reason that a researcher could not use both a between-subjects design and a within-subjects design to answer the same research question. In fact, professional researchers often take exactly this type of mixed methods approach.

Equity Activity: Applying Critical Theory to Experimental Research

A team of researchers is designing an experimental study to evaluate the effectiveness of a new disciplinary policy in reducing suspensions in middle schools. The study uses a **randomized controlled trial (RCT)**: 10 schools will implement the new policy, while 10 continue with their usual disciplinary practices.

The researchers plan to compare suspension rates across both groups after one academic year. The intervention is based on a behavioral system developed by a private educational consultancy, with minimal community input.



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodspych/?p=68#h5p-18>

License & Attribution

“Understanding Psychological Measurement” by Rebecca Anguiano is adapted from “[Experimental Research](#)“, “[Experimental Basics](#)“, and “[Experimental Design](#)” by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler, and Dana C. Leighton, which is licensed CC BY-NC-SA 4.0.

“Experimental Design” is licensed under CC BY-NC-SA 4.0.

18.

SINGLE-SUBJECT RESEARCH DESIGNS

Stephanie D'Costa

Learning Objectives

By the end of this chapter, you will be able to

1. Describe the basic elements of a single-subject research design.
2. Design simple single-subject studies using reversal and multiple-baseline designs.
3. Explain how single-subject research designs address the issue of internal validity.
4. Interpret the results of simple single-subject studies based on the visual inspection of

graphed data.

What Is Single-Subject Research?

Single-subject research is a type of quantitative research that involves studying a small number of people using methods that include an analysis of their interaction with their environment. Note that the term *single-subject* does not mean that only one participant is studied; it is more typical for there to be somewhere between two and 10 participants. (This is why single-subject research designs are sometimes called small- n designs, where n is the statistical symbol for the sample size.) Single-subject research can be contrasted with group research, which typically involves studying large numbers of participants and examining their behavior primarily in terms of group means, standard deviations, and so on. The majority of this textbook is devoted to understanding group research, which is the most common approach in psychology. But single-subject research is an important alternative, and it is the primary approach in some more applied areas of psychology.

Before continuing, it is important to distinguish single-subject research from qualitative case studies and other more

qualitative approaches that involve studying in detail a small number of participants. Qualitative case studies use in-depth analysis to interpret the meaning of what occurs during the case. More broadly speaking, qualitative research focuses on understanding people's subjective experience by observing behavior and collecting relatively unstructured data (e.g., detailed interviews) and analyzing those data using narrative rather than quantitative techniques. Single-subject research, in contrast, focuses on understanding objective behavior through experimental manipulation and control, collecting highly structured data, and analyzing those data quantitatively.

Assumptions of Single-Subject Research

Again, single-subject research involves studying a small number of participants and focusing intensively on the behavior of each one. But why take this approach instead of the group approach? There are several important assumptions underlying single-subject research, and it will help to consider them now.

First and foremost is the assumption that it is important to focus intensively on the behavior of individual participants. One reason for this is that group research can hide individual differences and generate results that do not represent the behavior of any individual. For example, a treatment that has a positive effect for half the people exposed to it but a negative

effect for the other half would, on average, appear to have no effect at all. Single-subject research, however, would likely reveal these individual differences. A second reason to focus intensively on individuals is that sometimes it is the behavior of a particular individual that is primarily of interest. A school psychologist, for example, might be interested in changing the behavior of a particular disruptive student. Although previous published research (both single-subject and group research) is likely to provide some guidance on how to do this, conducting a study on this student would be more direct and probably more effective.

A second assumption of single-subject research is that it is important to discover causal relationships through the manipulation of an independent variable, the careful measurement of a dependent variable, and the control of extraneous variables. For this reason, single-subject research is often considered a type of experimental research with good internal validity. Recall, for example, that Hall and his colleagues measured their dependent variable (studying) many times—first under a no-treatment control condition, then under a treatment condition (positive teacher attention), and then again under the control condition. Because there was a clear increase in studying when the treatment was introduced, a decrease when it was removed, and an increase when it was reintroduced, there is little doubt that the treatment was the cause of the improvement.

A third assumption of single-subject research is that it is

important to study strong and consistent effects that have biological or social importance. Applied researchers, in particular, are interested in treatments that have substantial effects on important behaviors and that can be implemented reliably in the real-world contexts in which they occur. This is sometimes referred to as **social validity** (Wolf, 1976)¹. The study by Hall and his colleagues, for example, had good social validity because it showed strong and consistent effects of positive teacher attention on a behavior that is of obvious importance to teachers, parents, and students. Furthermore, the teachers found the treatment easy to implement, even in their often-chaotic elementary school classrooms.

Who Uses Single-Subject Research?

Single-subject research has been around as long as the field of psychology itself. In the late 1800s, one of psychology's founders, Wilhelm Wundt, studied sensation and consciousness by focusing intensively on each of a small number of research participants. Herman Ebbinghaus's

1. Wolf, M. (1976). Social validity: The case for subjective measurement or how applied behavior analysis is finding its heart. *Journal of Applied Behavior Analysis*, 11, 203–214.

research on memory and Ivan Pavlov's research on classical conditioning are other early examples, both of which are still described in almost every introductory psychology textbook.

In the middle of the 20th century, B. F. Skinner clarified many of the assumptions underlying single-subject research and refined many of its techniques (Skinner, 1938)². He and other researchers then used it to describe how rewards, punishments, and other external factors affect behavior over time. This work was carried out primarily using nonhuman subjects—mostly rats and pigeons. This approach, which Skinner called the **experimental analysis of behavior**—remains an important subfield of psychology and continues to rely almost exclusively on single-subject research. For excellent examples of this work, look at any issue of the *Journal of the Experimental Analysis of Behavior*. By the 1960s, many researchers were interested in using this approach to conduct applied research primarily with humans—a subfield now called **applied behavior analysis** (Baer, Wolf, & Risley, 1968)³. Applied behavior analysis plays an especially important role in contemporary research on developmental disabilities, education, organizational behavior, and health,

2. Skinner, B. F. (1938). *The behavior of organisms: An experimental analysis*. New York, NY: Appleton-Century-Crofts.

3. Baer, D. M., Wolf, M. M., & Risley, T. R. (1968). Some current dimensions of applied behavior analysis. *Journal of Applied Behavior Analysis*, 1, 91–97.

among many other areas. Excellent examples of this work (including the study by Hall and his colleagues) can be found in the *Journal of Applied Behavior Analysis*.

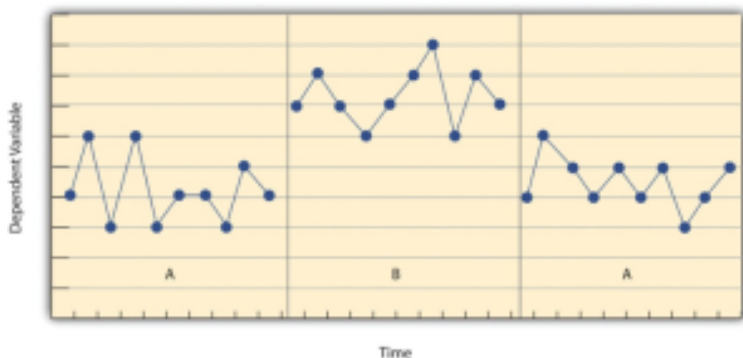
Although most contemporary single-subject research is conducted from the behavioral perspective, it can in principle be used to address questions framed in terms of any theoretical perspective. For example, a studying technique based on cognitive principles of learning and memory could be evaluated by testing it on individual high school students using the single-subject approach. The single-subject approach can also be used by clinicians who take any theoretical perspective—behavioral, cognitive, psychodynamic, or humanistic—to study processes of therapeutic change with individual clients and to document their clients' improvement (Kazdin, 1982)⁴.

General Features of Single-Subject Designs

Before looking at any specific single-subject research designs, it will be helpful to consider some features that are common to most of them. Many of these features are illustrated in Figure

4. Kazdin, A. E. (1982). *Single-case research designs: Methods for clinical and applied settings*. New York, NY: Oxford University Press.

18.1, which shows the results of a generic single-subject study. First, the dependent variable (represented on the y -axis of the graph) is measured repeatedly over time (represented by the x -axis) at regular intervals. Second, the study is divided into distinct phases, and the participant is tested under one condition per phase. The conditions are often designated by capital letters: A, B, C, and so on. Thus, Figure 18.1 represents a design in which the participant was tested first in one condition (A), then tested in another condition (B), and finally retested in the original condition (A). (This is called a reversal design and will be discussed in more detail shortly.)



*Figure 18.1 Results of a Generic Single-Subject Study
Illustrating Several Principles of Single-Subject Research
[\[Image Description\]](#)*

Another important aspect of single-subject research is that the change from one condition to the next does not usually occur after a fixed amount of time or number of observations. Instead, it depends on the participant's behavior. Specifically,

the researcher waits until the participant's behavior in one condition becomes fairly consistent from observation to observation before changing conditions. This is sometimes referred to as the **steady state strategy** (Sidman, 1960)⁵. The idea is that when the dependent variable has reached a steady state, then any change across conditions will be relatively easy to detect. Recall that we encountered this same principle when discussing experimental research more generally. The effect of an independent variable is easier to detect when the “noise” in the data is minimized.

Reversal Designs

The most basic single-subject research design is the **reversal design**, also called the **ABA design**. During the first phase, A, a **baseline** is established for the dependent variable. This is the level of responding before any treatment is introduced, and therefore, the baseline phase is a kind of control condition. When steady state responding is reached, phase B begins as the researcher introduces the treatment. There may be a period of adjustment to the treatment during which the behavior of interest becomes more variable and

5. Sidman, M. (1960). *Tactics of scientific research: Evaluating experimental data in psychology*. Boston, MA: Authors Cooperative.

begins to increase or decrease. Again, the researcher waits until that dependent variable reaches a steady state so that it is clear whether and how much it has changed. Finally, the researcher removes the treatment and again waits until the dependent variable reaches a steady state. This basic reversal design can also be extended with the reintroduction of the treatment (ABAB), another return to baseline (ABABA), and so on.

The study by Hall and his colleagues employed an ABAB reversal design. Figure 18.2 approximates the data for Robbie. The percentage of time he spent studying (the dependent variable) was low during the first baseline phase, increased during the first treatment phase until it leveled off, decreased during the second baseline phase, and again increased during the second treatment phase.

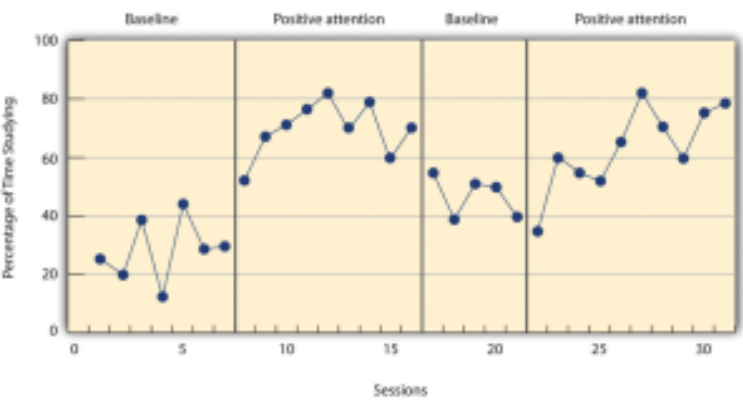


Figure 18.2 An Approximation of the Results for Hall and Colleagues' Participant Robbie in Their ABAB Reversal Design. [Image description]

Why is the reversal—the removal of the treatment—considered to be necessary in this type of design? Why use an ABA design, for example, rather than a simpler AB design? Notice that an AB design is essentially an interrupted time-series design applied to an individual participant. Recall that one problem with that design is that if the dependent variable changes after the treatment is introduced, it is not always clear that the treatment was responsible for the change. It is possible that something else changed at around the same time and that this extraneous variable is responsible for the change in the dependent variable. But if the dependent variable changes with the introduction of the treatment and then changes *back* with the removal of the treatment (assuming that the treatment does not create a permanent effect), it is much clearer that the treatment (and removal of the treatment) is the cause. In other words, the reversal greatly increases the internal validity of the study.

There are close relatives of the basic reversal design that allow for the evaluation of more than one treatment. In a **multiple-treatment reversal design**, a baseline phase is followed by separate phases in which different treatments are introduced. For example, a researcher might establish a baseline of studying behavior for a disruptive student (A), then introduce a treatment involving positive attention from the teacher (B), and then switch to a treatment involving taking a break (escaping a non-preferred situation) (C). The participant could then be returned to a baseline phase before

reintroducing each treatment—perhaps in the reverse order as a way of controlling for carryover effects. This particular multiple-treatment reversal design could also be referred to as an ABCACB design.

In an **alternating treatments design**, two or more treatments are alternated relatively quickly on a regular schedule. For example, positive attention for studying could be used one day and receiving a break when requested the next, and so on. Or one treatment could be implemented in the morning and another in the afternoon. The alternating treatments design can be a quick and effective way of comparing treatments, but only when the treatments are fast acting.

Multiple-Baseline Designs

There are two potential problems with the reversal design—both of which have to do with the removal of the treatment. One is that if a treatment is working, it may be unethical to remove it. For example, if a treatment seemed to reduce the incidence of self-injury in a child with an intellectual delay, it would be unethical to remove that treatment just to show that the incidence of self-injury increases. The second problem is that the dependent variable may not return to baseline when the treatment is removed. For example, when positive attention for studying is removed, a student might continue to study at an increased rate. This

could mean that the positive attention had a lasting effect on the student's studying, which, of course, would be good. But it could also mean that the positive attention was not really the cause of the increased studying in the first place. Perhaps something else happened at about the same time as the treatment—for example, the student's parents might have started rewarding him for good grades. One solution to these problems is to use a **multiple-baseline design**, which is represented in Figure 18.3. There are three different types of multiple-baseline designs, which we will now consider.

Multiple-Baseline Design Across Participants

In one version of the design, a baseline is established for each of several participants, and the treatment is then introduced for each one. In essence, each participant is tested in an AB design. The key to this design is that the treatment is introduced at a different *time* for each participant. The idea is that if the dependent variable changes when the treatment is introduced for one participant, it might be a coincidence. But if the dependent variable changes when the treatment is introduced for multiple participants—especially when the treatment is introduced at different times for the different participants—then it is unlikely to be a coincidence.

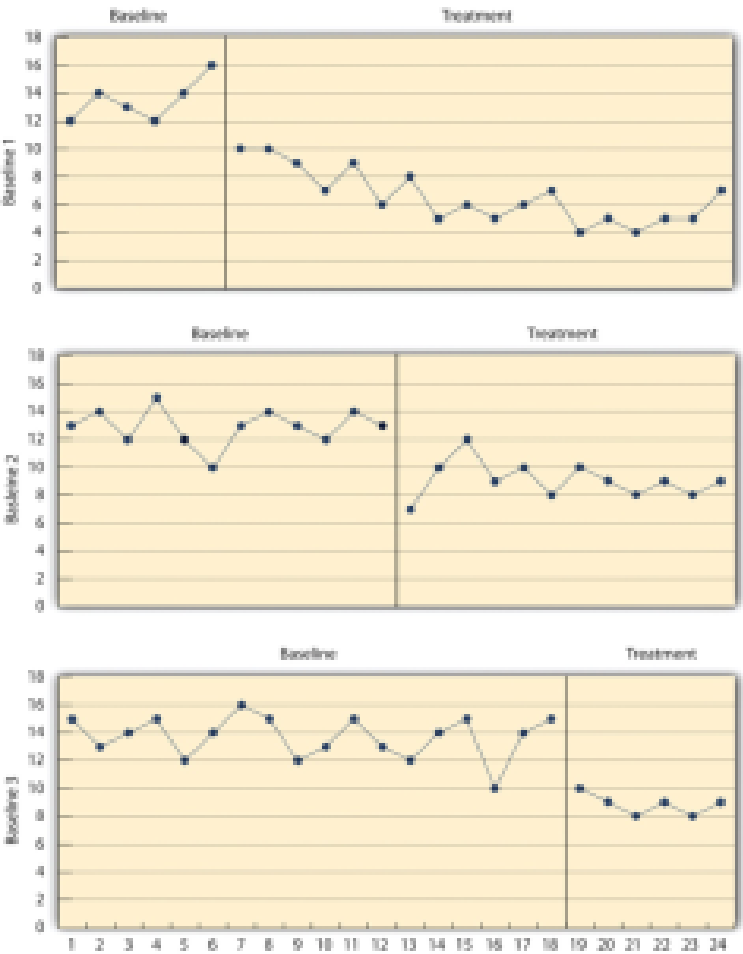


Figure 18.3 Results of a Generic Multiple-Baseline Study. The multiple baselines can be for different participants, dependent variables, or settings. The treatment is introduced at a different time on each baseline. [\[Image description\]](#)

As an example, consider a study by Scott Ross and Robert

Horner (Ross & Horner, 2009)⁶. They were interested in how a school-wide bullying prevention program affected the bullying behavior of particular problem students. At each of the three different schools, the researchers studied two students who had regularly engaged in bullying. During the baseline phase, they observed the students for 10-minute periods each day during lunch recess and counted the number of aggressive behaviors they exhibited toward their peers. After 2 weeks, they implemented the program at one school. After 2 more weeks, they implemented it at the second school. And after 2 more weeks, they implemented it at the third school. They found that the number of aggressive behaviors exhibited by each student dropped shortly after the program was implemented at the student's school. Notice that if the researchers had only studied one school or if they had introduced the treatment at the same time at all three schools, then it would be unclear whether the reduction in aggressive behaviors was due to the bullying program or something else that happened at about the same time it was introduced (e.g., a holiday, a television program, a change in the weather). But with their multiple-baseline design, this kind of coincidence would have to happen three separate times—a very unlikely occurrence—to explain their results.

6. Ross, S. W., & Horner, R. H. (2009). Bully prevention in positive behavior support. *Journal of Applied Behavior Analysis*, 42, 747–759.

Multiple-Baseline Design Across Behaviors

In another version of the multiple-baseline design, multiple baselines are established for the same participant but for different dependent variables, and the treatment is introduced at a different time for each dependent variable. Imagine, for example, a study on the effect of setting clear goals on the productivity of an office worker who has two primary tasks: making sales calls and writing reports. Baselines for both tasks could be established. For example, the researcher could measure the number of sales calls made and reports written by the worker each week for several weeks. Then the goal-setting treatment could be introduced for *one* of these tasks, and at a later time the same treatment could be introduced for the other task. The logic is the same as before. If productivity increases on one task after the treatment is introduced, it is unclear whether the treatment caused the increase. But if productivity increases on both tasks after the treatment is introduced—especially when the treatment is introduced at two different times—then it seems much clearer that the treatment was responsible.

Multiple-Baseline Design Across Settings

In yet a third version of the multiple-baseline design, multiple

baselines are established for the same participant but in different settings. For example, a baseline might be established for the amount of time a child spends reading during his free time at school and during his free time at home. Then a treatment such as positive attention might be introduced first at school and later at home. Again, if the dependent variable changes after the treatment is introduced in each setting, then this gives the researcher confidence that the treatment is, in fact, responsible for the change.

Data Analysis in Single-Subject Research

In addition to its focus on individual participants, single-subject research differs from group research in the way the data are typically analyzed. As we have seen throughout the book, group research involves combining data across participants. Group data are described using statistics such as means, standard deviations, correlation coefficients, and so on to detect general patterns. Finally, inferential statistics are used to help decide whether the result for the sample is likely to generalize to the population. Single-subject research, by contrast, relies heavily on a very different approach called **visual inspection**. This means plotting individual participants' data as shown throughout this chapter, looking carefully at those data, and making judgments about whether and to what extent the independent variable had an effect on

the dependent variable. Inferential statistics are typically not used.

In visually inspecting their data, single-subject researchers take several factors into account. One of them is changes in the **level** of the dependent variable from condition to condition. If the dependent variable is much higher or much lower in one condition than another, this suggests that the treatment had an effect. A second factor is **trend**, which refers to gradual increases or decreases in the dependent variable across observations. If the dependent variable begins increasing or decreasing with a change in conditions, then again this suggests that the treatment had an effect. It can be especially telling when a trend changes directions—for example, when an unwanted behavior is increasing during baseline but then begins to decrease with the introduction of the treatment. A third factor is **latency**, which is the time it takes for the dependent variable to begin changing after a change in conditions. In general, if a change in the dependent variable begins shortly after a change in conditions, this suggests that the treatment was responsible.

In the top panel of Figure 18.4, there are fairly obvious changes in the level and trend of the dependent variable from condition to condition. Furthermore, the latencies of these changes are short; the change happens immediately. This pattern of results strongly suggests that the treatment was responsible for the changes in the dependent variable. In the bottom panel of Figure 18.4, however, the changes in level are

fairly small. And although there appears to be an increasing trend in the treatment condition, it looks as though it might be a continuation of a trend that had already begun during baseline. This pattern of results strongly suggests that the treatment was not responsible for any changes in the dependent variable—at least not to the extent that single-subject researchers typically hope to see.

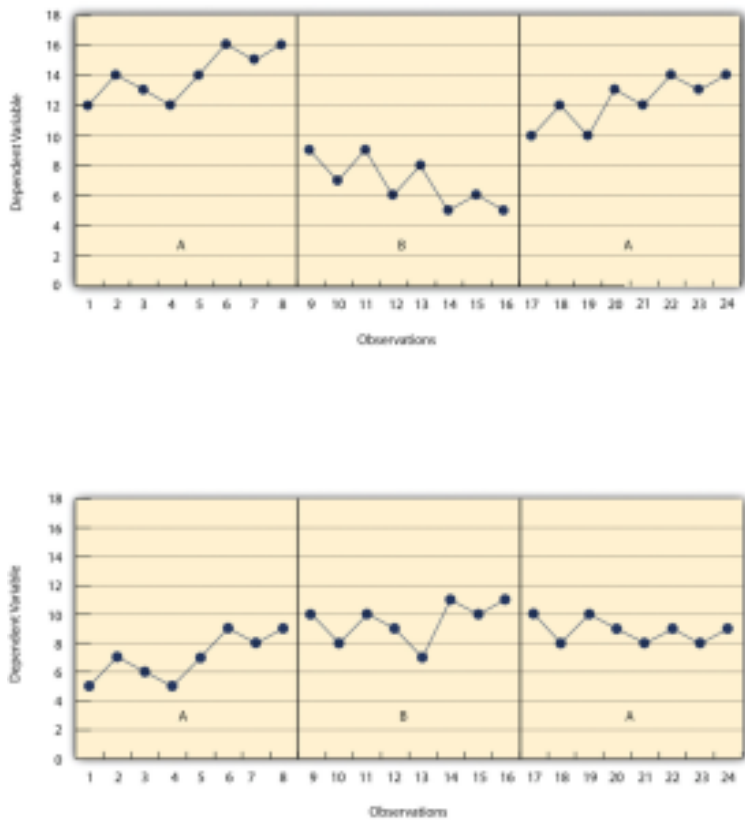


Figure 18.4 Results of a Generic Single-Subject Study Illustrating Level, Trend, and Latency. Visual inspection of the data suggests an effective treatment in the top panel but an ineffective treatment in the bottom panel. [\[Image description\]](#)

The results of single-subject research can also be analyzed using statistical procedures—and this is becoming more common. There are many different approaches, and single-subject researchers continue to debate which are the most

useful. One approach parallels what is typically done in group research. The mean and standard deviation of each participant's responses under each condition are computed and compared, and inferential statistical tests such as the t test or analysis of variance are applied (Fisch, 2001)⁷. (Note that averaging *across* participants is less common.) Another approach is to compute the **percentage of non-overlapping data** (PND) for each participant (Scruggs & Mastropieri, 2001)⁸. This is the percentage of responses in the treatment condition that are more extreme than the most extreme response in a relevant control condition. In the study of Hall and his colleagues, for example, all measures of Robbie's study time in the first treatment condition were greater than the highest measure in the first baseline, for a PND of 100%. The greater the percentage of non-overlapping data, the stronger the treatment effect. Still, formal statistical approaches to data analysis in single-subject research are generally considered a supplement to visual inspection, not a replacement for it.

Single Subject vs Group Design

7. Fisch, G. S. (2001). Evaluating data from behavioral analysis: Visual inspection or statistical models. *Behavioral Processes*, 54, 137–154.

8. Scruggs, T. E., & Mastropieri, M. A. (2001). How to summarize single-participant research: Ideas and applications. *Exceptionality*, 9, 227–244.

Debate

Single-subject research is similar to group research—especially experimental group research—in many ways. They are both quantitative approaches that try to establish causal relationships by manipulating an independent variable, measuring a dependent variable, and controlling extraneous variables. But there are important differences between these approaches too, and these differences sometimes lead to disagreements. It is worth addressing the most common points of disagreement between single-subject researchers and group researchers and how these disagreements can be resolved. As we will see, single-subject research and group research are probably best conceptualized as complementary approaches.

Data Analysis

One set of disagreements revolves around the issue of data analysis. Some advocates of group research worry that visual inspection is inadequate for deciding whether and to what extent a treatment has affected a dependent variable. One specific concern is that visual inspection is not sensitive enough to detect weak effects. A second is that visual inspection can be unreliable, with different researchers reaching different conclusions about the same set of data

(Danov & Symons, 2008)⁹. A third is that the results of visual inspection—an overall judgment of whether or not a treatment was effective—cannot be clearly and efficiently summarized or compared across studies (unlike the measures of relationship strength typically used in group research).

In general, single-subject researchers share these concerns. However, they also argue that their use of the steady state strategy, combined with their focus on strong and consistent effects, minimizes most of them. If the effect of a treatment is difficult to detect by visual inspection because the effect is weak or the data are noisy, then single-subject researchers look for ways to increase the strength of the effect or reduce the noise in the data by controlling extraneous variables (e.g., by administering the treatment more consistently). If the effect is still difficult to detect, then they are likely to consider it neither strong enough nor consistent enough to be of further interest. Many single-subject researchers also point out that statistical analysis is becoming increasingly common and that many of them are using this as a supplement to visual inspection—especially for the purpose of comparing results across studies (Scruggs & Mastropieri, 2001)¹⁰.

9. Danov, S. E., & Symons, F. E. (2008). A survey evaluation of the reliability of visual inspection and functional analysis graphs. *Behavior Modification*, 32, 828–839.

10. Scruggs, T. E., & Mastropieri, M. A. (2001). How to summarize single-participant research: Ideas and applications. *Exceptionality*, 9, 227–244.

Turning the tables, some advocates of single-subject research worry about the way that group researchers analyze their data. Specifically, they point out that focusing on group means can be highly misleading. Again, imagine that a treatment has a strong positive effect on half the people exposed to it and an equally strong negative effect on the other half. In a traditional between-subjects experiment, the positive effect on half the participants in the treatment condition would be statistically cancelled out by the negative effect on the other half. The mean for the treatment group would then be the same as the mean for the control group, making it seem as though the treatment had no effect when in fact it had a strong effect on every single participant!

But again, group researchers share this concern. Although they do focus on group statistics, they also emphasize the importance of examining distributions of individual scores. For example, if some participants were positively affected by a treatment and others negatively affected by it, this would produce a bimodal distribution of scores and could be detected by looking at a histogram of the data. The use of within-subjects designs is another strategy that allows group researchers to observe effects at the individual level and even to specify what percentage of individuals exhibit strong, medium, weak, and even negative effects. Finally, factorial designs can be used to examine whether the effects of an independent variable on a dependent variable differ in different groups of participants (introverts vs. extraverts).

External Validity

The second issue about which single-subject and group researchers sometimes disagree has to do with external validity—the ability to generalize the results of a study beyond the people and specific situation actually studied. In particular, advocates of group research point out the difficulty in knowing whether results for just a few participants are likely to generalize to others in the population. Imagine, for example, that in a single-subject study, a treatment has been shown to reduce self-injury for each of two children with intellectual disabilities. Even if the effect is strong for these two children, how can one know whether this treatment is likely to work for other children with intellectual delays?

Again, single-subject researchers share this concern. In response, they note that the strong and consistent effects they are typically interested in—even when observed in small samples—are likely to generalize to others in the population. Single-subject researchers also note that they place a strong emphasis on replicating their research results. When they observe an effect with a small sample of participants, they typically try to replicate it with another small sample—perhaps with a slightly different type of participant or under slightly different conditions. Each time they observe similar results, they rightfully become more confident in the generality of those results. Single-subject researchers can also point to the fact that the principles of classical and operant

conditioning—most of which were discovered using the single-subject approach—have been successfully generalized across an incredibly wide range of species and situations.

And, once again, turning the tables, single-subject researchers have concerns of their own about the external validity of group research. One extremely important point they make is that studying large groups of participants does not entirely solve the problem of generalizing to other *individuals*. Imagine, for example, a treatment that has been shown to have a small positive effect on average in a large group study. It is likely that although many participants exhibited a small positive effect, others exhibited a large positive effect, and still others exhibited a small negative effect. When it comes to applying this treatment to another large *group*, we can be fairly sure that it will have a small effect on average. But when it comes to applying this treatment to another *individual*, we cannot be sure whether it will have a small, a large, or even a negative effect. Another point that single-subject researchers make is that group researchers also face a similar problem when they study a single situation and then generalize their results to other situations. For example, researchers who conduct a study on the effect of cell phone use on drivers on a closed oval track probably want to apply their results to drivers in many other real-world driving situations. But notice that this requires generalizing from a single situation to a population of situations. Thus, the ability to generalize is based on much more than just the sheer number of participants one has

studied. It requires a careful consideration of the similarity of the participants *and* situations studied to the population of participants and situations to which one wants to generalize (Shadish, Cook, & Campbell, 2002)¹¹.

Single-Subject and Group Research as Complementary Methods

As with quantitative and qualitative research, it is probably best to conceptualize single-subject research and group research as complementary methods that have different strengths and weaknesses and that are appropriate for answering different kinds of research questions (Kazdin, 1982)¹². Single-subject research is particularly good for testing the effectiveness of treatments on individuals when the focus is on strong, consistent, and biologically or socially important effects. It is also especially useful when the behavior of particular individuals is of interest. Clinicians who work with

11. Shadish, W. R., Cook, T. D., & Campbell, D. T. (2002). *Experimental and quasi-experimental designs for generalized causal inference*. Boston, MA: Houghton Mifflin.

12. Kazdin, A. E. (1982). *Single-case research designs: Methods for clinical and applied settings*. New York, NY: Oxford University Press.

only one individual at a time may find that it is their only option for doing systematic quantitative research.

Group research, on the other hand, is ideal for testing the effectiveness of treatments at the group level. Among the advantages of this approach is that it allows researchers to detect weak effects, which can be of interest for many reasons. For example, finding a weak treatment effect might lead to refinements of the treatment that eventually produce a larger and more meaningful effect. Group research is also good for studying interactions between treatments and participant characteristics. For example, if a treatment is effective for those who are high in motivation to change and ineffective for those who are low in motivation to change, then a group design can detect this much more efficiently than a single-subject design. Group research is also necessary to answer questions that cannot be addressed using the single-subject approach, including questions about independent variables that cannot be manipulated (e.g., number of siblings, extraversion, culture).

Finally, it is important to understand that the single-subject and group approaches represent different research traditions. This factor is probably the most important one affecting which approach a researcher uses. Researchers in the experimental analysis of behavior and applied behavior analysis learn to conceptualize their research questions in ways that are amenable to the single-subject approach. Researchers in most other areas of psychology learn to conceptualize their

research questions in ways that are amenable to the group approach. At the same time, there are many topics in psychology in which research from the two traditions has informed each other and been successfully integrated. One example is research suggesting that both animals and humans have an innate “number sense”—an awareness of how many objects or events of a particular type have they have experienced without actually having to count them (Dehaene, 2011)¹³. Single-subject research with rats and birds and group research with human infants have shown strikingly similar abilities in those populations to discriminate small numbers of objects and events. This number sense—which probably evolved long before humans did—may even be the foundation of humans’ advanced mathematical abilities.

The Principle of Converging Evidence

Now that you have been introduced to many of the most commonly used research methods in psychology, it should be readily apparent that no design is perfect. Every research design has strengths and weaknesses. True experiments typically have

13. Dehaene, S. (2011). *The number sense: How the mind creates mathematics* (2nd ed.). New York, NY: Oxford.

high internal validity but may have problems with external validity, while non-experimental research (e.g., correlational research) often has good external validity but poor internal validity. Each study brings us closer to the truth, but no single study can ever be considered definitive. This is one reason why, in science, we say there is no such thing as scientific proof; there is only scientific evidence.

While the media will often try to reach strong conclusions on the basis of the findings of one study, scientists focus on evaluating a body of research. Scientists evaluate theories not by waiting for the perfect experiment but by looking at the overall trends in a number of partially flawed studies. The idea of **converging evidence** tells us to examine the pattern of flaws running through the research literature because the nature of this pattern can either support or undermine the conclusions we wish to draw. Suppose the findings from a number of different studies were largely consistent in supporting a particular conclusion. If all of the studies were flawed in a similar way, for example, if all of the studies were correlational and contained the third variable problem and the directionality problem, this would undermine confidence in the conclusions drawn because the consistency of the outcome may simply have resulted from a particular flaw that all of the studies shared. On the other hand, if all of the studies were flawed in different ways and the weakness of some of the studies were the strength of others (the low external validity of a true experiment was balanced by the high external validity of

a correlational study), then we could be more confident in our conclusions.

While there are fundamental tradeoffs in different research methods, the diverse set of approaches used by psychologists has complementary strengths that allow us to search for converging evidence. We can reach meaningful conclusions and come closer to understanding truth by examining a large number of different studies, each with different strengths and weaknesses. If the results of a large number of studies, all conducted using different designs, converge on the same conclusion, then our confidence in that conclusion can be increased dramatically. In science, we strive for progress, not perfection.

Equity Activity: Using Single Case Design to engage in race-conscious teacher coaching

Research has continued to demonstrate the ways that Black children are disproportionately impacted by discipline practices in K-12 education settings (Skiba et al., 2011)¹⁴ This is especially salient for

14. Skiba, R. J., Horner, R. H., Chung, C. G., Rausch, M. K., May, S. L., & Tobin, T.

those students who also qualify for special education services (Samimi et al., 2023) ¹⁵. Mossing and colleagues (2023) ¹⁶ used a single case design approach to understand if the teacher coaching process that specifically looks at addressing racial bias could interrupt this process. Their study took place in a small alternative school in the Midwest and had two teachers and 6 black students. The researchers coached the teachers using Mindful STOP ¹⁷ to help teachers regulate before engaging

(2011). Race is not neutral: A investigation of African American and Latino disproportionality in school discipline. *School Psychology Review*, 40(1), 85–107. <https://doi.org/10.1080/02796015.2011.12087730>

15. Samimi, C., Jefferson, N., Flanagan, S., & Anyon, Y. (2023). Intersections of disproportion: A critical quantitative examination of dis/ability and gender in Black students' school discipline outcomes. *The Urban Review*, 55(4), 456-475.
16. Mossing, K. W., Collins, T. A., Newman, D. S., King, K. A., & Pollard, J. F. (2024). Race-Conscious Teacher Coaching: Disrupting Implicit Bias and Improving Teachers' Relationships with Black Students. *Journal of Educational and Psychological Consultation*, 35(1), 114–141. <https://doi.org/10.1080/10474412.2024.2359673>
17. Cook, C. R., Duong, M. T., McIntosh, K., Fiat, A. E., Larson, M., Pullmann, M. D., McGinnis, J., & Bradshaw, C. (2018). Addressing discipline disparities for

with a student and reviewed a mindful reflection protocol¹⁸ at weekly coaching sessions. To determine if any changes occurred, trained observers went to the classroom for 20 minutes each day to document positive and negative communication. The following graph is for teacher 1's positive communication:

Black male students: Linking malleable root causes to feasible and effective practices. *School Psychology Review*, 47(2), 135–152. <https://doi.org/10.17105/SPR-2017-0026.V47-2>

18. Dray, B. J., & Wisneski, D. B. (2011). Mindful reflection as a process for developing culturally responsive practices. *Teaching Exceptional Children*, 44(1), 28–36. <https://doi.org/10.1177/004005991104400104>

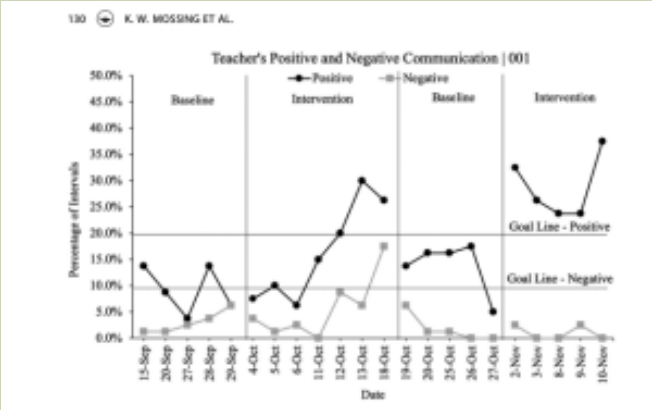


Figure 18.5. Teacher 1 communication. There was a physical altercation between students on 10/18. [Image Description]



An interactive HSP element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=74#h5p-19>

License & Attribution

“Single-Subject Research Design” by Stephanie D’Costa is adapted from “[Single-Subject Research](#)” by Rajiv S. Jhangiani; I-Chant A. Chiang; Carrie Cuttler; and Dana C. Leighton is licensed CC BY-NC-SA 3.0.

“Single-Subject Research Design” is licensed under CC BY-NC-SA 4.0.

Image Descriptions

Figure 18.1. Time-series chart with Time on the x-axis and Dependent Variable on the y-axis. The plot is divided by two vertical lines into three segments labeled beneath as A, B, and A. In the first A (baseline), blue dots fluctuate at lower values. In B (intervention), the dots shift upward with higher averages and more variability. In the final A (withdrawal), values fall toward the original baseline range, with small ups and downs. The pattern suggests the intervention increased the dependent

measure during phase B, with reduction after it was removed.

[\[Return to Figure 18.1\]](#)

Figure 18.2. A time-series line chart with tan background plots Percentage of Time Studying (y-axis 0–100) by Sessions (x-axis). The figure is divided by vertical lines into four labeled segments across time: Baseline, Positive attention, Baseline, Positive attention. Within each of three parallel series (separated horizontally by the vertical dividers), blue points fluctuate at lower levels during Baseline (roughly 10–40%) and shift upward during Positive attention (generally 50–90%), with some variability. The repeated increases when the intervention is introduced—and decreases when baseline returns—illustrate an experimental effect of positive attention on study time. [\[Return to Figure 18.2\]](#)

Figure 18.3. Three line graphs showing the results of a generic multiple-baseline study, in which different baselines are established and treatment is introduced to participants at different times. For Baseline 1, treatment is introduced one-quarter of the way into the study. The dependent variable ranges between 12 and 16 units during the baseline, but drops down to 10 units with treatment and mostly decreases until the end of the study, ranging between 4 and 10 units. For Baseline 2, treatment is introduced halfway through the study. The dependent variable ranges between 10 and 15 units during the baseline, then has a sharp decrease to 7 units when treatment is introduced. However, the dependent variable increases to 12 units soon after the drop and ranges between

8 and 10 units until the end of the study. For Baseline 3, treatment is introduced three-quarters of the way into the study. The dependent variable ranges between 12 and 16 units for the most part during the baseline, with one drop down to 10 units. When treatment is introduced, the dependent variable drops down to 10 units and then ranges between 8 and 9 units until the end of the study. [\[Return to Figure 18.3\]](#)

Figure 18.4. Two graphs showing the results of a generic single-subject study with an ABA design. In the first graph, under condition A, level is high and the trend is increasing. Under condition B, level is much lower than under condition A and the trend is decreasing. Under condition A again, level is about as high as the first time and the trend is increasing. For each change, latency is short, suggesting that the treatment is the reason for the change. In the second graph, under condition A, level is relatively low and the trend is increasing. Under condition B, level is a little higher than during condition A and the trend is increasing slightly. Under condition A again, level is a little lower than during condition B and the trend is decreasing slightly. It is difficult to determine the latency of these changes, since each change is rather minute, which suggests that the treatment is ineffective. [\[Return to Figure 18.4\]](#)

Figure 18.5. A time-series line chart with tan background plots Percentage of Time Studying (y-axis 0–100) by Sessions (x-axis). The figure is divided by vertical lines into four labeled segments across time: Baseline, Positive attention, Baseline,

Positive attention. Within each of three parallel series (separated horizontally by the vertical dividers), blue points fluctuate at lower levels during Baseline (roughly 10–40%) and shift upward during Positive attention (generally 50–90%), with some variability. The repeated increases when the intervention is introduced—and decreases when baseline returns—illustrate an experimental effect of positive attention on study time. [\[Return to Figure 18.5\]](#)

Media Attributions

- 10.1_10.2
- 10.2_10.3
- 10.3_10.4
- 10.4_10.5

19.

NON-EXPERIMENTAL RESEARCH

Mireille Ukeye

Learning Objectives

By the end of this chapter, you will be able to

1. Define non-experimental research, distinguish it clearly from experimental research, and give several examples.
2. Explain when a researcher might choose to conduct non-experimental research as opposed to experimental research.
3. Identify how non-experimental research supports ethical and inclusive study of diverse

populations.

What Is Non-Experimental Research?

Non-experimental research is research that lacks the manipulation of an independent variable. Rather than manipulating an independent variable, researchers conducting non-experimental research simply measure variables as they naturally occur (in the lab or the real world).

Most researchers in psychology consider the distinction between experimental and non-experimental research to be an extremely important one. This is because although experimental research can provide strong evidence that changes in an independent variable cause differences in a dependent variable, non-experimental research generally cannot. As we will see, however, this inability to make causal conclusions does not mean that non-experimental research is less important than experimental research. It is simply used in cases where experimental research is not able to be carried out.

When to Use Non-Experimental Research

Experimental research is appropriate when the researcher has a specific research question or hypothesis about a causal relationship between two variables, and it is possible, feasible, and ethical to manipulate the independent variable. It stands to reason, therefore, that non-experimental research is appropriate—even necessary—when these conditions are not met. There are many times in which non-experimental research is preferred, including when:

- the research question or hypothesis relates to a single variable rather than a statistical relationship between two variables (e.g., how accurate are people's first impressions?).
- the research question pertains to a non-causal statistical relationship between variables (e.g., is there a correlation between verbal intelligence and mathematical intelligence?).
- the research question is about a causal relationship, but the independent variable cannot be manipulated, or participants cannot be randomly assigned to conditions or orders of conditions for practical or ethical reasons (e.g., does damage to a person's hippocampus impair the formation of long-term memory traces?).
- the research question is broad and exploratory, or is

about what it is like to have a particular experience (e.g., what is it like to be a working mother diagnosed with depression?).

The choice between the experimental and non-experimental approaches is generally dictated by the nature of the research question. Recall the three goals of science are to describe, to predict, and to explain. If the goal is to explain and the research question pertains to causal relationships, then the experimental approach is typically preferred. If the goal is to describe or to predict, a non-experimental approach is appropriate. But the two approaches can also be used to address the same research question in complementary ways. For example, in Milgram's original (non-experimental) obedience study, he was primarily interested in one variable—the extent to which participants obeyed the researcher when he told them to shock the confederate—and he observed all participants performing the same task under the same conditions. However, Milgram subsequently conducted experiments to explore the factors that affect obedience. He manipulated several independent variables, such as the distance between the experimenter and the

participant, the participant and the confederate, and the location of the study (Milgram, 1974)¹.

Types of Non-Experimental Research

Non-experimental research falls into two broad categories: correlational research and observational research.

The most common type of non-experimental research conducted in psychology is correlational research. Correlational research is



considered non-experimental because it focuses on the statistical relationship between two variables but does not include the manipulation of an independent variable. More specifically, in **correlational research**, the researcher measures two variables with little or no attempt to control extraneous variables and then assesses the relationship between them. As an example, a researcher interested in the relationship between self-esteem and school achievement could collect data

1. Milgram, S. (1974). *Obedience to authority: An experimental view*. New York, NY: Harper & Row.

on students' self-esteem and their GPAs to see if the two variables are statistically related.

Observational research is non-experimental because it focuses on making observations of behavior in a natural or laboratory setting without manipulating anything. Milgram's original obedience study was non-experimental in this way. He was primarily interested in the extent to which participants obeyed the researcher when he told them to shock the confederate, and he observed all participants performing the same task under the same conditions. The variable was whether participants "remembered" having experienced mildly traumatic childhood events (e.g., getting lost in a shopping mall) that they had not actually experienced but that the researchers asked them about repeatedly. In this particular study, nearly a third of the participants "remembered" at least one event. (As with Milgram's original study, this study inspired several later experiments on the factors that affect false memories).

Cross-Sectional, Longitudinal, and Cross-Sequential Studies

When psychologists wish to study change over time (for example, when developmental

psychologists wish to study aging), they usually take one of three non-experimental approaches: cross-sectional, longitudinal, or cross-sequential.

Cross-sectional studies involve comparing two or more pre-existing groups of people (e.g., children at different stages of development). What makes this approach non-experimental is that there is no manipulation of an independent variable and no random assignment of participants to groups. Using this design, developmental psychologists compare groups of people of different ages (e.g., young adults spanning from 18-25 years of age versus older adults spanning 60-75 years of age) on various dependent variables (e.g., memory, depression, life satisfaction). Of course, the primary limitation of using this design to study the effects of aging is that differences between the groups other than age may account for differences in the dependent variable. For instance, differences between the groups may reflect the generation that people come from (a **cohort effect**) rather than a direct effect of age. For this reason, **longitudinal studies**, in which one group of people is followed over time as they

age, offer a superior means of studying the effects of aging. However, longitudinal studies are by definition more time-consuming and so require a much greater investment on the part of the researcher and the participants. A third approach, known as **cross-sequential studies**, combines elements of both cross-sectional and longitudinal studies. Rather than measuring differences between people in different age groups or following the same people over a long period of time, researchers adopting this approach choose a smaller period of time during which they follow people in different age groups. For example, they might measure changes over a ten-year period among participants who at the start of the study fall into the following age groups: 20 years old, 30 years old, 40 years old, 50 years old, and 60 years old. This design is advantageous because the researcher reaps the immediate benefits of being able to compare the age groups after the first assessment. Further, by following the different age groups over time, they can subsequently determine whether the original differences they found across the age

groups are due to true age effects or cohort effects.

Many observational research studies are more qualitative in nature. In qualitative research, the data are usually nonnumerical and therefore cannot be analyzed using statistical techniques. Rosenhan's observational study of the experience of people in psychiatric wards was primarily qualitative. The data were the notes taken by the "pseudopatients"—the people pretending to have heard voices—along with their hospital records. Rosenhan's analysis consists mainly of a written description of the experiences of the pseudopatients, supported by several concrete examples. To illustrate the hospital staff's tendency to "depersonalize" their patients, he noted, "Upon being admitted, I and other pseudopatients took the initial physical examinations in a semi-public room, where staff members went about their own business as if we were not there" (Rosenhan, 1973, p. 256)². Qualitative data has a separate set of analysis tools depending on the research question. For example, thematic analysis would focus on themes that emerge in the data, or

2. Rosenhan, D. L. (1973). On being sane in insane places. *Science*, 179, 250–258

conversation analysis would focus on the way the words were said in an interview or focus group.

Internal Validity

Internal validity is the extent to which the design of a study supports the conclusion that changes in the independent variable caused any observed differences in the dependent variable. Figure 19.1 shows how experimental, quasi-experimental, and non-experimental (correlational) research vary in terms of internal validity. Experimental research tends to be highest in internal validity because the use of manipulation (of the independent variable) and control (of extraneous variables) helps to rule out alternative explanations for the observed relationships. If the average score on the dependent variable in an experiment differs across conditions, it is quite likely that the independent variable is responsible for that difference. Non-experimental (correlational) research is lowest in internal validity because these designs fail to use manipulation or control. Quasi-experimental research falls in the middle because it contains some, but not all, of the features of a true experiment. For instance, it may fail to use random assignment to assign participants to groups or fail to use counterbalancing to control for potential order effects. Imagine, for example, that a researcher finds two similar schools, starts an anti-bullying program in one, and then finds fewer bullying incidents in that “treatment school” than in the

“control school.” While a comparison is being made with a control condition, the inability to randomly assign children to schools could still mean that students in the treatment school differed from students in the control school in some other way that could explain the difference in bullying (e.g., there may be a selection effect).

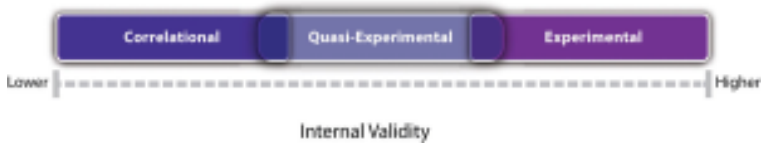


Figure 19.1. Internal Validity of Correlation, Quasi-Experimental, and Experimental Studies. Experiments are generally high in internal validity, quasi-experiments lower, and correlation (non-experimental) studies lower still.

Figure 19.1 shows that there is some overlap in the internal validity of experiments, quasi-experiments, and correlational (non-experimental) studies. For example, a poorly designed experiment that includes many confounding variables can be lower in internal validity than a well-designed quasi-experiment with no obvious confounding variables. Internal validity is also only one of several validities that one might consider.

Equity Activity: Survey on digital divide in rural communities

Researchers distributed a survey to residents in rural areas to assess internet access and its impact on education.

- Survey Topics: Internet availability, affordability, and perceived barriers to online learning.
- Findings: Many households lack access due to infrastructure gaps, and those with internet face high costs, limiting educational opportunities.
- Outcome: The findings are used to advocate for state and federal funding for broadband expansion.



An interactive HSP element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=77#h5p-20>

License & Attribution

“Non-Experimental Research” by Mireille Ukeye is adapted from “[Overview of Non-Experimental Research](#)” by Rajiv S. Jhangiani; I-Chant A. Chiang; Carrie Cuttler; and Dana C. Leighton is licensed CC BY 4.0.

“Overview of Non-Experimental Research” is licensed under CC BY-NC-SA 4.0.

Media Attributions

- [Internal Validity of Correlation, Quasi-Experimental, and Experimental Studies](#). © Rajiv S. Jhangiani; I-Chant A. Chiang; Carrie Cuttler; and Dana C. Leighton is licensed under a [CC BY-NC-SA \(Attribution NonCommercial ShareAlike\)](#) license

20.

CORRELATIONAL RESEARCH

Stephanie D'Costa

Learning Objectives

By the end of this chapter, you will be able to

1. Define correlational research and give several examples.
2. Explain why a researcher might choose to conduct correlational research rather than experimental research or another type of non-experimental research.
3. Interpret the strength and direction of different correlation coefficients.

4. Explain why correlation does not imply causation.

What Is Correlational Research?

Correlational research is a type of non-experimental research in which the researcher measures two variables (binary or continuous) and assesses the statistical relationship (i.e., the correlation) between them with little or no effort to control extraneous variables. There are many reasons that researchers interested in statistical relationships between variables would choose to conduct a correlational study rather than an experiment. The first is that they do not believe that the statistical relationship is a causal one or are not interested in causal relationships. Recall two goals of science are to describe and to predict, and the correlational research strategy allows researchers to achieve both of these goals. Specifically, this strategy can be used to describe the strength and direction of the relationship between two variables and if there is a relationship between the variables then the researchers can use scores on one variable to predict scores on the other.

Another reason that researchers would choose to use a correlational study rather than an experiment is that the statistical relationship of interest is thought to be causal, but

the researcher *cannot* manipulate the independent variable because it is impossible, impractical, or unethical. For example, while a researcher might be interested in the relationship between the frequency people use cannabis and their memory abilities, they cannot ethically manipulate the frequency at which people use cannabis. As such, they must rely on the correlational research strategy; they must simply measure the frequency that people use cannabis and measure their memory abilities using a standardized test of memory, and then determine whether the frequency people use cannabis is statistically related to memory test performance.

Correlation is also used to establish the reliability and validity of measurements. For example, a researcher might evaluate the validity of a brief extraversion test by administering it to a large group of participants along with a longer extraversion test that has already been shown to be valid. This researcher might then check to see whether participants' scores on the brief test are strongly correlated with their scores on the longer one. Neither test score is thought to cause the other, so there is no independent variable to manipulate. In fact, the terms *independent variable* and *dependent variable* do not apply to this kind of research.

Another strength of correlational research is that it is often higher in external validity than experimental research. Recall that there is typically a trade-off between internal validity and external validity. As greater controls are added to experiments, internal validity is increased but often at the expense of

external validity, as artificial conditions are introduced that do not exist in reality. In contrast, correlational studies typically have low internal validity because nothing is manipulated or controlled, but they often have high external validity. Since nothing is manipulated or controlled by the experimenter, the results are more likely to reflect relationships that exist in the real world.

Finally, extending upon this trade-off between internal and external validity, correlational research can help to provide converging evidence for a theory. If a theory is supported by a true experiment that is high in internal validity as well as by a correlational study that is high in external validity, then the researchers can have more confidence in the validity of their theory. As a concrete example, correlational studies establishing that there is a relationship between watching violent television and aggressive behavior have been complemented by experimental studies confirming that the relationship is a causal one (Bushman & Huesmann, 2001)¹.

1. Bushman, B. J., & Huesmann, L. R. (2001). Effects of televised violence on aggression. In D. Singer & J. Singer (Eds.), *Handbook of children and the media* (pp. 223–254). Thousand Oaks, CA: Sage.

Does Correlational Research Always Involve Quantitative Variables?

A common misconception among beginning researchers is that correlational research must involve two quantitative variables, such as scores on two extraversion tests or the number of daily hassles and number of symptoms people have experienced. However, the defining feature of correlational research is that the two variables are measured—neither one is manipulated—and this is true regardless of whether the variables are quantitative or categorical (See the chapter on [Variables](#) for a more detailed description).

Imagine, for example, that a researcher administers the Rosenberg Self-Esteem Scale to 50 American college students and 50 Japanese college students. Although this “feels” like a between-subjects experiment, it is a correlational study because the researcher did not manipulate the students’ nationalities. The same is true of the study by Cacioppo and Petty comparing college faculty and factory workers in terms of their need for cognition. It is a correlational study

because the researchers did not manipulate the participants' occupations.

Figure 20.1 shows data from a hypothetical study on the relationship between whether people make a daily list of things to do (a "to-do list") and stress. Notice that it is unclear whether this is an experiment or a correlational study because it is unclear whether the independent variable was manipulated. If the researcher randomly assigned some participants to make daily to-do lists and others not to, then it is an experiment. If the researcher simply asked participants whether they made daily to-do lists, then it is a correlational study. The distinction is important because if the study were an experiment, then it could be concluded that making the daily to-do lists reduced participants' stress. But if it were a correlational study, it could only be concluded that these variables are statistically related. Perhaps being stressed has a negative effect on people's ability to plan ahead (the directionality problem). Or perhaps people who are more conscientious are more likely to make to-do lists and less likely to be stressed (the third-variable problem). The crucial point is that what defines a

study as experimental or correlational is not the variables being studied, nor whether the variables are quantitative or categorical, nor the type of graph or statistics used to analyze the data. What defines a study is *how* the study is conducted.

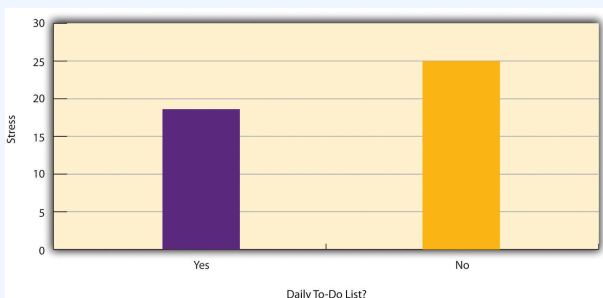


Figure 20.1. Results of a Hypothetical Study on Whether People Who Make Daily To-Do Lists Experience Less Stress Than People Who Do Not Make Such Lists [\[Image Description\]](#)

Data Collection in Correlational Research

Again, the defining feature of correlational research is that neither variable is manipulated. It does not matter how or

where the variables are measured. A researcher could have participants come to a laboratory to complete a computerized backward digit span task and a computerized risky decision-making task, and then assess the relationship between participants' scores on the two tasks. Or a researcher could go to a shopping mall to ask people about their attitudes toward the environment and their shopping habits, and then assess the relationship between these two variables. Both of these studies would be correlational because no independent variable is manipulated.

Correlations Between Quantitative Variables

Correlations between quantitative variables are often presented using **scatterplots**. Figure 20.2 shows some hypothetical data on the relationship between the amount of stress people are under and the number of physical symptoms they have. Each point in the scatterplot represents one person's score on both variables. For example, the circled point in Figure 20.2 represents a person whose stress score was 10 and who had three physical symptoms. Taking all the points into account, one can see that people under more stress tend to have more physical symptoms. This is a good example of a **positive relationship**, in which higher scores on one variable tend to be associated with higher scores on the other. In other words, they move in the same direction, either both up or both

down. A **negative relationship** is one in which higher scores on one variable tend to be associated with lower scores on the other. In other words, they move in opposite directions. There is a negative relationship between stress and immune system functioning, for example, because higher stress is associated with lower immune system functioning.

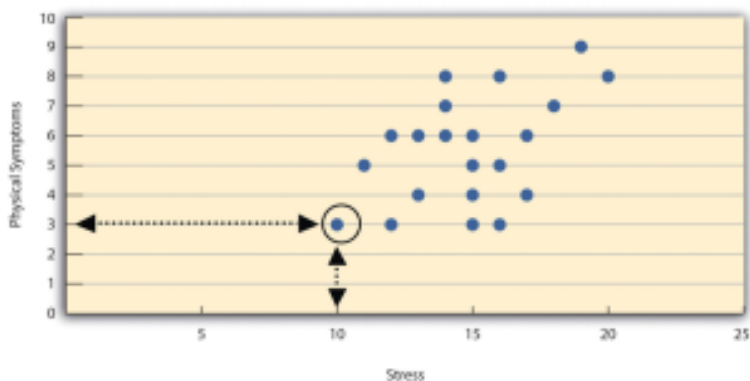


Figure 20.2 Scatterplot Showing a Hypothetical Positive Relationship Between Stress and Number of Physical Symptoms. The circled point represents a person whose stress score was 10 and who had three physical symptoms. Pearson's r for these data is $+.51$. [\[Image Description\]](#)

The strength of a correlation between quantitative variables is typically measured using a statistic called **Pearson's Correlation Coefficient (or Pearson's r)**. As Figure 20.3 shows, Pearson's r ranges from -1.00 (the strongest possible negative relationship) to $+1.00$ (the strongest possible positive relationship). A value of 0 means there is no relationship

between the two variables. When Pearson's r is 0, the points on a scatterplot form a shapeless "cloud." As its value moves toward -1.00 or $+1.00$, the points come closer and closer to falling on a single straight line. Correlation coefficients near $\pm.10$ are considered small, values near $\pm.30$ are considered medium, and values near $\pm.50$ are considered large. Notice that the sign of Pearson's r is unrelated to its strength. Pearson's r values of $+.30$ and $-.30$, for example, are equally strong; it is just that one represents a moderate positive relationship and the other a moderate negative relationship. With the exception of reliability coefficients, most correlations that we find in Psychology are small or moderate in size. Kristoffer Magnusson's [website on interpreting correlations](#) provides an excellent interactive visualization of correlations that permits you to adjust the strength and direction of a correlation while witnessing the corresponding changes to the scatterplot.

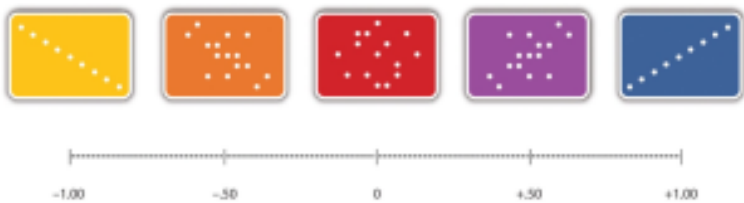


Figure 20.3 Range of Pearson's r , From -1.00 (Strongest Possible Negative Relationship), Through 0 (No Relationship), to $+1.00$ (Strongest Possible Positive Relationship) [\[Image Description\]](#)

There are two common situations in which the value of

Pearson's r can be misleading. Pearson's r is a good measure only for linear relationships, in which the points are best approximated by a straight line. It is not a good measure for nonlinear relationships, in which the points are better approximated by a curved line. Figure 20.4, for example, shows a hypothetical relationship between the amount of sleep people get per night and their level of depression. In this example, the line that best approximates the points is a curve—a kind of upside-down “U”—because people who get about eight hours of sleep tend to be the least depressed. Those who get too little sleep and those who get too much sleep tend to be more depressed. Even though Figure 20.4 shows a fairly strong relationship between depression and sleep, Pearson's r would be close to zero because the points in the scatterplot are not well fit by a single straight line. This means that it is important to make a scatterplot and confirm that a relationship is approximately linear before using Pearson's r . Nonlinear relationships are fairly common in psychology, but measuring their strength is beyond the scope of this book.

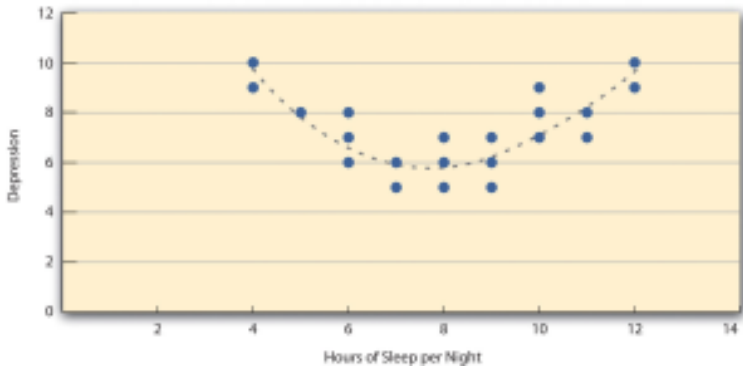


Figure 20.4 Hypothetical Nonlinear Relationship Between Sleep and Depression [\[Image Description\]](#)

The other common situations in which the value of Pearson's r can be misleading are when one or both of the variables have a limited range in the sample relative to the population. This problem is referred to as **restriction of range**. Assume, for example, that there is a strong negative correlation between people's age and their enjoyment of hip hop music as shown by the scatterplot in Figure 20.5. Pearson's r here is $-.77$. However, if we were to collect data only from 18- to 24-year-olds—represented by the shaded area of Figure 20.5—then the relationship would seem to be quite weak. In fact, Pearson's r for this restricted range of ages is 0. It is a good idea, therefore, to design studies to avoid restriction of range. For example, if age is one of your primary variables, then you can plan to collect data from people of a wide range of ages. Because restriction of range is not always anticipated or easily avoidable, however, it is good practice

to examine your data for possible restriction of range and to interpret Pearson's r in light of it. (There are also statistical methods to correct Pearson's r for restriction of range, but they are beyond the scope of this book).

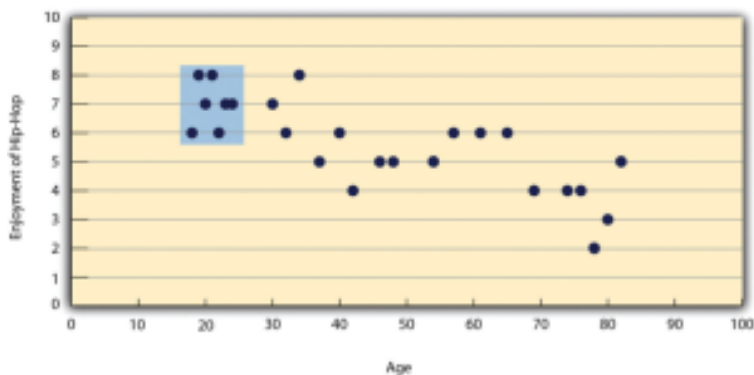


Figure 20.5 Hypothetical Data Showing How a Strong Overall Correlation Can Appear to Be Weak When One Variable Has a Restricted Range. The overall correlation here is $-.77$, but the correlation for the 18- to 24-year-olds (in the blue box) is 0. [\[Image Description\]](#)

Correlation Does Not Imply Causation

You have probably heard repeatedly that “Correlation does not imply causation.” An amusing example of this comes from a 2012 study that showed a positive correlation (Pearson's $r = 0.79$) between the per capita chocolate consumption of a nation and the number of Nobel prizes awarded to citizens of

that nation². It seems clear, however, that this does not mean that eating chocolate causes people to win Nobel prizes, and it would not make sense to try to increase the number of Nobel prizes won by recommending that parents feed their children more chocolate.

There are two reasons that correlation does not imply causation. The first is called the **directionality problem**. Two variables, X and Y , can be statistically related because X causes Y or because Y causes X . Consider, for example, a study showing that whether or not people exercise is statistically related to how happy they are—such that people who exercise are happier on average than people who do not. This statistical relationship is consistent with the idea that exercising causes happiness, but it is also consistent with the idea that happiness causes exercise. Perhaps being happy gives people more energy or leads them to seek opportunities to socialize with others by going to the gym. The second reason that correlation does not imply causation is called the **third-variable problem**. Two variables, X and Y , can be statistically related not because X causes Y , or because Y causes X , but because some third variable, Z , causes both X and Y . For example, the fact that nations that have won more Nobel prizes tend to have higher chocolate consumption probably reflects

2. Messerli, F. H. (2012). Chocolate consumption, cognitive function, and Nobel laureates. *New England Journal of Medicine*, 367, 1562-1564.

geography in that European countries tend to have higher rates of per capita chocolate consumption *and* invest more in education and technology (once again, per capita) than many other countries in the world. Similarly, the statistical relationship between exercise and happiness could mean that some third variable, such as physical health, causes both of the others. Being physically healthy could cause people to exercise and cause them to be happier. Correlations that are a result of a third variable are often referred to as **spurious correlations**.

[Tyler Vigen's website \[New Tab\]](#) has some excellent and amusing examples of spurious correlations. (Figure 20.6 provides one such example).

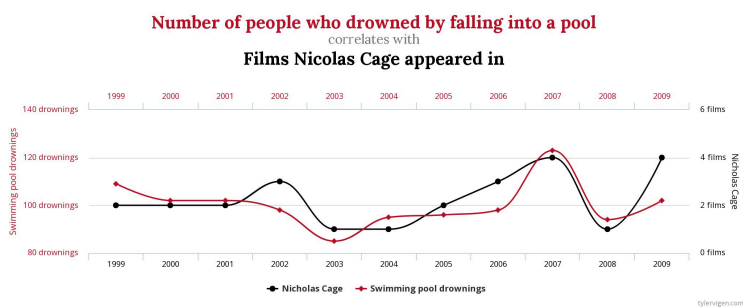


Figure 20.6. Example of a Spurious Correlation. [\[Image Description\]](#)

Equity Activity: Implications for correlation being seen as causation

Dr. Mueller's [Correlation or Causation? website](#) keeps track of news articles that make large claims based on correlational research. Many of the headlines suggest that a causal relationship has been demonstrated when a careful reading of the articles shows that it has not, because of the directionality and third-variable problems.



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=85#h5p-21>

License & Attribution

“Correlational Research” by Stephanie D’Costa is adapted from “[Correlational Research](#)” by Rajiv S. Jhangiani; I-Chant A. Chiang; Carrie Cuttler; and Dana C. Leighton is licensed CC BY-NC-SA 3.0.

“Correlational Research” is licensed under CC BY-NC-SA 4.0.

Image Descriptions

Figure 20.1. A vertical bar chart titled by its axes compares stress levels by whether someone keeps a Daily To-Do List. The y-axis is Stress (scale 0–30). Two bars are shown: Yes (purple) reaches about 18, and No (orange) reaches about 25. The difference suggests lower stress among people who use a daily to-do list. [\[Return to Figure 20.1\]](#)

Figure 20.2. Rectangular scatterplot with tan background. The x-axis is Stress (0–25) and the y-axis is Physical Symptoms (0–10). About twenty blue dots form an upward pattern—higher stress is associated with more symptoms. One

dot around Stress ≈ 10 , Symptoms ≈ 3 is highlighted with a circle. A long, dotted arrow extends left from this circled point along the same y-level, and a short, dotted arrow points down beneath it, indicating example movements (e.g., lowering stress and/or symptoms) relative to that observation. [\[Return to Figure 20.2\]](#)

Figure 20.3. A row of five colored panels illustrates correlation strength and direction: (1) yellow—points form a tight downward line (strong negative); (2) orange—looser downward trend (moderate negative); (3) red—cloud of points (near zero correlation); (4) purple—loose upward trend (moderate positive); (5) blue—tight upward line (strong positive). Below, a labeled axis reads -1.00 , -0.50 , 0 , $+0.50$, $+1.00$, matching the visual progression. [\[Return to Figure 20.3\]](#)

Figure 20.4. A tan-background scatterplot with Hours of Sleep per Night on the x-axis (0–14) and Depression on the y-axis (0–12). About twenty blue points form a curved pattern: higher depression scores near 4 hours and near 12 hours, with the lowest cluster around 7–8 hours. A dashed curve overlays the points, tracing the U-shape to highlight the curvilinear association between sleep duration and depression. [\[Return to Figure 20.4\]](#)

Figure 20.5. Tan scatterplot with Age on the x-axis (0–100) and Enjoyment of Hip-Hop on the y-axis (0–10). Points cluster at high enjoyment (6–8) for ages roughly 15–25, emphasized by a light blue rectangle. From about age 30

onward, points trend downward toward mid and lower enjoyment levels, with scattered values through ages 30–80. The pattern suggests a negative relationship between age and enjoyment of hip-hop. [\[Return to Figure 20.5\]](#)

Figure 20.6. A dual-axis line graph titled “Number of people who drowned by falling into a pool correlates with Films Nicolas Cage appeared in.” Years 1999–2009 run along the x-axis. The left y-axis (red) is Swimming pool drownings from 80 to 140 drownings; the right y-axis (black) is Nicolas Cage from 0 to 6 films. Two lines track together: a red line for drownings and a black line for Cage films. Both dip around 2003, rise to peaks in 2007 (≈ 125 drownings, ≈ 4 –5 films), drop sharply in 2008, and rise again in 2009. The legend labels Nicholas Cage (black) and Swimming pool drownings (red). The chart satirically illustrates a spurious correlation by visual similarity, not causation. [\[Return to Figure 20.6\]](#)

Media Attributions

- bd7cd73d89fce5f35f92e02c09c858c5
- [Scatterplot Showing a Hypothetical Positive Relationship Between Stress and Number of Physical Symptoms.](#) © Rajiv S. Jhangiani; I-Chant A. Chiang; Carrie Cuttler; and Dana C. Leighton is licensed under a [CC BY-NC-SA \(Attribution NonCommercial ShareAlike\)](#) license
- [Range of Pearson's r](#) © Rajiv S. Jhangiani; I-Chant A.

Chiang; Carrie Cuttler; and Dana C. Leighton is licensed under a [CC BY-NC-SA \(Attribution NonCommercial ShareAlike\)](#) license

- [Hypothetical Nonlinear Relationship Between Sleep and Depression](#) © Rajiv S. Jhangiani; I-Chant A. Chiang; Carrie Cuttler; and Dana C. Leighton is licensed under a [CC BY-NC-SA \(Attribution NonCommercial ShareAlike\)](#) license
- [Hypothetical Data Showing How a Strong Overall Correlation Can Appear to Be Weak When One Variable Has a Restricted Range](#) © Rajiv S. Jhangiani; I-Chant A. Chiang; Carrie Cuttler; and Dana C. Leighton is licensed under a [CC BY-NC-SA \(Attribution NonCommercial ShareAlike\)](#) license
- [Nicholas Cage and Pool Drownings](#) © Tyler Viegen is licensed under a [CC BY \(Attribution\)](#) license

21.

QUASI-EXPERIMENTAL RESEARCH

Stephanie D'Costa

Learning Objectives

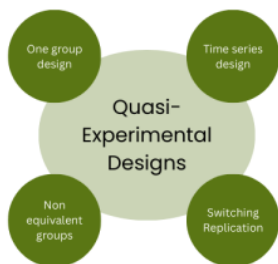
By the end of this chapter, you will be able to

1. Explain what quasi-experimental research is and distinguish it clearly from both experimental and correlational research.
2. Describe three different types of one-group quasi-experimental designs.
3. Describe the different types of non-equivalent groups quasi-experimental designs.
4. Identify the threats to internal validity

associated with each of these designs.

The prefix *quasi* means “resembling.” Thus, quasi-experimental research is research that resembles experimental research but is not true experimental research. Recall that with a true between-groups experiment, random assignment to conditions is used to ensure the groups are equivalent, and with a true within-subjects design, counterbalancing is used to guard against order effects. Quasi-experiments are missing one of these safeguards. Although an independent variable is manipulated, either a control group is missing or participants are not randomly assigned to conditions (Cook & Campbell, 1979)¹.

1. Cook, T. D., & Campbell, D. T. (1979). *Quasi-experimentation: Design & analysis issues in field settings*. Boston, MA: Houghton Mifflin.



Because the independent variable is manipulated before the dependent variable is measured, quasi-experimental research eliminates the directionality problem associated with non-experimental research.

But because either counterbalancing techniques are not used or participants are not randomly assigned to conditions—making it likely that there are other differences between conditions—quasi-experimental research does not eliminate the prob-

lem of confounding variables. In terms of internal validity, therefore, quasi-experiments are generally somewhere between non-experimental studies and true experiments.

Quasi-experiments are most likely to be conducted in field settings in which random assignment is difficult or impossible. They are often conducted to evaluate the effectiveness of a treatment—perhaps a type of psychotherapy or an educational intervention. There are many different kinds of quasi-experiments, but we will discuss just a few of the most common ones in this chapter.

One-Group Designs

One-Group Posttest Only Design

In a **one-group posttest only design**, a treatment is implemented (or an independent variable is manipulated), and then a dependent variable is measured once after the treatment is implemented. Imagine, for example, a researcher who is interested in the effectiveness of an anti-drug education program on elementary school students' attitudes toward illegal drugs. The researcher could implement the anti-drug program, and then immediately after the program ends, the researcher could measure students' attitudes toward illegal drugs.

This is the weakest type of quasi-experimental design. A major limitation of this design is the lack of a control or comparison group. There is no way to determine what the attitudes of these students would have been if they hadn't completed the anti-drug program. Despite this major limitation, results from this design are frequently reported in the media and are often misinterpreted by the general population. For instance, advertisers might claim that 80% of women noticed their skin looked brighter after using Brand X cleanser for a month. If there is no comparison group, then this statistic means little to nothing.

One-Group Pretest-Posttest Design

In a **one-group pretest-posttest design**, the dependent variable is measured once before the treatment is implemented and once after it is implemented. Let's return to the example of a researcher who is interested in the effectiveness of an anti-drug education program on elementary school students' attitudes toward illegal drugs. The researcher could measure the attitudes of students at a particular elementary school during one week, implement the anti-drug program during the next week, and finally, measure their attitudes again the following week. The pretest-posttest design is much like a within-subjects experiment in which each participant is tested first under the control condition and then under the treatment condition. It is unlike a within-subjects experiment, however, in that the order of conditions is not counterbalanced because it typically is not possible for a participant to be tested in the treatment condition first and then in an "untreated" control condition.

If the average posttest score is better than the average pretest score (e.g., attitudes toward illegal drugs are more negative after the anti-drug educational program), then it makes sense to conclude that the treatment might be responsible for the improvement. Unfortunately, one often cannot conclude this with a high degree of certainty because there may be other explanations for why the posttest scores may have changed. These alternative explanations pose threats to internal validity.

One alternative explanation goes under the name of **history**. Other things might have happened between the pretest and the posttest that caused a change from pretest to posttest. Perhaps an anti-drug program aired on television, and many of the students watched it, or perhaps a celebrity died of a drug overdose, and many of the students heard about it.

Another alternative explanation goes under the name of **maturation**. Participants might have changed between the pretest and the posttest in ways that they were going to anyway because they are growing and learning. If it were a year-long anti-drug program, participants might become less impulsive or better reasoners, and this might be responsible for the change in their attitudes toward illegal drugs.

Another threat to the internal validity of one-group pretest-posttest designs is **testing**, which refers to when the act of measuring the dependent variable during the pretest affects participants' responses at posttest. For instance, completing the measure of attitudes towards illegal drugs may have had an effect on those attitudes. Simply completing this measure may have inspired further thinking and conversations about illegal drugs that then produced a change in posttest scores.

Similarly, **instrumentation** can be a threat to the internal validity of studies using this design. Instrumentation refers to when the basic characteristics of the measuring instrument change over time. When human observers are used to measure behavior, they may, over time, gain skill, become fatigued, or change the standards on which observations are based. So

participants may have taken the measure of attitudes toward illegal drugs very seriously during the pretest when it was novel, but then they may have become bored with the measure at posttest and been less careful in considering their responses.

Another alternative explanation for a change in the dependent variable in a pretest-posttest design is **regression to the mean**. This refers to the statistical fact that an individual who scores extremely high or extremely low on a variable on one occasion will tend to score less extremely on the next occasion. For example, a bowler with a long-term average of 150 who suddenly bowls a 220 will almost certainly score lower in the next game. Her score will “regress” toward her mean score of 150. Regression to the mean can be a problem when participants are selected for further study *because* of their extreme scores. Imagine, for example, that only students who scored especially high on the test of attitudes toward illegal drugs (those with extremely favorable attitudes toward drugs) were given the anti-drug program and then were retested. Regression to the mean all but guarantees that their scores will be lower at the posttest even if the training program has no effect.

A closely related concept—and an extremely important one in psychological research—is **spontaneous remission**. This is the tendency for many medical and psychological problems to improve over time without any form of treatment. The common cold is a good example. If one were to measure symptom severity in 100 common cold sufferers today, give

them a bowl of chicken soup every day, and then measure their symptom severity again in a week, they would probably be much improved. This does not mean that the chicken soup was responsible for the improvement, however, because they would have been much improved without any treatment at all. The same is true of many psychological problems. A group of severely depressed people today is likely to be less depressed on average in 6 months. In reviewing the results of several studies of treatments for depression, researchers Michael Posternak and Ivan Miller found that participants in waitlist control conditions improved an average of 10 to 15% before they received any treatment at all (Posternak & Miller, 2001)². Thus, one must generally be very cautious about inferring causality from pretest-posttest designs.

A common approach to ruling out the threats to internal validity described above is by revisiting the research design to include a control group, one that does not receive the treatment effect. A control group would be subject to the same threats from history, maturation, testing, instrumentation, regression to the mean, and spontaneous remission, and so would allow the researcher to measure the actual effect of the

2. Posternak, M. A., & Miller, I. (2001). Untreated short-term course of major depression: A meta-analysis of studies using outcomes from studies using wait-list control groups. *Journal of Affective Disorders*, 66, 139–146.

treatment (if any). Of course, including a control group would mean that this is no longer a one-group design.

Does Psychotherapy Work?

Early studies on the effectiveness of psychotherapy tended to use pretest-posttest designs. In a classic 1952 article, researcher Hans Eysenck summarized the results of 24 such studies showing that about two thirds of patients improved between the pretest and the posttest (Eysenck, 1952)³. But Eysenck also compared these results with archival data from state hospital and insurance company records showing that similar patients recovered at about the same rate *without* receiving psychotherapy. This parallel suggested to Eysenck that the improvement that patients showed in the pretest-posttest studies might be no more than spontaneous remission. Note that Eysenck did not conclude that psychotherapy was ineffective. He merely concluded that there was no evidence that it was, and he wrote of “the

3. Eysenck, H. J. (1952). The effects of psychotherapy: An evaluation. *Journal of Consulting Psychology*, 16, 319–324.

necessity of properly planned and executed experimental studies into this important field” (p. 323). You can read the entire article here on the [Classics in the History of Psychology website](#).

Fortunately, many other researchers took up Eysenck’s challenge, and by 1980 hundreds of experiments had been conducted in which participants were randomly assigned to treatment and control conditions, and the results were summarized in a classic book by Mary Lee Smith, Gene Glass, and Thomas Miller (Smith, Glass, & Miller, 1980)⁴. They found that overall psychotherapy was quite effective, with about 80% of treatment participants improving more than the average control participant. Subsequent research has focused more on the conditions under which different types of psychotherapy are more or less effective.

4. Smith, M. L., Glass, G. V., & Miller, T. I. (1980). *The benefits of psychotherapy*. Baltimore, MD: Johns Hopkins University Press.

Interrupted Time-Series Design

A variant of the pretest-posttest design is the **interrupted time-series design**. A time-series is a set of measurements taken at intervals over a period of time. For example, a manufacturing company might measure its workers' productivity each week for a year. In an interrupted time-series design, a time-series like this one is "interrupted" by a treatment. In one classic example, the treatment was the reduction of the work shifts in a factory from 10 hours to 8 hours (Cook & Campbell, 1979)⁵. Because productivity increased rather quickly after the shortening of the work shifts, and because it remained elevated for many months afterward, the researcher concluded that the shortening of the shifts caused the increase in productivity. Notice that the interrupted time-series design is like a pretest-posttest design in that it includes measurements of the dependent variable both before and after the treatment. It is unlike the pretest-posttest design, however, in that it includes multiple pretest and posttest measurements.

Figure 21.1 shows data from a hypothetical interrupted time-series study. The dependent variable is the number of student absences per week in a research methods course. The

5. Cook, T. D., & Campbell, D. T. (1979). *Quasi-experimentation: Design & analysis issues in field settings*. Boston, MA: Houghton Mifflin.

treatment is that the instructor begins publicly taking attendance each day so that students know that the instructor is aware of who is present and who is absent. The top panel of Figure 21.1 shows how the data might look if this treatment worked. There is a consistently high number of absences before the treatment, and there is an immediate and sustained drop in absences after the treatment. The bottom panel of Figure 21.1 shows how the data might look if this treatment did not work. On average, the number of absences after the treatment is about the same as the number before. This figure also illustrates an advantage of the interrupted time-series design over a simpler pretest-posttest design. If there had been only one measurement of absences before the treatment at Week 7 and one afterward at Week 8, then it would have looked as though the treatment were responsible for the reduction. The multiple measurements both before and after the treatment suggest that the reduction between Weeks 7 and 8 is nothing more than normal week-to-week variation.

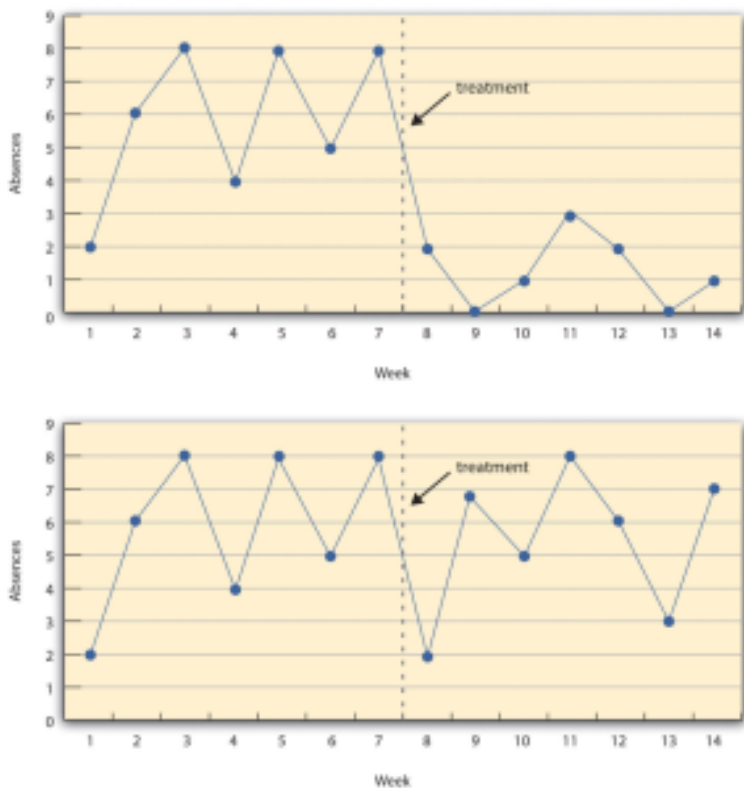


Figure 21.1. A Hypothetical Interrupted Time-Series Design. The top panel shows data that suggest that the treatment caused a reduction in absences. The bottom panel shows data that suggest that it did not. [\[Image description\]](#)

Non-Equivalent Groups Designs

Recall that when participants in a between-subjects experiment are randomly assigned to conditions, the resulting groups are likely to be quite similar. In fact, researchers consider them to be equivalent. When participants are not

randomly assigned to conditions, however, the resulting groups are likely to be dissimilar in some ways. For this reason, researchers consider them to be non-equivalent. A **non-equivalent groups design**, then, is a between-subjects design in which participants have not been randomly assigned to conditions. There are several types of non-equivalent groups designs we will consider.

Posttest Only Non-Equivalent Groups Design

The first non-equivalent groups design we will consider is the **posttest only non-equivalent groups design**. In this design, participants in one group are exposed to a treatment, a non-equivalent group is not exposed to the treatment, and then the two groups are compared. Imagine, for example, a researcher who wants to evaluate a new method of teaching fractions to third graders. One way would be to conduct a study with a treatment group consisting of one class of third-grade students and a control group consisting of another class of third-grade students. This design would be a non-equivalent groups design because the students are not randomly assigned to classes by the researcher, which means there could be important differences between them. For example, the parents of higher-achieving or more motivated students might have been more likely to request that their children be assigned to Ms. Williams's class. Or the principal might have assigned the

“troublemakers” to Mr. Jones’s class because he is a stronger disciplinarian. Of course, the teachers’ styles and even the classroom environments might be very different and might cause different levels of achievement or motivation among the students. If at the end of the study there was a difference in the two classes’ knowledge of fractions, it might have been caused by the difference between the teaching methods—but it might have been caused by any of these confounding variables.

Of course, researchers using a posttest only non-equivalent groups design can take steps to ensure that their groups are as similar as possible. In the present example, the researcher could try to select two classes at the same school, where the students in the two classes have similar scores on a standardized math test and the teachers are the same sex, are close in age, and have similar teaching styles. Taking such steps would increase the internal validity of the study because it would eliminate some of the most important confounding variables. But without true random assignment of the students to conditions, there remains the possibility of other important confounding variables that the researcher was not able to control.

Pretest-Posttest Non-Equivalent Groups Design

Another way to improve upon the posttest only non-equivalent groups design is to add a pretest. In the **pretest-posttest non-equivalent groups design**, there is a treatment

group that is given a pretest, receives a treatment, and then is given a posttest. But at the same time, there is a non-equivalent control group that is given a pretest, does not receive the treatment, and then is given a posttest. The question, then, is not simply whether participants who receive the treatment improve, but whether they improve *more* than participants who do not receive the treatment.

Imagine, for example, that students in one school are given a pretest on their attitudes toward drugs, then are exposed to an anti-drug program, and finally, are given a posttest. Students in a similar school are given the pretest, not exposed to an anti-drug program, and finally, are given a posttest. Again, if students in the treatment condition become more negative toward drugs, this change in attitude could be an effect of the treatment, but it could also be a matter of history or maturation. If it really is an effect of the treatment, then students in the treatment condition should become more negative than students in the control condition. But if it is a matter of history (e.g., news of a celebrity drug overdose) or maturation (e.g., improved reasoning), then students in the two conditions would be likely to show similar amounts of change. This type of design does not completely eliminate the possibility of confounding variables, however. Something could occur at one of the schools but not the other (e.g., a student drug overdose), so students at the first school would be affected by it while students at the other school would not.

Returning to the example of evaluating a new measure of

teaching third graders, this study could be improved by adding a pretest of students' knowledge of fractions. The changes in scores from pretest to posttest would then be evaluated and compared across conditions to determine whether one group demonstrated a bigger improvement in knowledge of fractions than another. Of course, the teachers' styles and even the classroom environments might still be very different and might cause different levels of achievement or motivation among the students that are independent of the teaching intervention. Once again, differential history also represents a potential threat to internal validity. If asbestos is found in one of the schools causing it to be shut down for a month then this interruption in teaching could produce a difference across groups on posttest scores.

If participants in this kind of design are randomly assigned to conditions, it becomes a true between-groups experiment rather than a quasi-experiment. In fact, it is the kind of experiment that Eysenck called for—and that has now been conducted many times—to demonstrate the effectiveness of psychotherapy.

Interrupted Time-Series Design with Non-Equivalent Groups

One way to improve upon the interrupted time-series design is to add a control group. The **interrupted time-series design with non-equivalent groups** involves taking a set of

measurements at intervals over a period of time, both before and after an intervention of interest in two or more non-equivalent groups. Once again, consider the manufacturing company that measures its workers' productivity each week for a year before and after reducing work shifts from 10 hours to 8 hours. This design could be improved by locating another manufacturing company who does not plan to change their shift length and using them as a non-equivalent control group. If productivity increased rather quickly after the shortening of the work shifts in the treatment group, but productivity remained consistent in the control group, then this provides better evidence for the effectiveness of the treatment.

Similarly, in the example of examining the effects of taking attendance on student absences in a research methods course, the design could be improved by using students in another section of the research methods course as a control group. If a consistently higher number of absences was found in the treatment group before the intervention, followed by a sustained drop in absences after the treatment, while the non-equivalent control group showed consistently high absences across the semester, then this would provide superior evidence for the effectiveness of the treatment in reducing absences.

Pretest-Posttest Design with Switching Replication

Some of these non-equivalent control group designs can be

further improved by adding a switching replication. Using a **pretest-posttest design with switching replication design**, non-equivalent groups are administered a pretest of the dependent variable, then one group receives a treatment while a non-equivalent control group does not receive a treatment, the dependent variable is assessed again, and then the treatment is added to the control group, and finally the dependent variable is assessed one last time.

As a concrete example, let's say we wanted to introduce an exercise intervention for the treatment of depression. We recruit one group of patients experiencing depression and a non-equivalent control group of students experiencing depression. We first measure depression levels in both groups, and then we introduce the exercise intervention to the patients experiencing depression, but we hold off on introducing the treatment to the students. We then measure depression levels in both groups. If the treatment is effective, we should see a reduction in the depression levels of the patients (who received the treatment) but not in the students (who have not yet received the treatment). Finally, while the group of patients continues to engage in the treatment, we would introduce the treatment to the students with depression. Now and only now should we see the students' levels of depression decrease.

One of the strengths of this design is that it includes a built-in replication. In the example given, we would get evidence for the efficacy of the treatment in two different samples (patients and students). Another strength of this design is that it

provides more control over history effects. It becomes rather unlikely that some outside event would perfectly coincide with the introduction of the treatment in the first group and with the delayed introduction of the treatment in the second group. For instance, if a change in the weather occurred when we first introduced the treatment to the patients, and this explained their reductions in depression the second time that depression was measured, then we would see depression levels decrease in both groups. Similarly, the switching replication helps to control for maturation and instrumentation. Both groups would be expected to show the same rates of spontaneous remission of depression, and if the instrument for assessing depression happened to change at some point in the study, the change would be consistent across both groups. Of course, demand characteristics, placebo effects, and experimenter expectancy effects can still be problems. But they can be controlled for using some of the methods described in Chapter 5.

Switching Replication with Treatment Removal Design

In a basic pretest-posttest design with switching replication, the first group receives a treatment, and the second group receives the same treatment a little bit later on (while the initial group continues to receive the treatment). In contrast, in a **switching replication with treatment removal design**,

the treatment is removed from the first group when it is added to the second group. Once again, let's assume we first measure the depression levels of patients with depression and students with depression. Then we introduce the exercise intervention to only the patients. After they have been exposed to the exercise intervention for a week, we assess depression levels again in both groups. If the intervention is effective, then we should see depression levels decrease in the patient group but not the student group (because the students haven't received the treatment yet). Next, we would remove the treatment from the group of patients with depression. So we would tell them to stop exercising. At the same time, we would tell the student group to start exercising. After a week of the students exercising and the patients not exercising, we would reassess depression levels. Now, if the intervention is effective, we should see that the depression levels have decreased in the student group but that they have increased in the patient group (because they are no longer exercising).

Demonstrating a treatment effect in two groups staggered over time and demonstrating the reversal of the treatment effect after the treatment has been removed can provide strong evidence for the efficacy of the treatment. In addition to providing evidence for the replicability of the findings, this design can also provide evidence for whether the treatment continues to show effects after it has been withdrawn.

Equity Activity: Addressing immigration stress

You are a researcher working with a local school district who wants to provide mental health services to their Latine immigrant students who have been reporting high levels of stress after anti-immigrant sentiment has been occurring across the country. You provide students with the [Hispanic Stress Inventory- Adolescent Version](#) as a pre-test and then provide an 8-week mental health intervention. At the conclusion of the intervention, you give the students the HSI-A as a post-test. When you analyze the data, you find that students report significantly lower stress levels after the completion of the intervention.



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=88#h5p-22>

License & Attribution

“Quasi Experimental Research” by Stephanie D’Costa is adapted from “[Quasi-Experimental Research](#)” by Rajiv S. Jhangiani; I-Chant A. Chiang; Carrie Cuttler; and Dana C. Leighton is licensed CC BY-NC-SA 3.0.

“Quasi Experimental Research” is licensed under CC BY-NC-SA 4.0.

Image Descriptions

Figure 21.1. The figure contains two tan line charts labeled Week (x-axis 1–14) and Absences (y-axis 0–9). In both, a vertical dashed line at week 8 is annotated “treatment.” Top panel: Before week 8, absences vary between about 2 and 8. After treatment, absences fall sharply—0 at weeks 9–10, then 1–3 with one week at 0—indicating a strong positive effect. Bottom panel: Absences before week 8 fluctuate between roughly 2 and 8. After treatment, the same up-and-down

pattern continues with no downward shift, indicating little or no effect. [\[Return to Figure 21.1\]](#)

Media Attributions

- Quasi-Experimental Designs
- 8.1_7.3

22.

QUALITATIVE RESEARCH

Rebecca Anguiano

Learning Objectives

By the end of this chapter, you will be able to

1. List several ways in which qualitative research differs from quantitative research in psychology.
2. Describe the strengths and weaknesses of qualitative research in psychology compared with quantitative research.
3. Give examples of qualitative research in psychology.

What Is Qualitative Research?

This textbook is primarily about **quantitative research**, in part because most studies conducted in psychology are quantitative in nature. Quantitative researchers typically start with a focused research question or hypothesis, collect a small amount of numerical data from a large number of individuals, describe the resulting data using statistical techniques, and draw general conclusions about some large population. Although this method is by far the most common approach to conducting empirical research in psychology, there is an important alternative called **qualitative research**. Qualitative research originated in the disciplines of anthropology and sociology but is now used to study psychological topics as well. Qualitative researchers generally begin with a less focused research question, collect large amounts of relatively “unfiltered” data from a relatively small number of individuals, and describe their data using nonstatistical techniques, such as grounded theory, thematic analysis, critical discourse analysis, or interpretative phenomenological analysis. They are usually less concerned with drawing general conclusions about human behavior than with understanding in detail the *experience* of their research participants.

Consider, for example, a study by researcher Per Lindqvist and his colleagues, who wanted to learn how the families of teenage suicide victims cope with their loss (Lindqvist,

Johansson, & Karlsson, 2008)¹. They did not have a specific research question or hypothesis, such as, What percentage of family members join suicide support groups? Instead, they wanted to understand the variety of reactions that families had, with a focus on what it is like from *their* perspectives. To address this question, they interviewed the families of 10 teenage suicide victims in their homes in rural Sweden. The interviews were relatively unstructured, beginning with a general request for the families to talk about the victim and ending with an invitation to talk about anything else that they wanted to tell the interviewer. One of the most important themes that emerged from these interviews was that even as life returned to “normal,” the families continued to struggle with the question of why their loved one committed suicide. This struggle appeared to be especially difficult for families in which the suicide was most unexpected.

The Purpose of Qualitative Research

Again, this textbook is primarily about quantitative research in

-
1. Lindqvist, P., Johansson, L., & Karlsson, U. (2008). In the aftermath of teenage suicide: A qualitative study of the psychosocial consequences for the surviving family members. *BMC Psychiatry*, 8, 26. Retrieved from <http://www.biomedcentral.com/1471-244X/8/26>

psychology. The strength of quantitative research is its ability to provide precise answers to specific research questions and to draw general conclusions about human behavior. This method is how we know that people have a strong tendency to obey authority figures, for example, and that female undergraduate students are not substantially more talkative than male undergraduate students. But while quantitative research is good at providing precise answers to specific research questions, it is not nearly as good at *generating* novel and interesting research questions. Likewise, while quantitative research is good at drawing general conclusions about human behavior, it is not nearly as good at providing detailed descriptions of the behavior of particular groups in particular situations. And quantitative research is not very good at communicating what it is actually like to be a member of a particular group in a particular situation.

But the relative weaknesses of quantitative research are the relative strengths of qualitative research. Qualitative research can help researchers to generate new and interesting research questions and hypotheses. The research of Lindqvist and colleagues, for example, suggests that there may be a general relationship between how unexpected a suicide is and how consumed the family is with trying to understand why the teen committed suicide. This relationship can now be explored using quantitative research. But it is unclear whether this question would have arisen at all without the researchers sitting down with the families and listening to what they

themselves wanted to say about their experience. Qualitative research can also provide rich and detailed descriptions of human behavior in the real-world contexts in which it occurs. Among qualitative researchers, this depth is often referred to as “thick description” (Geertz, 1973)². Similarly, qualitative research can convey a sense of what it is actually like to be a member of a particular group or in a particular situation—what qualitative researchers often refer to as the “lived experience” of the research participants. Lindqvist and colleagues, for example, describe how all the families spontaneously offered to show the interviewer the victim’s bedroom or the place where the suicide occurred—revealing the importance of these physical locations to the families. It seems unlikely that a quantitative study would have discovered this detail.

2. Geertz, C. (1973). *The interpretation of cultures*. New York, NY: Basic Books.

Table 22.1. Some Contrasts Between Qualitative and Quantitative Research

Qualitative	Quantitative
1. In-depth information about relatively few people	1. Less depth of information with larger samples
2. Conclusions are based on interpretations drawn by the investigator	2. Conclusions are based on statistical analyses
3. Broader and exploratory	3. Specific and focused

Data Collection and Analysis in Qualitative Research

Data collection approaches in qualitative research are quite varied and can involve naturalistic observation, participant observation, archival data, artwork, and many other things. But one of the most common approaches, especially for psychological research, is to conduct **interviews**. Interviews in qualitative research can be unstructured—consisting of a small number of general questions or prompts that allow participants to talk about what is of interest to them—or structured, where there is a strict script that the interviewer does not deviate from. Most interviews are in between the two and are called semi-structured interviews, where the researcher has a few consistent questions and can follow up by asking more detailed questions about the topics that come up. Such interviews can be lengthy and detailed, but they are usually

conducted with a relatively small sample. The unstructured interview was the approach used by Lindqvist and colleagues in their research on the families of suicide victims because the researchers believed that families, not researchers, should get to decide what and how much to disclose about such a sensitive topic.

Another approach used in qualitative research involves small groups of people who participate together in interviews focused on a particular topic or issue, known as **focus groups**. The interaction among participants in a focus group can sometimes bring out more information than can be learned in a one-on-one interview. The use of focus groups has become a standard technique in business and industry among those who want to understand consumer tastes and preferences. The content of all focus group interviews is usually recorded and transcribed to facilitate later analyses. From an equity lens, focus groups can be especially valuable because they allow participants—particularly those from marginalized or underrepresented groups—to build on one another's experiences, validate shared realities, and raise issues that researchers may not have anticipated or included in individually structured interviews. However, we also know from social psychology that group dynamics are often at play in any group, including focus groups, and it is useful to be aware of those possibilities. For example, the desire to be liked by others can lead participants to provide inaccurate answers that they believe will be perceived favorably by the other

participants. The same may be said for personality characteristics. For example, highly extraverted participants can sometimes dominate discussions within focus groups.

Data Analysis in Qualitative Research

Although quantitative and qualitative research generally differ along several important dimensions (e.g., the specificity of the research question, the type of data collected), it is the method of data analysis that distinguishes them more clearly than anything else. To illustrate this idea, imagine a team of researchers that conducts a series of unstructured interviews with people recovering from alcohol use disorder to learn about the role of their religious faith in their recovery. Although this project sounds like qualitative research, imagine further that once they collect the data, they code the data in terms of how often each participant mentions God (or a “higher power”), and they then use descriptive and inferential statistics to find out whether those who mention God more often are more successful in abstaining from alcohol. Now it sounds like quantitative research. In other words, the quantitative-qualitative distinction depends more on what researchers do with the data they have collected than with why or how they collected the data.

But what does qualitative data analysis look like? Just as there are many ways to collect data in qualitative research,

there are many ways to analyze data. Here we focus on one general approach called **grounded theory** (Glaser & Strauss, 1967)³. This approach was developed within the field of sociology in the 1960s and has gradually gained popularity in psychology. Remember that in quantitative research, it is typical for the researcher to start with a theory, derive a hypothesis from that theory, and then collect data to test that specific hypothesis. In qualitative research using grounded theory, researchers start with the data and develop a theory or an interpretation that is “grounded in” those data. They do this analysis in stages. First, they identify ideas that are repeated throughout the data. Then they organize these ideas into a smaller number of broader themes. Finally, they write a **theoretical narrative**—an interpretation of the data in terms of the themes that they have identified. This theoretical narrative focuses on the subjective experience of the participants and is usually supported by many direct quotations from the participants themselves. The end product of a grounded theory study is a unified theory or explanation about why and how something happens.

As an example, consider a study by researchers Laura Abrams and Laura Curran, who used the grounded theory approach to study the experience of postpartum depression

3. Glaser, B. G., & Strauss, A. L. (1967). *The discovery of grounded theory: Strategies for qualitative research*. Chicago, IL: Aldine.

symptoms among low-income mothers (Abrams & Curran, 2009)⁴. Their data were the result of unstructured interviews with 19 participants. Table 22.2 shows the five broad themes the researchers identified and the more specific repeating ideas that made up each of those themes. In their research report, they provide numerous quotations from their participants, such as this one from “Destiny:”

Well, just recently my apartment was broken into and the fact that his Medicaid for some reason was cancelled so a lot of things was happening within the last two weeks all at one time. So that in itself I don’t want to say almost drove me mad but it put me in a funk....Like I really was depressed. (p. 357)

Their theoretical narrative focused on the participants’ experience of their symptoms, not as an abstract “affective disorder” but as closely tied to the daily struggle of raising children alone under often difficult circumstances.

4. Abrams, L. S., & Curran, L. (2009). “And you’re telling me not to stress?” A grounded theory study of postpartum depression symptoms among low-income mothers. *Psychology of Women Quarterly*, 33, 351–362.

Table 22.2. Themes and Repeating Ideas in a Study of Postpartum Depression Among Low-Income Mothers. Based on Research by Abrams and Curran (2009).

Theme	Repeating ideas
Ambivalence	“I wasn’t prepared for this baby,” “I didn’t want to have any more children.”
Caregiving overload	“Please stop crying,” “I need a break,” “I can’t do this anymore.”
Juggling	“No time to breathe,” “Everyone depends on me,” “Navigating the maze.”
Mothering alone	“I really don’t have any help,” “My baby has no father.”
Real-life worry	“I don’t have any money,” “Will my baby be OK?” “It’s not safe here.”

The Quantitative-Qualitative “Debate”

Given their differences, it may come as no surprise that quantitative and qualitative research in psychology and related fields do not coexist in complete harmony. Some quantitative researchers criticize qualitative methods on the grounds that they lack objectivity, are difficult to evaluate in terms of reliability and validity, and do not allow generalization to people or situations other than those actually studied. At the same time, some qualitative researchers criticize quantitative methods on the grounds that they overlook the richness of

human behavior and experience and instead answer simple questions about easily quantifiable variables.

In general, however, qualitative researchers are well aware of the issues of objectivity, reliability, validity, and generalizability. In fact, they have developed a number of frameworks for addressing these issues (which are beyond the scope of our discussion). And in general, quantitative researchers are well aware of the issue of oversimplification. They do not believe that all human behavior and experience can be adequately described in terms of a small number of variables and the statistical relationships among them. Instead, they use simplification as a strategy for uncovering general principles of human behavior.

Many researchers from both the quantitative and qualitative camps now agree that the two approaches can and should be combined into what has come to be called **mixed-methods research** (Todd, Nerlich, McKeown, & Clarke, 2004)⁵. (In fact, the studies by Lindqvist and colleagues and by Abrams and Curran both combined quantitative and qualitative approaches.) One approach to combining quantitative and qualitative research is to use qualitative research for hypothesis generation and quantitative research

5. Todd, Z., Nerlich, B., McKeown, S., & Clarke, D. D. (2004) *Mixing methods in psychology: The integration of qualitative and quantitative methods in theory and practice*. London, UK: Psychology Press.

for hypothesis testing. Again, while a qualitative study might suggest that families who experience an unexpected suicide have more difficulty resolving the question of why, a well-designed quantitative study could test a hypothesis by measuring these specific variables in a large sample. A second approach to combining quantitative and qualitative research is referred to as **triangulation**. The idea is to use both quantitative and qualitative methods simultaneously to study the same general questions and to compare the results. If the results of the quantitative and qualitative methods converge on the same general conclusion, they reinforce and enrich each other. If the results diverge, then they suggest an interesting new question: Why do the results diverge, and how can they be reconciled? Although we discuss triangulation between quantitative and qualitative data, triangulation can also take place between two (or more) different sources of qualitative data (such as interviews combined with archival analysis).

Using qualitative research can often help clarify quantitative results via triangulation. Trenor, Yu, Waight, Zerda, and Sha (2008)⁶ investigated the experience of female engineering students at a university. In the first phase, female

6. Trenor, J.M., Yu, S.L., Waight, C.L., Zerda, K.S & Sha T.-L. (2008). The relations of ethnicity to female engineering students' educational experiences and college and career plans in an ethnically diverse learning environment. *Journal of Engineering Education*, 97(4), 449-465.

engineering students were asked to complete a survey, where they rated a number of their perceptions, including their sense of belonging. Their results were compared across the student ethnicities, and statistically, the various ethnic groups showed no differences in their ratings of their sense of belonging. One might look at that result and conclude that ethnicity does not have anything to do with one's sense of belonging. However, in the second phase, the authors also conducted interviews with the students, and in those interviews, many minority students reported how the diversity of cultures at the university enhanced their sense of belonging. Without the qualitative component, we might have drawn the wrong conclusion about the quantitative results.

This example shows how qualitative and quantitative research work together to help us understand human behavior. Some researchers have characterized qualitative research as best for identifying behaviors or the phenomenon, whereas quantitative research is best for understanding meaning or identifying the mechanism. However, Bryman (2012)⁷ argues for breaking down the divide between these arbitrarily different ways of investigating the same questions.

7. Bryman, A. (2012). *Social Research Methods*, 4th ed. Oxford: OUP.

Equity Activity: Centering voices through qualitative research

Title: “Whose Story Gets Told?” – Using Qualitative Methods to Understand Educational Barriers

Objective: To help students understand how qualitative methods (e.g., interviews, focus groups, narrative analysis) can amplify the lived experiences of marginalized communities and challenge dominant assumptions in research and policy.

Scenario: Your research team is exploring barriers to academic success in a large urban high school where dropout rates are disproportionately high among Black, Latinx, and newcomer immigrant students. Previous research in this district has focused mostly on test scores and attendance data to explain these trends.

You want to use qualitative methods to understand **what students themselves** say about their school experiences, including how school climate, racism, language access, mental health, or neighborhood safety affect their engagement and sense of belonging.

Part 1: Research Design Challenge (Discussion)

In your group, design a qualitative study that centers the voices of the most affected students. Address the following:

1. **Research Questions:** What do you want to understand that cannot be captured by numbers alone?
(e.g., “How do Latinx students describe the impact of language barriers on their academic identity?”)
2. **Participants:** Who will you interview or observe? How will you ensure inclusivity across gender, language, disability, and other identities?
3. **Methods:** What qualitative methods will you use and why?
(e.g., one-on-one interviews for depth, focus groups, etc.)
4. **Analysis:** How will you analyze the data to

surface themes that reflect students' real concerns, not just your assumptions?

5. **Impact:** How will you **share the findings** with the students and community in a way that supports advocacy or change?

Part 2: Critical Reflection (Written Response)

After designing your study, reflect on these questions:



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=90#h5p-23>

Key Takeaways:

- **Qualitative research** is not just about collecting stories—it’s about listening for patterns of oppression and resilience.
- When done ethically, it helps **amplify voices** that are often left out of decision-making.
- This approach aligns with equity-focused research by uncovering lived realities and co-creating knowledge with communities.

License & Attribution

“Qualitative Research” by Rebecca Anguiano is adapted from “[Qualitative Research](#)” by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler, and Dana C. Leighton, which is licensed CC BY-NC-SA 4.0.

“Qualitative Research” is licensed under CC BY-NC-SA 4.0.

23.

OVERVIEW OF SURVEY RESEARCH

Stephanie D'Costa

Learning Objectives

By the end of this chapter, you will be able to

1. Define what survey research is, including its two important characteristics.
2. Describe several different ways that survey research can be used and give some examples.

What Is Survey Research?

Survey research is a quantitative and qualitative method with two important characteristics. First, the variables of interest are measured using self-reports (using questionnaires or interviews). In essence, survey researchers ask their participants (who are often called **respondents** in survey research) to report directly on their own thoughts, feelings, and behaviors. Second, considerable attention is paid to the issue of sampling. In particular, survey researchers have a strong preference for large samples because they provide the most accurate estimates of what is true in the population. In fact, survey research may be the only approach in psychology in which random sampling is routinely used. Beyond these two characteristics, almost anything goes in survey research. Surveys can be long or short. They can be conducted in person, by telephone, through the mail, or over the Internet. They can be about voting intentions, consumer preferences, social attitudes, health, or anything else that it is possible to ask people about and receive meaningful answers. Although survey data are often analyzed using statistics, there are many questions that lend themselves to more qualitative analysis.

Most survey research is non-experimental. It is used to describe single variables (e.g., the percentage of voters who prefer one presidential candidate or another, the prevalence of schizophrenia in the general population, etc.) and also to assess statistical relationships between variables (e.g., the relationship

between income and health). But surveys can also be used within experimental research. The study by Lerner and her colleagues¹ is a good example. Their use of self-report measures and a large national sample identifies their work as survey research. But their manipulation of an independent variable (anger vs. fear) to assess its effect on a dependent variable (risk judgments) also identifies their work as experimental.

History and Uses of Survey Research

Survey research may have its roots in English and American “social surveys” conducted around the turn of the 20th century by researchers and reformers who wanted to document the extent of social problems such as poverty (Converse, 1987)². By the 1930s, the US government was conducting surveys to document economic and social conditions in the country. The need to draw conclusions about the entire population helped spur advances in sampling

-
1. Lerner, J. S., Gonzalez, R. M., Small, D. A., & Fischhoff, B. (2003). Effects of fear and anger on perceived risks of terrorism: A national field experiment. *Psychological Science*, 14, 144–150.
 2. Converse, J. M. (1987). *Survey research in the United States: Roots and emergence, 1890–1960*. Berkeley, CA: University of California Press.

procedures. At about the same time, several researchers who had already made a name for themselves in market research, studying consumer preferences for American businesses, turned their attention to election polling. A watershed event was the presidential election of 1936 between Alf Landon and Franklin Roosevelt. A magazine called *Literary Digest* conducted a survey by sending ballots (which were also subscription requests) to millions of Americans. Based on this “straw poll,” the editors predicted that Landon would win in a landslide. At the same time, the new pollsters were using scientific methods with much smaller samples to predict just the opposite—that Roosevelt would win in a landslide. In fact, one of them, George Gallup, publicly criticized the methods of *Literary Digest* before the election and all but guaranteed that his prediction would be correct. And of course, it was, demonstrating the effectiveness of careful survey methodology (We will consider the reasons that Gallup was right later in this chapter). Gallup’s demonstration of the power of a carefully crafted survey led to the first national election survey by the Survey Research Center at the University of Michigan, in 1948. This work eventually became the [American National Election Studies](#) as a collaboration of Stanford University and the University of Michigan, and these studies continue today.

From market research and election polling, survey research made its way into several academic fields, including political science, sociology, and public health—where it continues to be one of the primary approaches to collecting new data.

Beginning in the 1930s, psychologists made important advances in questionnaire design, including techniques that are still used today, such as the Likert scale. (See “What Is a Likert Scale?” in Section 7.2 “Constructing Survey Questionnaires”.) Survey research has a strong historical association with the social psychological study of attitudes, stereotypes, and prejudice. Early attitude researchers were also among the first psychologists to seek larger and more diverse samples than the convenience samples of university students that were routinely used in psychology (and still are).

Survey research continues to be important in psychology today. For example, survey data have been instrumental in estimating the prevalence of various mental disorders and identifying statistical relationships among those disorders and with various other factors. The [National Comorbidity Survey](#) is a large-scale mental health survey conducted in the United States. In just one part of this survey, nearly 10,000 adults were given a structured mental health interview in their homes in 2002 and 2003. Table 23.1 presents results on the lifetime prevalence of some anxiety, mood, and substance use disorders. (Lifetime prevalence is the percentage of the population that develops the problem sometime in their lifetime.) Obviously, this kind of information can be of great use both to basic researchers seeking to understand the causes and correlates of mental disorders as well as to clinicians and policymakers who need to understand exactly how common these disorders are.

Table 23.1. Some *Lifetime Prevalence Results from the National Comorbidity Survey

Disorder	Total	Female	Male
Generalized anxiety disorder	5.7	7.1	4.2
Obsessive-compulsive disorder	2.3	3.1	1.6
Major depressive disorder	16.9	20.2	13.2
Bipolar disorder	4.4	4.5	4.3
Alcohol abuse	13.2	7.5	19.6
Drug abuse	8.0	4.8	11.6

**The lifetime prevalence of a disorder is the percentage of people in the population who develop that disorder at any time in their lives.*

Survey research can even be used as a data collection method within experimental research to test specific hypotheses about causal relationships between variables. Such studies, when conducted on large and diverse samples, can be a useful supplement to laboratory studies conducted on university students. Survey research is thus a flexible approach that can be used to study a variety of basic and applied research questions.

Constructing Surveys

Types of Items

Questionnaire items can be either open-ended or closed-ended.

Open-ended items simply ask a question and allow participants to answer in whatever way they choose. The following are examples of open-ended questionnaire items.

- “What is the most important thing to teach children to prepare them for life?”
- “Please describe a time when you were discriminated against because of your age.”
- “Is there anything else you would like to tell us about?”

Open-ended items are useful when researchers do not know how participants might respond or when they want to avoid influencing their responses. Open-ended items are more qualitative in nature, so they tend to be used when researchers have more vaguely defined research questions—often in the early stages of a research project. Open-ended items are relatively easy to write because there are no response options to worry about. However, they take more time and effort on the part of participants, and they are more difficult for the researcher to analyze because the answers must be transcribed, coded, and submitted to some form of qualitative analysis,

such as content analysis. Another disadvantage is that respondents are more likely to skip open-ended items because they take longer to answer. It is best to use open-ended questions when the answer is uncertain or for quantities that can easily be converted to categories later in the analysis.

Closed-ended items ask a question and provide a set of response options for participants to choose from. Example questions could be:

How old are you?

- _____ Under 18
- _____ 18 to 34
- _____ 35 to 49
- _____ 50 to 70
- _____ Over 70

On a scale of 0 (no pain at all) to 10 (worst pain ever experienced), how much pain are you in right now?

Have you ever in your adult life been depressed for a period of 2 weeks or more? Yes No

Closed-ended items are used when researchers have a good idea of the different responses that participants might make.

They are more quantitative in nature, so they are also used when researchers are interested in a well-defined variable or construct, such as participants' level of agreement with some statement, perceptions of risk, or frequency of a particular behavior. Closed-ended items are more difficult to write because they must include an appropriate set of response options. However, they are relatively quick and easy for participants to complete. They are also much easier for researchers to analyze because the responses can be easily converted to numbers and entered into a spreadsheet. For these reasons, closed-ended items are much more common.

All closed-ended items include a set of response options from which a participant must choose. For categorical variables like sex, race, or political party preference, the categories are usually listed and participants choose the one (or ones) to which they belong. For quantitative variables, a rating scale is typically provided. A **rating scale** is an ordered set of responses that participants must choose from. The number of response options on a typical rating scale ranges from three to 11—although five and seven are probably most common. Five-point scales are best for unipolar scales where only one construct is tested, such as frequency (Never, Rarely, Sometimes, Often, Always). Seven-point scales are best for bipolar scales where there is a dichotomous spectrum, such as liking (Like very much, Like somewhat, Like slightly, Neither like nor dislike, Dislike slightly, Dislike somewhat, Dislike very much). For bipolar questions, it is useful to offer an earlier

question that branches them into an area of the scale; if asking about liking ice cream, first ask “Do you generally like or dislike ice cream?” Once the respondent chooses like or dislike, refine it by offering them relevant choices from the seven-point scale. Branching improves both reliability and validity (Krosnick & Berent, 1993)³. Although you often see scales with numerical labels, it is best to only present verbal labels to the respondents but convert them to numerical values in the analyses. Avoid partial labels or length or overly specific labels. In some cases, the verbal labels can be supplemented with (or even replaced by) meaningful graphics.

What Is a Likert Scale?

In reading about psychological research, you are likely to encounter the term *Likert scale*. Although this term is sometimes used to refer to almost any rating scale (e.g., a 0-to-10 life satisfaction scale), it has a much more precise meaning.

In the 1930s, researcher Rensis Likert (pronounced

3. Krosnick, J.A. & Berent, M.K. (1993). Comparisons of party identification and policy preferences: The impact of survey question format. *American Journal of Political Science*, 27(3), 941-964.

LICK-ert) created a new approach for measuring people's attitudes (Likert, 1932)⁴. It involves presenting people with several statements—including both favorable and unfavorable statements—about some person, group, or idea. Respondents then express their agreement or disagreement with each statement on a 5-point scale: *Strongly Agree, Agree, Neither Agree nor Disagree, Disagree, Strongly Disagree*. Numbers are assigned to each response and then summed across all items to produce a score representing the attitude toward the person, group, or idea. For items that are phrased in an opposite direction (e.g., negatively worded statements instead of positively worded statements), reverse coding is used so that the numerical scoring of statements also runs in the opposite direction. The entire set of items came to be called a Likert scale.

Thus, unless you are measuring people's attitude toward something by assessing their level of agreement with several statements about it, it is

4. Likert, R. (1932). A technique for the measurement of attitudes. *Archives of Psychology*, 140, 1–55.

best to avoid calling it a Likert scale. You are probably just using a “rating scale.”

Context Effects on Survey Responses

It is important to note that there are potential **context effects** that may impact how participants respond to your survey. They are called context effects because they are not related to the content of the item but to the *context* in which the item appears (Schwarz & Strack, 1990)⁵. For example, there is an **item-order effect** when the order in which the items are presented affects people’s responses. One item can change how participants interpret a later item or change the information that they retrieve to respond to later items. For example, researcher Fritz Strack and his colleagues asked college students about both their general life satisfaction and their dating frequency (Strack, Martin, & Schwarz, 1988)⁶. When the life

5. Schwarz, N., & Strack, F. (1990). Context effects in attitude surveys: Applying cognitive theory to social research. In W. Stroebe & M. Hewstone (Eds.), *European review of social psychology* (Vol. 2, pp. 31–50). Chichester, UK: Wiley.

6. Strack, F., Martin, L. L., & Schwarz, N. (1988). Priming and communication: The social determinants of information use in judgments of life satisfaction. *European Journal of Social Psychology*, 18, 429–442.

satisfaction item came first, the correlation between the two was only $-.12$, suggesting that the two variables are only weakly related. But when the dating frequency item came first, the correlation between the two was $+.66$, suggesting that those who date more have a strong tendency to be more satisfied with their lives. Reporting the dating frequency first made that information more accessible in memory, so that they were more likely to base their life satisfaction rating on it.

The response options provided can also have unintended effects on people's responses (Schwarz, 1999)⁷. For example, when people are asked how often they are "really irritated" and given response options ranging from "less than once a year" to "more than once a month," they tend to think of major irritations and report being irritated infrequently. But when they are given response options ranging from "less than once a day" to "several times a month," they tend to think of minor irritations and report being irritated frequently. People also tend to assume that middle response options represent what is normal or typical. So if they think of themselves as normal or typical, they tend to choose middle response options. For example, people are likely to report watching more television when the response options are centered on a middle option of 4 hours than when centered on a middle option of 2 hours.

7. Schwarz, N. (1999). Self-reports: How the questions shape the answers. *American Psychologist*, 54, 93–105

To mitigate against order effects, rotate questions and response items when there is no natural order. Counterbalancing or randomizing the order of presentation of the questions in online surveys are good practices for survey questions and can reduce response order effects that show that among undecided voters, the first candidate listed in a ballot receives a 2.5% boost simply by virtue of being listed first⁸!

Conducting the Survey

The four main ways to conduct surveys are through in-person interviews, by telephone, through the mail, and over the internet. As with other aspects of survey design, the choice depends on both the researcher's goals and the budget. In-person interviews have the highest response rates and provide the closest personal contact with respondents. Personal contact can be important, for example, when the interviewer must see and make judgments about respondents, as is the case with some mental health interviews. But in-person interviewing is by far the most costly approach. Telephone surveys have lower response rates and still provide some personal contact with respondents. They can also be costly but

8. Miller, J.M. & Krosnick, J.A. (1998). The impact of candidate name order on election outcomes. *Public Opinion Quarterly*, 62(3), 291-330.

are generally less so than in-person interviews. Traditionally, telephone directories have provided fairly comprehensive sampling frames. However, this trend is less true today as more people choose to only have cell phones and do not install landlines that would be included in telephone directories. Mail surveys are less costly still but generally have even lower response rates—making them most susceptible to non-response bias.

Not surprisingly, internet surveys are becoming more common. They are increasingly easy to construct and use (see “Online Survey Creation”). Although initial contact can be made by mail with a link provided to the survey, this approach does not necessarily produce higher response rates than an ordinary mail survey. A better approach is to make initial contact by email with a link directly to the survey. This approach can work well when the population consists of the members of an organization who have known email addresses and regularly use them (e.g., a university community). For other populations, it can be difficult or impossible to find a comprehensive list of email addresses to serve as a sampling frame. Alternatively, a request to participate in the survey with a link to it can be posted on websites known to be visited by members of the population. But again, it is very difficult to get anything approaching a random sample this way because the members of the population who visit the websites are likely to be different from the population as a whole. However, internet survey methods are in rapid

development. Because of their low cost and because more people are online than ever before, internet surveys are likely to become the dominant approach to survey data collection in the near future.

Finally, it is important to note that some of the concerns that people have about collecting data online (e.g., that internet-based findings differ from those obtained with other methods) have been found to be myths. Table 23.2 (adapted from Gosling, Vazire, Srivastava, & John, 2004)⁹ addresses three such preconceptions about data collected in web-based studies:

9. Gosling, S. D., Vazire, S., Srivastava, S., & John, O. P. (2004). Should we trust web-based studies? A comparative analysis of six preconceptions about internet questionnaires. *American Psychologist*, 59(2), 93-104.

Table 23.2. Some Preconceptions and Findings Pertaining to Web-based Studies

Preconception	Finding
Internet samples are not demographically diverse	Internet samples are more diverse than traditional samples in many domains, although they are not completely representative of the population
Internet samples are maladjusted, socially isolated, or depressed	Internet users do not differ from nonusers on markers of adjustment and depression
Internet-based findings differ from those obtained with other methods	Evidence so far suggests that internet-based findings are consistent with findings based on traditional methods (e.g., on self-esteem, personality), but more data are needed.

Online Survey Creation

There are now several online tools for creating online questionnaires. After a questionnaire is created, a link to it can then be emailed to potential respondents or embedded in a web page. The following websites are among those that offer free accounts. Although the free accounts limit the number of questionnaire items and the number of respondents, they can be useful for doing small-scale surveys and for practicing the principles of

good questionnaire construction. Here are some commonly used online survey tools:

- [SurveyMonkey](#)
- [PsyToolkit](#) (free, noncommercial, and does many experimental paradigms)
- [Qualtrics](#)
- [PsycData](#)

A small note of caution: the data from US survey software are held on US servers and are subject to seizure as granted through the Patriot Act. To avoid infringing on any rights, the following is a list of online survey sites that are hosted in Canada:

- [Fluid Surveys \[New Tab\]](#)
- [Simple Survey \[New Tab\]](#)
- [Lime Survey \[New Tab\]](#)

There are also survey sites hosted in other countries outside of North America.

Another new tool for survey researchers is [Mechanical Turk \(MTurk\) \[New Tab\]](#), created by Amazon.com. Originally created for simple usability testing, MTurk has a database of over 500,000

workers from over 190 countries¹⁰. You can put simple tasks (for example, different question wording to test your survey items), set parameters as your sample frame dictates, and deploy your experiment at a very low cost (for example, a few cents for less than 5 minutes). MTurk has been lauded as an inexpensive way to gather high-quality data (Buhrmester, Kwang, & Gosling, 2011)¹¹.

-
10. Natala@aws. (2011, January 26). Re: MTurk CENSUS: About how many workers were on Mechanical Turk in 2010? Message posted to Amazon Web Services Discussion Forums. Retrieved from <https://forums.aws.amazon.com/thread.jspa?threadID=58891>
 11. Buhrmester, M., Kwang, T., & Gosling, S.D. (2011). Amazon's Mechanical Turk: A new source of inexpensive, yet high quality, data? *Perspectives on Psychological Science*, 6(1), 3-5.

Equity Activity: Survey to understand teenage screen time



One or more interactive elements has been excluded from this version of the text. You can view them online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=92#oembed-1>

Video 23.1. [*How to Help Teens Build Healthy Video Habits*](#) by [*American Psychological Association*](#)

After watching the above video, consider how a survey researcher might go about exploring this topic of teenagers' video habits. Practice crafting survey questions to explore this topic from multiple perspectives:



An interactive H5P element has been excluded from this version of the text. You can view it online here:

[https://smc.pressbooks.pub/
critresearchmethodpsych/?p=92#h5p-24](https://smc.pressbooks.pub/critresearchmethodpsych/?p=92#h5p-24)

License & Attribution

“Overview of Survey Research” by Stephanie D’Costa is adapted from the section “[Constructing Surveys](#)“, “[Conducting Surveys](#)” and “[Survey Research](#)” by Rajiv S. Jhangiani; I-Chant A. Chiang; Carrie Cuttler; and Dana C. Leighton is licensed CC BY-NC-SA 3.0.

“Overview of Survey Research” is licensed under CC BY-NC-SA 4.0.

PART V

STATISTICS

24.

INTRODUCTION TO STATISTICS: DESCRIPTIVE & INFERENCEAL

Rebecca Anguiano

Learning Objectives

By the end of this chapter, you will be able to

1. Differentiate between descriptive and inferential statistics and explain the purposes each serves in research.
2. Identify and describe common types of

descriptive statistics, including measures of center, variability, shape, and position.

3. Evaluate how statistical summaries and inferences can reinforce deficit-based or vulnerability narratives, particularly when context and structural factors are ignored.
4. Apply a critical research methods perspective to question assumptions about objectivity, accuracy, and neutrality in statistical knowledge

Statistics shape how we understand the world—from scientific research and public policy to advertising, health recommendations, and everyday decision making. This chapter introduces statistics as both a set of tools for summarizing and analyzing data and a way of thinking critically about numerical claims. You will learn the difference between **descriptive statistics**, which organize and describe patterns in data, and **inferential statistics**, which allow researchers to draw conclusions about populations from samples. Alongside these foundational concepts, the chapter emphasizes a **critical research methods perspective**, encouraging you to question how statistics are produced,

interpreted, and used, and to consider whose experiences are represented—or overlooked—by the numbers. By the end of the chapter, you will be better equipped to interpret statistical information thoughtfully, recognize common pitfalls in reasoning, and use statistics responsibly in both scientific and everyday contexts.

What Are Statistics?

Statistics include numerical facts and figures. For instance:

- The largest earthquake measured 9.2 on the Richter scale.
- Men are at least 10 times more likely than women to commit murder.
- One in every eight South Africans is HIV positive.
- By the year 2050, there will be 12 people aged 65 and over for every new baby born.

The study of statistics involves math and relies upon calculations of numbers. But it also relies heavily on how the numbers are chosen and how the statistics are interpreted. For example, consider the following three scenarios and the interpretations based on the presented statistics. You will find that the numbers may be right, but the interpretation may be wrong. Try to identify a major flaw with each interpretation before we describe it.

1. A new advertisement for Ben & Jerry's ice cream introduced in late May of last year resulted in a 30% increase in ice cream sales for the following three months. Thus, the advertisement was effective.
Major flaw: Ice cream consumption generally increases in the months of June, July, and August, regardless of advertisements. This effect is called a history effect and leads people to interpret outcomes as the result of one variable when another variable (in this case, one having to do with the passage of time) is actually responsible.
2. The more churches in a city, the more crime there is. Thus, churches lead to crime.
Major flaw: Both increased churches and increased crime rates can be explained by larger populations. In bigger cities, there are both more churches and more crime. This problem, which we will discuss in more detail in the chapter on sampling, refers to the third-variable problem. Namely, a third variable can cause both situations; however, people erroneously believe that there is a causal relationship between the two primary variables rather than recognizing that a third variable can cause both.
3. Seventy-five percent more interracial marriages are occurring this year than 25 years ago. Thus, our society accepts interracial marriages.
Major flaw: We don't have the information we need. What is the rate at which marriages are occurring?

Suppose only 1% of marriages 25 years ago were interracial, and so now 1.75% of marriages are interracial (1.75 is 75% higher than 1). But this latter number is hardly evidence suggesting the acceptability of interracial marriages. In addition, the statistic provided does not rule out the possibility that the number of interracial marriages has seen dramatic fluctuations over the years, and this year is not the highest. Again, there is simply not enough information to understand fully the impact of the statistics.

As a whole, these examples show that statistics are not only facts and figures; they are something more than that. In the broadest sense, “statistics” refers to a range of techniques and procedures for analyzing, interpreting, displaying, and making decisions based on **data**.

Why Do We Study Statistics?

Virtually every student of the behavioral sciences takes some form of statistics class. This is because statistics is how we communicate in science. It serves as the link between a research idea and usable conclusions. Without statistics, we would be unable to interpret the massive amounts of information contained in data. Even small datasets contain hundreds—if not thousands—of numbers, each representing a specific observation we made. Without a way to organize these

numbers into a more interpretable form, we would be lost, having wasted the time and money of our participants, ourselves, and the communities we serve.

Beyond its use in science, however, there is a more personal reason to study statistics. Like most people, you probably feel that it is important to “take control of your life.” But what does this mean? Partly, it means being able to properly evaluate the data and claims that bombard you every day. If you cannot distinguish good from faulty reasoning, then you are vulnerable to manipulation and to decisions that are not in your best interest. Statistics provides tools that you need in order to react intelligently to information you hear or read. In this sense, statistics is one of the most important things that you can study.

To be more specific, here are some claims that we have heard on several occasions. (We are not saying that each one of these claims is true!)

- Four out of five dentists recommend Dentine.
- Almost 85% of lung cancers in men and 45% in women are tobacco-related.
- Condoms are effective 94% of the time.
- People tend to be more persuasive when they look others directly in the eye and speak loudly and quickly.
- Women make 75 cents to every dollar a man makes when they work the same job.
- A surprising new study shows that eating egg whites can

increase one's life span.

- People predict that it is very unlikely there will ever be another baseball player with a batting average over 400.
- There is an 80% chance that in a room full of 30 people at least two people will share the same birthday.
- 79.48% of all statistics are made up on the spot.

All of these claims are statistical in character. We suspect that some of them sound familiar; if not, we bet that you have heard other claims like them. Notice how diverse the examples are. They come from psychology, health, law, sports, business, etc. Indeed, data and data interpretation show up in discourse from virtually every facet of contemporary life.

Statistics are often presented in an effort to add credibility to an argument or advice. You can see this by paying attention to television advertisements. Many of the numbers thrown about in this way do not represent careful statistical analysis. They can be misleading and push you into decisions that you might find cause to regret. For these reasons, learning about statistics is a long step toward taking control of your life. (It is not, of course, the only step needed to do so.) The purpose of this course, beyond preparing you for a career in psychology, is to help you learn statistical essentials. It will make you into an intelligent consumer of statistical claims.

You can take the first step right away. To be an intelligent consumer of statistics, your first reflex must be to question the statistics you encounter. The British Prime Minister Benjamin

Disraeli is quoted by Mark Twain as having said, “There are three kinds of lies—lies, damned lies, and statistics.” This quote reminds us why it is so important to understand statistics. So let us invite you to reform your statistical habits from now on. No longer will you blindly accept numbers or findings. Instead, you will begin to think about the numbers, their sources, and most importantly, the procedures used to generate them.

The above section puts an emphasis on defending ourselves against fraudulent claims wrapped up as statistics, but let us look at a more positive note. Just as important as detecting the deceptive use of statistics is the appreciation of the proper use of statistics. You must also learn to recognize statistical evidence that supports a stated conclusion. Statistics are all around you, sometimes used well, sometimes not. We must learn how to distinguish the two cases.

A Critical Perspective on Statistics

Statistics are powerful tools for summarizing data and drawing conclusions, but they are not neutral. The way statistics are collected, analyzed, and presented can shape how groups are understood—sometimes reinforcing narratives of **vulnerability, deficiency, or pathology**, particularly for marginalized populations. A critical research methods perspective asks not only *what* the statistics show, but *how* and *why* they are produced, and *whose interests* they serve.

Types of Statistical Analyses

Now that we understand the nature of our data, let's turn to the types of statistics we can use to interpret them. There are two types of statistics: **descriptive and inferential**.

Descriptive Statistics

Descriptive statistics are numbers that are used to summarize and describe data. The word “data” refers to the information that has been collected from an experiment, a survey, a historical record, etc. (By the way, data is plural. One piece of information is called a datum.) If we are analyzing birth certificates, for example, a descriptive statistic might be the percentage of certificates issued in New York State or the average age of the mother. Any other number we choose to compute also counts as a descriptive statistic for the data from which the statistic is computed. Several descriptive statistics are often used at one time to give a full picture of the data.

Descriptive statistics are just descriptive. They do not involve generalizing beyond the data at hand. Generalizing from our data to another set of cases is the business of inferential statistics, which you'll be studying in another section. Here we focus on (mere) descriptive statistics.

Some descriptive statistics are shown in Table 24.1. The table shows the average salaries for various occupations in the United States in 1999. Descriptive statistics like these offer

insight into American society. It is interesting to note, for example, that we pay the people who educate our children and who protect our citizens a great deal less than we pay people who take care of our feet or our teeth.

Table 24.1. Average Salaries for Various U.S. Occupations in 1999.

Occupation	Salary
Pediatricians	\$112,760
Dentists	\$106,130
Podiatrists	\$100,090
Physicists	\$76,140
Architects	\$53,410
School, clinical, and counseling psychologists	\$49,720
Flight attendants	\$47,910
Elementary school teachers	\$39,560
Police officers	\$38,710
Floral designers	\$18,980

For more descriptive statistics, consider Table 24.2. It shows the number of unmarried men per 100 unmarried women in U.S. metro areas in 1990. From this table we see that men outnumber women most in Jacksonville, North Carolina, and women outnumber men most in Sarasota, Florida. You can see that descriptive statistics can be useful if we are looking for an opposite-sex partner! (These data come from the [Information Please Almanac](#).)

Table 24.2. Number of Unmarried Men per 100 Unmarried Women in U.S. Metro Areas in 1990.
Note: Unmarried includes never–married, widowed, and divorced persons, 15 years or older.

Cities with Mostly Men	Men per 100 Women	Cities with Mostly Women	Men per 100 Women
1. Jacksonville, North Carolina	224	1. Sarasota, Florida	66
2. Killeen–Temple, Texas	123	2. Bradenton, Florida	68
3. Fayetteville, North Carolina	118	3. Altoona, Pennsylvania	69
4. Brazoria, Texas	117	4. Springfield, Illinois	70
5. Lawton, Oklahoma	116	5. Jacksonville, Tennessee	70
6. State College, Pennsylvania	113	6. Gadsden, Alabama	70
7. Clarksville–Hopkinsville, Tennessee–Kentucky	113	7. Wheeling, West Virginia–Ohio	70

Cities with Mostly Men	Men per 100 Women	Cities with Mostly Women	Men per 100 Women
8. Anchorage, Alaska	112	8. Charleston, West Virginia	71
9. Salinas–Seaside–Monterey, California	112	9. St. Joseph, Missouri	71
10. Bryan–College Station, Texas	111	10. Lynchburg, Virginia	71

These descriptive statistics may make us ponder why the numbers are so disparate in these cities. One potential explanation, for instance, as to why there are more women in Florida than men may involve the fact that elderly individuals tend to move down to the Sarasota region and that women tend to outlive men. Thus, more women might live in Sarasota than men. However, in the absence of proper data, this is only speculation.

You probably know that descriptive statistics are central to the world of sports. Every sporting event produces numerous statistics, such as the shooting percentage of players on a basketball team. For the Olympic marathon (a foot race of 26.2 miles), we possess data that cover more than a century of competition. (The first modern Olympics took place in 1896.)

[Table 24.3](#) and [Table 24.4](#) show the winning times for women and men, respectively. (Women have only been allowed to compete since 1984.)

Table 24.3. Women’s Winning Olympic Marathon Times, 1984–2004.

Year	Winner	Country	Time
1984	Joan Benoit	United States	2:24:52
1988	Rosa Mota	Portugal	2:25:40
1992	Valentina Yegorova	Unified Team	2:32:41
1996	Fatuma Roba	Ethiopia	2:26:05
2000	Naoko Takahashi	Japan	2:23:14
2004	Mizuki Noguchi	Japan	2:26:20

***Table 24.4. Men's Winning Olympic Marathon Times,
1896–2004.***

Year	Winner	Country	Time
1896	Spyridon Louis	Greece	2:58:50
1900	Michel Théato	France	2:59:45
1904	Thomas Hicks	United States	3:28:53
1906	Billy Sherring	Canada	2:51:23
1908	Johnny Hayes	United States	2:55:18
1912	Kenneth McArthur	South Africa	2:36:54
1920	Hannes Kolehmainen	Finland	2:32:35
1924	Albin Stenroos	Finland	2:41:22
1928	Boughera El Ouafi	France	2:32:57
1932	Juan Carlos Zabala	Argentina	2:31:36
1936	Sohn Kee-chung	Japan	2:29:19
1948	Delfo Cabrera	Argentina	2:34:51

Year	Winner	Country	Time
1952	Emil Zátopek	Czechoslovakia	2:23:03
1956	Alain Mimoun	France	2:25:00
1960	Abebe Bikila	Ethiopia	2:15:16
1964	Abebe Bikila	Ethiopia	2:12:11
1968	Mamo Wolde	Ethiopia	2:20:26
1972	Frank Shorter	United States	2:12:19
1976	Waldemar Cierpinski	East Germany	2:09:55
1980	Waldemar Cierpinski	East Germany	2:11:03
1984	Carlos Lopes	Portugal	2:09:21
1988	Gelindo Bordin	Italy	2:10:32
1992	Hwang Young-cho	South Korea	2:13:23
1996	Josia Thugwane	South Africa	2:12:36
2000	Gezahegne Abera	Ethiopia	2:10.10

Year	Winner	Country	Time
2004	Stefano Baldini	Italy	2:10:55

There are many descriptive statistics that we can compute from the data in these tables. To gain insight into the improvement in speed over the years, let us divide the men's times into two pieces, namely, the first 13 races (up to 1952) and the second 13 (starting from 1956). The mean winning time for the first 13 races is 2 hours, 44 minutes, and 22 seconds (written 2:44:22). The mean winning time for the second 13 races is 2:13:18. This is quite a difference (over half an hour). Does this prove that the fastest men are running faster? Or is the difference just due to chance, no more than what often emerges from chance differences in performance from year to year? We can't answer this question with descriptive statistics alone. All we can affirm is that the two means are "suggestive."

Examining [Table 24.3](#) and [Table 24.4](#) leads to many other questions. We note that Takahashi (the lead female runner in 2000) would have beaten the male runner in 1956 and all male runners in the first 12 marathons. This fact leads us to ask whether the gender gap will close or remain constant. When we look at the times within each gender, we also wonder how far they will decrease (if at all) in the next century of the Olympics. Might we one day witness a sub-2-hour marathon?

The study of statistics can help you make reasonable guesses about the answers to these questions.

To summarize, there are a variety of ways that statisticians use to describe a data set:

- Exploratory data analysis uses graphs and numerical summaries to describe the variables and their relationship to each other. You will learn how to organize a data set by grouping the data into intervals called classes and forming a *frequency distribution*.
- Center of the Data uses *measures of central tendency* to locate the middle or center of a distribution where most of the data tends to be concentrated. The three best-known measures of central tendency used by statisticians are discussed later in this section of the text: the mean, median, and mode.
- Spread of the Data uses *measures of variability* to describe how far apart data points lie from each other and from the center of the distribution. The section on variability discusses the four most common measures of variability: range, interquartile range, variance, and standard deviation.
- Shape of the Data can be either *symmetrical* (the normal distribution) or *skewed* (positive or negative). The section on probability and the normal distribution reveals how the shape of the distribution can easily be discernible through graphs.

- Fractiles or Quantiles of the Data uses *measures of position* that partition, or divide, an ordered data set into equal parts: first, second, and third quartile; percentiles; and the standard score (z-score). A fractile is a point where a specified proportion of the data lies below that point. Section 2.5 discusses how fractiles are used to specify the position of a data point within a data set.

It is also important to differentiate what we use to describe populations vs. what we use to describe samples. A population is described by a **parameter**; the parameter is the true value of the descriptive in the population, but one that we can never know for sure. For example, the Bureau of Labor Statistics reports that the average hourly wage of chefs is \$23.87. However, even if this number were computed using information from every single chef in the United States (making it a parameter), it would quickly become slightly off as one chef retires and a new chef enters the job market. Additionally, as noted above, there is virtually no way to collect data from every single person in a population. In order to understand a variable, we estimate the population parameter using a sample statistic. Here, the term statistic refers to the specific number we compute from the data (e.g., the average), not the field of statistics. A **sample statistic** is an estimate of the true population parameter, and if our sample is

representative of the population, then the statistic is considered to be a good estimator of the parameter.

Even the best sample will be somewhat off from the full population, earlier referred to as sampling bias, and as a result, there will always be a tiny discrepancy between the parameter and the statistic we use to estimate it. This difference is known as sampling error, and, as we will see throughout the course, understanding sampling error is the key to understanding statistics. Every observation we make about a variable, be it a full research study or observing an individual's behavior, is incapable of being completely representative of all possibilities for that variable. Knowing where to draw the line between an unusual observation and a true difference is what statistics is all about.

Descriptive Statistics and Deficit Narratives

Descriptive statistics often highlight differences between groups, such as disparities in income, health outcomes, educational attainment, or crime rates. While these summaries may be numerically accurate, they can be misleading when interpreted without context. For example, reporting lower average test scores or higher rates of illness in a particular group may implicitly frame that group as lacking ability, motivation, or resilience. What is often missing from such descriptions are the **structural and historical conditions**—such as

discrimination, unequal access to resources, environmental exposures, or policy decisions—that help produce these patterns.

In this way, descriptive statistics can contribute to **deficit-based narratives** that locate problems within groups rather than in the systems that shape their lived experiences. The statistics themselves do not explain *why* differences exist, yet they are frequently interpreted as evidence of inherent weakness or failure. Understanding this is important for utilizing and interpreting descriptive statistics responsibly.

Inferential Statistics

Descriptive statistics are wonderful at telling us what our data look like. However, what we often want to understand is how our data behave. What variables are related to other variables? Under what conditions will the value of a variable change? Are two groups different from each other, and if so, are people within each group different or similar? These are the questions answered by **inferential statistics**, and inferential statistics are how we generalize from our sample back up to our population.

For example, we will learn how to use a t-statistic to determine whether people change over time when enrolled in an intervention. We will also use an F-statistic to determine if we can predict future values of a variable based on current known values of a variable. There are many types of inferential statistics, each allowing us insight into a different behavior of

the data we collect. This course will only touch on a small subset (or a sample) of them, but the principles we learn along the way will make it easier to learn new tests, as most inferential statistics follow the same structure and format.

A Note about Statistical Software

Many pieces of technology support statistical analysis and quantitative data analysis done by psychologists. Commonly used technologies include the proprietary Statistical Package for the Social Sciences (SPSS), the free and open-source tool JASP, and the programming language R. Several of the figures used in this text were generated using JASP, but providing an overview or introduction to these technologies is outside the scope of this work. Instruction manuals can be found on the [JASP website](#).

Mathematical Notation

As noted earlier, statistics is not math. It does, however, use math as a tool. Many statistical formulas involve summing numbers. Fortunately, there is a convenient notation for expressing summation. This section covers the basics of this summation notation.

Let's say we have a variable X that represents the weights (in grams) of 4 grapes:

Table 24.5. Weight of Grapes

Grape	X
Grape 1	4.6
Grape 2	5.1
Grape 3	4.9
Grape 4	4.4

We label the weight of Grape 1 as $\mathbf{X_1}$, of Grape 2 as $\mathbf{X_2}$, etc. The following formula means to sum up the weights of the four grapes:

$$\sum_{i=1}^4 X_i$$

The Greek letter $\sum$ indicates summation. The “i = 1” at the bottom indicates that the summation is to start with $\mathbf{X_1}$, and the 4 at the top indicates that the summation will end with $\mathbf{X_4}$. The $\mathbf{X_i}$ indicates that X is the variable to be summed as i goes from 1 to 4. Therefore,

$$\sum_{i=1}^4 X_i = X_1 + X_2 + X_3 + X_4 = 4.6 + 5.1 + 4.9 + 4.4 = 19$$

The symbol

$$\sum_{i=1}^3 X_i$$

indicates that only the first 3 scores are to be summed. The index variable i goes from 1 to 3.

When all the scores of a variable (such as X) are to be summed, it is often convenient to use the following abbreviated notation:

$$\sum X$$

Thus, when no values of i are shown, it means to sum all the values of X .

Many formulas involve squaring numbers before they are summed. This is indicated as

$$\sum X^2 = 4.6^2 + 5.1^2 + 4.9^2 + 4.4^2 = 21.16 + 26.01 + 24.01 + 19.36 = 90.54$$

Notice that:

$$(\sum x)^2 \neq \sum x^2$$

because the expression on the left means to sum up all the values of X and then square the sum ($19^2 = 361$), whereas the expression on the right means to square the numbers and then sum the squares (90.54, as shown).

Some formulas involve the sum of cross products. Below are the data for variables X and Y . The cross products (XY) are shown in the third column. The sum of the cross products is $3 + 4 + 21 = 28$.

Table 24.6. Values of X and Y and Their Cross Products

X	Y	XY
1	3	3
2	2	4
3	7	21

In summation notation, this is written as:

$$\sum XY = 28$$

Inferential Statistics and the Illusion of Objectivity

Inferential statistics extend findings from samples to populations, often using concepts such as statistical significance, confidence intervals, and effect sizes. Although these tools are presented as objective and precise, they rely on assumptions about sampling, measurement, and model fit that are not always met in real-world research—especially when studying marginalized populations.

When samples are unrepresentative or when key variables (such as racism, disability access, immigration status, or caregiving labor) are excluded or poorly measured, inferential statistics can produce conclusions that appear authoritative but obscure important realities. Statistical significance may be

interpreted as “truth,” even when the underlying data fail to capture the complexity of people’s lives. In this sense, inferential statistics can give a **false sense of certainty**, reinforcing claims that are incomplete or even inaccurate.

Challenging the Myth That Statistics Always Provide Accurate Knowledge

A critical approach challenges the common assumption that statistics automatically produce accurate or unbiased knowledge. As discussed in the chapter on [QuantCrit](#) earlier in this textbook, numbers do not speak for themselves; they are the result of human decisions about **what to measure, whom to include, how to categorize people, and which comparisons matter**. Categories such as race, gender, disability, or socioeconomic status are often treated as fixed variables, even though they are socially constructed and context-dependent.

This does not mean that statistics are useless or inherently misleading. Rather, it means that statistical knowledge is **partial and situated**. Responsible use of statistics requires acknowledging uncertainty, questioning assumptions, and resisting interpretations that individualize or naturalize inequality.

Equity Activity: Descriptive Statistics and data disaggregation

In addition to using the above statistical procedures, the data disaggregation process can be applied to identify equity gaps. Disaggregation means breaking down data into smaller groupings, such as income, gender, and racial/ethnic groupings. Disaggregating data can reveal deprivations or inequalities that may not be fully reflected in aggregated data.

The National Center for Mental Health Promotion and Youth Violence Prevention provides two examples of the importance of disaggregating data into smaller subpopulations (National Center Brief 2012)¹. One area where data disaggregation is commonly used is to show disproportionate minority contact, such as the number of times a minority youth is involved with the court system. In fact, the Office of Juvenile Justice and Delinquency

1. National Center Brief (2012). The Importance of Disaggregating Student Data. The National Center for Mental Health Promotion and Youth Violence Prevention, April.

Prevention (OJJDP) uses a specific indicator (Relative Rate Index) to show if there are differences in arrest rates or court sentences, for example, between racial/ethnic groups that are not explained by simple differences in population numbers. A similar step was taken by the Department of Health and Human Services (HHS) as part of the Affordable Care Act. Disaggregated data can be used to see if there are meaningful differences by subpopulations in who is accessing mental services and what treatments are successful.

In descriptive statistics, a particular group of interest (or target group) can be disaggregated by certain characteristics, such as race, ethnicity, gender, age, socio-economic status, disability, education level, employment in different sectors (e.g., health care, biotechnology, cybersecurity), salary levels, and other different factors. Disaggregating data is viewed as a critical first step while embarking on the equity-minded journey. According to the Annie E. Casey Foundation (2020), “disaggregating data and presenting it in a meaningful way can help bring

attention and commitment to the solving of social and racial equity problems.”²



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=95#h5p-25>

Practice Problems

Short Answer Reflections



An interactive H5P element has been excluded from this version of the text. You

2. Annie E. Casey Foundation (2020). By the Numbers: A Race for Results Case Study-Using Disaggregated Data to Inform Policies, Practices, and Decision-Making. Baltimore, MD.

can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=95#h5p-26>

Test Your Knowledge



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=95#h5p-27>

License & Attribution

“Descriptive and Inferential Statistics” by Rebecca Anguiano is adapted from

[“Introduction”](#) by Linda R. Cote Ph.D.; Rupa G. Gordon Ph.D., Chrislyn E. Randell Ph.D., Judy Schmitt, and Helena Marvin, which is licensed CC BY-NC-SA 4.0 and [“Descriptive Statistics”](#) by Lynette H. Bikos, which is licensed by CC BY-NC-SA.

“Descriptive and Inferential Statistics” is licensed under CC BY-NC-SA 4.0.

25.

FREQUENCY DISTRIBUTIONS

Makenzie O'Neil

Learning Objectives

By the end of this chapter, you will be able to

1. Create frequency distribution tables and graphs to describe and display the distribution of a variable.
2. Interpret frequency distributions in order to interpret and draw reasonable conclusions about various datasets.

As described in the previous chapter, **Descriptive statistics**

refers to a set of techniques for summarizing and displaying data. Let us assume here that the data are quantitative and consist of scores on one or more variables for each of several study participants. Although in most cases the primary research question will be about one or more statistical relationships between variables, it is also important to describe each variable individually. For this reason, over the next few chapters, we will begin by looking at some of the most common techniques for describing single variables.

The Distribution of a Variable

Every variable has a **distribution**, which is the way the scores are distributed across the levels of that variable. For example, in a sample of 100 university students, the distribution of the variable “number of siblings” might be such that 10 of them have no siblings, 30 have one sibling, 40 have two siblings, and so on. In the same sample, the distribution of the variable “sex” might be such that 44 have a score of “male” and 56 have a score of “female.”

Frequency Tables

One way to display the distribution of a variable is in a **frequency table**. Table 25.1, for example, is a frequency table showing a hypothetical distribution of scores on the

Rosenberg Self-Esteem Scale for a sample of 40 college students. The first column lists the values of the variable—the possible scores on the Rosenberg scale—and the second column lists the frequency of each score. This table shows that there were three students who had self-esteem scores of 24, five who had self-esteem scores of 23, and so on. From a frequency table like this, one can quickly see several important aspects of a distribution, including the range of scores (from 15 to 24), the most and least common scores (22 and 17, respectively), and any extreme scores that stand out from the rest.

Table 25.1. Frequency Table Showing a Hypothetical Distribution of Scores on the Rosenberg Self-Esteem Scale

Self-esteem	Frequency
24	3
23	5
22	10
21	8
20	5
19	3
18	3
17	0
16	2
15	1

There are a few other points worth noting about frequency tables. First, the levels listed in the first column usually go from the highest at the top to the lowest at the bottom, and they usually do not extend beyond the highest and lowest scores in the data. For example, although scores on the Rosenberg scale can vary from a high of 30 to a low of 0, Table 25.1 only includes levels from 24 to 15 because that range includes all the scores in this particular data set. Second, when there are many different scores across a wide range of values, it is often better to create a grouped frequency table, in which the first column lists ranges of values and the second column lists the frequency of scores in each range. Table 25.2, for example, is a grouped frequency table showing a hypothetical distribution of simple reaction times for a sample of 20 participants. In a grouped frequency table, the ranges must all be of equal width, and there are usually between five and 15 of them. Finally, frequency tables can also be used for categorical variables, in which case the levels are category labels. The order of the category labels is somewhat arbitrary, but they are often listed from the most frequent at the top to the least frequent at the bottom.

Table 25.2. A Grouped Frequency Table Showing a Hypothetical Distribution of Reaction Times

Reaction time (ms)	Frequency
241–260	1
221–240	2
201–220	2
181–200	9
161–180	4
141–160	2

Frequency Graphs

Pie Charts

In a pie chart, each category on a scale is represented by a slice of the pie. The area of the slice is proportional to the percentage of responses in the category. Pie charts are effective for displaying the proportion of responses for a small number of categories. For instance, the pie chart shown in Figure 25.1 represents data from 500 iMac customers, categorized as a previous Mac owner, a previous Windows owner, or a new computer purchaser.

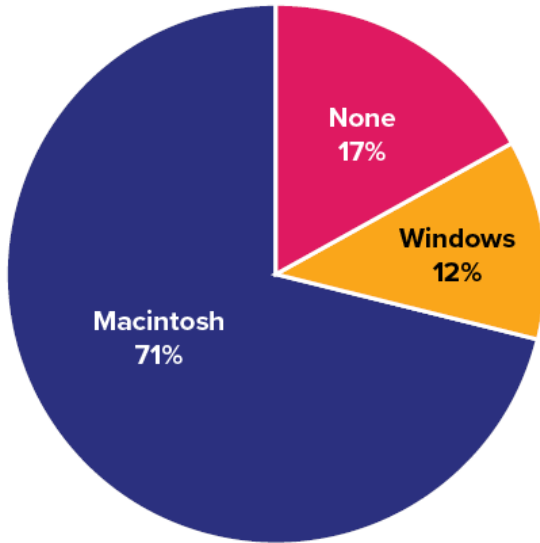


Figure 25.1. Pie chart showing the percentage of iMac purchasers who previously owned a Macintosh computer, a Windows computer, or no computer.

Pie charts are not recommended, however, when you have a large number of categories. Additionally, if a pie chart is based on a small number of observations, it can be misleading to label the pie slices with percentages. For example, if just 5 people had been interviewed by Apple Computers, and 3 were former Windows users, it would be misleading to display a pie chart with the Windows slice showing 60%. With so few people interviewed, such a large percentage of Windows users might easily have occurred since chance can cause large errors with small samples. In this case, it is better to alert the user of the pie chart to the actual numbers involved. The slices should

therefore be labeled with the actual frequencies observed (e.g., 3) instead of with percentages.

Bar Charts

Bar charts can also be used to represent frequencies of different categories. A bar chart of the iMac purchases is shown in Figure 25.2. Frequencies are shown on the y-axis, and the type of computer previously owned is shown on the x-axis. The bars should not be touching, which indicates separate categories on the x-axis. Typically, the y-axis shows the number of observations in each category rather than the percentage of observations in each category, as is typical in pie charts.

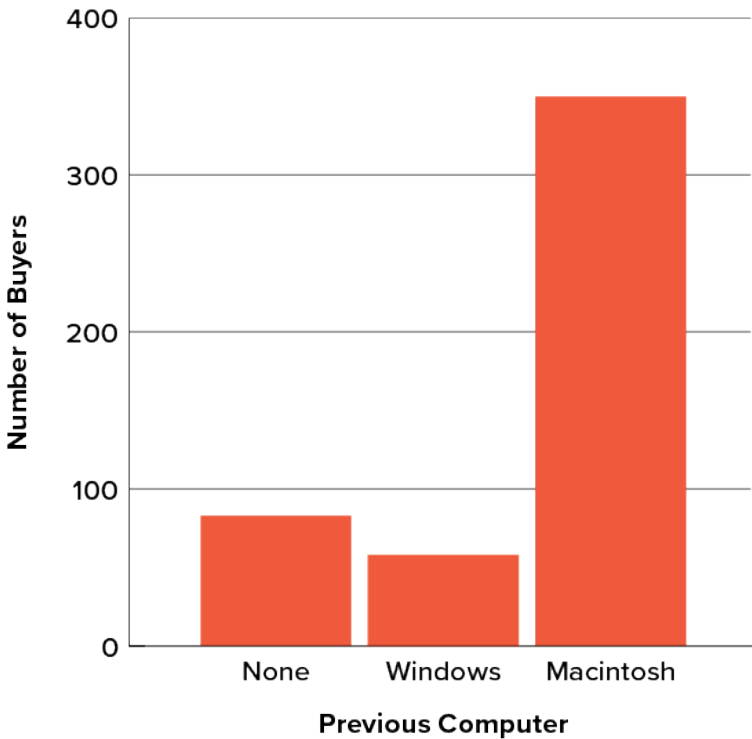


Figure 25.2. Bar chart of iMac purchases as a function of previous computer ownership. [[Image Description](#)]

Histograms

A **histogram** is a graphical display of a distribution used for quantitative (interval and ratio) data. It presents the same information as a frequency table, but in a way that is even quicker and easier to grasp. The histogram in Figure 25.3 presents the distribution of self-esteem scores in Table 25.1. The x -axis of the histogram represents the variable, and the

y -axis represents frequency. Above each level of the variable on the x -axis is a vertical bar that represents the number of individuals with that score. When the variable is quantitative, as in this example, there is usually no gap between the bars. When the variable is categorical, however, a bar graph is used, and there is usually a small gap between the bars. (The gap at 17 in this histogram reflects the fact that there were no scores of 17 in this data set.)

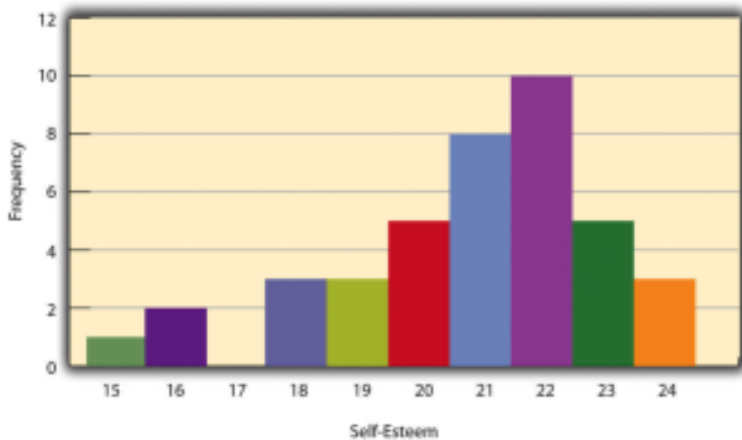


Figure 25.3. Histogram Showing the Distribution of Self-Esteem Scores Presented in Table 25.1 [\[Image Description\]](#)

Similar to a frequency table, when there are many different scores across a wide range of values, it is often better to create a grouped histogram, in which the y -axis still represents frequency, but the x -axis represents a range of values for each bar. Each bar represents a range of scores broken into intervals,

called class intervals. This can be seen in Figure 25.4, where the data represent the scores of 642 students on a psychology test ranging from 46 to 167; each class interval represents a range of 10 points on the scale, each class interval represents a range of 10 points on the scale.

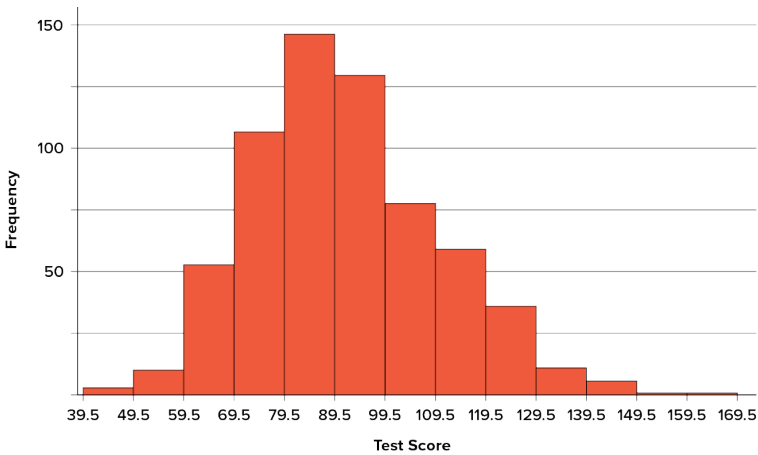


Figure 25.4. Histogram of scores on a psychology test.
[\[Image Description\]](#)

Frequency Polygons

Frequency polygons are a graphical device for understanding the shapes of distributions. They serve the same purpose as histograms, but are especially helpful for comparing sets of data. Frequency polygons are also a good choice for displaying cumulative frequency distributions.

To create a frequency polygon:

- Start just as for histograms, by choosing a class interval.
- Draw an x-axis representing the values of the scores in your data. Mark the middle of each class interval with a tick mark, and label it with the middle value represented by the class.
- Draw the y-axis to indicate the frequency of each class. Place a point in the middle of each class interval at the height corresponding to its frequency.
- Finally, connect the points. You should include one class interval below the lowest value in your data and one above the highest value. The graph will then touch the x-axis on both sides.

The frequency distribution of 642 psychology test scores, shown in Figure 25.4, was used to create the frequency polygon shown in Figure 25.5.

The first label on the x-axis is 35. This represents an interval extending from 29.5 to 39.5. Since the lowest test score is 46, this interval has a frequency of 0. The point labeled 45 represents the interval from 39.5 to 49.5. There are three scores in this interval. There are 147 scores in the interval that surrounds 85. You can easily discern the shape of the distribution from Figure 25.5. Most of the scores are between 65 and 115.

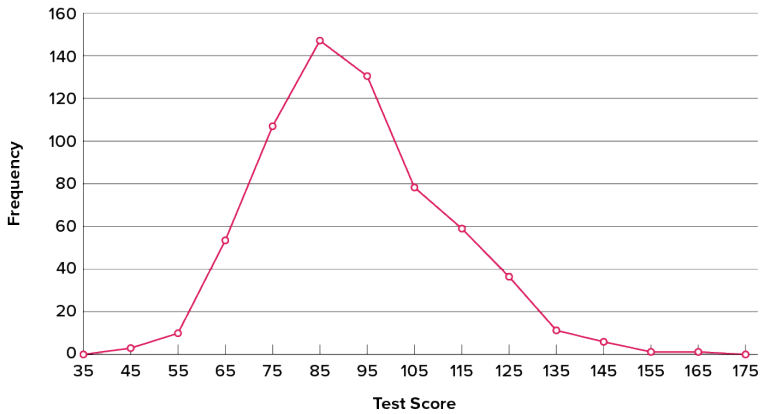


Figure 25.5. Frequency polygon for the psychology test scores. [\[Image Description\]](#)

Frequency polygons are useful for comparing distributions. This is achieved by overlaying the frequency polygons drawn for different datasets. Figure 25.6 provides an example. The data come from a task in which the goal is to move a computer cursor to a target on the screen as fast as possible. In 20 of the trials, the target was a small rectangle; in the other 20, the target was a large rectangle. Time to reach the target was recorded on each trial. The two distributions (one for each target) are plotted together in Figure 25.6. The figure shows that, although there is some overlap in times, it generally took longer to move the cursor to the small target than to the large one.

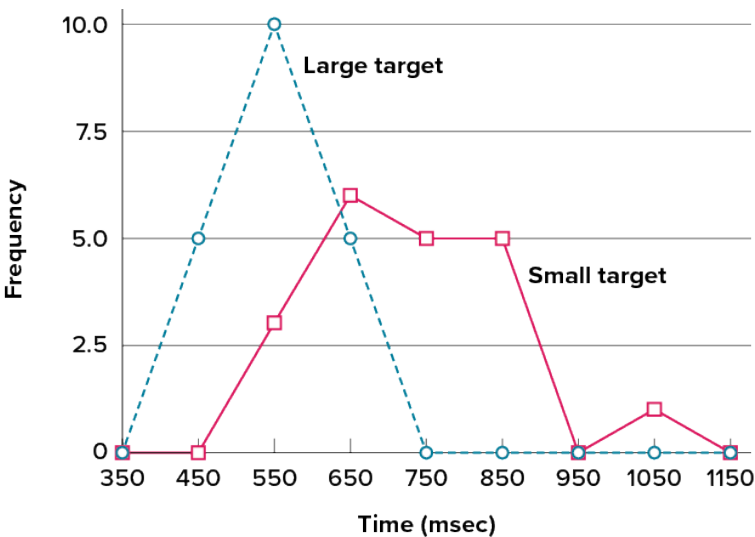


Figure 25.6. Overlaid frequency polygons for the cursor task. [\[Image Description\]](#)

Distribution Shapes

When the distribution of a quantitative variable is displayed in a histogram, it has a shape. The shape of the distribution of self-esteem scores in Figure 25.4 is typical. There is a peak somewhere near the middle of the distribution and “tails” that taper in either direction from the peak. The distribution of Figure 25.4 is unimodal, meaning it has one distinct peak, but distributions can also be bimodal, meaning they have two distinct peaks. Figure 25.7, for example, shows a hypothetical bimodal distribution of scores on the Beck Depression

Inventory. Distributions can also have more than two distinct peaks, but these are relatively rare in psychological research.

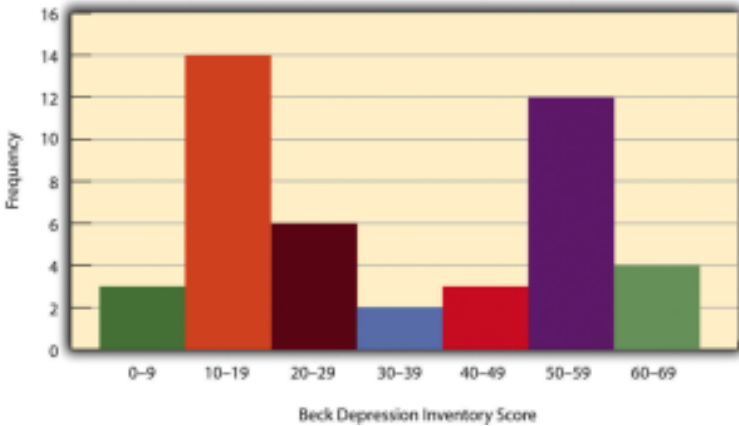


Figure 25.7. Histogram Showing a Hypothetical Bimodal Distribution of Scores on the Beck Depression Inventory.
[\[Image description\]](#)

Another characteristic of the shape of a distribution is whether it is symmetrical or skewed. The distribution in the center of Figure 25.8 is **symmetrical**. Its left and right halves are mirror images of each other. The distribution on the left is negatively **skewed**, with its peak shifted toward the upper end of its range and a relatively long negative tail. The distribution on the right is positively skewed, with its peak toward the lower end of its range and a relatively long positive tail.



Figure 25.8. Histograms Showing Negatively Skewed, Symmetrical, and Positively Skewed Distributions [Image Description]

An **outlier** is an extreme score that is much higher or lower than the rest of the scores in the distribution. Sometimes outliers represent truly extreme scores on the variable of interest. For example, on the Beck Depression Inventory, a single clinically depressed person might be an outlier in a sample of otherwise happy and high-functioning peers. However, outliers can also represent errors or misunderstandings on the part of the researcher or participant, equipment malfunctions, or similar problems. We will say more about how to interpret outliers and what to do about them in the next chapter.

Equity Activity: Income inequality in urban schools

A university has collected data on student participation in **extracurricular activities** (e.g., clubs, sports teams, community service, etc.). The data shows the number of students from different

ethnic groups who participate in extracurricular activities. The university is concerned about potential disparities in involvement across different ethnic groups.

Below are two pie charts showing the number of students enrolled at the university and the number of students who participate in extracurricular activities, broken down by ethnic group:

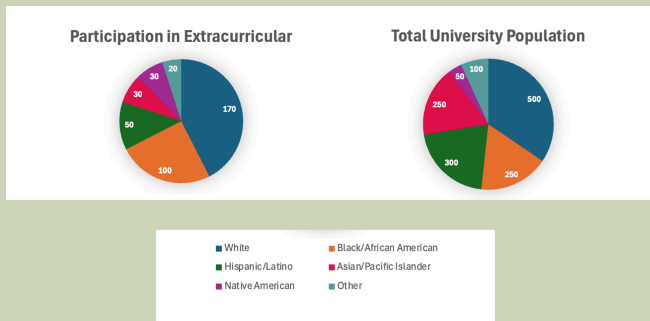


Figure 25.9. Extracurricular Participation by Ethnic Group
[\[Image Description\]](#)



An interactive HSP element has been excluded from this version of the text. You can view it online [here](#):

[https://smc.pressbooks.pub/
critresearchmethodpsych/?p=106#h5p-28](https://smc.pressbooks.pub/critresearchmethodpsych/?p=106#h5p-28)

Online Descriptive Statistics

Although many researchers use commercially available software such as SPSS and Excel to analyze their data, there are several free online analysis tools that can also be extremely useful. Many allow you to enter or upload your data and then make one click to conduct several descriptive statistical analyses. Among them are the following.

[Rice Virtual Lab in Statistics](#)

[VassarStats](#)

[Bright Stat](#)

For a more complete list, see the [Statpages website](#).

License & Attribution

“Frequency Distributions” by Makenzie O’Neil is adapted from “[Describing Data Using Distributions & Graphs](#)” by Linda R. Cote Ph.D.; Rupa G. Gordon Ph.D., Chrislyn E. Randell Ph.D., Judy Schmitt, and Helena Marvin, which is licensed CC BY-NC-SA 4.0.

“Frequency Distributions” is licensed under CC BY-NC-SA 4.0.

Image Descriptions

Figure 25.2. Vertical bar chart with Number of Buyers on the y-axis (0–400) and Previous Computer on the x-axis (categories: None, Windows, Macintosh). Bars show approximate counts: None $\approx$ 80, Windows $\approx$ 60, Macintosh $\approx$ 350. Macintosh dominates, with far more buyers than the other categories. [[Return to Figure 25.2](#)]

Figure 25.3. A histogram shows Self-Esteem on the x-axis (scores 15 to 24) and Frequency on the y-axis (0–12). Bars rise

from low counts at 15–18, increase around 19–21, peak near 22 (≈ 11 cases), then decline at 23–24 (mid-single-digit counts). The distribution is unimodal and clustered in the low-20s.

[\[Return to Figure 25.3\]](#)

Figure 25.4. Vertical histogram with Test Score on the x-axis (bins from about 40 to 170) and Frequency on the y-axis (up to 150). Bars are low at the left (≈ 40 –60), rise through 60–80, peak near 80–90 at roughly 140–150 cases, then steadily decline across 90–120 and taper to small counts from 130 to 160+. The distribution is unimodal and positively skewed (right-skewed). [\[Return to Figure 25.4\]](#)

Figure 25.5. Line graph (frequency polygon) with Test Score on the x-axis (35–175) and Frequency on the y-axis (0–160). Open-circle points connect to form a sharp rise from near zero at 35–55, increasing to about 50 at 65, 105 at 75, and peaking around 150 at 85. The line then declines: ~ 130 at 95, ~ 80 at 105, ~ 58 at 115, ~ 36 at 125, ~ 10 at 135, and approaches zero by 155–175. The distribution is unimodal and positively skewed (right-skewed). [\[Return to Figure 25.5\]](#)

Figure 25.6. Line chart with Time (msec) on the x-axis (350–1150) and Frequency on the y-axis (0–10). Two polygons are plotted: a teal dashed line with open circles labeled Large target and a magenta solid line with open squares labeled Small target. The large-target distribution rises from 350 ms to a peak frequency of 10 at ~ 550 ms, then drops to zero by 750 ms. The small-target distribution begins near zero at 450 ms, climbs to ~ 6 at 650 ms, stays around 5 through

750–850 ms, and then declines, with a small bump near 1050 ms. Overall, times are faster (shorter) for the large target and slower for the small target. [[Return to Figure 25.6](#)]

Figure 25.7. Histogram-style bar chart with Beck Depression Inventory Score bins on the x-axis (0–9, 10–19, 20–29, 30–39, 40–49, 50–59, 60–69) and Frequency on the y-axis (0–16). Approximate counts by bin: 0–9 $\approx$ 3, 10–19 $\approx$ 14 (largest), 20–29 $\approx$ 6, 30–39 $\approx$ 1, 40–49 $\approx$ 3, 50–59 $\approx$ 12 (second largest), 60–69 $\approx$ 3. The distribution is bimodal, concentrated in the 10–19 and 50–59 ranges. [[Return to Figure 25.7](#)]

Figure 25.8. Side-by-side silhouettes show distribution shapes. Left (green) has a long left tail and a peak toward the right and is negatively skewed. Center (blue) is balanced with a central peak and is symmetrical. Right (orange) has a long right tail and a peak toward the left and is positively skewed. [[Return to Figure 25.8](#)]

Figure 25.9. Side-by-side pie charts with the headings “Participation in Extracurricular” (left) and “Total University Population” (right). A legend is located below maps colors: White (dark blue), Black/African American (orange), Hispanic/Latino (green), Asian/Pacific Islander (red), Native American (purple), Other (teal). The pie graph entitled Participation in Extracurricular counts by slice are White 170, Black/African American 100, Hispanic/Latino 50, Asian/Pacific Islander 30, Native American 30, Other 20. The pie graph entitled Total University Population counts by slice are White 500, Black/African American 250, Hispanic/Latino

300, Asian/Pacific Islander 250, Native American 50, Other
100. [[Return to Figure 25.9](#)]

26.

CENTRAL TENDENCY

Mireille Ukeye and Makenzie O'Neil

Learning Objectives

By the end of this chapter, you will be able to

1. Define and distinguish between the mean, median, and mode, and explain the appropriate contexts in which each measure of central tendency is most informative.
2. Interpret how different distributions influence the relationship between measures of central tendency.
3. Apply knowledge of central tendency to real-world equity issues by analyzing how

statistical misrepresentation can obscure disparities.

Measures of Central Tendency

It is useful to be able to describe the characteristics of a distribution more precisely. Here we look at how to do this in terms of one important characteristic: their central tendency.

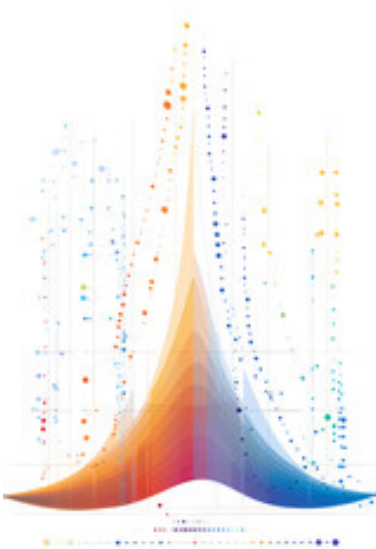
Central Tendency

The **central tendency** of a distribution is its middle—the point around which the scores in the distribution tend to cluster. (Another term for central tendency is *average*.) Looking back at [Figure 25.1 \[New Tab\]](#), for example, we can see that the self-esteem scores tend to cluster around the values of 20 to 22. Here we will consider the three most common measures of central tendency: the mean, the median, and the mode.

The **mean** of a distribution (symbolized M) is the sum of the scores divided by the number of scores. It is an average. As a formula, it looks like this:

$$M = \frac{\sum X}{N}$$

In this formula, the symbol Σ (the Greek letter sigma) is the summation sign and means to sum across the values of the variable X . N represents the number of scores. The mean is by far the most common measure of central tendency, and there are some good reasons for this. It usually provides a good indication of the central tendency of a distribution, and it is easily understood by most people. In addition, the mean has statistical properties that make it especially useful in doing inferential statistics.



An alternative to the mean is the **median**. The median is the middle score in the sense that half the scores in the distribution are less than it and half are greater than it. The simplest way to find the median is to organize the scores from lowest to highest and locate the score in the middle. Consider, for example, the following set of seven scores:

8 4 12 14 3 2 3

To find the median, simply rearrange the scores from lowest to highest and locate the one in the middle.

2 3 3 **4** 8 12 14

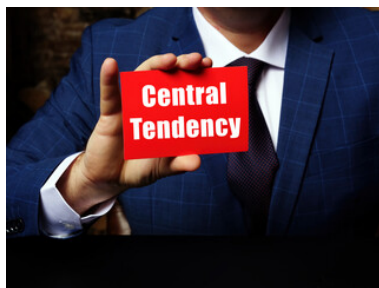
In this case, the median is 4 because there are three scores

lower than 4 and three scores higher than 4. When there is an even number of scores, there are two scores in the middle of the distribution, in which case the median is the value halfway between them. For example, if we were to add a score of 15 to the preceding data set, there would be two scores (both 4 and 8) in the middle of the distribution, and the median would be halfway between them (6).

One final measure of central tendency is the mode. The **mode** is the most frequent score in a distribution. In the self-esteem distribution presented in [Table 25.1](#) [New Tab] and [Figure 12.1](#) [New Tab] for example, the mode is 22. More students had that score than any other. The mode is the only measure of central tendency that can also be used for categorical variables.

In a distribution that is both unimodal and symmetrical, the mean, median, and mode will be very close to each other at the peak of the distribution.

In a bimodal or asymmetrical distribution, the mean, median, and mode can be quite different. In a bimodal distribution, the mean and median will tend to be between the peaks, while the mode will be at the tallest peak. In a skewed distribution, the mean will differ from the median in the direction of the skew (i.e., the direction of the longer tail). For highly skewed distributions,



the mean can be pulled so far in the direction of the skew that it is no longer a good measure of the central tendency of that distribution. Imagine, for example, a set of four simple reaction times of 200, 250, 280, and 250 milliseconds (ms). The mean is 245 ms. But the addition of one more score of 5,000 ms—perhaps because the participant was not paying attention—would raise the mean to 1,445 ms. Not only is this measure of central tendency greater than 80% of the scores in the distribution, but it also does not seem to represent the behavior of anyone in the distribution very well. This is why researchers often prefer the median for highly skewed distributions (such as distributions of reaction times).

Keep in mind, though, that you are not required to choose a single measure of central tendency in analyzing your data. Each one provides slightly different information, and all of them can be useful.

Percentile Ranks

In many situations, it is useful to have a way to describe the location of an individual score within its distribution. One approach is the percentile rank. The **percentile rank** of a score is the percentage of scores in the distribution that are lower than that score. Consider, for example, the distribution in Table 25.1. For any score in the distribution, we can find its percentile rank by counting the number of scores in the distribution that are lower than that score and converting that

number to a percentage of the total number of scores. Notice, for example, that five of the students represented by the data in Table 25.1 had self-esteem scores of 23. In this distribution, 32 of the 40 scores (80%) are lower than 23. Thus, each of these students has a percentile rank of 80. (It can also be said that they scored “at the 80th percentile.”) Percentile ranks are often used to report the results of standardized tests of ability or achievement. If your percentile rank on a test of verbal ability were 40, for example, this would mean that you scored higher than 40% of the people who took the test.

Equity Activity: Income inequality in urban schools

A researcher analyzes teacher salaries across urban and suburban districts.

- Findings: The mean salary in urban districts is equal to that in suburban districts. However, the median reveals that salaries in suburban districts are higher than those in urban districts, as the mean in urban districts was skewed by highly paid administrators.

This example illustrates how outliers affect

measures of central tendency and how choosing the wrong measure could misrepresent the data.



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=108#h5p-29>

License & Attribution

“Central Tendency” by Mireille Ukeye and Makenzie O’Neil is adapted from “[Describing Single Variables](#)” by Rajiv S. Jhangiani; I-Chant A. Chiang; Carrie Cuttler; and Dana C. Leighton is licensed CC BY 4.0.

“Describing Single Variables” is licensed under CC BY-NC-SA 4.0.

27.

VARIABILITY

Rebecca Anguiano

Learning Objectives

By the end of this chapter, you will be able to

1. Define variability and its purpose in statistics.
2. Compute measures of variability such as range, interquartile range, standard deviation, and variance to draw conclusions about data sets.

Spread and Variability

Variability refers to how “spread out” a group of scores is.

To see what we mean by spread out, consider the graphs in Figure 27.1. These graphs represent the scores on two quizzes. The mean score for each quiz is 7.0. Despite the equality of means, you can see that the distributions are quite different. Specifically, the scores on Quiz 1 are more densely packed, and those on Quiz 2 are more spread out. The differences among students were much greater on Quiz 2 than on Quiz 1.

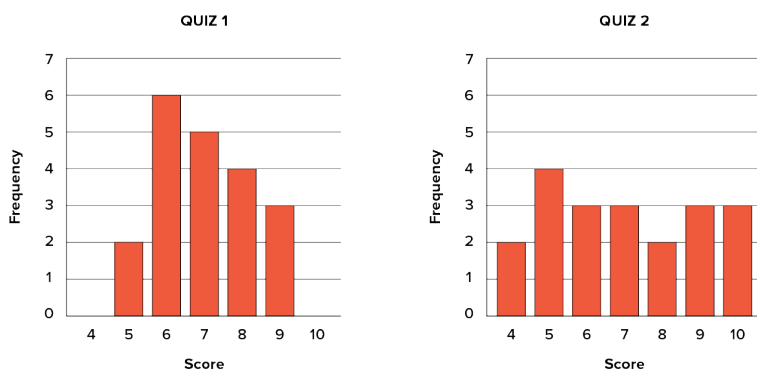


Figure 27.1. Bar charts of Quizzes 1 and 2. [\[Image Description\]](#)

The terms variability, spread, and dispersion are synonyms and refer to how spread out a distribution is. Just as in the section on central tendency, where we discussed measures of the center of a distribution of scores, in this section, we will discuss measures of the variability of a distribution. There are three frequently used measures of variability: range, variance, and standard deviation. In the next few paragraphs, we will look at each of these measures of variability in more detail.

Range

The range is the simplest measure of variability to calculate, and one you have probably encountered many times in your life. The **range** is simply the highest score minus the lowest score. Let's take a few examples. What is the range of the following group of numbers: 10, 2, 5, 6, 7, 3, 4? Well, the highest number is 10, and the lowest number is 2, so $10 - 2 = 8$. The range is 8. Let's take another example. Here's a dataset with 10 numbers: 99, 45, 23, 67, 45, 91, 82, 78, 62, 51. What is the range? The highest number is 99 and the lowest number is 23, so $99 - 23 = 76$; the range is 76. Now, consider the two quizzes shown in Figure 27.1. On Quiz 1, the lowest score is 5 and the highest score is 9. Therefore, the range is 4. The range on Quiz 2 was larger: the lowest score was 4, and the highest score was 10. Therefore, the range is 6.

The problem with using range is that it is extremely sensitive to outliers, and one number far away from the rest of the data will greatly alter the value of the range. For example, in the set of numbers 1, 3, 4, 4, 5, 8, and 9, the range is 8 ($9 - 1$). However, if we add a single person whose score is nowhere close to the rest of the scores, say, 20, the range more than doubles from 8 to 19.

Interquartile Range

The **interquartile range (IQR)** is the range of the middle 50% of the scores in a distribution and is sometimes used to

communicate where the bulk of the data in the distribution are located. It is computed as follows:

$$\text{IQR} = 75\text{th percentile} - 25\text{th percentile}$$

For Quiz 1, the 75th percentile is 8 and the 25th percentile is 6. The interquartile range is therefore 2. For Quiz 2, which has greater spread, the 75th percentile is 9, the 25th percentile is 5, and the interquartile range is 4. Recall that in the discussion of box plots, the 75th percentile was called the upper hinge, and the 25th percentile was called the lower hinge. Using this terminology, the interquartile range is referred to as the H-spread.

Sum of Squares

Variability can also be defined in terms of how close the scores in the distribution are to the middle of the distribution. Using the mean as the measure of the middle of the distribution, we can see how far, on average, each data point is from the center. The data from Quiz 1 are shown in [Table 27.1](#).

There are a few things to note about how [Table 27.1](#) is formatted. The raw data scores (X) are always placed in the left-most column. This column is then summed at the bottom (ΣX) to facilitate calculating the mean by dividing the sum of X values by the number of scores in the table (N). The mean score is 7.0 ($\Sigma X/N = 140/20 = 7.0$). Once you have the mean, you can easily work your way down the second column calculating the deviation scores ($X - M$), representing how far

each score deviates from the mean, here calculated as the score (X value) minus 7. This column is also summed and has a very important property: it will always sum to 0, or close to zero if you have rounding error due to many decimal places ($\Sigma(X - M) = 0$). This step is used as a check on your math to make sure you haven't made a mistake. If this column sums to 0, you can move on to filling in the third column, which is composed of the squared deviation scores. The deviation scores are squared to remove negative values and appear in the third column $(X - M)^2$. When these values are summed, you have the sum of the squared deviations, or the **sum of squares** (SS), calculated with the formula $SS = \sum (X - M)^2$.

Table 27.1. Calculation of Variance for Quiz 1 Scores

X	X – M	(X – M) ²
9	2	4
9	2	4
9	2	4
8	1	1
8	1	1
8	1	1
8	1	1
7	0	0
7	0	0
7	0	0
7	0	0
7	0	0

X	X – M	(X – M) ²
6	–1	1
6	–1	1
6	–1	1
6	–1	1
6	–1	1
6	–1	1
5	–2	4
5	–2	4
$\sum X = 140$		
$(\sum X)^2 = 19,600$		
$\sum (X - M) = 0$		
$\sum (X - M)^2 = 30$		

$SS = 4 + 4 + 4 + 1 + 1 + 1 + 1 + 0 + 0 + 0 + 0 + 0 + 1 + 1 + 1 + 1 + 1 + 1 + 4 + 4 = 30$

The preceding formula is called the definitional formula, as it shows the logic behind the sum of squared deviations calculation. As mentioned earlier, there can be rounding errors in calculating the deviation scores. Also, when the set of scores

is large, calculating the deviation scores, squaring the scores, and then summing those values can be tedious. To simplify the sum of squares calculation, the computational formula is used instead. The computational formula is as follows:

$$SS = \sum X^2 - \frac{(\sum X)^2}{N}$$

The last column in [Table 27.1](#) represents the X values squared and then summed— $\sum X^2$. At the bottom of the first column, the $\sum X$ value is squared— $(\sum X)^2$. These are the values used in the computational formula for the sum of squares. As you can see in the calculation below, the SS value is the same for both the definitional formula and the computational formula:

$$SS = \sum X^2 - \frac{(\sum X)^2}{N} = 1,010 - \frac{19,600}{20} = 30$$

As we will see, the sum of squares appears again and again in different formulas—it is a very important value, and using the $\mathbf{X}$ and $\mathbf{X}^2$ columns in this table makes it simple to calculate the SS without error.

Variance

Now that we have the sum of squares calculated, we can use it to compute our formal measure of average distance from the mean—the variance. The variance is defined as the average squared difference of the scores from the mean. We square the deviation scores because, as we saw in the second column of

[Table 27.1](#), the sum of raw deviations is always 0, and there's nothing we can do mathematically without changing that.

The population parameter for variance is σ^2 (“sigma-squared”) and is calculated as:

$$\sigma^2 = \frac{SS}{N}$$

We can use the value we previously calculated for SS in the numerator, then simply divide that value by **N** to get the variance. If we assume that the values in [Table 27.1](#) represent the full population, then we can take our value of the sum of squares and divide it by **N** to get our population variance:

$$\sigma^2 = \frac{30}{20} = 1.5$$

So, on average, scores in this population are 1.5 squared units away from the mean. This measure of spread exhibits much more robustness (a term used by statisticians to mean resilience or resistance to outliers) than the range, so it is a much more useful value to compute. Additionally, as we will see in future chapters, variance plays a central role in inferential statistics.

The sample statistic used to estimate the variance is s^2 (“s-squared”):

$$s^2 = \frac{SS}{N - 1} = \frac{SS}{df}$$

This formula is very similar to the formula for the population variance with one change: we now divide by $N - 1$ instead of N . The value $N - 1$ has a special name: the degrees of freedom (abbreviated as df). You don't need to understand

in depth what degrees of freedom are (essentially, they account for the fact that we have to use a sample statistic to estimate the mean [M] before we estimate the variance) in order to calculate variance, but knowing that the denominator is called df provides a nice shorthand for the variance formula:

$$\frac{SS}{df}$$

Going back to the values in Table 27.1 and treating those scores as a sample, we can estimate the sample variance as:

$$s^2 = \frac{30}{20 - 1} = 1.58$$

Notice that this value is slightly larger than the one we calculated when we assumed these scores were the full population. This is because our value in the denominator is slightly smaller, making the final value larger. In general, as your sample size N gets bigger, the effect of subtracting 1 becomes less and less. Comparing a sample size of 10 to a sample size of 1000; $10 - 1 = 9$, or 90% of the original value, whereas $1000 - 1 = 999$, or 99.9% of the original value. Thus, larger sample sizes will bring the estimate of the sample variance closer to that of the population variance. This is a key idea and principle in statistics that we will see over and over again: larger sample sizes better reflect the population.

Standard Deviation

The standard deviation is simply the square root of the variance. This is a useful and interpretable statistic because

taking the square root of the variance (recalling that variance is the average squared difference) puts the standard deviation back into the original units of the measure we used. Thus, when reporting descriptive statistics in a study, scientists virtually always report mean and standard deviation. Standard deviation is therefore the most commonly used measure of spread for our purposes, representing the average distance of the scores from the mean.

The population parameter for standard deviation is σ (“sigma”), which, intuitively, is the square root of the variance parameter σ^2 (occasionally, the symbols work out nicely that way). The formula is simply the formula for variance under a square root sign:

$$\sigma = \sqrt{\frac{SS}{N}} = \sqrt{\frac{30}{20}} = \sqrt{1.5} = 1.22$$

The sample statistic follows the same conventions and is given as s in mathematical formulas. (Note that in American Psychological Association [APA] format for reporting results, sample standard deviation is reported using the abbreviation SD.)

$$s = \sqrt{\frac{SS}{df}} = \sqrt{\frac{30}{20-1}} = \sqrt{1.58} = 1.26$$

The standard deviation is an especially useful measure of variability when the distribution is normal or approximately normal because the proportion of the distribution within a given number of standard deviations from the mean can be calculated. For example, 68% of the distribution is within one

standard deviation (above and below) of the mean and approximately 95% of the distribution is within two standard deviations of the mean, as shown in Figure 27.2. Therefore, if you had a normal distribution with a mean of 50 and a standard deviation of 10, then 68% of the distribution would be between $50 - 10 = 40$ and $50 + 10 = 60$. Similarly, about 95% of the distribution would be between $50 - 2 \times 10 = 30$ and $50 + 2 \times 10 = 70$.

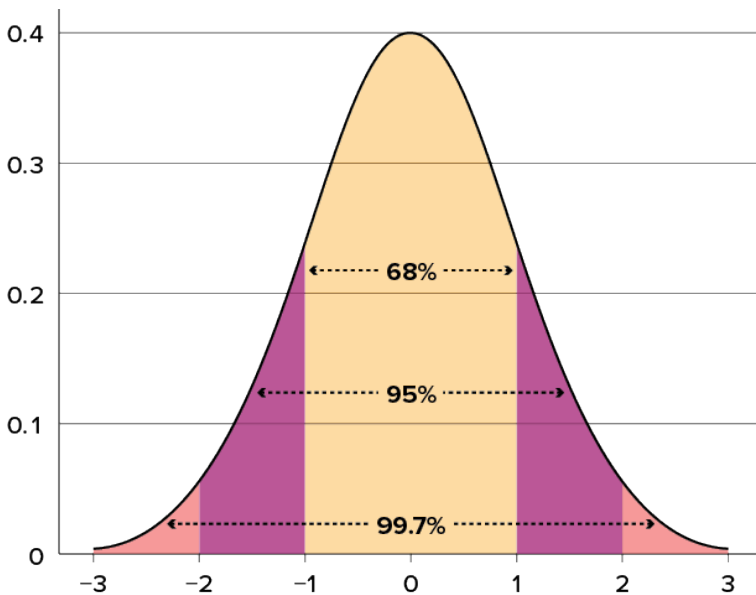


Figure 27.2. Percentages of the Normal Distribution. [\[Image Description\]](#)

Figure 27.3 shows two normal distributions. The red (left-most) distribution has a mean of 40 and a standard deviation

of 5; the blue (right-most) distribution has a mean of 60 and a standard deviation of 10. For the red distribution, 68% of the distribution is between 45 and 55; for the blue distribution, 68% is between 50 and 70. Notice that as the standard deviation gets smaller, the distribution becomes much narrower, regardless of where the center of the distribution (mean) is. Figure 27.4 presents several more examples of this effect.

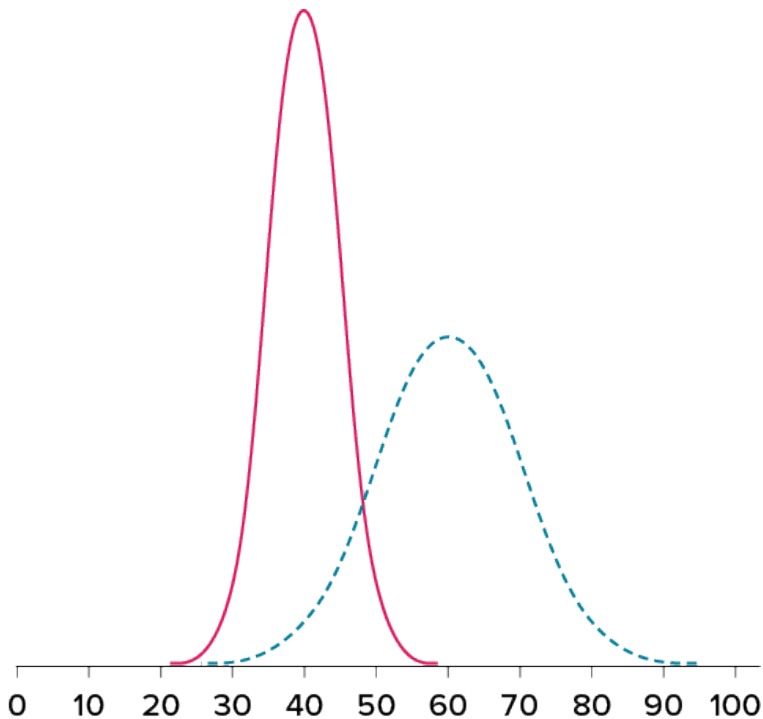


Figure 27.3. Normal Distributions with Standard Deviations of 5 and 10. [\[Image Description\]](#)

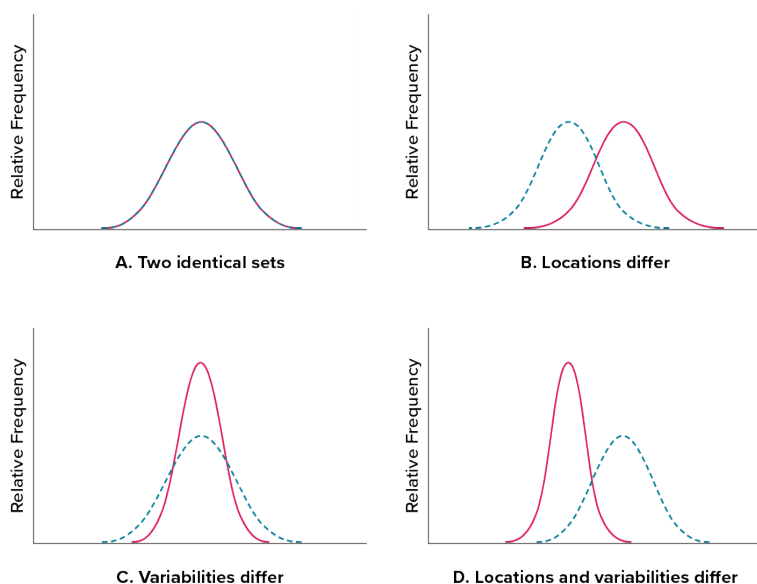


Figure 27.4. Differences Between Two Datasets. [\[Image Description\]](#)

Equity Activity: Understanding variability in educational outcomes

A school district is investigating whether students in different neighborhoods have equal access to high-quality education. They collect standardized math test scores from **three middle schools** serving distinct neighborhoods:

- **School A:** Affluent neighborhood
- **School B:** Mixed-income neighborhood
- **School C:** Low-income neighborhood

Each school tested 100 students. The **mean** math score and **standard deviation (SD)** of scores are shown below:

Table 27.2. Mean and Standard Deviations of Three

School	Mean Score	St
School A	82	5
School B	75	10
School C	71	18



An interactive H5P element has been excluded from this version of the text. You can view it online here:

[https://smc.pressbooks.pub/
critresearchmethodpsych/?p=116#h5p-30](https://smc.pressbooks.pub/critresearchmethodpsych/?p=116#h5p-30)

Practice Problems

Test Your Knowledge



An interactive H5P element has been excluded from this version of the text. You can view it online here:

[https://smc.pressbooks.pub/
critresearchmethodpsych/?p=116#h5p-31](https://smc.pressbooks.pub/critresearchmethodpsych/?p=116#h5p-31)

License & Attribution

“Variability” by Rebecca Anguiano is adapted from “[Measures of Central Tendency and Spread](#)” by Linda R. Cote Ph.D.; Rupa G. Gordon Ph.D., Chrislyn E. Randell Ph.D., Judy Schmitt, and Helena Marvin, which is licensed CC BY-NC-SA 4.0.

“Variability” is licensed under CC BY-NC-SA 4.0.

Image Descriptions

Figure 27.1. Two bar charts compare score frequency (y-axis 0–7) by Score (x-axis 4–10). QUIZ 1 counts: 5=2, 6=6 (highest), 7=5, 8=4, 9=3; 4 and 10=0. QUIZ 2 counts: 4=2, 5=4, 6=3, 7=3, 8=2, 9=3, 10=3. Quiz 1 is tightly centered around 6–8; Quiz 2 is more even across scores. [[Return to Figure 27.1](#)]

Figure 27.2. A bell-shaped normal distribution is centered at 0 with x-axis ticks from -3 to $+3$ standard deviations. Shading highlights coverage: the middle band from -1 to $+1$ SD

(orange) is labeled 68%; the wider band from -2 to $+2$ SD (orange plus purple) is labeled 95%; nearly the whole curve from -3 to $+3$ SD (orange, purple, and red tails) is labeled 99.7%. Horizontal dotted arrows mark each percentage span. [[Return to Figure 27.2](#)]

Figure 27.3. Line chart with x-axis from 0–100 shows two bell curves. A magenta solid curve is tall and narrow, centered around ~ 42 , indicating a smaller standard deviation. A teal dashed curve is shorter and wider, centered around ~ 60 , indicating a larger standard deviation. The curves overlap modestly between about 48–55. The figure contrasts equal-shape distributions with different spreads and means. [[Return to Figure 27.3](#)]

Figure 27.4. Grid of four panels titled A–D with y-axis “Relative Frequency.” Each panel shows a magenta solid curve and a teal dashed curve. A. Two identical sets: curves overlap completely (same mean and variability). B. Locations differ: curves have similar width but distinct centers (different means). C. Variabilities differ: curves share the same center; magenta is narrow/tall, teal is wider/shorter (different standard deviations). D. Locations and variabilities differ: curves have different centers and spreads. The figure illustrates how distributions can vary by mean and/or variability. [[Return to Figure 27.4](#)]

28.

THE STANDARD NORMAL DISTRIBUTION AND Z-SCORES

Makenzie O'Neil

Learning Objectives

By the end of this chapter, you will be able to

1. Articulate the basic components of the Standard Normal Distribution.
2. Calculate Z-scores.
3. Describe what information a Z-score provides about a given value and the distribution.
4. Compare scores across distributions through

the process of standardization/Z-scores.

We now understand how to describe and present our data visually and numerically. These simple tools and the principles behind them will help you interpret information presented to you and understand the basics of a variable. Moving forward, we now turn our attention to how scores within a distribution are related to one another, how to precisely describe a score's location within the distribution, and how to compare scores from different distributions.

Normal Distributions

The normal distribution is the most important and most widely used distribution in statistics. It is sometimes called the “bell curve,” although the tonal qualities of such a bell would be less than pleasing. It is also called the “Gaussian curve” or Gaussian distribution after the mathematician Karl Friedrich Gauss.

Strictly speaking, it is not correct to talk about “the normal distribution” since there are many normal distributions. Normal distributions can differ in their means and in their standard deviations. Figure 28.1 shows three normal distributions. The blue (left-most) distribution has a mean of

-3 and a standard deviation of 0.5 , the distribution in red (the middle distribution) has a mean of 0 and a standard deviation of 1 , and the black (right-most) distribution has a mean of 2 and a standard deviation of 3 . These, as well as all other normal distributions, are symmetric with relatively more values at the center of the distribution and relatively few in the tails. What is consistent about all normal distributions is the shape and the proportion of scores within a given distance along the x -axis. We will focus on the standard normal distribution (also known as the unit normal distribution), which has a mean of 0 and a standard deviation of 1 (i.e., the red distribution in Figure 28.1).

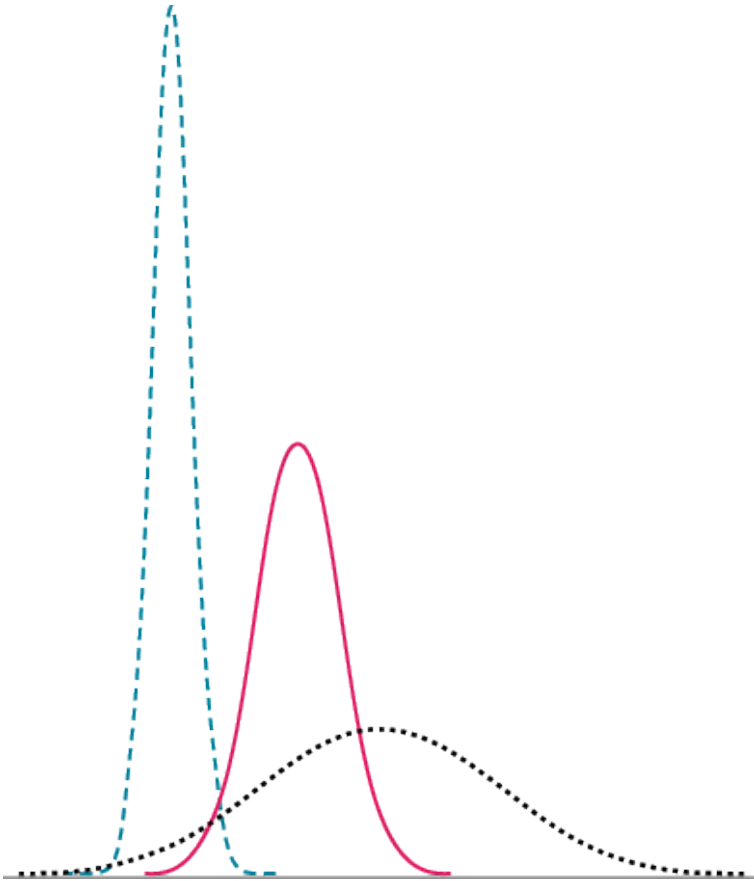


Figure 28.1. Normal Distributions Differing in Mean and Standard Deviation. [\[Image Description\]](#)

Seven features of normal distributions

are listed below.

1. Normal distributions are symmetric around their mean.
2. The mean, median, and mode of a normal distribution are equal.
3. The area under the normal curve is equal to 1.0.
4. Normal distributions are denser in the center and less dense in the tails.
5. Normal distributions are defined by two parameters, the mean (μ) and the standard deviation (σ).
6. 68% of the area of a normal distribution is within one standard deviation of the mean.
7. Approximately 95% of the area of a normal distribution is within two standard deviations of the mean.

These properties enable us to use the normal distribution to understand how scores relate to one another within and across a distribution. But first, we need to learn how to calculate

the standardized score that makes up a standard normal distribution.

Z-Scores

A z-score is a standardized version of a raw score (x) that gives information about the relative location of that score within its distribution. The formula for converting a raw score into a z-score is

$$z = \frac{x - \mu}{\sigma}$$

for values from a population and

$$z = \frac{x - M}{s}$$

for values from a sample.

As you can see, z-scores combine information about where the distribution is located (the mean/center) with how wide the distribution is (the standard deviation/spread) to interpret a raw score (x). Specifically, z-scores will tell us how far the score is away from the mean in units of standard deviations and in what direction.

The value of a z-score has two parts: the sign (positive or negative) and the magnitude (the actual number). The sign of the z-score tells you in which half of the distribution the z-score falls: a positive sign (or no sign) indicates that the score is above the mean and on the right-hand side or upper end of the distribution, and a negative sign tells you the score is below the mean and on the left-hand side or lower end of

the distribution. The magnitude of the number tells you, in units of standard deviations, how far away the score is from the center or mean. The magnitude can take on any value between negative and positive infinity, but for reasons we will see soon, they generally fall between -3 and 3 .

Let's look at some examples. A z-score value of -1.0 tells us that this z-score is 1 standard deviation (because of the magnitude 1.0) below (because of the negative sign) the mean. Similarly, a z-score value of 1.0 tells us that this z-score is 1 standard deviation above the mean. Thus, these two scores are the same distance away from the mean but in opposite directions. A z-score of -2.5 is two-and-a-half standard deviations below the mean and is therefore farther from the center than both of the previous scores, and a z-score of 0.25 is closer than all of the ones before. In a later chapter, we will learn to formalize the distinction between what we consider "close to" the center or "far from" the center. For now, we will use a rough cut-off of 1.5 standard deviations in either direction as the difference between close scores (those within 1.5 standard deviations or between $z = -1.5$ and $z = 1.5$) and extreme scores (those farther than 1.5 standard deviations—below $z = -1.5$ or above $z = 1.5$).

We can also convert raw scores into z-scores to get a better idea of where in the distribution those scores fall. Let's say we get a score of 68 on an exam. We may be disappointed to have scored so low, but perhaps it was just a very hard exam. Having information about the distribution of all scores in the class

would be helpful to put some perspective on ours. We find out that the class got an average score of 54 with a standard deviation of 8. To find out our relative location within this distribution, we simply convert our test score into a z-score.

$$z = \frac{X - \mu}{\sigma} = \frac{68 - 54}{8} = 1.75$$

We find that we are 1.75 standard deviations above the average, above our rough cut-off for close and far. Suddenly, our 68 is looking pretty good!

Figure 28.2 shows both the raw score and the z-score on their respective distributions. Notice that the red line indicating where each score lies is in the same relative spot for both. This is because transforming a raw score into a z-score does not change its relative location; it only makes it easier to know precisely where it is.

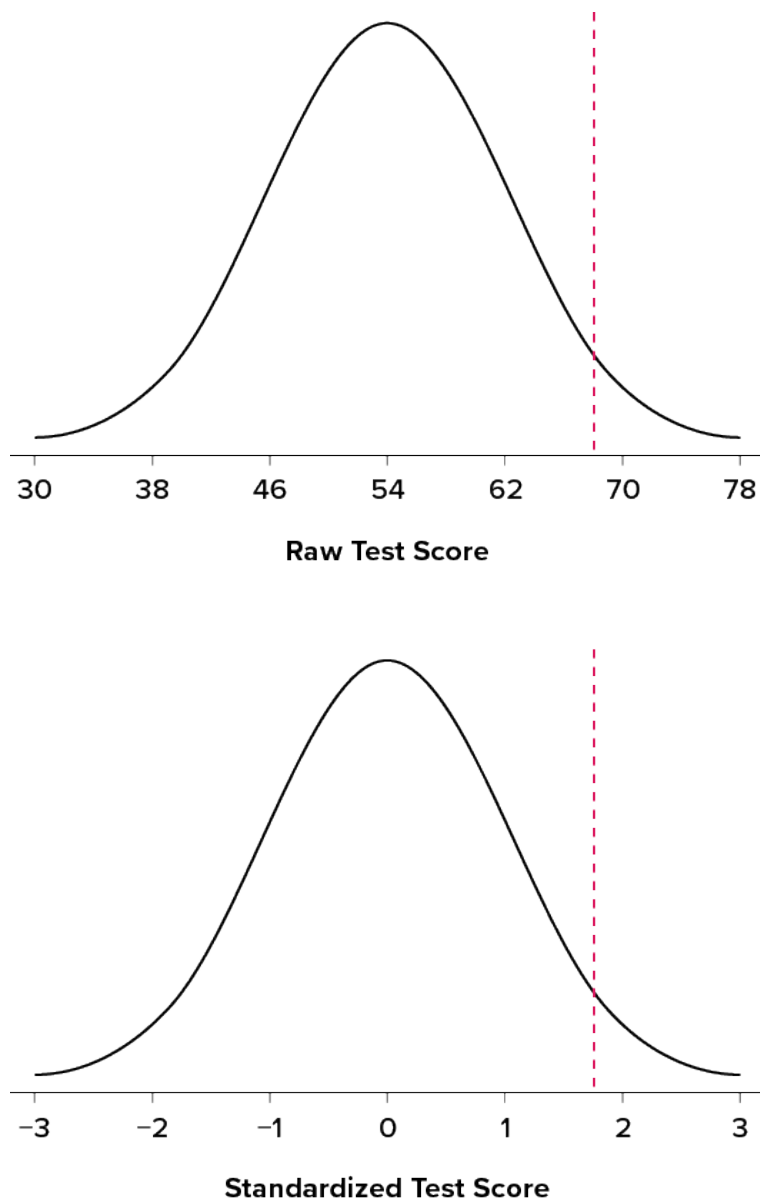


Figure 28.2. Raw and Standardized Versions of a Single Score. [\[Image Description\]](#)

Z-Scores are also useful for comparing scores from different distributions. Let's say we take the SAT and score 501 on both the math and critical reading sections. Does that mean we did equally well on both? Scores on the math portion are distributed normally with a mean of 511 and a standard deviation of 120, so our z-score on the math section is

$$z_{\text{math}} = \frac{501 - 511}{120} = -0.08$$

which is just slightly below average (note the use of "math" as a subscript; subscripts are used when presenting multiple versions of the same statistic in order to know which one is which and have no bearing on the actual calculation). The critical reading section has a mean of 495 and a standard deviation of 116, so

$$z_{\text{CR}} = \frac{501 - 495}{116} = 0.05$$

So even though we were almost exactly average on both tests, we did a little bit better on the critical reading portion relative to other people.

Finally, z-scores are incredibly useful if we need to combine information from different measures that are on different scales. Let's say we give a set of employees a series of tests on things like job knowledge, personality, and leadership. We may want to combine these into a single score we can use to rate employees for development or promotion, but look what happens when we take the average of raw scores from different scales, as shown in Table 28.1.

Table 28.1. Raw Test Scores on Different Scales (Ranges in Parentheses).

Employee	Job Knowledge (0–100)	Personality (1–5)	Leadership (1–5)	Average
Employee 1	98	4.2	1.1	34.43
Employee 2	96	3.1	4.5	34.53
Employee 3	97	2.9	3.6	34.50

Because the job knowledge scores were so big and the scores were so similar, they overpowered the other scores and removed almost all variability in the average. However, if we standardize these scores into z-scores, our averages retain more variability, and it is easier to assess differences between employees, as shown in Table 28.2.

Table 28.2. Standardized Scores.

Employee	Job Knowledge (0–100)	Personality (1–5)	Leadership (1–5)	Average
Employee 1	1.00	1.14	–1.12	0.34
Employee 2	–1.00	–0.43	0.81	–0.20
Employee 3	0.00	–0.71	0.30	–0.14

Setting the Scale of a Distribution

Another convenient characteristic of z-scores is that they can be converted into any “scale” that we would like. Here, the term scale means how far apart the scores are (their spread) and where they are located (their central tendency). This can be very useful if we don’t want to work with negative numbers or if we have a specific range we would like to present. The formulas for transforming z to x are:

$$x = z\sigma + \mu$$

for a population and

$$x = zs + M$$

for a sample. Notice that these are just simple

rearrangements of the original formulas for calculating z from raw scores.

Let's say we create a new measure of intelligence, and initial calibration finds that our scores have a mean of 40 and a standard deviation of 7. Three people who have scores of 52, 43, and 34 want to know how well they did on the measure. We can convert their raw scores into z -scores:

$$z = \frac{52 - 40}{7} = 1.71$$

$$z = \frac{43 - 40}{7} = 0.43$$

$$z = \frac{34 - 40}{7} = -0.80$$

A problem is that these new z -scores aren't exactly intuitive for many people. We can give people information about their relative location in the distribution (for instance, the first person scored well above average), or we can translate these z -scores into the more familiar metric of IQ scores, which have a mean of 100 and a standard deviation of 15:

$$\text{IQ} = 1.71(15) + 100 = 125.65$$

$$\text{IQ} = 0.43(15) + 100 = 106.45$$

$$\text{IQ} = -0.80(15) + 100 = 88$$

Z-Scores and the Area under the Curve

Z-Scores and the standard normal distribution go hand-in-hand. A z-score will tell you exactly where in the standard normal distribution a value is located, and any normal distribution can be converted into a standard normal distribution by converting all of the scores in the distribution into z-scores, a process known as **standardization**.

Standard deviations can be used to divide the normal distribution: 68% of the distribution falls within 1 standard deviation of the mean, 95% within (roughly) 2 standard deviations, and 99.7% within 3 standard deviations. Because z-scores are in units of standard deviations, this means that 68% of scores fall between $z = -1.0$ and $z = 1.0$, and so on. We call this 68% (or any percentage we have based on our z-scores) the proportion of the **area under the curve**. Any area under the curve is bounded by (defined by, delineated by, etc.) by a single z-score or pair of z-scores.

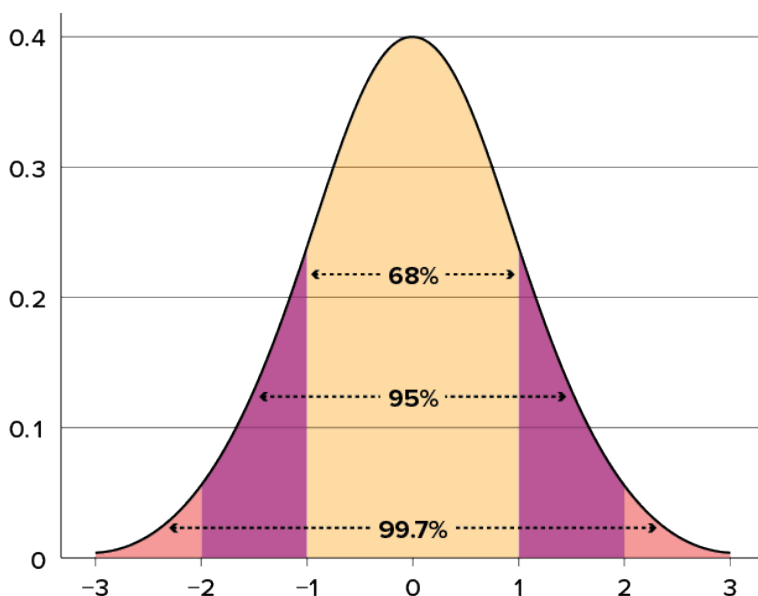


Figure 28.3. Percentages of the Normal Distribution. [\[Image Description\]](#)

An important property to point out here is that, by virtue of the fact that the total area under the curve of a distribution is always equal to 1.0 (see [section on Normal Distributions](#) at the beginning of this chapter), these areas under the curve can be added together or subtracted from 1 to find the proportion in other areas. For example, we know that the area between $z = -1.0$ and $z = 1.0$ (i.e., within one standard deviation of the mean) contains 68% of the area under the curve, which can be represented in decimal form as .6800. (To change a percentage to a decimal, simply move the decimal point 2 places to the left.) Because the total area under the curve is equal to 1.0, that

means that the proportion of the area outside $z = -1.0$ and $z = 1.0$ is equal to $1.0 - .6800 = .3200$ or 32% (see Figure 28.3). This area is called the **area in the tails of the distribution**. Because this area is split between two tails and because the normal distribution is symmetrical, each tail has exactly one-half, or 16%, of the area under the curve.

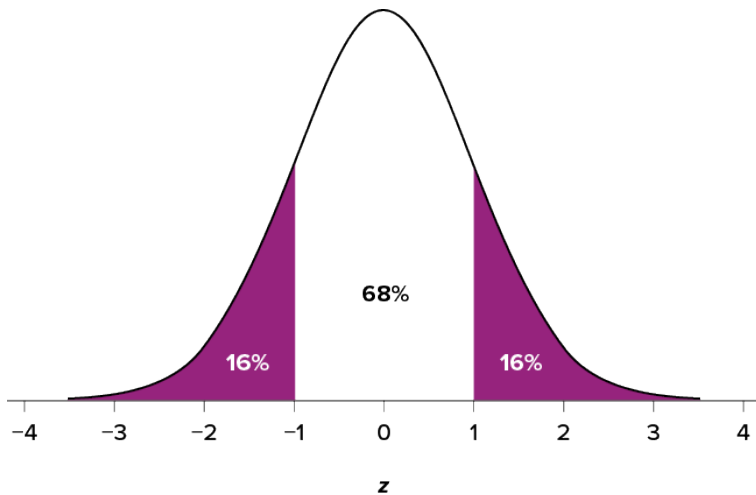


Figure 28.4. Shaded areas represent the area under the curve in the tails. [\[Image Description\]](#)

We will have much more to say about this concept in the coming chapters. As it turns out, this is a quite powerful idea that enables us to make statements about how likely an outcome is and what that means for research questions we would like to answer and hypotheses we would like to test.

But first, we need to make a brief foray into some ideas about probability.

Equity Activity: Gender pay disparity

A company is investigating potential gendered pay disparities among employees from two different groups: Engineers (primarily men at this company) and Sales (primarily women at this company). The company collects salary data for a sample of employees from each of these two groups, and finds that Engineers make an average of \$85,000 while Salespeople make an average of \$78,000 per year. The company claims that any pay disparity is not rooted in gender discrimination, but from a difference in industry standards for compensation. An industry analysis finds that Engineers on average make $\mu=\$80,000$ ($\sigma=10$) per year, and Salespeople make $\mu=\$75,000$ ($\sigma=12$). The goal is to determine whether Salespeople are paid equitably relative to Engineers across both industries, using **z-scores** to compare salaries within each group.



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=122#h5p-32>

Practice Problems

Short Answer Reflections



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=122#h5p-33>

Test Your Knowledge



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=122#h5p-34>

License & Attribution

“The Standard Normal Distribution and Z-scores” by Makenzie O’Neil is adapted from “[Z-scores and the Standard Normal Distribution](#)” by Linda R. Cote Ph.D.; Rupa G. Gordon Ph.D., Chrislyn E. Randell Ph.D., Judy Schmitt, and Helena Marvin, which is licensed CC BY-NC-SA 4.0.

“The Standard Normal Distribution and Z-scores” is licensed under CC BY-NC-SA 4.0.

Image Descriptions

Figure 28.1. The graphic shows three overlapping normal distribution curves without labeled axes. A tall, very narrow dashed teal curve peaks far left, indicating the smallest mean and the smallest standard deviation. A solid magenta curve peaks to the right of the teal one and is moderately narrow, showing a higher mean and a moderate standard deviation. A dotted black curve peaks furthest to the right and spreads the widest, representing the largest mean and the largest standard deviation. The arrangement illustrates how changing the mean shifts a normal curve left or right and how increasing standard deviation flattens and widens it. [[Return to Figure 28.1](#)]

Figure 28.2. The figure shows two normal-distribution curves with the same shape. The upper plot is labeled Raw Test Score with an axis from 30 to 78; a vertical dashed red line marks a raw score of 68, which lies well above the center peak (mean ≈ 54). The lower plot is labeled Standardized Test Score with an axis from -3 to 3 ; a vertical dashed red line marks $z \approx 1.75$, aligning horizontally with the raw 68 above to indicate that the raw score's distance from the mean equals 1.75 standard deviations. [[Return to Figure 28.2](#)]

Figure 28.3. A bell-shaped normal distribution is centered at 0 with x-axis ticks from -3 to $+3$ standard deviations. Shading highlights coverage: the middle band from -1 to $+1$ SD (orange) is labeled 68%; the wider band from -2 to $+2$ SD (orange plus purple) is labeled 95%; nearly the whole curve

from -3 to $+3$ SD (orange, purple, and red tails) is labeled 99.7%. Horizontal dotted arrows mark each percentage span. [[Return to Figure 28.3](#)]

Figure 28.4. A bell-shaped normal distribution is plotted over the horizontal z -scale from -4 to 4 . The middle band from $z = -1$ to $z = 1$ is unshaded and marked “68%,” indicating the proportion within one standard deviation of the mean ($z = 0$). The left and right tails beyond $z = -1$ and $z = 1$ are shaded purple; each tail is labeled “16%,” showing the remaining area split equally on both sides. The x -axis is labeled “ z .” [[Return to Figure 28.4](#)]

29.

PROBABILITY AND THE NORMAL DISTRIBUTION

Makenzie O'Neil

Learning Objectives

By the end of this chapter, you will be able to

1. Calculate basic probability.
2. Use the Unit Normal Table to identify proportions of data under a normal distribution.

Probability can seem like a daunting topic for many students. In a mathematical statistics course, this might be true, as the meaning and purpose of probability get obscured and

overwhelmed by equations and theory. In this chapter, we will focus only on the principles and ideas necessary to lay the groundwork for future inferential statistics. We accomplish this by quickly tying the concepts of probability to what we already know about normal distributions and z-scores.

What Is Probability?

When we speak of the probability of something happening, we are talking about how likely it is that “thing” will happen based on the conditions present. For instance, what is the probability that it will rain? That is, how likely do we think it is that it will rain today under the circumstances or conditions today? To define or understand the conditions that might affect how likely it is to rain, we might look out the window and say, “It’s sunny outside, so it’s not very likely that it will rain today.” Stated using probability language: given that it is sunny outside, the probability of rain is low. “Given” is the word we use to state what the conditions are. As the conditions change, so does the probability. Thus, if it were cloudy and windy outside, we might say, “Given the current weather conditions, there is a high probability that it is going to rain.”

In these examples, we spoke about whether or not it is going to rain. Raining is an example of an event, which is the catch-all term we use to talk about any specific thing happening; it is a generic term that we specified to mean “rain” in exactly the

same way that “conditions” is a generic term that we specified to mean “sunny” or “cloudy and windy.”

It should also be noted that the terms “low” and “high” are relative and vague, and they will likely be interpreted differently by different people (in other words: given how vague the terminology was, the probability of different interpretations is high). Most of the time, we try to use more precise language or, even better, numbers to represent the probability of our event. Regardless, the basic structure and logic of our statements are consistent with how we speak about probability using numbers and formulas.

Let’s look at a slightly deeper example. Say we have a regular, six-sided die (note that die is singular and dice is plural) and want to know how likely it is that we will roll a 1. That is, what is the probability of rolling a 1, given that the die is not weighted (which would introduce what we call a bias, though that is beyond the scope of this chapter). We could roll the die and see if it is a 1 or not, but that won’t tell us about the probability; it will only tell us a single result. We could also roll the die hundreds or thousands of times, recording each outcome and seeing what the final list looks like, but this is time-consuming, and rolling a die that many times may lead down a dark path to gambling or, worse, playing Dungeons & Dragons. What we need is a simple equation that represents what we are looking for and what is possible.

To calculate the probability of an event, which here is defined as rolling a 1 on an unbiased die, we need to know

two things: how many outcomes satisfy the criteria of our event (stated differently, how many outcomes would count as what we are looking for) and the total number of outcomes possible. In our example, only a single outcome, rolling a 1, will satisfy our criteria, and there are a total of six possible outcomes (rolling a 1, rolling a 2, rolling a 3, rolling a 4, rolling a 5, and rolling a 6). Thus, the probability of rolling a 1 on an unbiased die is 1 in 6 or $1/6$. Put into an equation using generic terms, we get:

$$\text{Probability of an event} = \frac{\text{Number of outcomes that satisfy our criteria}}{\text{Total number of possible outcomes}}$$

We can also use $P()$ as shorthand for probability and A as shorthand for an event:

$$P(A) = \frac{\text{Number of outcomes that count as } A}{\text{Total number of possible outcomes}}$$

Using this equation, let's now calculate the probability of rolling an even number on this die:

$$P(\text{even number}) = \frac{2, 4, \text{ or } 6}{1, 2, 3, 4, 5, \text{ or } 6} = \frac{3}{6} = \frac{1}{2}$$

So, we have a 50% chance of rolling an even number with this die. The principles laid out here operate under a certain set of conditions and can be elaborated into ideas that are complex yet powerful and elegant. However, such extensions are not necessary for a basic understanding of statistics, so we will end our discussion on the math of probability here. Now, let's turn back to more familiar topics.

Probability in Graphs and Distributions

We will see shortly that the normal distribution is the key to how probability works for our purposes. To understand exactly how, let's first look at a simple, intuitive example using pie charts.

Probability in Pie Charts

Recall that a pie chart represents how frequently a category was observed and that all slices of the pie chart add up to 100%, or 1. This means that if we randomly select an observation from the data used to create the pie chart, the probability of it taking on a specific value is exactly equal to the size of that category's slice in the pie chart.

Take, for example, the pie chart in Figure 29.1 representing the favorite sports of 100 people. If you put this pie chart on a dart board and aimed blindly (assuming you are guaranteed to hit the board), the likelihood of hitting the slice for any given sport would be equal to the size of that slice. So, the probability of hitting the baseball slice is the highest at 36%. The probability is equal to the proportion of the chart taken up by that section.

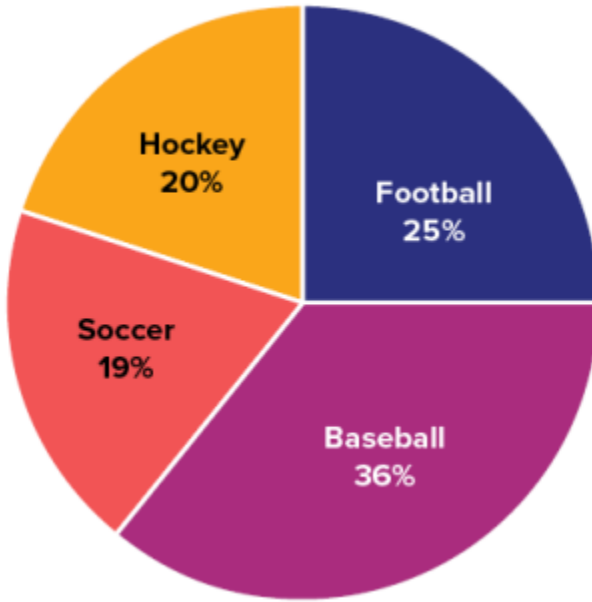


Figure 29.1. Favorite Sports. [\[Image Description\]](#)

We can also add slices together. For instance, maybe we want to know the probability of finding someone whose favorite sport is usually played on grass. The outcomes that satisfy this criterion are baseball, football, and soccer. To get the probability, we simply add their slices together to see what proportion of the area of the pie chart is in that region: $36\% + 25\% + 20\% = 81\%$. We can also add sections together even if they do not touch. If we want to know the likelihood that someone's favorite sport is not called football somewhere in the world (i.e., baseball and hockey), we can add those slices even though they aren't adjacent or contiguous in the chart itself: $36\% + 20\% = 56\%$. We are able to do all of this because

(1) the size of the slice corresponds to the area of the chart taken up by that slice, (2) the percentage for a specific category can be represented as a decimal (this step was skipped for ease of explanation above), and (3) the total area of the chart is equal to 100% or 1.0, which makes the size of the slices interpretable.

Probability in Normal Distributions

If the language at the end of the last section sounded familiar, that's because it's exactly the language used in the previous chapter to describe the normal distribution. Recall that the normal distribution has an area under its curve that is equal to 1 and that it can be split into sections by drawing a line through it that corresponds to a given z -score. Because of this, we can interpret areas under the normal curve as probabilities that correspond to z -scores.

First, let's look at the area between $z = -1.00$ and $z = 1.00$ presented in Figure 29.2. We were told earlier that this region contains 68% of the area under the curve. Thus, if we randomly chose a z -score from all possible z -scores, there is a 68% chance that it will be between $z = -1.00$ and $z = 1.00$ because those are the z -scores that satisfy our criteria.

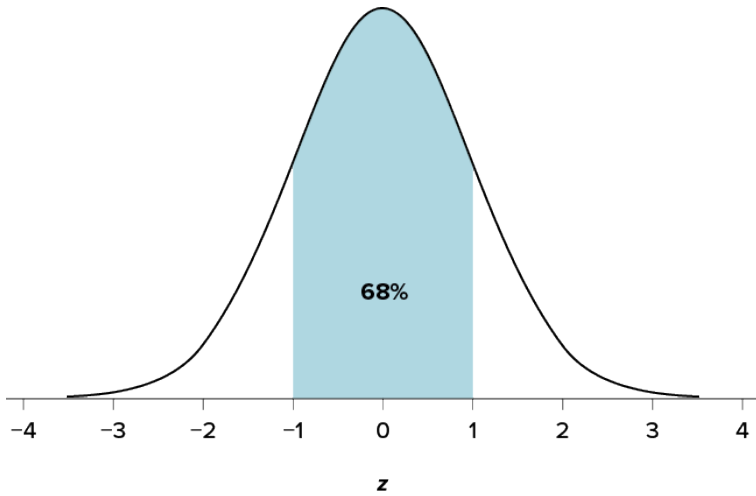


Figure 29.2. There is a 68% chance of selecting a z score from the blue-shaded region. [\[Image Description\]](#)

Just like a pie chart is broken up into slices by drawing lines through it, we can also draw a line through the normal distribution to split it into sections. Take a look at the normal distribution in Figure 29.3, which has a line drawn through it at $z = 1.25$. This line creates two sections of the distribution: the smaller section called the tail and the larger section called the body. Differentiating between the body and the tail does not depend on which side of the distribution the line is drawn. All that matters is the relative size of the pieces: bigger is always body.

As you can see, we can break up the normal distribution into 3 pieces (lower tail, body, and upper tail) as in Figure 29.2 or into 2 pieces (body and tail) as in Figure 29.3. We

can then find the proportion of the area in the body and tail based on where the line was drawn (i.e., at what z-score). Mathematically, this is done using calculus. Fortunately, the exact values are given to you in the Standard Normal Distribution Table, also known as the z-table. A portion of this table is shown in Figure 29.1. (The entire table appears in [Appendix A](#).) Using the z-values in the table (A), we can find the area under the normal curve in any body (B), tail (C), or combination of tails, as well as the proportion between z and the mean (D).

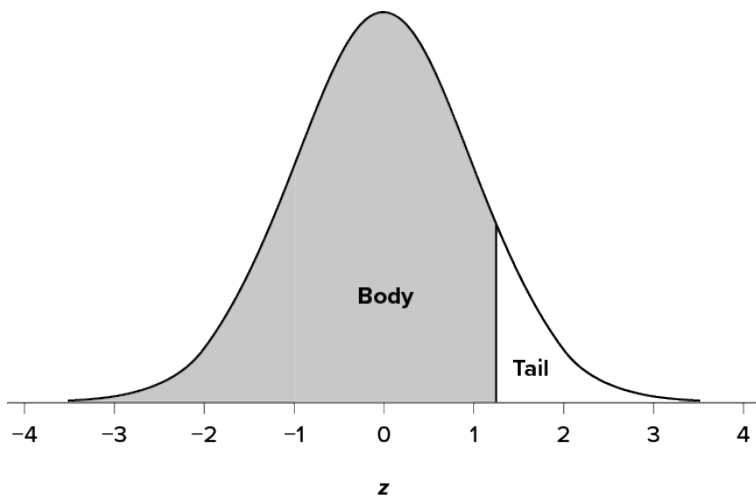


Figure 29.3. Body and Tail of the Normal Distribution. [\[Image Description\]](#)

For example, suppose we want to find the area in the body for

a z-score of 1.62. As shown in Figure 29.4, the row for 1.62 corresponds with a value of .9474 for the proportion in the body of the distribution. Thus, the odds of randomly selecting someone with a z-score less than (to the left of) $z = 1.62$ is 94.74% because that is the proportion of the area taken up by values that satisfy our criteria.

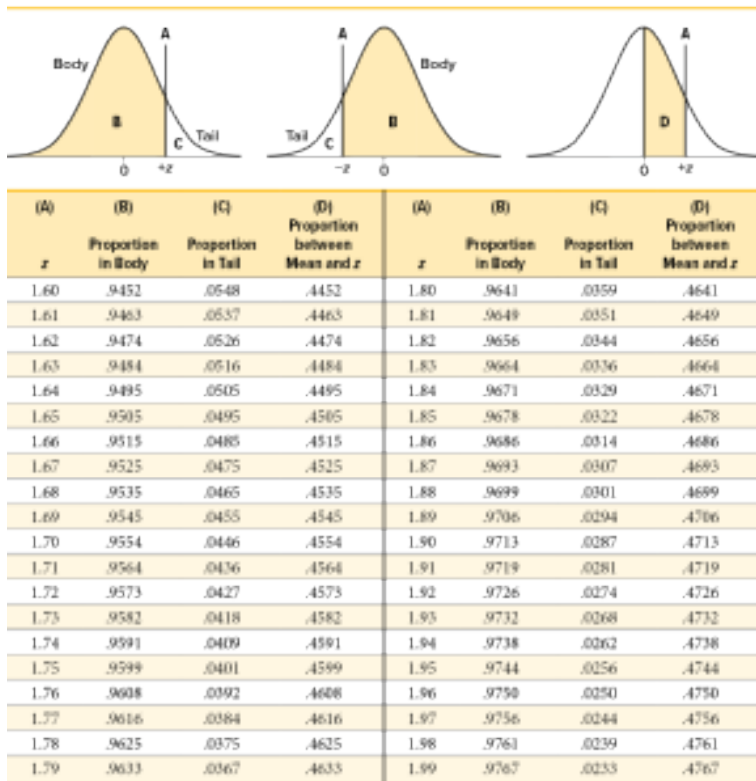


Figure 29.4. Standard Normal Distribution Table (z Table)
[\[Image Description and accessible version in Appendix A – opens in New Tab\]](#)

The z-table only presents the area in the body for positive z-scores because the normal distribution is symmetrical. Thus, the area in the body of $z = 1.62$ is equal to the area in the body for $z = -1.62$, though now—as illustrated in the middle distribution at the top of Figure 29.4—the body will be the shaded area to the right of z . (When in doubt, drawing out your distribution and shading the area you need to find will always help.) Because the total area under the normal curve is always equal to 1.00, the area in the tail (Column C) is simply the area in the body (Column B) subtracted from 1.00 ($1.00 - .9474 = .0526$).

Let's look at another example. This time, let's find the area corresponding to z-scores more extreme than $z = -1.96$ and $z = 1.96$. That is, let's find the area in the tails of the distribution for values less than $z = -1.96$ (farther negative and therefore more extreme) and greater than $z = 1.96$ (farther positive and therefore more extreme). This region is illustrated in Figure 29.5.

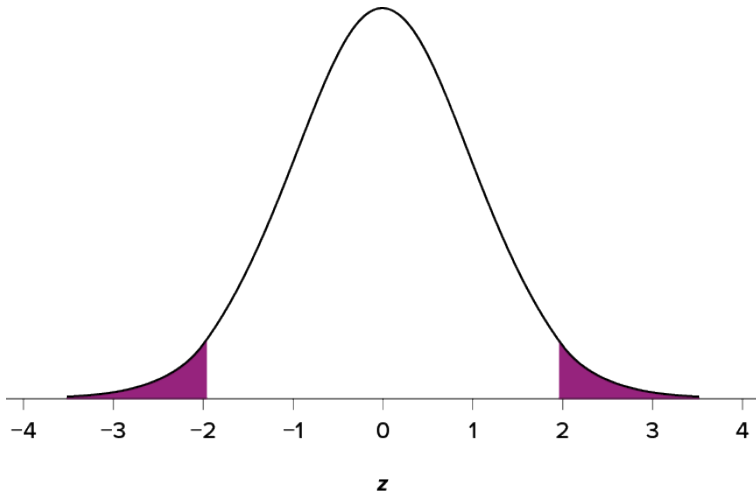


Figure 29.5. Area in the Tails Beyond $z = -1.96$ and $z = 1.96$.
[\[Image Description\]](#)

Let's start with the tail for $z = 1.96$. If we go to the z -table (Figure 29.4) we will find that the area in the tail to the right of $z = 1.96$ is equal to .0250. Because the normal distribution is symmetrical, the area in the tail for $z = -1.96$ is the exact same value, .0250. Finally, to get the total area in the shaded region, we simply add the areas together to get .0500. Thus, there is a 5% chance of randomly getting a value more extreme than $z = -1.96$ or $z = 1.96$.

Finally, we can find the area between two z -scores by shading and subtracting. Figure 29.6 shows the area between $z = 0.50$ and $z = 1.50$. Because this is a subsection of a body (rather than just a body or a tail), we must first find the larger of the two bodies, in this case, the body for $z = 1.50$, and

subtract the smaller of the two bodies, or the body for $z = 0.50$. Aligning the distributions vertically, as in Figure 29.6, makes this clearer. From the complete z-table in [Appendix A](#), we see that the area in the body for $z = 1.50$ is .9332, and the area in the body for $z = 0.50$ is .6915. Subtracting these gives us $.9332 - .6915 = .2417$.

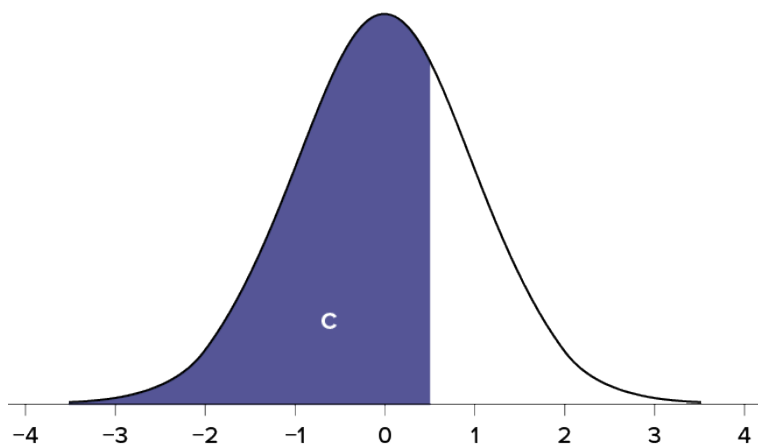
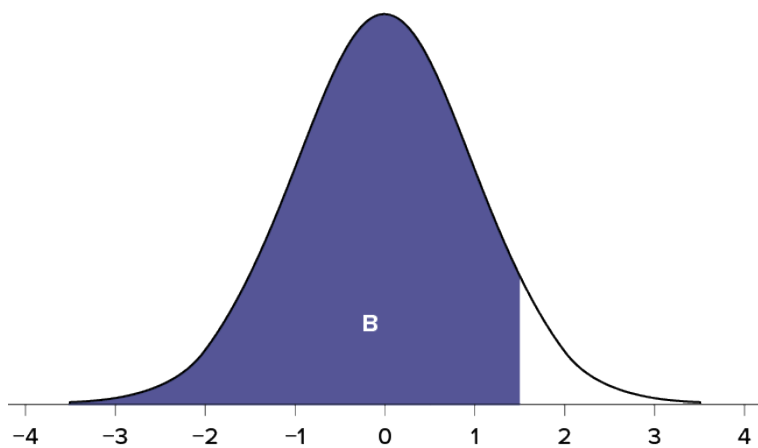
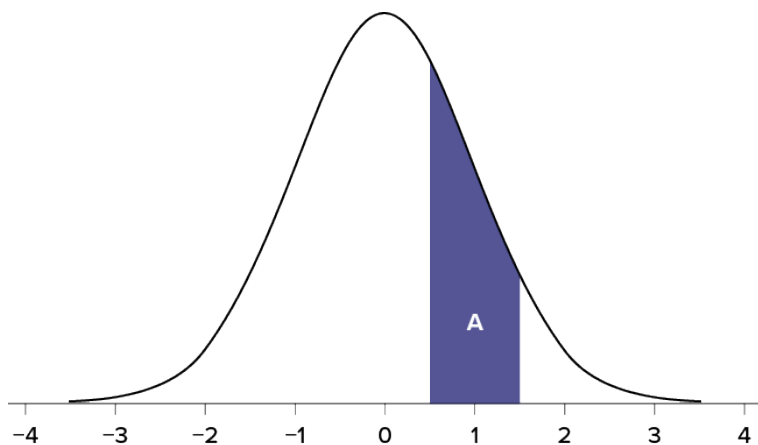


Figure 29.6. Area Between $z = 0.50$ and 1.50 (A), Along With the Corresponding Areas in the Body (B and C). [\[Image Description\]](#)

Probability: The Bigger Picture

The concepts and ideas presented in this chapter are likely not intuitive at first. Probability is a tough topic for everyone, but the tools it gives us are incredibly powerful and enable us to do amazing things with data analysis. They are the heart of how inferential statistics work.

To summarize, the probability that an event happens is the number of outcomes that qualify as that event (i.e., the number of ways the event could happen) compared to the total number of outcomes (i.e., how many things are possible). This extends to graphs like a pie chart, where the biggest slices take up more of the area and are therefore more likely to be chosen at random. This idea then brings us back around to our normal distribution, which can also be broken up into regions or areas, each of which is bounded by one or two z-scores and corresponds to all z-scores in that region. The probability of randomly getting one of those z-scores in the specified region can then be found on the Standard Normal Distribution Table. Thus, the larger the region, the more likely an event is, and vice versa. Because the tails of the distribution are, by definition, smaller, and we go farther out into the tail, the

likelihood or probability of finding a result out in the extremes becomes small.

Equity Activity: Student grading equity

A university in the United States is evaluating whether its grading system is equitable for students from two different backgrounds: (1) “legacy” college students, who have direct relatives that graduated from the university, and (2) first-generation college students, who are the first in their family to attend a university. A group of students from each of these groups took the same standardized exam, and their scores followed normal distributions:

- **Group X (“legacy” college students):** Mean = 78, Standard Deviation = 8
- **Group Y (first-generation college students):** Mean = 72, Standard Deviation = 10

—



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=130#h5p-35>

Practice Problems

Short Answer Reflections



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=130#h5p-36>

Test Your Knowledge



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=130#h5p-37>

License & Attribution

“Probability & the Normal Distribution” by Makenzie O’Neil is adapted from “[Probability](#)” by Linda R. Cote Ph.D.; Rupa G. Gordon Ph.D., Chrislyn E. Randell Ph.D., Judy Schmitt, and Helena Marvin, which is licensed CC BY-NC-SA 4.0.

“Probability & the Normal Distribution” is licensed under CC BY-NC-SA 4.0.

Image Descriptions

Figure 29.1. A four-slice pie chart shows survey preferences for sports. The largest slice (magenta) is Baseball – 36%. A navy slice is Football – 25%. An orange slice is Hockey – 20%. A red slice is Soccer – 19%. Each segment is labeled with the sport and percentage inside the slice. [[Return to Figure 29.1](#)]

Figure 29.2. A standard normal distribution is shown with

z-scores on the x-axis from -4 to 4 . The region within one standard deviation of the mean (from $z = -1$ to $z = +1$) is filled in light blue and marked “68%,” indicating that about 68% of values fall in this interval. [[Return to Figure 29.2](#)]

Figure 29.3. A standard normal distribution is plotted with z on the x-axis (-4 to 4). The region from the far left up to a vertical line a little past $z = 1$ is filled gray and labeled Body. The narrow portion to the right of that line is unlabeled except for Tail, illustrating the distinction between the bulk of the distribution and the extreme right-tail area beyond a cutoff (critical z). [[Return to Figure 29.3](#)]

Figure 29.4. A row of three beige, bell-curve diagrams on a black background illustrates how areas are labeled on a z -distribution. Left panel: the area from the mean ($z = 0$) to a cutoff left of center is shaded and labeled B—the “body” between the mean and a negative z . Middle panel: the area from the mean to a cutoff right of center is shaded and labeled B—the “body” between the mean and a positive z . Right panel: only the extreme right tail beyond a positive z is shaded and labeled D—the tail probability to the right of z . [[Return to Figure 29.4](#)]

Figure 29.5. A bell-shaped standard normal distribution is plotted with the horizontal axis labeled z from -4 to $+4$. Purple shading marks the two extreme regions: left of $z \approx -2$ and right of $z \approx +2$. The central area between -2 and $+2$ is unshaded, indicating that only the tails (the most extreme

values in both directions) are highlighted. [[Return to Figure 29.5](#)]

Figure 29.6. A vertical stack of three standard normal distribution plots (x-axis labeled z , roughly -4 to $+4$). In panel A (top), only the band between $z = 0.5$ and $z = 1.5$ on the right side is shaded. In panel B (middle), the entire region from the far left tail up to $z = 1.5$ is shaded. In panel C (bottom), the region from the far left tail up to $z = 0.5$ is shaded. Each panel highlights a different probability region under the same bell curve. [[Return to Figure 29.6](#)]

Media Attributions

- Favorite Sports Pie Chart © Judy Schmitt is licensed under a [CC BY-NC-SA \(Attribution NonCommercial ShareAlike\)](#) license
- z curves © Judy Schmitt adapted by Theresa Huff is licensed under a [CC BY-NC-SA \(Attribution NonCommercial ShareAlike\)](#) license

30.

SAMPLING DISTRIBUTIONS

Mireille Ukeye

Learning Objectives

By the end of this chapter, you will be able to

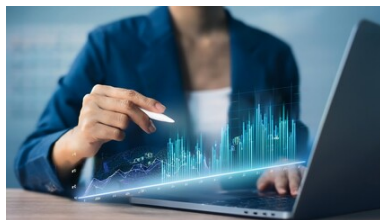
1. Calculate z-scores, understand the basic features of a standardized normal distribution, and apply standardization across different distributions to compare scores and draw informed conclusions.
2. Explore strategies to promote inclusivity and cultural responsiveness in both educational and research practices.

People, Samples, and Populations

Much of statistical analysis involves working with individual scores that are organized into samples, which are ideally representative of a larger population. Understanding where a particular score falls within a distribution can be achieved using z -scores, which indicate how far a score is from the mean in standard deviation units. This concept also connects to probability, helping us determine the likelihood of observing scores above or below a certain value within the distribution.

Inherent in this work is the notion that an individual score will differ from the mean, which we quantify as a z -score. All of the individual scores will differ from the mean in different amounts and different directions, which is natural and expected. We quantify these differences as variance and standard deviation. Measures of spread and the idea of variability in observations are key principles in inferential statistics. We know that any observation, whether it is a single score, a set of scores, or a particular descriptive statistic, will differ from the center of whatever distribution it belongs in.

This is equally true of things outside of statistics and formatting data collection and analysis. Some days you hear your alarm and wake up easily,



but other days you need to hit snooze a few (dozen) times. Some days, traffic is light, but other days it is very heavy. Some classes you are able to focus, pay attention, and take good notes, but other days, you find yourself zoning out the entire time. Each individual observation is an insight but is not, by itself, the entire story, and it takes an extreme deviation from what we expect for us to think that something strange is going on. Being a little sleepy is normal, but being completely unable to get out of bed might indicate that we are sick. Light traffic is a good thing, but almost no cars on the road might make us think we forgot it is Saturday. Zoning out occasionally is fine, but if we cannot focus at all, we might be in a stats class rather than a fun one.

All of these principles carry forward from scores within samples to samples within populations. Just like an individual score will differ from its mean, an individual sample mean will differ from the true population mean. Sampling error is an incredibly important principle. We know ahead of time that if we collect data and compute a sample, the observed value of that sample will be at least slightly off from what we expect it to be based on our supposed population mean; this is natural and expected. However, if our sample mean is extremely different from what we expect based on the population mean, there may be something going on.

The Sampling Distribution of Sample Means

To see how we use sampling error, we will learn about a new, theoretical distribution known as the sampling distribution. In the same way that we can gather a lot of individual scores and put them together to form a distribution with a center and spread, if we were to take many samples, all of the same size, and calculate the mean of each of those, we could put those means together to form a distribution. This new distribution is, intuitively, known as the distribution of sample means. It is one example of what we call a sampling distribution, which can be formed from a set of any statistic, such as a mean, a test statistic, or a correlation coefficient. For our purposes, understanding the distribution of sample means will be enough to see how all other sampling distributions work to enable and inform our inferential analyses, so these two terms will be used interchangeably from here on out. Let's take a deeper look at some of its characteristics.

The sampling distribution of sample means can be described by its shape, center, and spread, just like any of the other distributions we have worked with. The shape of our sampling distribution is normal: a bell-shaped curve with a single peak and two tails extending symmetrically in either direction. The center of the sampling distribution of sample means—which is, itself, the mean or average of the means—is the true population mean, μ . This will sometimes be written

as μ_M to denote it as the mean of the sample means. The spread of the sampling distribution is called the standard error, the quantification of sampling error, denoted σ_M . The formula for standard error is:

$$\sigma_M = \frac{\sigma}{\sqrt{n}}$$

Notice that the sample size is in this equation. The sampling distribution refers to samples of a specific size. That is, all sample means must be calculated from samples of the same size n , such as $n = 10$, $n = 30$, or $n = 100$. This sample size refers to how many people or observations are in each individual sample, not how many samples are used to form the sampling distribution. This is because the sampling distribution is a theoretical distribution, not one we will ever actually calculate or observe. Figure 30.1 displays the principles stated here in graphical form.

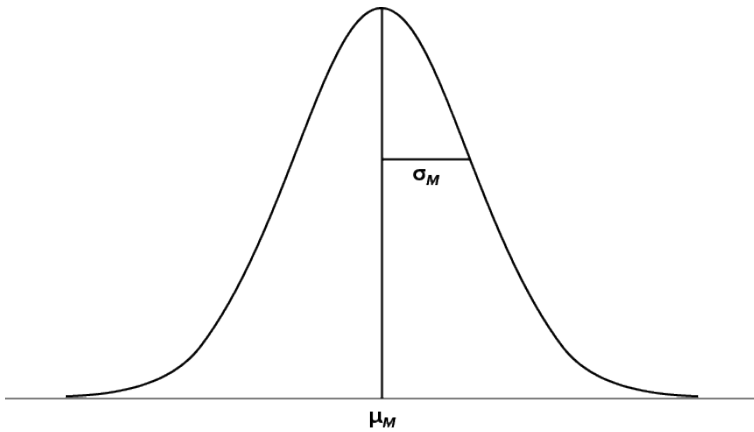


Figure 30.1. The Sampling Distribution of Sample Means.
[\[Image Description\]](#)

Two Important Axioms

We just learned that the sampling distribution is theoretical: we never actually see it. If that is true, then how can we know it works? How can we use something we don't see? The answer lies in two very important mathematical facts: the central limit theorem and the law of large numbers. We will not go into the math behind how these statements were derived, but knowing what they are and what they mean is important to understanding why inferential statistics work and how we can draw conclusions about a population based on information gained from a single sample.

Central Limit Theorem

The central limit theorem states:

For samples of a single size n , drawn from a population with a given mean μ and variance s^2 , the sampling distribution of sample means will have a mean $\mu_M = \mu$ and variance $\sigma_X^2 = \frac{\sigma^2}{n}$. This distribution will approach normality as n increases.

From this, we are able to find the standard deviation of our sampling distribution, the standard error. As you can see, just like any other standard deviation, the standard error is simply the square root of the variance of the distribution.

The last sentence of the central limit theorem states that the sampling distribution will be more normal as the sample size of the samples used to create it increases. What this means is that bigger samples will create a more normal distribution, so we are better able to use the techniques we developed for normal distributions and probabilities. So, how large is large enough? In general, a sampling distribution will be normal if either of two characteristics is true: (1) the population from which the samples are drawn is normally distributed or (2) the sample size is equal to or greater than 30. This second criterion is very important because it enables us to use methods developed for normal distributions even if the true population distribution is skewed.

Law of Large Numbers

The law of large numbers simply states that as our sample size increases, the probability that our sample mean is an accurate representation of the true population mean also increases. It is the formal mathematical way to state that larger samples are more accurate.

The law of large numbers is related to the central limit theorem, specifically the formulas for variance and standard error. Notice that the sample size appears in the denominators of those formulas. A larger denominator in any fraction means that the overall value of the fraction gets smaller (i.e., $1/2 = 0.50$, $1/3 = 0.33$, $1/4 = 0.25$, and so on). Thus, larger sample sizes will create smaller standard errors. We already know that standard error is the spread of the sampling distribution and that a smaller spread creates a narrower distribution. Therefore, larger sample sizes create narrower sampling distributions, which increases the probability that a sample mean will be close to the center and decreases the probability that it will be in the tails. This is illustrated in Figure 30.2 and Figure 30.3.

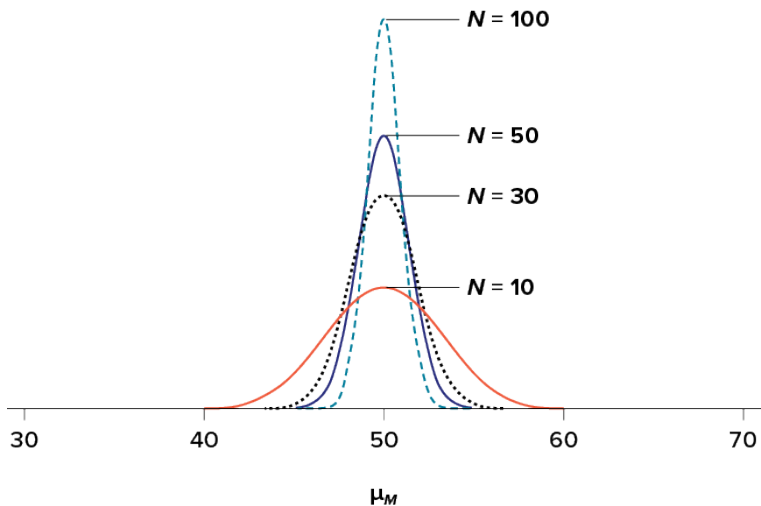


Figure 30.2. Sampling Distributions from the Same Population with $m = 50$ and $s = 10$ but Different Sample Sizes ($N = 10$, $N = 30$, $N = 50$, $N = 100$). [\[Image Description\]](#)

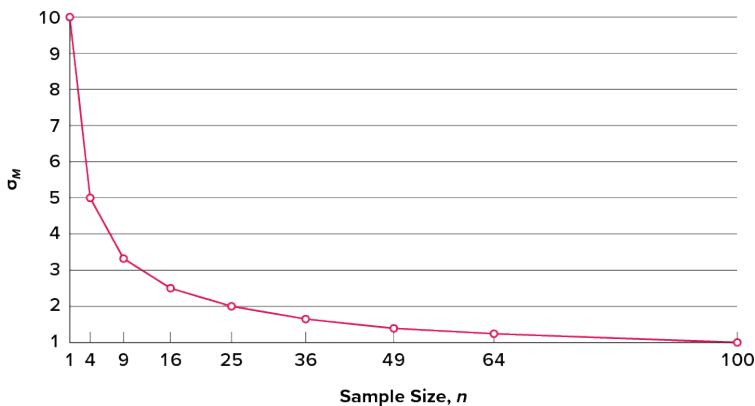


Figure 30.3. Relationship Between Sample Size and Standard Error for a Constant $s = 10$. [\[Image Description\]](#)

Using Standard Error for Probability

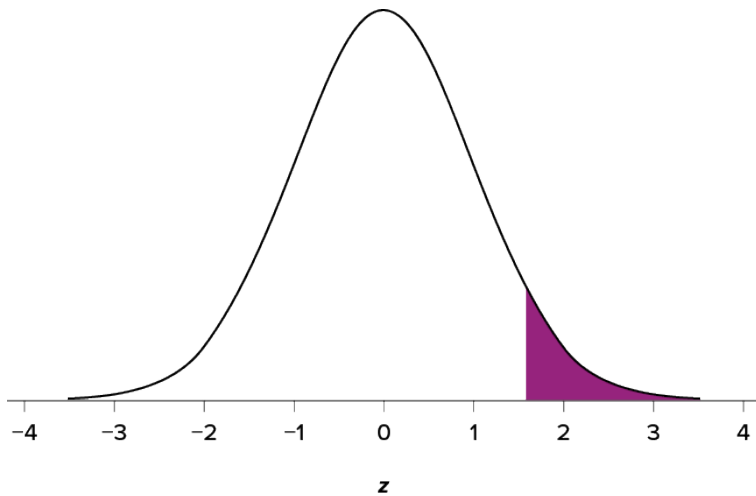
In this chapter, we saw that we can use z-scores to split up a normal distribution and calculate the proportion of the area under the curve in one of the new regions, giving us the probability of randomly selecting a z-score in that range. We can follow the exact sample process for sample means, converting them into z-scores and calculating probabilities. The only difference is that instead of dividing a raw score by the standard deviation, we divide the sample mean by the standard error.

$$z = \frac{M - \mu}{\sigma_M} = \frac{M - \mu}{\sigma/\sqrt{n}}$$

Let's say we are drawing samples from a population with a mean of 50 and a standard deviation of 10 (the same values used in Figure 30.2). What is the probability that we get a random sample of size 10 with a mean greater than or equal to 55? That is, for $n = 10$, what is the probability that $M \geq 55$? First, we need to convert this sample mean score into a z-score:

$$z = \frac{55 - 50}{10/\sqrt{10}} = \frac{5}{3.16} = 1.58$$

Now we need to shade the area under the normal curve corresponding to scores greater than $z = 1.58$, as in Figure 30.4.



1.58 (image description available)" width="647"
 height="422"> Figure 30.4. Area Under the Curve Greater
 Than $z = 1.58$. [\[Image Description\]](#)

Now we go to our z -table and find that the area to the left of $z = 1.58$ is .9429. Finally, because we need the area to the right (per our shaded diagram), we simply subtract this from 1 to get $1.00 - .9429 = .0571$. So, the probability of randomly drawing a sample of 10 people from a population with a mean of 50 and a standard deviation of 10 whose sample mean is 55 or more is $p = .0571$, or 5.71%. Notice that we are talking about means that are 55 or more. That is because, strictly speaking, it's impossible to calculate the probability of a score taking on exactly 1 value since the "shaded region" would just be a line with no area to calculate.

Now let's do the same thing but assume that instead of only

having a sample of 10 people, we took a sample of 50 people.

First, we find z :

$$z = \frac{55 - 50}{\frac{10}{\sqrt{50}}} = \frac{5}{1.41} = 3.55$$

Then we shade the appropriate region of the normal distribution, as shown in Figure 30.5.

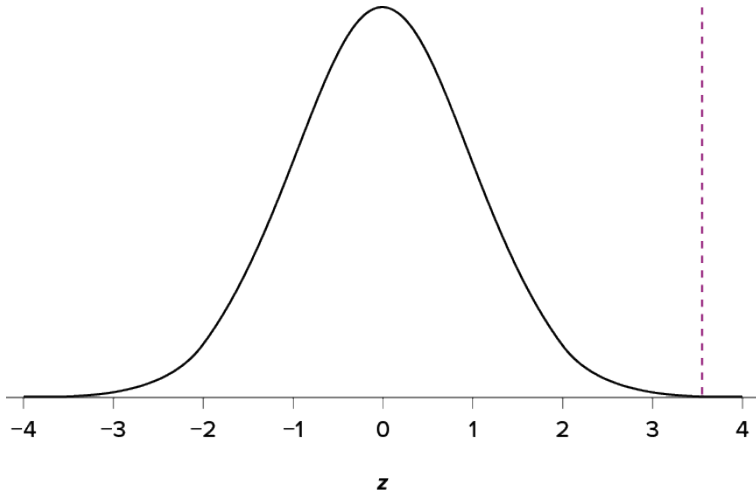


Figure 30.5. Area Under the Curve Greater Than $z = 3.55$.
[\[Image Description\]](#)

Notice that no region of Figure 30.5 appears to be shaded. That is because the area under the curve that far out into the tail is so small that it can't even be seen (the red line has been added to show exactly where the region starts). Thus, we already know that the probability must be smaller for $N = 50$

than $N = 10$ because the size of the area (the proportion) is much smaller.

We run into a similar issue when we try to find $z = 3.55$ on our Standard Normal Distribution Table. The table only goes up to 3.09 because everything beyond that is almost 0 and changes so little that it's not worth printing values. The closest we can get is subtracting the largest value, .9990, from 1 to get .001. We know that, technically, the actual probability is smaller than this (since 3.55 is farther into the tail than 3.09), so we say that the probability is $p < .001$, or less than 0.1%.

This example shows what an impact sample size can have. From the same population, looking for exactly the same thing, changing only the sample size took us from roughly a 5% chance (or about 1/20 odds) to a less than 0.1% chance (or less than 1 in 1000). As the sample size n increased, the standard error decreased, which in turn caused the value of z to increase, which finally caused the p value to decrease. You can think of this relationship like gears: turning the first gear (sample size) clockwise causes the next gear (standard error) to turn counterclockwise, which causes the third gear (z) to turn clockwise, which finally causes the last gear (probability) to turn counterclockwise. All of these pieces fit together, and the relationships will always be the same:

$$n \uparrow \quad \sigma_M \downarrow \quad z \uparrow \quad p \downarrow$$

Let's look at this one more way. For the same population of sample size 50 and standard deviation 10, what proportion of

sample means fall between 47 and 53 if they are of sample size 10 and sample size 50?

We'll start again with $n = 10$. Converting 47 and 53 into z scores, we get $z = -0.95$ and $z = 0.95$, respectively. From our z -table, we find that the proportion between these two scores is .6578 (the process here is left off for the student to practice converting M to z and z to proportions). So, 65.78% of the sample means of a sample size of 10 will fall between 47 and 53. For $n = 50$, our z -scores for 47 and 53 are ± 2.13 , which gives us a proportion of the area as .9668, almost 97%! Shaded regions for each of these sampling distributions are displayed in Figure 30.6. The sampling distributions are shown on the original scale, rather than as z -scores, so you can see the effect of the shading and how much of the body falls into the range, which is marked off with thin dotted lines.

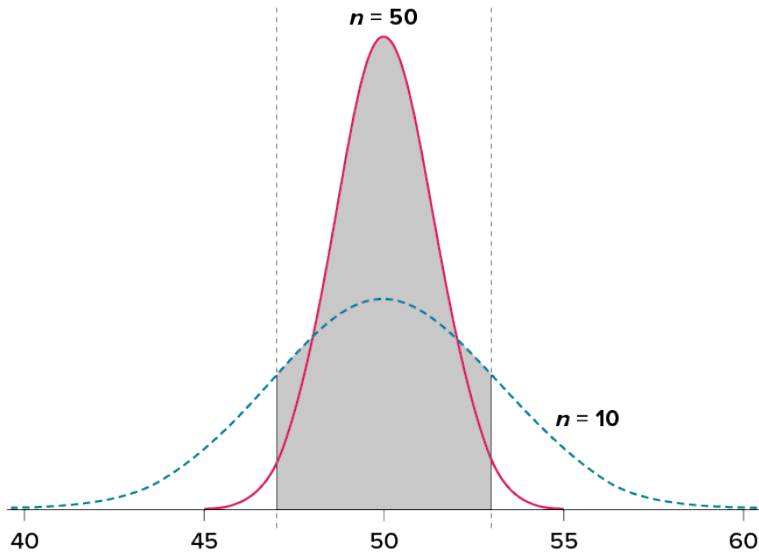


Figure 30.6. Areas Between 47 and 53 for Sampling Distributions of $n = 10$ and $n = 50$. [\[Image Description\]](#)

Sampling Distribution, Probability, and Inference

We've seen how we can use the standard error to determine probability based on our normal curve. We can think of the standard error as how much we would naturally expect our statistic—be it a mean or some other statistic—to vary. In our formula for z based on a sample mean, the numerator ($M - \mu$) is what we call an observed effect. That is, it is what we observe in our sample mean versus what we expected based on the population from which that sample mean was calculated.

Because the sample mean will naturally move around due to sampling error, our observed effect will also change naturally. In the context of our formula for z , then, our standard error is how much we would naturally expect the observed effect to change. Changing by a little is completely normal, but changing by a lot might indicate something is going on. This is the basis of inferential statistics and the logic behind hypothesis testing.

Equity Activity: Equity in AP courses

In a school district, data is being collected to examine the representation of students from historically marginalized communities in advanced placement (AP) courses. The district randomly selects 10 students from each high school in the district to calculate the percentage of students from underrepresented racial, ethnic, or socioeconomic backgrounds enrolled in AP courses. This process is repeated across several samples, creating a sampling distribution of the mean percentage of marginalized students in AP courses.

The sampling distribution reveals the variability in the percentage of underrepresented students across

different schools in the district. Some schools show higher percentages of marginalized students in AP courses, while others have lower percentages. By analyzing this data, the district can identify which schools may be providing better access to advanced academic opportunities and which schools may have systemic barriers preventing equitable access. This analysis helps to guide targeted inclusive initiatives aimed at increasing the participation of historically marginalized students in advanced coursework, ensuring that all students, regardless of their background, have equal opportunities for academic success.



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=138#h5p-38>

Practice Problems

Short Answer Reflections



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=138#h5p-39>

Test Your Knowledge



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=138#h5p-40>

License & Attribution

“Sampling Distributions” by Mireille Ukeye is adapted from “[Sampling Distributions](#)” by Linda R. Cote Ph.D.; Rupa G. Gordon Ph.D.; Chrislyn E. Randell Ph.D.; Judy Schmitt; and Helena Marvin is licensed CC BY 4.0.

“Sampling Distributions” is licensed under CC BY-NC-SA 4.0.

Image Descriptions

Figure 30.1. A normal distribution is drawn with a vertical line at its peak marking the mean of the sampling distribution, labeled μ_m on the x-axis. From that vertical line, a short horizontal line extends rightward into the curve and is labeled σ_m , indicating the standard deviation of the sampling distribution (the standard error). No numerical scale is shown. [[Return to Figure 30.1](#)]

Figure 30.2. A horizontal axis labeled μ_m runs from about 40 to 60. Over it are four normal curves that all peak at ~ 50 .

The widest, lowest red curve is labeled $N = 10$. A taller, narrower black dotted curve is $N = 30$. An even narrower solid blue curve is $N = 50$. The narrowest, tallest dashed teal curve is $N = 100$. The figure illustrates that larger sample sizes produce sampling distributions with smaller spread (lower standard error) around the same mean. [[Return to Figure 30.2](#)]

Figure 30.3. A Cartesian plot has x-axis “Sample Size, n ” with tick labels 1, 4, 9, 16, 25, 36, 49, 64, 100 and y-axis “ σ_m ” from 1 to 10. A pink line with open circles starts at $\sigma_m \approx 10$ when $n=1$, drops sharply to ≈ 5 at $n=4$, ≈ 3.2 at $n=9$, ≈ 2.5 at $n=16$, ≈ 2.0 at $n=25$, ≈ 1.6 at $n=36$, ≈ 1.3 at $n=49$, ≈ 1.2 at $n=64$, and approaches ≈ 1.0 at $n=100$. The curve illustrates the inverse relationship $\sigma_m = \sigma/\sqrt{n}$: increasing sample size reduces standard error, with diminishing returns at larger n . [[Return to Figure 30.3](#)]

Figure 30.4. A standard normal distribution is plotted with horizontal axis labeled z from -4 to 4 . A vertical cutoff near $z = 1.58$ marks the start of the shaded region. The area to the right of this cutoff (the right tail) is filled in purple, representing the probability of obtaining a z -score greater than 1.58 , which is approximately 0.057 . [[Return to Figure 30.4](#)]

Figure 30.5. A standard normal distribution is plotted with the horizontal axis labeled z from -4 to 4 . A vertical magenta dashed line at $z = 3.55$ marks the threshold. The region to the right of this line (not shaded) represents the extreme right tail of the distribution, whose area is approximately 0.0002 .

(0.02%), indicating how rare z-scores above 3.55 are. [[Return to Figure 30.5](#)]

Figure 30.6. A horizontal axis from 40 to 60 shows sample-mean values. Two normal curves are drawn with the same center at 50: a narrow magenta solid curve labeled “ $n = 50$ ” and a wider teal dashed curve labeled “ $n = 10$.” Vertical dashed lines at approximately 47 and 53 bracket an interval around the mean. The region between these lines is shaded only under the $n = 50$ curve, illustrating that with the larger sample size (smaller standard error) more of the distribution’s area lies between 47 and 53 compared with the smaller sample size. [[Return to Figure 30.6](#)]

31.

INTRODUCTION TO HYPOTHESIS TESTING USING Z-SCORES

Makenzie O'Neil

Learning Objectives

By the end of this chapter, you will be able to

1. Articulate the logic of hypothesis testing.
2. Describe the four steps of hypothesis testing.
3. Conduct a hypothesis test using Z-scores.
4. Compare the difference between Type I and Type II Errors.

This chapter lays out the basic logic and process of hypothesis testing. We will perform z tests using data from a sample mean to make an inference about a population.

Logic and Purpose of Hypothesis Testing

A hypothesis is a prediction that is tested in a research study. The statistician R. A. Fisher explained the concept of hypothesis testing with a story of a lady tasting tea. Here we will present an example based on James Bond, who insisted that martinis should be shaken rather than stirred. Let's consider a hypothetical experiment to determine whether Mr. Bond can tell the difference between a shaken martini and a stirred martini. Suppose we gave Mr. Bond a series of 16 taste tests. In each test, we flipped a fair coin to determine whether to stir or shake the martini. Then we presented the martini to Mr. Bond and asked him to decide whether it was shaken or stirred. Let's say Mr. Bond was correct on 13 of the 16 taste tests. Does this prove that Mr. Bond has at least some ability to tell whether the martini was shaken or stirred?

This result does not prove that he does; it could be that he was just lucky and guessed right 13 out of 16 times. But how plausible is the explanation that he was just lucky? To assess its plausibility, we determine the probability that someone who was just guessing would be correct 13/16 times or more. This probability can be computed to be .0106. This is a pretty low

probability, and therefore, someone would have to be very lucky to be correct 13 or more times out of 16 if they were just guessing. So either Mr. Bond was very lucky, or he can tell whether the drink was shaken or stirred. The hypothesis that he was guessing is not proven false, but considerable doubt is cast on it. Therefore, there is strong evidence that Mr. Bond can tell whether a drink was shaken or stirred.

Example: Physicians' Reactions

Let's consider another example. The case study Physicians' Reactions sought to determine whether physicians spend less time with obese patients. Physicians were sampled randomly, and each was shown a chart of a patient complaining of a migraine headache. They were then asked to estimate how long they would spend with the patient. The charts were identical except that for half the charts, the patient was obese, and for the other half, the patient was of average weight. The chart a particular physician viewed was determined

randomly. 33 physicians viewed charts of average-weight patients, and 38 physicians viewed charts of obese patients.

The mean time physicians reported that they would spend with obese patients was 24.7 minutes, as compared to a mean of 31.4 minutes for normal-weight patients. How might this difference between means have occurred? One possibility is that physicians were influenced by the weight of the patients. On the other hand, perhaps by chance, the physicians who viewed charts of the obese patients tend to see patients for less time than the other physicians. Random assignment of charts does not ensure that the groups will be equal in all respects other than the chart they viewed. In fact, it is certain the groups differed in many ways by chance. The two groups could not have exactly the same mean age (if measured precisely enough, such as in days). Perhaps a physician's age affects how long the physician sees patients. There are innumerable differences between the groups that could affect how long they view patients. With this in mind, is it plausible that these chance differences are responsible for the difference in times?

To assess the plausibility of the hypothesis that the difference in mean times is due to chance, we compute the probability of getting a difference as large or larger than the observed difference ($31.4 - 24.7 = 6.7$ minutes) if the difference were, in fact, due solely to chance. Using methods presented in later chapters, this probability can be computed to be .0057. Since this is such a low probability, we have confidence that the difference in times is due to the patient's weight and is not due to chance.

The Probability Value

It is very important to understand precisely what the probability values mean. In the James Bond example, the computed probability of .0106 is the probability he would be correct on 13 or more taste tests (out of 16) if he were just guessing. It is easy to mistake this probability of .0106 as the probability that he cannot tell the difference. This is not at all what it means. The probability of .0106 is the probability of a specific outcome (13 or more out of 16), assuming a particular circumstance is true (James Bond was only guessing). It is not the probability that the circumstance is true. In statistics, it is conventional to refer to these possible circumstances as hypotheses since they are hypothesized circumstances. Using

this terminology, the *probability value* is the probability of an outcome given the hypothesis. It is not the probability of the hypothesis given the outcome.

This is not to say that we ignore the probability of the hypothesis. If the probability of the outcome given the hypothesis is sufficiently low, we have evidence that the hypothesis is false. However, we do not compute the probability that the hypothesis is false. In the James Bond example, the hypothesis is that he cannot tell the difference between shaken and stirred martinis. The probability value is low (.0106), thus providing evidence that he can tell the difference. However, we have not computed the probability that he can tell the difference.

The Null Hypothesis

The hypothesis that an apparent effect is due to chance is called the null hypothesis, written H_0 (“H-naught”). In the [Physicians’ Reactions](#) example, the null hypothesis is that in the population of physicians, the mean time expected to be spent with obese patients is equal to the mean time expected to be spent with average-weight patients. This null hypothesis can be written as:

$$H_0 : \mu_{\text{obese}} - \mu_{\text{average}} = 0$$

The null hypothesis in a correlational study of the relationship between high school grades and college grades

would typically be that the population correlation is 0. This can be written as

$$H_0 : \rho = 0$$

where ρ (Greek letter “rho”) is the population correlation, which we will cover in detail in later chapters.

Although the null hypothesis is usually that the value of a parameter is 0, there are occasions in which the null hypothesis is a value other than 0. For example, if we are working with mothers in the U.S. whose children are at risk of low birth weight, we can use 7.47 pounds, the average birth weight in the U.S., as our null value and test for differences against that.

For now, we will focus on testing a value of a single mean against what we expect from the population. Using birth weight as an example, our null hypothesis takes the form:

$$H_0 : \mu = 7.47$$

The number on the right-hand side is our null hypothesis value, which is informed by our research question. Notice that we are testing the value for μ , the population parameter, not the sample statistic **M**. This is for two reasons: (1) once we collect data, we know what the value of **M** is—it’s not a mystery or a question, it is observed and used for the second reason, which is (2) we are interested in understanding the population, not just our sample.

Keep in mind that the null hypothesis is typically the opposite of the researcher’s hypothesis. In the [Physicians’ Reactions](#) study, the researchers hypothesized that physicians would expect to spend less time with obese patients. The null

hypothesis that the two types of patients are treated identically is put forward with the hope that it can be discredited and therefore rejected. If the null hypothesis were true, a difference as large as or larger than the sample difference of 6.7 minutes would be very unlikely to occur. Therefore, the researchers rejected the null hypothesis of no difference and concluded that in the population, physicians intend to spend less time with obese patients.

Typically, the null hypothesis is the idea that nothing is going on: there is no effect of our treatment, no relationship between our variables, and no difference in our sample mean from what we expected about the population mean. This is always our baseline starting assumption, and it is what we seek to reject. If we are trying to treat depression, we want to find a difference in average symptoms between our treatment and control groups. If we are trying to predict job performance, we want to find a relationship between conscientiousness and evaluation scores. However, until we have evidence against it, we must use the null hypothesis as our starting point.

The Alternative Hypothesis

If the null hypothesis is rejected, then we will need some other explanation, which we call the alternative hypothesis, H_A or H_1 . The alternative hypothesis is simply the reverse of the null hypothesis, and there are three options, depending on where we expect the difference to lie. Thus, our alternative hypothesis

is the mathematical way of stating our research question. If we expect our obtained sample mean to be above or below the null hypothesis value, which we call a directional hypothesis, then our alternative hypothesis takes the form

$$H_A : \mu > 7.47 \text{ or } H_A : \mu < 7.47$$

based on the research question itself. We should only use a directional hypothesis if we have good reason, based on prior observations or research, to suspect a particular direction. When we do not know the direction, such as when we are entering a new area of research, we use a non-directional alternative:

$$H_A : \mu \neq 7.47$$

We will set different criteria for rejecting the null hypothesis based on the directionality (greater than, less than, or not equal to) of the alternative. To understand why, we need to see where our criteria come from and how they relate to z-scores and distributions.

Critical Values, p Values, and Significance Level

A low probability value casts doubt on the null hypothesis. How low must the probability value be in order to conclude that the null hypothesis is false? Although there is clearly no right or wrong answer to this question, it is conventional to conclude the null hypothesis is false if the probability value is less than .05. More conservative researchers conclude the null

hypothesis is false only if the probability value is less than .01. When a researcher concludes that the null hypothesis is false, the researcher is said to have rejected the null hypothesis. The probability value below which the null hypothesis is rejected is called the α level or simply α (“alpha”). It is also called the significance level. If α is not explicitly specified, assume that $\alpha = .05$.

The significance level is a threshold we set before collecting data in order to determine whether or not we should reject the null hypothesis. We set this value beforehand to avoid biasing ourselves by viewing our results and then determining what criteria we should use. If our data produce values that meet or exceed this threshold, then we have sufficient evidence to reject the null hypothesis; if not, we fail to reject the null (we never “accept” the null).

There are two criteria we use to assess whether our data meet the thresholds established by our chosen significance level, and they both have to do with our discussions of probability and distributions. Recall that probability refers to the likelihood of an event, given some situation or set of conditions. In hypothesis testing, that situation is the assumption that the null hypothesis value is the correct value, or that there is no effect. The value laid out in H_0 is our condition under which we interpret our results. To reject this assumption, and thereby reject the null hypothesis, we need results that would be very unlikely if the null hypothesis were true. Now, recall that values of z that fall in the tails of the standard normal

distribution represent unlikely values. That is, the proportion of the area under the curve as extreme as z —or more extreme than z —is very small as we get into the tails of the distribution. Our significance level corresponds to the area in the tail that is exactly equal to α . If we use our normal criterion of $\alpha = .05$, then 5% of the area under the curve becomes what we call the rejection region (also called the critical region) of the distribution. This is illustrated in Figure 31.1. The shaded rejection region takes up 5% of the area under the curve. Any result that falls in that region is sufficient evidence to reject the null hypothesis.

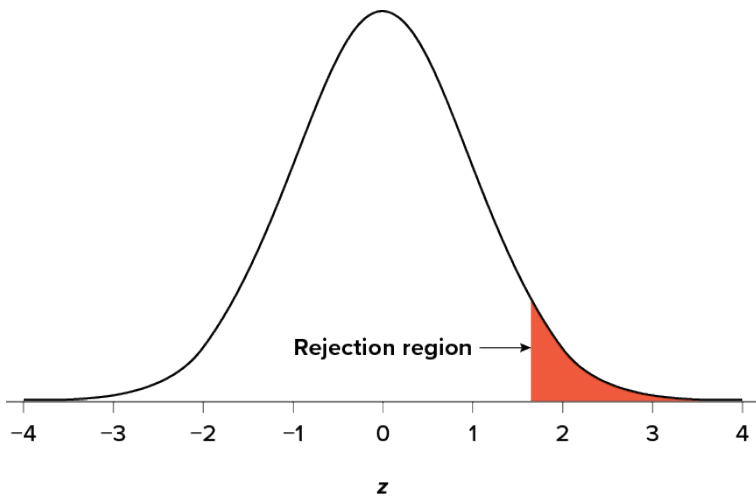


Figure 31.1. The Rejection Region for a One-Tailed Test.
[\[Image Description\]](#)

The rejection region is bounded by a specific z value, as is

any area under the curve. In hypothesis testing, the value corresponding to a specific rejection region is called the critical value, z_{crit} (“z crit”), or z^* (hence the other name “critical region”). Finding the critical value works exactly the same as finding the z-score corresponding to any area under the curve in a normal distribution. If we go to the normal table, we will find that the z-score corresponding to 5% of the area under the curve is equal to 1.645 ($z = 1.64$ corresponds to .0505 and $z = 1.65$ corresponds to .0495, so .05 is exactly in between them) if we go to the right and -1.645 if we go to the left. The direction must be determined by your alternative hypothesis, and drawing and shading the distribution is helpful for keeping directionality straight.

Suppose, however, that we want to do a non-directional test. We need to put the critical region in both tails, but we don’t want to increase the overall size of the rejection region (for reasons we will see later). To do this, we simply split it in half so that an equal proportion of the area under the curve falls in each tail’s rejection region. For $\alpha = .05$, this means 2.5% of the area is in each tail, which, based on the z-table, corresponds to critical values of $z^* = \pm 1.96$. This is shown in Figure 31.2.

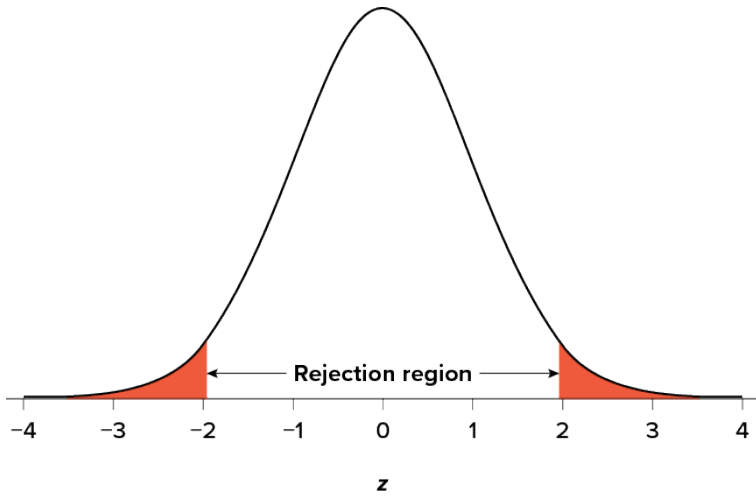


Figure 31.2. Two-tailed Rejection Region. [\[Image Description\]](#)

Thus, any z -score falling outside ± 1.96 (greater than 1.96 in absolute value) falls in the rejection region. When we use z -scores in this way, the obtained value of z (sometimes called *z obtained* and abbreviated z_{obt}) is something known as a test statistic, which is simply an inferential statistic used to test a null hypothesis. The formula for our z -statistic has not changed:

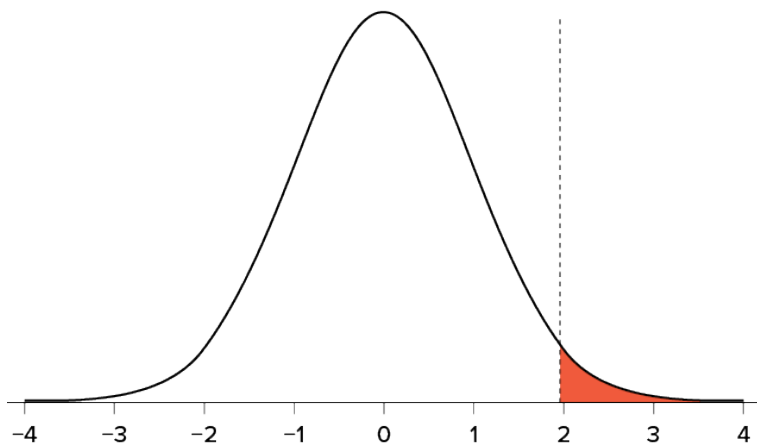
$$z = \frac{M - \mu}{\frac{\sigma}{\sqrt{n}}}$$

To formally test our hypothesis, we compare our obtained z -statistic to our critical z -value. If $z_{\text{obt}} > z_{\text{crit}}$, that means it falls in the rejection region (to see why, draw a line for $z = 2.5$

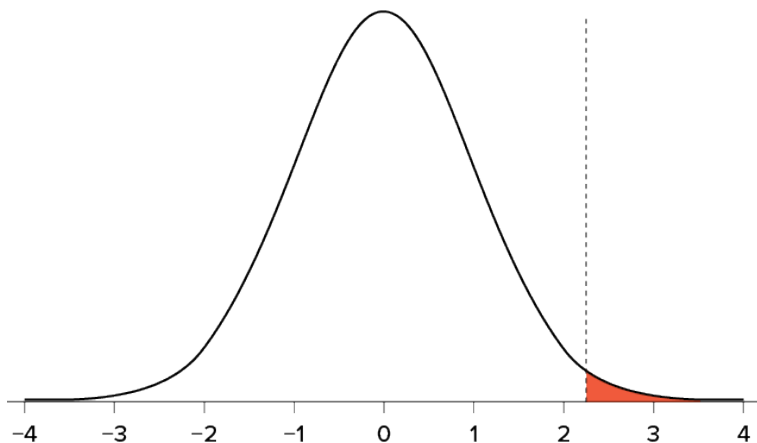
on Figure 31.1 or Figure 31.2), and so we reject H_0 . If $z_{\text{obt}} < z_{\text{crit}}$, we fail to reject. Remember that as z gets larger, the corresponding area under the curve beyond z gets smaller. Thus, the proportion, or p-value, will be smaller than the area for α , and if the area is smaller, the probability gets smaller. Specifically, the probability of obtaining that result, or a more extreme result, under the condition that the null hypothesis is true gets smaller.

The z-statistic is very useful when we are doing our calculations by hand. However, when we use computer software, it will report to us a p-value, which is simply the proportion of the area under the curve in the tails beyond our obtained z-statistic. We can directly compare this p-value to α to test our null hypothesis: if $p < \alpha$, we reject H_0 , but if $p > \alpha$, we fail to reject. Note also that the reverse is always true. If we use critical values to test our hypothesis, we will always know if p is greater than or less than α . If we reject, we know that $p < \alpha$ because the obtained z-statistic falls farther out into the tail than the critical z value that corresponds to α , so the proportion (p-value) for that z-statistic will be smaller. Conversely, if we fail to reject, we know that the proportion will be larger than α because the z-statistic will not be as far into the tail. This is illustrated for a one-tailed test in Figure 31.3.

Rejection region for $\alpha = .05$, $z^* = 1.96$



Shaded p value for $z_{\text{obt}} = 2.25$; reject H_0



Shaded p value for $z_{\text{obt}} = 1.25$; fail to reject H_0

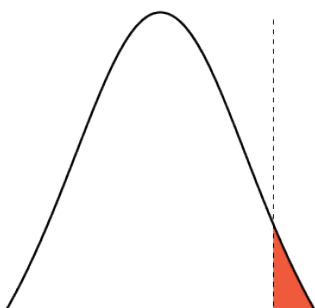


Figure 31.3. Relationship Between α , z_{obt} , and p . [Image Description]

When the null hypothesis is rejected, the effect is said to have statistical significance or be statistically significant. For example, in the [Physicians' Reactions](#) case study, the probability value is .0057. Therefore, the effect of obesity is statistically significant, and the null hypothesis that obesity makes no difference is rejected. It is important to keep in mind that statistical significance means only that the null hypothesis of exactly no effect is rejected; it does not mean that the effect is important, which is what “significant” usually means. When an effect is significant, you can have confidence that the effect is not exactly zero. Finding that an effect is significant does not tell you about how large or important the effect is.

Do not confuse statistical significance with practical significance. A small effect can be highly significant if the sample size is large enough.

Why does the word “significant” in the phrase “statistically significant” mean something so different from other uses of the word? Interestingly, this is because the meaning of “significant” in everyday language has changed. It turns out that when the procedures for hypothesis testing were developed, something was “significant” if it signified something. Thus, finding that an effect is statistically significant signifies that the effect is real and not due to chance.

Over the years, the meaning of “significant” changed, leading to the potential misinterpretation.

The Hypothesis Testing Process

A Four-Step Procedure

The process of testing hypotheses follows a simple four-step procedure. This process will be what we use for the remainder of the textbook and course, and although the hypothesis and statistics we use will change, this process will not.

Step 1: State the Hypotheses

Your hypotheses are the first thing you need to lay out. Otherwise, there is nothing to test! You have to state the null hypothesis (which is what we test) and the alternative hypothesis (which is what we expect). These should be stated mathematically as they were presented above and in words, explaining in normal English what each one means in terms of the research question.

Step 2: Find the Critical Values

Next, we formally lay out the criteria we will use to test our hypotheses. There are two pieces of information that inform our critical values: α , which determines how much of the

area under the curve comprises our rejection region, and the directionality of the test, which determines where the region will be.

Step 3: Calculate the Test Statistic and Effect Size

Once we have our hypotheses and the standards we use to test them, we can collect data and calculate our test statistic—in this case, z . This step is where the vast majority of differences in future chapters will arise: different tests used for different data are calculated in different ways, but the way we use and interpret them remains the same. As part of this step, we will also calculate effect size to better quantify the magnitude of the difference between our groups. Although effect size is not considered part of hypothesis testing, reporting it as part of the results is the approved convention.

Step 4: Make the Decision

Finally, once we have obtained our test statistic, we can compare it to our critical value and decide whether we should reject or fail to reject the null hypothesis. When we do this, we must interpret the decision in relation to our research question, stating what we concluded, what we based our conclusion on, and the specific statistics we obtained.

Example A: Movie Popcorn

Let's see how hypothesis testing works in action by working through an example. Say that a movie theater owner likes to keep a very close eye on how much popcorn goes into each bag sold, so he knows that the average bag has 8 cups of popcorn and that this varies a little bit, about half a cup. That is, the known population mean is $\mu = 8.00$ and the known population standard deviation is $s = 0.50$. The owner wants to make sure that the newest employee is filling bags correctly, so over the course of a week, he randomly assesses 25 bags filled by the employee to test for a difference ($N = 25$). He doesn't want bags over-filled or under-filled, so he looks for differences in both directions. This scenario has all of the information we need to begin our hypothesis testing procedure.

Step 1: State the Hypotheses

Our manager is looking for a difference in the mean weight of popcorn bags compared to the population

mean of 8. We will need both a null and an alternative hypothesis written both mathematically and in words. We'll always start with the null hypothesis:

H_0 : There is no difference in the weight of popcorn bags from this employee

Notice that we phrase the hypothesis in terms of the population parameter μ , which in this case would be the true average weight of bags filled by the new employee. Our assumption of no difference, the null hypothesis, is that this mean is exactly the same as the known population mean value we want it to match, 8.00. Now let's do the alternative:

H_A : There is a difference in the weight of popcorn bags from this employee

In this case, we don't know if the bags will be too full or not full enough, so we do a two-tailed alternative hypothesis that there is a difference.

Step 2: Find the Critical Values

Our critical values are based on two things: the directionality of the test and the level of significance. We decided in Step 1 that a two-tailed test is the appropriate directionality. We were given no information about the level of significance, so we assume that $\alpha = .05$ is what we will use. As stated

earlier in the chapter, the critical values for a two-tailed z-test at $\alpha = .05$ are $z^* = \pm 1.96$. This will be the criteria we use to test our hypothesis. We can now draw out our distribution, as shown in Figure 31.4, so we can visualize the rejection region and make sure it makes sense.

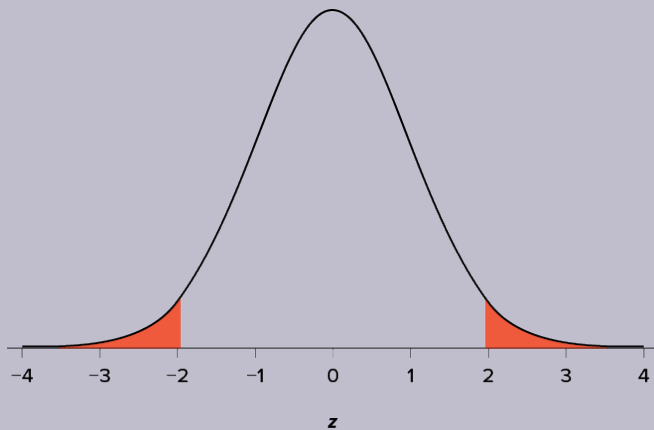


Figure 31.4. Rejection Region for $z^* = \pm 1.96$. [\[Image Description\]](#)

Step 3: Calculate the Test Statistic and Effect Size

Now we come to our formal calculations. Let's say that the manager collects data and finds that the average weight of this employee's popcorn bags is

$M = 7.75$ cups. We can now plug this value, along with the values presented in the original problem, into our equation for z :

$$z = \frac{7.75 - 8.00}{\frac{0.50}{\sqrt{25}}} = \frac{-0.25}{0.10} = -2.50$$

So our test statistic is $z = -2.50$, which we can draw onto our rejection region distribution as shown in Figure 31.5.

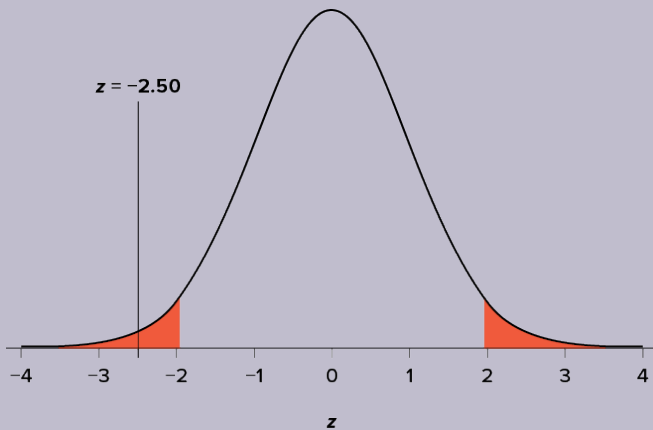


Figure 31.5. Test Statistic Location. [\[Image Description\]](#)

Effect Size

When we reject the null hypothesis, we are stating

that the difference we found was statistically significant, but we have mentioned several times that this tells us nothing about practical significance. To get an idea of the actual size of what we found, we can compute a new statistic called an effect size. Effect size gives us an idea of how large, important, or meaningful a statistically significant effect is. For mean differences like we calculated here, our effect size is Cohen's d :

$$d = \frac{M - \mu}{\sigma}$$

This is very similar to our formula for z , but we no longer take into account the sample size (since overly large samples can make it too easy to reject the null). Cohen's d is interpreted in units of standard deviations, just like z . For our example:

$$d = \frac{7.75 - 8.00}{0.50} = \frac{-0.25}{0.50} = 0.50$$

Cohen's d is interpreted as small, moderate, or large. Specifically, $d = 0.20$ is small, $d = 0.50$ is moderate, and $d = 0.80$ is large. Obviously, values can fall in between these guidelines, so we should use our best judgment and the context of the problem to make our final interpretation of size. Our effect size

happens to be exactly equal to one of these, so we say that there is a moderate effect.

Effect sizes are incredibly useful and provide important information and clarification that overcomes some of the weaknesses of hypothesis testing. Any time you perform a hypothesis test, whether statistically significant or not, you should always calculate and report effect size.

Step 4: Make the Decision

Looking at Figure 31.5, we can see that our obtained z-statistic falls in the rejection region. We can also directly compare it to our critical value: in terms of absolute value, $-2.50 > -1.96$, so we reject the null hypothesis. We can now write our conclusion:

Reject H_0 . Based on the sample of 25 bags, we can conclude that the average popcorn bag from this employee is smaller ($M = 7.75$ cups) than the average weight of popcorn bags at this movie theater, and the effect size was moderate, $z = -2.50$, $p < .05$, $d = 0.50$.

When we write our conclusion, we write out the words to communicate what it actually means, but we also include the average sample size we

calculated (the exact location doesn't matter, just somewhere that flows naturally and makes sense), the z-statistic and p-value, and the effect size. We don't know the exact p-value, but we do know that because we rejected the null, it must be less than α .

Example B: Office Temperature

Let's do another example to solidify our understanding. Let's say that the office building you work in is supposed to be kept at 74 degrees Fahrenheit during the summer months but is allowed to vary by 1 degree in either direction. You suspect that, as a cost-saving measure, the temperature was secretly set higher. You set up a formal way to test your hypothesis.

Step 1: State the Hypotheses

You start by laying out the null hypothesis:

H_0 : There is no difference in the average building temperature

$$H_0 : \mu = 74$$

Next, you state the alternative hypothesis. You have reason to suspect a specific direction of change, so you make a one-tailed test:

H_A : The average building temperature is higher than claimed

$$H_A : \mu > 74$$

Step 2: Find the Critical Values

You know that the most common level of significance is $\alpha = .05$, so you keep that the same and know that the critical value for a one-tailed z test is $z^* = 1.645$. To keep track of the directionality of the test and rejection region, you draw out your distribution as shown in Figure 31.6.

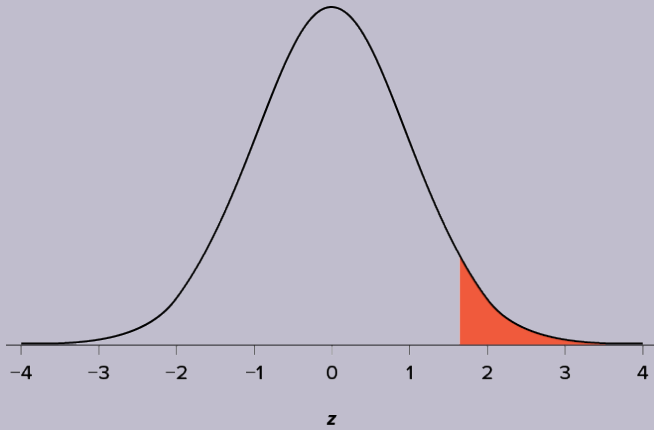


Figure 31.6. Rejection Region. [\[Image Description\]](#)

Step 3: Calculate the Test Statistic and Effect Size

Now that you have everything set up, you spend one week collecting temperature data:

Table 31.1.
Five-day
Temperatures

Day	Temp
Monday	77
Tuesday	76
Wednesday	74
Thursday	78
Friday	78

You calculate the average of these scores to be $M = 76.6$ degrees. You use this to calculate the test statistic, using $\mu = 74$ (the supposed average temperature), $s = 1.00$ (how much the temperature should vary), and $n = 5$ (how many data points you collected):

$$z = \frac{76.60 - 74.00}{\frac{1.00}{\sqrt{5}}} = \frac{2.60}{0.45} = 5.78$$

This value falls so far into the tail that it cannot even be plotted on the distribution (Figure 31.7)! Because the result is significant, you also calculate an effect size:

$$d = \frac{76.60 - 74.00}{1.00} = \frac{2.60}{1.00} = 2.60$$

The effect size you calculate is definitely large, meaning someone has some explaining to do!

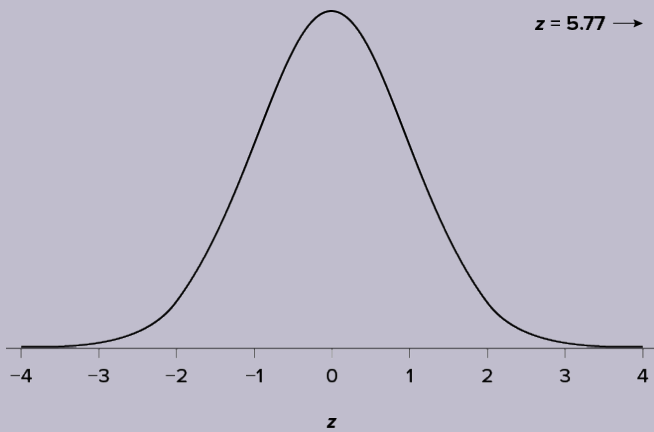


Figure 31.7. Obtained *z* Statistic. [\[Image Description\]](#)

Step 4: Make the Decision

You compare your obtained *z*-statistic, $z = 5.77$, to

the critical value, $z^* = 1.645$, and find that $z > z^*$. Therefore, you reject the null hypothesis, concluding: Reject H_0 . Based on 5 observations, the average temperature ($M = 76.6$ degrees) is statistically significantly higher than it is supposed to be, and the effect size was large, $z = 5.77$, $p < .05$, $d = 2.60$.

Example C: Different Significance Level

Finally, let's take a look at an example phrased in generic terms, rather than in the context of a specific research question, to see the individual pieces one more time. This time, however, we will use a stricter significance level, $\alpha = .01$, to test the hypothesis.

Step 1: State the Hypotheses

We will use 60 as an arbitrary null hypothesis value:

H_0 : The average score does not differ from the population

$$H_0) : \mu = 60$$

We will assume a two-tailed test:

H_A : The average score does differ

$$H_A : \mu \neq 60$$

Step 2: Find the Critical Values

We have seen the critical values for z-tests at $\alpha = .05$ levels of significance several times. To find the values for $\alpha = .01$, we will go to the [Standard Normal Distribution Table](#) [New Tab] and find the z-score cutting off .005 (.01 divided by 2 for a two-tailed test) of the area in the tail, which is $z^* = \pm 2.575$. Notice that this cutoff is much higher than it was for $\alpha = .05$. This is because we need much less of the area in the tail, so we need to go very far out to find the cutoff. As a result, this will require a much larger effect or a much larger sample size in order to reject the null hypothesis.

Step 3: Calculate the Test Statistic and Effect Size

We can now calculate our test statistic. We will use $s = 10$ as our known population standard deviation and the following data to calculate our sample mean:

$$\{61, 65, 58, 54, 60, 62, 61, 59, 61, 63\}$$

The average of these scores is $M = 60.40$. From this we calculate our z-statistic as:

$$z = \frac{60.40 - 60.00}{\frac{10.00}{\sqrt{10}}} = \frac{0.40}{3.16} = 0.13$$

The Cohen's d effect size calculation is:

$$d = \frac{M - \mu}{\sigma} = \frac{60.40 - 60.00}{10.00} = \frac{0.40}{10.00} = 0.04$$

Step 4: Make the Decision

Our obtained z-statistic, $z = 0.13$, is very small. It is much less than our critical value of 2.575. Thus, this time, we fail to reject the null hypothesis. Our conclusion would look something like:

Fail to reject H_0 . Based on the sample of 10 scores, we cannot conclude that there is an effect causing

the mean ($M = 60.40$) to be statistically significantly different from 60.00, $z = 0.13$, $p > .01$, $d = 0.04$, and the effect size supports this interpretation.

Notice two things about the end of the conclusion. First, we wrote that p is greater than instead of p is less than, like we did in the previous two examples. This is because we failed to reject the null hypothesis. We don't know exactly what the p -value is, but we know it must be larger than the α level we used to test our hypothesis. Second, we used .01 instead of the usual .05, because this time we tested at a different level. The number you compare to the p -value should always be the significance level you test at.

Other Considerations in Hypothesis Testing

There are several other considerations we need to keep in mind when performing hypothesis testing.

Errors in Hypothesis Testing

In the [Physicians' Reactions](#) case study, the probability value

associated with the significance test is .0057. Therefore, the null hypothesis was rejected, and it was concluded that physicians intend to spend less time with obese patients. Despite the low probability value, it is possible that the null hypothesis of no true difference between obese and average-weight patients is true and that the large difference between sample means occurred by chance. If this is the case, then the conclusion that physicians intend to spend less time with obese patients is in error. This type of error is called a **Type I error**. More generally, a Type I error occurs when a significance test results in the rejection of a true null hypothesis.

By one common convention, if the probability value is below .05, then the null hypothesis is rejected. Another convention, although slightly less common, is to reject the null hypothesis if the probability value is below .01. The threshold for rejecting the null hypothesis is called the α level or simply α . It is also called the significance level. As discussed in the introduction to hypothesis testing, it is better to interpret the probability value as an indication of the weight of evidence against the null hypothesis than as part of a decision rule for making a reject or do-not-reject decision. Therefore, keep in mind that rejecting the null hypothesis is not an all-or-nothing decision.

The Type I error rate is affected by the α level: the lower the α level, the lower the Type I error rate. It might seem that α is the probability of a Type I error. However, this is not correct.

Instead, α is the probability of a Type I error given that the null hypothesis is true. If the null hypothesis is false, then it is impossible to make a Type I error.

The second type of error that can be made in significance testing is failing to reject a false null hypothesis. This kind of error is called a **Type II error**. When a statistical test is not significant, it means that the data do not provide strong evidence that the null hypothesis is false. Lack of significance does not support the conclusion that the null hypothesis is true. Therefore, a researcher should not make the mistake of incorrectly concluding that the null hypothesis is true when a statistical test was not significant. Instead, the researcher should consider the test inconclusive. Contrast this with a Type I error in which the researcher erroneously concludes that the null hypothesis is false when, in fact, it is true.

A Type II error can only occur if the null hypothesis is false. If the null hypothesis is false, then the probability of a Type II error is called β (“beta”). The probability of correctly rejecting a false null hypothesis equals $1 - \beta$ and is called **statistical power**. Power is simply our ability to correctly detect an effect that exists. It is influenced by the size of the effect (larger effects are easier to detect), the significance level we set (making it easier to reject the null makes it easier to detect an effect, but increases the likelihood of a Type I error), and the sample size used (larger samples make it easier to reject the null).

Misconceptions in Hypothesis Testing

Misconceptions about significance testing are common. This section lists three important ones.

1. **Misconception:** The probability value (p-value) is the probability that the null hypothesis is false.
 - **Proper interpretation:** The probability value (p-value) is the probability of a result as extreme or more extreme given that the null hypothesis is true. It is the probability of the data given the null hypothesis. It is not the probability that the null hypothesis is false.
2. **Misconception:** A low probability value indicates a large effect.
 - **Proper interpretation:** A low probability value indicates that the sample outcome (or an outcome more extreme) would be very unlikely if the null hypothesis were true. A low probability value can occur with small effect sizes, particularly if the sample size is large.
3. **Misconception:** A non-significant outcome means that the null hypothesis is probably true.

- **Proper interpretation:** A non-significant outcome means that the data do not conclusively demonstrate that the null hypothesis is false.

Equity Activity: Home internet access

A school district is studying whether students who lack reliable home internet access have lower average exam scores compared to the general student population. The district reports that the average exam score for all students is 85 with a standard deviation of 15 ($\mu=85, \sigma=15$). One school surveys 49 students who report having unreliable home internet access, and their average exam score is 81.2. The school wants to test whether these students perform significantly worse than the general student population. Using $\alpha = 0.05$ determine if students with unreliable home internet access score significantly lower on exams compared to the overall student body.



An interactive H5P element has been

 excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=148#h5p-41>

Practice Problems

Short Answer Reflections



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=148#h5p-42>

Test Your Knowledge



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=148#h5p-43>

License & Attribution

“Introduction to Hypothesis Testing Using Z-Scores” by Makenzie O’Neil is adapted from [“Introduction to Hypothesis Testing”](#) by Linda R. Cote Ph.D.; Rupa G. Gordon Ph.D., Chrislyn E. Randell Ph.D., Judy Schmitt, and Helena Marvin, which is licensed CC BY-NC-SA 4.0.

“Introduction to Hypothesis Testing Using Z-Scores” is licensed under CC BY-NC-SA 4.0.

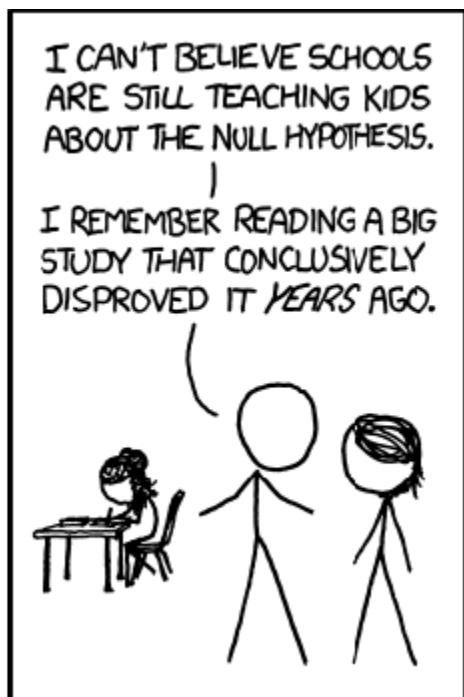


Figure 31.8. Null Hypothesis [\[Image Description\]](#)

Image Descriptions

Figure 31.1. A standard normal (bell) curve is centered at 0 on a horizontal axis labeled z (range -4 to 4). The extreme right tail, beginning around $z \approx 1.9$ – 2.0 , is shaded orange and an arrow points to it with the label “Rejection region.” This illustrates a right-tailed hypothesis test where very large

positive z-scores fall in the critical region (e.g., $\alpha \approx .05$ for a cutoff near $z = 1.96$). [[Return to Figure 31.1](#)]

Figure 31.2. A standard normal (bell) curve is centered at 0 on a horizontal axis labeled z (-4 to 4). The left and right tails are shaded orange beginning around $z = -2$ and $z = +2$, with a double-arrow label “Rejection region” spanning between them. This illustrates a two-tailed hypothesis test where extreme negative or positive z-scores (approximately beyond ± 1.96 for $\alpha = .05$) fall in the critical regions. [[Return to Figure 31.2](#)]

Figure 31.3. A vertical stack of three normal-distribution plots illustrates how α , the obtained z , and the p-value relate in a right-tailed hypothesis test. Top panel—critical region. Title: “Rejection region for $\alpha = .05$, $z^* = 1.96$.” A dashed vertical line at $z = 1.96$ marks the critical value. The small region under the curve to the right of 1.96 is shaded, indicating the rejection region whose total area equals $\alpha = .05$. Middle panel— $z_o \diamondsuit_t = 2.25$. Caption: “Shaded p value for $z_o \diamondsuit_t = 2.25$; reject H_0 .” A dashed line at $z = 2.25$ sits right of 1.96 . The area to the right of 2.25 is shaded; it is smaller than $\alpha = .05$, so the panel labels this case reject H_0 . Bottom panel— $z_o \diamondsuit_t = 1.25$. Caption: “Shaded p value for $z_o \diamondsuit_t = 1.25$; fail to reject H_0 .” A dashed line at $z = 1.25$ lies left of 1.96 . The area to the right of 1.25 is shaded and larger than $\alpha = .05$, so the panel notes fail to reject H_0 . Overall, the figure shows: when $z_o \diamondsuit_t > 1.96$, the p-value (right-tail area) is less than $.05$ and we reject H_0 ; when $z_o \diamondsuit_t < 1.96$, the p-value exceeds $.05$ and we do not reject H_0 . [[Return to Figure 31.3](#)]

Figure 31.4. A standard normal distribution is plotted with z on the horizontal axis. Two small red areas are shaded: one in the left tail starting at $z = -1.96$ and one in the right tail starting at $z = +1.96$. These symmetric shaded regions represent the rejection regions for a two-tailed test at $\alpha = .05$ (approximately .025 in each tail). The center of the curve at $z = 0$ is unshaded. [[Return to Figure 31.4](#)]

Figure 31.5. A standard normal distribution is shown with z on the horizontal axis. Small red areas are shaded in both tails beginning around $z \approx \pm 2$ (the two-tailed critical regions for $\alpha \approx .05$). A labeled vertical line at $z = -2.50$ sits to the left of -2 , clearly within the left shaded tail, indicating the test statistic falls in the rejection region (so H_0 would be rejected). [[Return to Figure 31.5](#)]

Figure 31.6. A standard normal distribution is plotted with z on the horizontal axis. The area to the right of $z = 1.645$ is filled in red, indicating the one-tailed critical region for $\alpha \approx .05$. The rest of the curve (from the left tail up through the mean at 0 and out to 1.645) is unshaded. This diagram shows that test statistics greater than 1.645 fall in the rejection region for a right-tailed z test. [[Return to Figure 31.6](#)]

Figure 31.7. A standard normal distribution is shown with z on the horizontal axis (ticks -4 to $+4$). The bell curve is centered at 0. In the upper-right corner, a label and right-pointing arrow read “ $z = 5.77$,” indicating an obtained z -score that lies well beyond the right tail of the graph (off the visible

scale) signaling an extremely rare outcome under the standard normal model. [[Return to Figure 31.7](#)]

Figure 31.8. A black-and-white comic shows two stick figures talking while a third sits at a desk writing. The dialogue text above them reads: “I can’t believe schools are still teaching kids about the null hypothesis. I remember reading a big study that conclusively disproved it years ago.” The joke is that saying the null hypothesis was “disproved” by one study misunderstands what a null hypothesis is. [[Return to Figure 31.8](#)]

32.

T-TESTS

Stephanie D'Costa

Learning Objectives

By the end of this chapter, you will be able to

1. Calculate t-scores
2. Differentiate between independent and dependent sample t-tests.
3. Discriminate between statistical significance and failure to reject the null hypothesis

As we have seen throughout this book, many studies in psychology focus on the difference between two means. The most common null hypothesis test for this type of statistical

relationship is the ***t*-test**. In this section, we look at three types of *t*-tests that are used for slightly different research designs: the one-sample *t*-test, the dependent-samples *t*-test, and the independent-samples *t*-test. You may have already taken a course in statistics, but we will refresh your statistical knowledge.

One-Sample *t*-Test

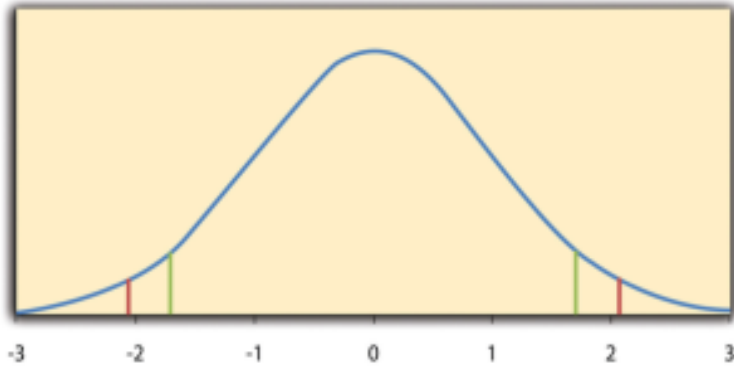
The **one-sample *t*-test** is used to compare a sample mean (M) with a hypothetical population mean (μ_0) that provides some interesting standard of comparison. In the null hypothesis, we assume that the population mean (μ) is equal to a known or expected value (μ_0). The alternative hypothesis is that the mean for the population is different from the hypothetical population mean: $\mu \neq \mu_0$. To decide between these two hypotheses, we need to find the probability of obtaining the sample mean (or one more extreme) if the null hypothesis were true. But finding this *p*-value requires first computing a test statistic called *t*. (A **test statistic** is a statistic that is computed only to help find the *p*-value.) The formula for *t* is as follows:

$$t = \frac{M - \mu_0}{\left(\frac{SD}{\sqrt{N}} \right)}$$

Again, M is the sample mean and μ_0 is the hypothetical

population mean of interest. SD is the sample standard deviation, and N is the sample size.

The reason the t -statistic (or any test statistic) is useful is that we know how it is distributed when the null hypothesis is true. As shown in Figure 32.1, this distribution is unimodal and symmetrical, and it has a mean of 0. Its precise shape depends on a statistical concept called the degrees of freedom, which for a one-sample t -test is $N - 1$. (There are 24 degrees of freedom for the distribution shown in Figure 32.1.) The important point is that knowing this distribution makes it possible to find the p -value for any t -score. Consider, for example, a t -score of 1.50 based on a sample of 25. The probability of a t -score at least this extreme is given by the proportion of t -scores in the distribution that are at least this extreme. For now, let us define *extreme* as being far from zero in either direction. Thus, the p -value is the proportion of t -scores that are 1.50 or above *or* that are -1.50 or below—a value that turns out to be .14.



*Figure 32.1 Distribution of t -Scores (With 24 Degrees of Freedom) When the Null Hypothesis Is True. The red vertical lines represent the two-tailed critical values, and the green vertical lines the one-tailed critical values when $\alpha = .05$.
[\[Image Description\]](#)*

Fortunately, we do not have to deal directly with the distribution of t -scores. If we were to enter our sample data and hypothetical mean of interest into one of the online statistical tools or into a program like SPSS (Excel does not have a one-sample t -test function), the output would include both the t -score and the p -value. At this point, the rest of the procedure is simple. If p is equal to or less than .05, we reject the null hypothesis and conclude that the population mean differs from the hypothetical mean of interest. If p is greater than .05, we retain the null hypothesis and conclude that there is not enough evidence to say that the population mean differs from the hypothetical mean of interest. (Again, technically, we conclude only that we do not have enough evidence to conclude that it *does* differ.)

If we were to compute the t -score by hand, we could use a table like Table 32.1 to make the decision. This table does not provide actual p -values. Instead, it provides the **critical values** of t for different degrees of freedom (df) when α is .05. For now, let us focus on the two-tailed critical values in the last column of the table. Each of these values should be interpreted as a pair of values: one positive and one negative. For example, the two-tailed critical values when there are 24 degrees of freedom are 2.064 and -2.064 . These are represented by the red vertical lines in Figure 32.1. The idea is that any t -score below the lower critical value (the left-hand red line in Figure 32.1) is in the lowest 2.5% of the distribution, while any t -score above the upper critical value (the right-hand red line) is in the highest 2.5% of the distribution. Therefore, any t -score beyond the critical value in *either* direction is in the most extreme 5% of t -scores when the null hypothesis is true and has a p -value less than .05. Thus, if the t -score we compute is beyond the critical value in either direction, then we reject the null hypothesis. If the t -score we compute is between the upper and lower critical values, then we retain the null hypothesis.

Table 32.1 Table of Critical Values of t When $\alpha = .05$

df	One-tailed Critical Value	Two-tailed Critical Value
3	2.353	3.182
4	2.132	2.776
5	2.015	2.571
6	1.943	2.447
7	1.895	2.365
8	1.860	2.306
9	1.833	2.262
10	1.812	2.228
11	1.796	2.201
12	1.782	2.179
13	1.771	2.160
14	1.761	2.145
15	1.753	2.131
16	1.746	2.120
17	1.740	2.110
18	1.734	2.101
19	1.729	2.093
20	1.725	2.086
21	1.721	2.080
22	1.717	2.074

23	1.714	2.069
24	1.711	2.064
25	1.708	2.060
30	1.697	2.042
35	1.690	2.030
40	1.684	2.021
45	1.679	2.014
50	1.676	2.009
60	1.671	2.000
70	1.667	1.994
80	1.664	1.990
90	1.662	1.987
100	1.660	1.984

Thus far, we have considered what is called a **two-tailed test**, where we reject the null hypothesis if the t -score for the sample is extreme in either direction. This test makes sense when we believe that the sample mean might differ from the hypothetical population mean, but we do not have good reason to expect the difference to go in a particular direction. But it is also possible to do a **one-tailed test**, where we reject the null hypothesis only if the t -score for the sample is extreme in one direction that we specify before collecting the data. This test makes sense when we have good reason to expect

the sample mean will differ from the hypothetical population mean in a particular direction.

Here is how it works. Each one-tailed critical value in Table 32.1 can again be interpreted as a pair of values: one positive and one negative. A t -score below the lower critical value is in the lowest 5% of the distribution, and a t -score above the upper critical value is in the highest 5% of the distribution. For 24 degrees of freedom, these values are -1.711 and 1.711 . (These are represented by the green vertical lines in Figure 32.1.) However, for a one-tailed test, we must decide before collecting data whether we expect the sample mean to be lower than the hypothetical population mean, in which case we would use only the lower critical value, or we expect the sample mean to be greater than the hypothetical population mean, in which case we would use only the upper critical value. Notice that we still reject the null hypothesis when the t -score for our sample is in the most extreme 5% of the t -scores we would expect if the null hypothesis were true—so α remains at .05. We have simply redefined *extreme* to refer only to one tail of the distribution. The advantage of the one-tailed test is that critical values are less extreme. If the sample mean differs from the hypothetical population mean in the expected direction, then we have a better chance of rejecting the null hypothesis. The disadvantage is that if the sample mean differs from the hypothetical population mean in the unexpected direction, then there is no chance at all of rejecting the null hypothesis.

Example One-Sample t -Test

Imagine that a health psychologist is interested in the accuracy of university students' estimates of the number of calories in a chocolate chip cookie. He shows the cookie to a sample of 10 students and asks each one to estimate the number of calories in it. Because the actual number of calories in the cookie is 250, this is the hypothetical population mean of interest (μ_0). The null hypothesis is that the mean estimate for the population (μ) is 250. Because he has no real sense of whether the students will underestimate or overestimate the number of calories, he decides to do a two-tailed test. Now, imagine further that the participants' actual estimates are as follows:

250, 280, 200, 150, 175, 200, 200, 220, 180, 250

The mean estimate for the sample (M) is 212.00 calories, and the standard deviation (SD) is 39.17. The health psychologist can now compute the t -score for his sample:

$$t = \frac{212 - 250}{\left(\frac{39.17}{\sqrt{10}} \right)} = -3.07$$

If he enters the data into one of the online analysis tools or uses SPSS, it would also tell him that the two-tailed p -value for this t -score (with $10 - 1 = 9$ degrees of freedom) is .013. Because this is less than .05, the health psychologist would reject the null hypothesis and conclude that university students tend to underestimate the number of calories in a chocolate chip cookie. If he computes the t -score by hand, he could look at Table 32.1 and see that the critical value of t for a two-tailed test with 9 degrees of freedom is ± 2.262 . The fact that his t -score was more extreme than this critical value would tell him that his p -value is less than .05 and that he should reject the null hypothesis. Using APA style, these results would be reported as follows: $t(9) = -3.07, p = .01$. Note that the t and p are italicized, the degrees of freedom appear in brackets with no decimal remainder, and the values of t and p are rounded to two decimal places.

Finally, if this researcher had gone into this study with good reason to expect that university students

underestimate the number of calories, then he could have done a one-tailed test instead of a two-tailed test. The only thing this decision would change is the critical value, which would be -1.833 . This slightly less extreme value would make it a bit easier to reject the null hypothesis. However, if it turned out that university students overestimate the number of calories—no matter how much they overestimate it—the researcher would not have been able to reject the null hypothesis.

The Dependent-Samples t -Test

The **dependent-samples t -test** (sometimes called the paired-samples t -test) is used to compare two means for the same sample tested at two different times or under two different conditions. This comparison is appropriate for pretest-posttest designs or within-subjects experiments. The null hypothesis is that the means at the two times or under the two conditions are the same in the population. The alternative hypothesis is that they are not the same. This test can also be one-tailed if the researcher has good reason to expect the difference to go in a particular direction.

It helps to think of the dependent-samples t -test as a special case of the one-sample t -test. However, the first step in the

dependent-samples *t*-test is to reduce the two scores for each participant to a single **difference score** by taking the difference between them. At this point, the dependent-samples *t*-test becomes a one-sample *t*-test on the difference scores. The hypothetical population mean (μ_0) of interest is 0 because this is what the mean difference score would be if there were no difference on average between the two times or two conditions. We can now think of the null hypothesis as being that the mean difference score in the population is 0 ($\mu_0 = 0$) and the alternative hypothesis as being that the mean difference score in the population is not 0 ($\mu_0 \neq 0$).

Example Dependent-Samples *t*-Test

Imagine that the health psychologist now knows that people tend to underestimate the number of calories in junk food and has developed a short training program to improve their estimates. To test the effectiveness of this program, he conducts a pretest-posttest study in which 10 participants estimate the number of calories in a chocolate chip cookie before the training program and then again

afterward. Because he expects the program to increase the participants' estimates, he decides to do a one-tailed test. Now, imagine further that the pretest estimates are

230, 250, 280, 175, 150, 200, 180, 210, 220, 190

and that the posttest estimates (for the same participants in the same order) are

250, 260, 250, 200, 160, 200, 200, 180, 230, 240

The difference scores, then, are as follows:

20, 10, -30, 25, 10, 0, 20, -30, 10, 50

Note that it does not matter whether the first set of scores is subtracted from the second or the second from the first, as long as it is done the same way for all participants. In this example, it makes sense to subtract the pretest estimates from the posttest estimates so that positive difference scores mean that the estimates went up after the training, and negative difference scores mean the estimates went down.

The mean of the difference scores is 8.50 with a standard deviation of 27.27. The health psychologist can now compute the t -score for his sample as follows:

$$t = \frac{8.5 - 0}{\left(\frac{27.27}{\sqrt{10}} \right)} = 1.11$$

If he enters the data into one of the online analysis tools or uses Excel or SPSS, it would tell him that the one-tailed p -value for this t -score (again with $10 - 1 = 9$ degrees of freedom) is .148. Because this is greater than .05, he would retain the null hypothesis and conclude that the training program does not significantly increase people's calorie estimates. If he were to compute the t -score by hand, he could look at Table 32.1 and see that the critical value of t for a one-tailed test with 9 degrees of freedom is 1.833. (It is positive this time because he was expecting a positive mean difference score.) The fact that his t -score was less extreme than this critical value would tell him that his p -value is greater than .05 and that he should fail to reject the null hypothesis.

The Independent-Samples t -Test

The **independent-samples t -test** is used to compare the means of two separate samples (M_1 and M_2). The two samples might have been tested under different conditions in a between-subjects experiment, or they could be pre-existing

groups in a cross-sectional design (e.g., women and men, extraverts and introverts). The null hypothesis is that the means of the two populations are the same: $\mu_1 = \mu_2$. The alternative hypothesis is that they are not the same: $\mu_1 \neq \mu_2$. Again, the test can be one-tailed if the researcher has good reason to expect the difference to go in a particular direction.

The t -statistic here is a bit more complicated because it must take into account two sample means, two standard deviations, and two sample sizes. The formula is as follows:

$$t = \frac{M_1 - M_2}{\sqrt{\frac{SD^2_1}{n_1} + \frac{SD^2_2}{n_2}}}$$

Notice that this formula includes squared standard deviations (the variances) that appear inside the square root symbol. Also, lowercase n_1 and n_2 refer to the sample sizes in the two groups or conditions (as opposed to capital N , which generally refers to the total sample size). The only additional thing to know here is that there are $N - 2$ degrees of freedom for the independent-samples t -test.

Example Independent-Samples t -Test

Now the health psychologist wants to compare the calorie estimates of people who regularly eat junk food with the estimates of people who rarely eat junk food. He believes the difference could come out in either direction, so he decides to conduct a two-tailed test. He collects data from a sample of eight participants who eat junk food regularly and seven participants who rarely eat junk food. The data are as follows:

Junk food eaters: 180, 220, 150, 85, 200, 170,
150, 190

Non-junk food eaters: 200, 240, 190, 175, 200,
300, 240

The mean for the non-junk food eaters is 220.71 with a standard deviation of 41.23. The mean for the junk food eaters is 168.12 with a standard deviation of 42.66. He can now compute his t -score as follows:

$$t = \frac{220.71 - 168.12}{\sqrt{\frac{41.23^2}{8} + \frac{42.66^2}{7}}} = 2.42$$

If he enters the data into one of the online analysis tools or uses Excel or SPSS, it would tell him that the two-tailed p -value for this t -score (with $15 - 2 = 13$ degrees of freedom) is .015. Because this p -value is less than .05, the health psychologist would reject the null hypothesis and conclude that people who eat junk food regularly make lower calorie estimates than people who eat it rarely. If he were to compute the t -score by hand, he could look at Table 32.1 and see that the critical value of t for a two-tailed test with 13 degrees of freedom is ± 2.160 . The fact that his t -score was more extreme than this critical value would tell him that his p -value is less than .05 and that he should reject the null hypothesis.

The t -Statistic

We were introduced to hypothesis testing using the z -statistic for sample means in previous chapters. This was a useful way to link the material and ease us into the new way of looking at data, but it isn't a very common test because it relies on knowing the population's standard deviation, s , which is rarely

going to be the case. Instead, we will estimate that parameter s using the sample statistic s in the same way that we estimate μ using $\mathbf{M}$ (μ will still appear in our formulas because we suspect something about its value, and that is what we are testing). Our new statistic is called t , and for testing one population mean using a single sample (called a one-sample t -test), it takes the form:

$$t = \frac{M - \mu}{s_M} = \frac{M - \mu}{\frac{s}{\sqrt{n}}}$$

Notice that t looks almost identical to z ; this is because they test the exact same thing: the value of a sample mean compared to what we expect of the population. The only difference is that the standard error is now denoted $\mathbf{S_M}$ to indicate that we use the sample statistic for standard deviation, s , instead of the population parameter s . The process of using and interpreting the standard error and the full test statistic remains exactly the same.

We have also learned that the formulas for sample standard deviation and population standard deviation differ by one key factor: the denominator for the parameter is $\mathbf{N}$ but the denominator for the statistic is $\mathbf{N - 1}$, also known as degrees of freedom, $\mathbf{df}$. Because we are using a new measure of spread, we can no longer use the standard normal distribution and the z -table to find our critical values. For t -tests, we will use the t -distribution and t -table to find these values.

The t -distribution, like the standard normal distribution, is symmetric and normally distributed with a mean of 0 and

standard error (as the measure of standard deviation for sampling distributions) of 1. However, because the calculation of standard error uses degrees of freedom, there will be a different t distribution for every degree of freedom. Luckily, they all work exactly the same, so in practice, this difference is minor.

Figure 32.2 shows four curves: a normal distribution curve labeled z , and three t -distribution curves for 2, 10, and 30 degrees of freedom. Two things should stand out: First, for lower degrees of freedom (e.g., 2), the tails of the distribution are much fatter, meaning a larger proportion of the area under the curve falls in the tail. This means that we will have to go farther out into the tail to cut off the portion corresponding to 5% or $\alpha = .05$, which will in turn lead to higher critical values. Second, as the degrees of freedom increase, we get closer and closer to the z curve. Even the distribution with $df = 30$, corresponding to a sample size of just 31 people, is nearly indistinguishable from z . In fact, a t -distribution with infinite degrees of freedom (theoretically, of course) is exactly the standard normal distribution. Because of this, the bottom row of the t -table also includes the critical values for z tests at the specific significance levels. Even though these curves are very close, it is still important to use the correct table and critical values, because small differences can add up quickly.

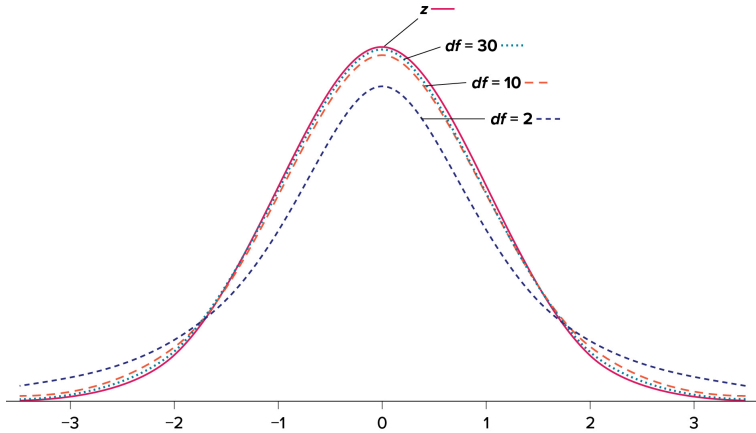


Figure 32.2. *Distributions Comparing Effects of Degrees of Freedom.* [\[Image Description\]](#)

The t -distribution table lists critical values for one- and two-tailed tests at several levels of significance, arranged into columns. The rows of the t -table list degrees of freedom up to $df = 100$ in order to use the appropriate distribution curve. It does not, however, list all possible degrees of freedom in this range, because that would take too many rows. Above $df = 40$, the rows jump in increments of 10. If a problem requires you to find critical values and the exact degrees of freedom is not listed, you always round down to the next smallest number. For example, if you have 48 people in your sample, the degrees of freedom are $N - 1 = 48 - 1 = 47$; however, 47 doesn't appear on our table, so we round down and use the critical values for $df = 40$, even though 50 is closer. We do this because it avoids inflating Type I error (false positives) by using criteria that are too lax.

Hypothesis Testing with t

Hypothesis testing with the t -statistic works exactly the same way as z tests did, following the four-step process of (1) stating the hypotheses, (2) finding the critical values, (3) computing the test statistic and effect size, and (4) making the decision.

Example Oil Change Speed

We will work through an example: Let's say that you move to a new city and find an auto shop to change your oil. Your old mechanic did the job in about 30 minutes (although you never paid close enough attention to know how much that varied), and you suspect that your new shop takes much longer. After four oil changes, you think you have enough evidence to demonstrate this.

Step 1: State the Hypotheses

Our hypotheses for one-sample t -tests are identical to those we used for z tests. We still state the null and alternative hypotheses mathematically in terms

of the population parameter and which are written out in readable English. For our example:

H_0 : There is no difference in the average time to change a car's oil

$$H_0 : \mu = 30$$

H_A : This shop takes longer to change oil than your old mechanic

$$H_A : \mu > 30$$

Step 2: Find the Critical Values

As noted above, our critical values still delineate the area in the tails under the curve corresponding to our chosen level of significance. Because we have no reason to change significance levels, we will use $\alpha = .05$, and because we suspect a direction of effect, we have a one-tailed test. To find our critical values for t , we need to add one more piece of information: the degrees of freedom. For this example:

$$df = N - 1 = 4 - 1 = 3$$

Going to our t -table, a portion of which is found in Table 32.2a, we locate the column corresponding to our one-tailed significance level of .05 and find where it intersects with the row for 3 degrees of freedom. As we can see in Table 32.2b, our critical

value is $t^* = 2.353$. (See table above for all critical values.)

Table 32.2a. *t* Distribution Table (*t*-Table) with Proportions

					.05
df	.25	.20	.15	.10	

Table 32.2b. *t* Distribution Table (*t*-Table) with Proportion (*a*) in Two Tails Combined

df	.50	.40	.30	.20	.10	.05	.02	.01	.001
1	1.000	1.376	1.963	3.078	6.314	12.706	31.821	63.657	636.578
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	31.600
3	0.765	1.078	1.250	1.638	2.353	3.182	4.541	5.841	12.924
4	0.741	1.041	1.190	1.533	2.132	2.776	3.747	4.604	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	6.869

We can then shade this region on our *t* distribution to visualize our rejection region (Figure 32.3).

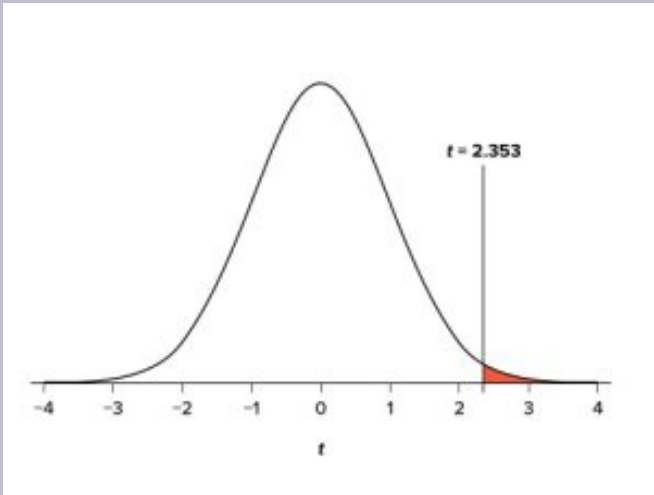


Figure 32.3. Rejection Region. [\[Image Description\]](#)

Step 3: Calculate the Test Statistic and Effect Size

The four wait times you experienced for your oil changes at the new shop were 46 minutes, 58 minutes, 40 minutes, and 71 minutes. We will use these to calculate **M** and **s** by first filling in the sum of squares in Table 32.3.

Table 32.3. Sum of Squares.

X	X – M	(X – M)²
46	-7.75	60.06
58	4.25	18.06
40	-13.75	189.06
71	17.25	297.56
Σ = 215	Σ = 0	Σ = 564.74

After filling in the first row to get $\Sigma X = 215$, we find that the mean is $M = 53.75$ (215 divided by sample size 4), which allows us to fill in the rest of the table to get our sum of squares $SS = 564.74$, which we then plug into the formula for standard deviation:

$$s = \sqrt{\frac{\sum(X - M)^2}{N - 1}} = \sqrt{\frac{SS}{df}} = \sqrt{\frac{564.74}{3}} = 13.72$$

Next, we take this value and plug it into the formula for standard error:

$$s_M = \frac{s}{\sqrt{n}} = \frac{13.72}{2} = 6.86$$

And, finally, we put the standard error, sample mean, and null hypothesis value into the formula for our test statistic t :

$$t = \frac{M - \mu}{s_M} = \frac{53.75 - 30}{6.86} = \frac{23.75}{6.68} = 3.46$$

This may seem like a lot of steps, but it is really just taking our raw data to calculate one value at a time and carrying that value forward into the next equation: data → sample size/degrees of freedom → mean → sum of squares → standard deviation → standard error → test statistic. At each step, we simply match the symbols of what we just calculated to where they appear in the next formula to make sure we are plugging everything in correctly.

Next, we need to calculate an effect size, which is still Cohen's d , but now we use s in place of σ :

$$d = \frac{M - \mu}{s} = \frac{53.75 - 30.00}{13.72} = 1.73$$

This is a large effect. It should also be noted that for some things, like the minutes in our current example, we can also interpret the magnitude of the difference we observed (23 minutes and 45 seconds)

as an indicator of importance, since time is a familiar metric.

Step 4: Make the Decision

Now that we have our critical value and test statistic, we can make our decision using the same criteria we used for a z-test. Our obtained t -statistic was $t = 3.46$ and our critical value was $t^* = 2.353$: $t > t^*$, so we reject the null hypothesis and conclude:

Based on our four oil changes, the new mechanic takes longer on average ($M = 53.75$, $SD = 13.72$) to change oil than our old mechanic, and the effect size was large, $t(3) = 3.46$, $p < .05$, $d = 1.74$.

Notice that we also include the degrees of freedom in parentheses next to t . Figure 32.4 shows the output from JASP.

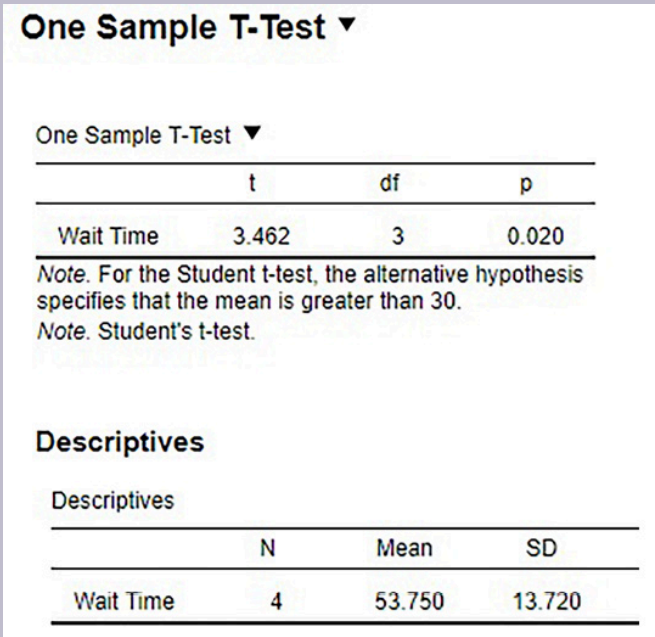


Figure 32.4. Output from JASP for the one-sample t-test described in this example. The output provides the t-value (3.462), degrees of freedom (3), and the exact p-value (.020, which is less than .05). Note that the mean (53.750) and standard deviation for the sample are also provided (13.720). Based on our four oil changes, the new mechanic takes longer on average ($M = 53.75$, $SD = 13.72$) to change oil than our old mechanic, $t(3) = 3.46$, $p = .02$. Note: Cohen's d is not shown as JASP does not provide that value for a one-sample t-test. [\[Image Description\]](#)

Confidence Intervals

Up to this point, we have learned how to estimate the population parameter for the mean using sample data and a sample statistic. From one point of view, this makes sense: we have one value for our parameter, so we use a single value (called a point estimate) to estimate it. However, we have seen that all statistics have sampling error and that the value we find for the sample mean will bounce around based on the people in our sample, simply due to random chance. Thinking about estimation from this perspective, it would make more sense to take that error into account rather than relying just on our point estimate. To do this, we calculate what is known as a confidence interval.

A confidence interval starts with our point estimate and then creates a range of scores considered plausible based on our standard deviation, our sample size, and the level of confidence with which we would like to estimate the parameter. This range, which extends equally in both directions away from the point estimate, is called the margin of error. We calculate the margin of error by multiplying our two-tailed critical value by our standard error:

$$\text{Margin of error} = t^* \left(\frac{s}{\sqrt{n}} \right)$$

One important consideration when calculating the margin of error is that it can only be calculated using the critical value for a two-tailed test. This is because the margin of error moves

away from the point estimate in both directions, so a one-tailed value does not make sense.

The critical value we use will be based on a chosen level of confidence, which is equal to $1 - \alpha$. Thus, a 95% level of confidence corresponds to $\alpha = .05$. Thus, at the .05 level of significance, we create a 95% confidence interval. How to interpret that is discussed further on.

Once we have our margin of error calculated, we add it to our point estimate for the mean to get an upper bound to the confidence interval and subtract it from the point estimate for the mean to get a lower bound for the confidence interval:

$$\text{Upper bound} = M + \text{Margin of error}$$

$$\text{Lower bound} = M - \text{Margin of error}$$

or simply:

$$\text{Confidence interval} = M \pm t^* \left(\frac{s}{\sqrt{n}} \right)$$

To write out a confidence interval, we always use round brackets (i.e., parentheses) and put the lower bound, a comma, and the upper bound:

$$\text{Confidence interval} = (\text{Lower bound}, \text{Upper bound})$$

Let's see what this looks like with some actual numbers by taking our oil change data and using it to create a 95% confidence interval, estimating the average length of time it takes at the new mechanic. We already found that our average was $M = 53.75$ and our standard error was $S_M = 6.86$. We also found a critical value to test our hypothesis, but remember that we were testing a one-tailed hypothesis, so that critical value

won't work. To see why that is, look at the column headers on the t -table. The column for one-tailed $\alpha = .05$ is the same as a two-tailed $\alpha = .10$. If we used the old critical value, we'd actually be creating a 90% confidence interval ($1.00 - 0.10 = 0.90$, or 90%). To find the correct value, we use the column for two-tailed $\alpha = .05$ and, again, the row for 3 degrees of freedom, to find $t^* = 3.182$.

Now we have all the pieces we need to construct our confidence interval:

$$\begin{aligned}
 95\% \text{ CI} &= 53.75 \pm 3.182(6.86) \\
 \text{Upper bound} &= 53.75 + 3.182(6.86) \\
 &= 53.75 + 21.83 \\
 &= 75.58 \\
 \text{Lower bound} &= 53.75 - 3.182(6.86) \\
 &= 53.75 - 21.83 \\
 &= 31.92 \\
 95\% \text{ CI} &= (31.92, 75.58)
 \end{aligned}$$

So we find that our 95% confidence interval runs from 31.92 minutes to 75.58 minutes, but what does that actually mean? The range (31.92, 75.58) represents values of the mean that we consider reasonable or plausible based on our observed data. It includes our point estimate of the mean, $M = 53.75$, in the center, but it also has a range of values that could also have been the case based on what we know about how much these scores vary (i.e., our standard error).

It is very tempting to also interpret this interval by saying that we are 95% confident that the true population mean falls within the range (31.92, 75.58), but this is not true. The

reason it is not true is that phrasing our interpretation this way suggests that we have firmly established an interval and the population mean does or does not fall into it, suggesting that our interval is firm and the population mean will move around. However, the population mean is an absolute that does not change; it is our interval that will vary from data collection to data collection, even taking into account our standard error. The correct interpretation, then, is that we are 95% confident that the range (31.92, 75.58) brackets the true population mean. This is a very subtle difference, but it is an important one.

Hypothesis Testing with Confidence Intervals

As a function of how they are constructed, we can also use confidence intervals to test hypotheses. However, we are limited to testing two-tailed hypotheses only, because of how the intervals work, as discussed above.

Once a confidence interval has been constructed, using it to test a hypothesis is simple. If the range of the confidence interval brackets (or contains, or is around) the null hypothesis value, we fail to reject the null hypothesis. If it does not bracket the null hypothesis value (i.e., if the entire range is above the null hypothesis value or below it), we reject the null hypothesis. The reason for this is clear if we think about what a confidence interval represents. Remember: a confidence

interval is a range of values that we consider reasonable or plausible based on our data. Thus, if the null hypothesis value is in that range, then it is a value that is plausible based on our observations. If the null hypothesis is plausible, then we have no reason to reject it. Thus, if our confidence interval brackets the null hypothesis value, thereby making it a reasonable or plausible value based on our observed data, then we have no evidence against the null hypothesis and fail to reject it. However, if we build a confidence interval of reasonable values based on our observations and it does not contain the null hypothesis value, then we have no empirical (observed) reason to believe the null hypothesis value and therefore reject the null hypothesis.

Example: Friendliness

You hear that the national average on a measure of friendliness is 38 points. You want to know if people in your community are more or less friendly than people nationwide, so you collect data from 30 random people in town to look for a difference. We'll

follow the same four-step hypothesis-testing procedure as before.

Step 1: State the Hypotheses

We will start by laying out our null and alternative hypotheses:

H_0 : There is no difference in how friendly the local community is compared to the national average.

$$H_0 : \mu = 38$$

H_A : There is a difference in how friendly the local community is compared to the national average.

$$H_A : \mu \neq 38$$

Step 2: Find the Critical Values

We need our critical values in order to determine the width of our margin of error. We will assume a significance level of $\alpha = .05$ (which will give us a 95% CI). From the t -table, a two-tailed critical value at $\alpha = .05$ with 29 degrees of freedom ($N - 1 = 30 - 1 = 29$) is $t^* = 2.045$.

Step 3: Calculate the Confidence Interval

Now we can construct our confidence interval. After we collect our data, we find that the average person in our community scored 39.85, or $M = 39.85$, and our standard deviation was $s = 5.61$. First, we need to use this standard deviation, plus our sample size of $N = 30$, to calculate our standard error:

$$s_M = \frac{s}{\sqrt{n}} = \frac{5.61}{5.48} = 1.02$$

Now we can put that value, our point estimate for the sample mean, and our critical value from Step 2 into the formula for a confidence interval:

$$95\% \text{ CI} = 39.85 \pm 2.045(1.02)$$

$$\begin{aligned} \text{Upper bound} &= 39.85 + 2.045(1.02) \\ &= 39.85 + 2.09 \\ &= 41.94 \end{aligned}$$

$$\begin{aligned} \text{Lower bound} &= 39.85 - 2.045(1.02) \\ &= 39.85 - 2.09 \\ &= 37.76 \end{aligned}$$

$$95\% \text{ CI} = (37.76, 41.94)$$

Step 4: Make the Decision

Finally, we can compare our confidence interval to our null hypothesis value. The null value of 38 is higher than our lower bound of 37.76 and lower than our upper bound of 41.94. Thus, the confidence interval brackets our null hypothesis value, and we fail to reject the null hypothesis:

Fail to reject H_0 . Based on our sample of 30 people, our community is not different in average friendliness ($M = 39.85$, $SD = 5.61$) than the nation as a whole, 95% CI = (37.76, 41.94).

Note that we don't report a test statistic or p -value because that is not how we tested the hypothesis, but we do report the value we found for our confidence interval.

An important characteristic of hypothesis testing is that both methods will always give you the same result. That is because both are based on the standard error and critical values in their calculations. To check this, we can calculate a t -statistic for the example above and find it to be $t = 1.81$, which is smaller than our critical value of 2.045 and fails to reject the null hypothesis.

Confidence Intervals Using z

Confidence intervals can also be constructed using z -score criteria if one knows the population standard deviation. The format, calculations, and interpretation are all exactly the same, only replacing t^* with z^* and $\mathbf{S_M}$ with σ_M .

Equity Activity: Global Warming- examining the data

You hear a lot of talk about increasing global temperature, so you decide to see for yourself if there has been an actual change in recent years. You know that the average land temperature from 1951-1980 was 8.79 degrees Celsius. You find annual average temperature data from 1981-2017 and decide to construct a 99% confidence interval (because you want to be as sure as possible and look for differences in both directions, not just one) using this data to test for a difference from the previous average.

Table 32.4. Average Temperature from 1981-2017

Year	Temp
1981	9.301
1982	8.788
1983	9.173
1984	8.824
1985	8.799
1986	8.985
1987	9.141
1988	9.345
1989	9.076
1990	9.378

Year	Temp
1991	9.336
1992	8.974
1993	9.008
1994	9.175
1995	9.484
1996	9.168
1997	9.326
1998	9.660
1999	9.406
2000	9.332

Year	Temp
2001	9.542
2002	9.695
2003	9.649
2004	9.451
2005	9.829
2006	9.662
2007	9.876
2008	9.581
2009	9.657
2010	9.828

Year	Temp
2011	9.650
2012	9.635
2013	9.753
2014	9.714
2015	9.962
2016	10.160
2017	10.049



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodspsysh/?p=155#h5p-44>

Practice Problems

Short Answer Reflections



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=155#h5p-45>

Test Your Knowledge



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=155#h5p-46>

License & Attribution

“T-Tests” by Stephanie D’Costa is adapted from [“Some Basic Null Hypothesis Tests”](#) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler, and Dana C. Leighton is licensed CC BY 3.0; [“One Sample t-tests”](#) by Lynette H. Bikos is licensed CC BY-NC-SA 4.0;

Table 32.2a and b are adapted from [“Table t”](#) by Jsmura/Wikimedia Commons, [CC BY-SA 4.0](#).

“T-Tests” is licensed under CC BY-NC-SA 4.0.

Image Descriptions

Figure 32.1. A bell-shaped normal distribution curve on a beige/cream background. The horizontal axis is labeled from -3 to 3, representing standard deviations from the mean. Two vertical red lines mark the critical values at approximately -2 and +2 on the x-axis, defining the rejection regions in both tails of the distribution. These shaded tail regions represent

the critical areas for a two-tailed hypothesis test. The curve is symmetrical around zero, with the peak at the center. [[Return to Figure 32.1](#)]

Figure 32.2 A graph comparing four distribution curves centered at zero with x-axis ranging from -3 to 3. The curves shown are: 1) A solid pink/magenta line representing the standard normal (z) distribution; 2) A blue dotted line showing the t -distribution with $df = 30$, which closely overlaps the z -distribution; 3) A red dashed line showing the t -distribution with $df = 10$, slightly flatter than the z -distribution; and 4) A navy blue dashed line showing the t -distribution with $df = 2$, noticeably flatter and with heavier tails. All curves are symmetrical and bell-shaped. As degrees of freedom increase, the t -distributions become more peaked and approach the normal distribution. The t -distribution with $df = 2$ has the flattest peak and the heaviest tails, while the t -distribution with $df = 30$ is nearly indistinguishable from the normal distribution. [[Return to Figure 32.2](#)]

Figure 32.3. A bell-shaped t -distribution curve centered at zero with x-axis labeled from -4 to 4. A vertical line marks the critical value at $t = 2.353$ on the right side of the distribution. The area to the right of this critical value (the right tail) is shaded in orange/red, representing the rejection region for a right-tailed hypothesis test. This shaded area corresponds to the alpha level (significance level) of the test. The curve is symmetrical around zero but only the right tail is highlighted as the critical region. [[Return to Figure 32.3](#)]

Figure 32.4. Statistical output showing results of a One Sample T-Test and descriptive statistics for Wait Time data. The One Sample T-Test table shows: Variable: Wait Time, t-statistic: 3.462, degrees of freedom (df): 3, p-value: 0.020. A note indicates that for the Student t-test, the alternative hypothesis specifies that the mean is greater than 30, and identifies this as a Student's t-test. The Descriptives table shows: Variable: Wait Time, Sample size (N): 4, Mean: 53.750, Standard deviation (SD): 13.720. With a p-value of 0.020 (less than the typical significance level of 0.05), there is statistically significant evidence that the mean wait time is greater than 30 minutes. [[Return to Figure 32.4](#)]

Media Attributions

- [Rejection Region t2.353](#) © [Judy Schmitt](#) is licensed under a [CC BY-NC-SA \(Attribution NonCommercial ShareAlike\)](#) license

33.

ANALYSIS OF VARIANCE

Rebecca Anguiano

Learning Objectives

By the end of this chapter, you will be able to

1. Identify when to use an Analysis of Variance (ANOVA) vs. a T-test.
2. Calculate ANOVAs.
3. Discriminate between statistical significance and failure to reject the null hypothesis.
4. Apply a critical lens to the use and interpretation of ANOVAs.

Analysis of variance (ANOVA) serves the same purpose as the t -tests we learned in the chapter of t -tests: it tests for

differences in group means. ANOVA is more flexible in that it can handle any number of groups, unlike t -tests, which are limited to two groups (independent samples) or two time points (dependent samples). Thus, the purpose and interpretation of ANOVA will be the same as it was for t -tests, as will the hypothesis-testing procedure. However, ANOVA will, at first glance, look much different from a mathematical perspective, although, as we will see, the basic logic behind the test statistic for ANOVA is actually the same. This chapter will describe the general design of ANOVA, with a focus on calculating the independent samples one-way ANOVA, which is an extension of the independent samples t -test, where three or more different groups are compared on a single independent (or grouping) variable.

Observing and Interpreting Variability

We have seen time and again that scores, be they individual data or group means, will differ naturally. Sometimes this is due to random chance, and other times it is due to actual differences. Our job as scientists, researchers, and data analysts is to determine if the observed differences are systematic and meaningful (via a hypothesis test) and, if so, what is causing those differences. Through this, it becomes clear that, although we are usually interested in the mean or average score, it is the variability in the scores that is key.

Take a look at Figure 33.1, which shows scores for many people on a test of skill used as part of a job application. The x-axis has each individual person, in no particular order, and the y-axis contains the score each person received on the test. As we can see, the job applicants differed quite a bit in their performance, and understanding why that is the case would be extremely useful information. However, there’s no interpretable pattern in the data, especially because we only have information on the test, not on any other variable (remember that the x-axis here only shows individual people and is not ordered or interpretable).

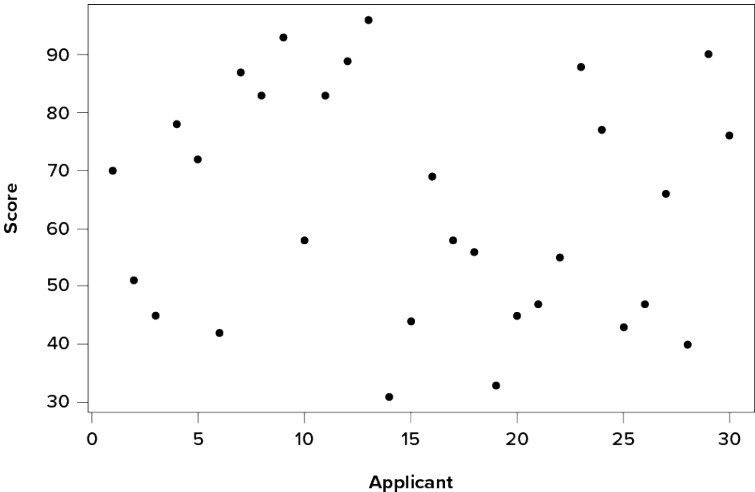


Figure 33.1. Scores on a Job Test. [\[Image Description\]](#)

Our goal is to explain this variability that we are seeing in the dataset. Let’s assume that, as part of the job application procedure, we also collected data on the highest degree each

applicant earned. With knowledge of what the job requires, we could sort our applicants into three groups: applicants who have a college degree related to the job, applicants who have a college degree that is not related to the job, and applicants who did not earn a college degree. This is a common way that job applicants are sorted, and we can use ANOVA to test if these groups are actually different. Figure 33.2 presents the same job applicant scores, but now they are color coded by group membership (i.e., which group they belong in). Now that we can differentiate between applicants this way, a pattern starts to emerge: applicants with a relevant degree (coded red) tend to be near the top, applicants with no college degree (coded black) tend to be near the bottom, and applicants with an unrelated degree (coded green) tend to fall into the middle. However, even within these groups, there is still some variability, as shown in Figure 33.2.

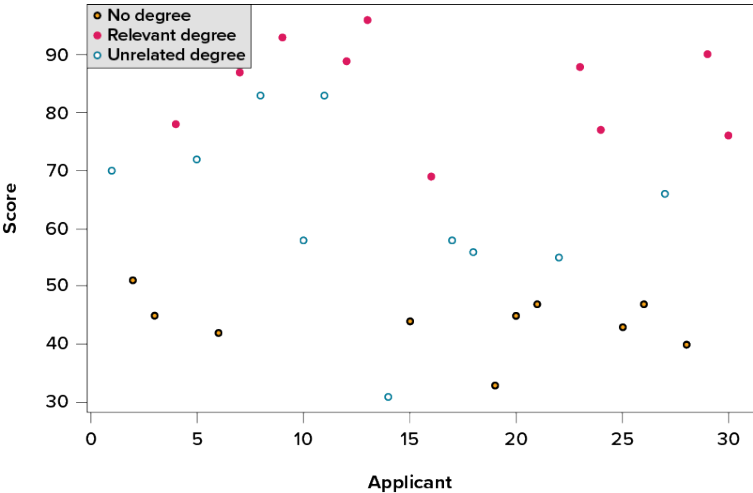


Figure 33.2. Applicant Scores Coded by Degree Earned.
[\[Image Description\]](#)

This pattern is even easier to see when the applicants are sorted and organized into their respective groups, as shown in Figure 33.3.

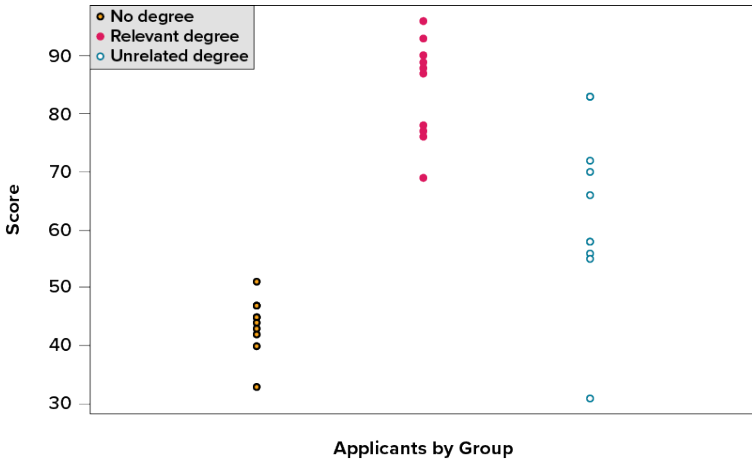


Figure 33.3. Applicant Scores by Group. [\[Image Description\]](#)

Now that we have our data visualized into an easily interpretable format, we can clearly see that our applicants' scores differ largely along group lines. Those applicants who do not have a college degree received the lowest scores, those who had a degree relevant to the job received the highest scores, and those who did have a degree but one that is not related to the job tended to fall somewhere in the middle. Thus, we have systematic variability between our groups.

We can also clearly see that within each group, our applicants' scores differed from one another. Those applicants without a degree tended to score very similarly, since the scores are clustered close together. Our group of applicants with relevant degrees varied a little bit more than that, and our group of applicants with unrelated degrees varied quite a bit. It may be that there are other factors that cause the observed

score differences within each group, or they could just be due to random chance. Because we do not have any other explanatory data in our dataset, the variability we observe within our groups is considered random error, with any deviations between a person and that person's group mean caused only by chance. Thus, we have unsystematic (random) variability within our groups.

The process and analyses used in ANOVA will take these two sources of variability (systematic variability between groups and random error within groups, or how much groups differ from each other and how much people differ within each group) and compare them to one another to determine if the groups have any explanatory value in our outcome variable. By doing this, we will test for statistically significant differences between the group means, just like we did for t -tests. We will go step by step to break down the math to see how ANOVA actually works.

Critical Perspective: Variability, Structure, and Inequality

While we often describe within-group variability as “random error,” it is important to recognize that this variability may not be random in a social sense. Differences within groups can reflect unequal access to resources, varied life experiences, exposure to discrimination, or institutional constraints that are not measured in the dataset. When such factors are

unobserved, they are statistically treated as error, even though they may be systematic and socially patterned.

Similarly, between-groups variability should not be interpreted as evidence that group membership itself causes differences. In many applied contexts, group membership reflects structural position rather than inherent characteristics. ANOVA allows us to observe patterns in variability, but it does not explain *why* those patterns exist. A critical interpretation requires researchers to consider what social, economic, or institutional forces may be producing the observed differences

Sources of Variability

ANOVA is all about looking at the different sources of variability (i.e., the reasons that scores differ from one another) in a dataset. Fortunately, the way we calculate these sources of variability takes a very familiar form: the sum of squares. Before we get into the calculations themselves, we must first lay out some important terminology and notation.

In ANOVA, we are working with two variables: a grouping or explanatory variable and a continuous outcome variable. The grouping variable is our predictor (it predicts or explains the values in the outcome variable) or, in experimental terms, our independent variable, and is made up of k groups, with k being any whole number 2 or greater. That is, ANOVA requires two or more groups to work, and it is usually conducted with three or more. In ANOVA, we refer to groups

as levels, so the number of levels is just the number of groups, which again is k . In the above example, our grouping variable was education, which had 3 levels, so $k = 3$. When we report any descriptive value (e.g., mean, sample size, standard deviation) for a specific group, we will use a subscript 1... k to denote which group it refers to. For example, if we have three groups and want to report the standard deviation s for each group, we would report them as s_1 , s_2 , and s_3 .

Our second variable is our outcome variable. This is the variable on which people differ, and we are trying to explain or account for those differences based on group membership. In the example above, our outcome was the score each person earned on the test. Our outcome variable will still use X for scores as before. When describing the outcome variable using means, we will use subscripts to refer to specific, individual group means. So, if we have $k = 3$ groups, our means will be **M_1** , **M_2** , and **M_3** . We will also have a single mean representing the average of all participants across all groups. This is known as the grand mean, and we use the symbol **M_G** . These different means—the individual group means and the overall grand mean—will be how we calculate our sums of squares.

Finally, we now have to differentiate between several different sample sizes. Our data will now have sample sizes for each group, and we will denote these with a lowercase n and a subscript, just like with our other descriptive statistics: n_1 , n_2 , and n_3 . We also have the overall sample size in our dataset, and

we will denote this with a capital N. The total sample size is just the group sample sizes added together.

Between-Groups Sum of Squares

One source of variability we identified in Figure 33.3 of the above example was differences or variability between the groups. That is, the groups clearly had different average levels. The variability arising from these differences is known as between-groups variability, and between-groups sum of squares is used to calculate between-groups variability.

Our calculations for sums of squares in ANOVA will take on the same form as they did for regular calculations of variance. Each observation, in this case the group means, is compared to the overall mean, in this case the grand mean, to calculate a deviation score. These deviation scores are squared so that they do not cancel each other out and sum to zero. The squared deviations are then added up, or summed. There is, however, one small difference. Because each group mean represents a group composed of multiple people, before we sum the deviation scores, we must multiply them by the number of people within that group. Incorporating this, we find our equation for the between-groups sum of squares to be:

$$SS_B = \sum n_j(M_j - M_G)^2$$

The subscript j refers to the “ j^{th} ” group where $j = 1 \dots k$ to keep track of which group mean and sample size we are

working with. As you can see, the only difference between this equation and the familiar sum of squares for variance is that we are adding in the sample size. Everything else logically fits together in the same way.

Within-Groups Sum of Squares

The other source of variability in the figures—within-groups variability—comes from differences that occur within each group. That is, each individual deviates a little bit from their respective group mean, just like the group means differed from the grand mean. We therefore label this source the within-groups variance. Because we are trying to account for variance based on group-level means, any deviation from the group means indicates an inaccuracy or error. Thus, our within-groups variability represents our error in ANOVA.

The formula for this sum of squares is again going to take on the same form and logic. What we are looking for is the distance between each individual person and the mean of the group to which they belong. We calculate this deviation score, square it so that they can be added together, then sum all of them into one overall value:

$$SS_W = \sum (X_{ij} - M_j)^2$$

In this instance, because we are calculating this deviation score for each individual person, there is no need to multiply by how many people we have. The subscript j again represents a group, and the subscript i refers to a specific person. So, X_{ij}

is read as “the i^{th} person of the j^{th} group.” It is important to remember that the deviation score for each person is only calculated relative to their group mean; do not calculate these scores relative to the other group means.

Total Sum of Squares

The total sum of squares can also be computed as a check for our calculations of between-groups and within-groups sums of squares. The calculation for this score is exactly the same as it would be if we were calculating the overall variance in the dataset (because that’s what we are interested in explaining) without worrying about or even knowing about the groups into which our scores fall:

$$SS_T = \sum (X_i - M_G)^2$$

We can see that our total sum of squares is just each individual score minus the grand mean. As with our within-groups sum of squares, we are calculating a deviation score for each individual person, so we do not need to multiply anything by the sample size; that is only done for a between-groups sum of squares.

An important feature of the sums of squares in ANOVA is that they all fit together. We could work through the algebra to demonstrate that if we added together the formulas for SS_B and SS_W , we would end up with the formula for SS_T . That is:

$$SS_T = SS_B + SS_W$$

This will prove to be very convenient, because if we know

the values of any two of our sums of squares, it is very quick and easy to find the value of the third. It is also a good way to check calculations: if you calculate each SS by hand, you can make sure that they all fit together as shown above, and if not, you know that you made a math mistake somewhere.

We can see from the above formulas that calculating an ANOVA by hand from raw data can take a very, very long time. For this reason, you will not be required to calculate the SS values by hand, but you should still take the time to understand how they fit together and what each one represents to ensure you understand the analysis itself.

ANOVA Table

All of our sources of variability fit together in meaningful, interpretable ways, as we saw above, and the easiest way to show these relationships is to organize them in a table. The ANOVA table (Table 33.1) shows how we calculate the df, MS, and F values. The first column of the ANOVA table, labeled “Source,” indicates which of our sources of variability we are using: between groups (B), within groups (W), or total (T). The second column, labeled “SS,” contains our values for the sum of squared deviations, also known as the sum of squares, that we learned to calculate above.

Table 33.1. ANOVA Table.

Source	SS	df	MS	F
Between	SS_B	$k - 1$	$\frac{SS_B}{df_B}$	$\frac{MS_B}{MS_W}$
Within	SS_W	$N - k$	$\frac{SS_W}{df_W}$	
Total	SS_T	$N - 1$		

As noted previously, calculating these by hand takes too long, so the formulas are not presented in Table 33.1. However, remember that SS_T is the sum of SS_B and SS_W , in case you are only given two SS values and need to calculate the third.

The next column, labeled “df,” is our degrees of freedom. As with the sums of squares, there is a different df for each group, and the formulas are presented in the table. Total degrees of freedom is calculated by subtracting 1 from the overall sample size (N). (Remember, the capital N in the df calculations refers to the overall sample size, not a specific group sample size.) Notice that df_T , just like for total sums of squares, is the Between (df_B) and Within (df_W) rows added together. If you take $N - k + k - 1$, then the “ $- k$ ” and “ $+ k$ ” portions will cancel out, and you are left with $N - 1$. This is a convenient way to quickly check your calculations.

The third column, labeled “MS,” shows our mean squared deviation for each source of variance. A mean square is just

another way to say variability and is calculated by dividing the sum of squares by its corresponding degrees of freedom. Notice that we show this in the ANOVA table for the Between row and the Within row, but not for the Total row. There are two reasons for this. First, our Total mean square would just be the variance in the full dataset (put together the formulas to see this for yourself), so it would not be new information. Second, the mean square values for Between and Within would not add up to equal the Total mean square because they are divided by different denominators. This is in contrast to the first two columns, where the Total row was both the conceptual total (i.e., the overall variance and degrees of freedom) and the literal total of the other two rows.

The final column in the ANOVA table, labeled “F,” is our test statistic for ANOVA. The F statistic, just like a t - or z -statistic, is compared to a critical value to see whether we can reject or fail to reject a null hypothesis. Thus, although the calculations look different for ANOVA, we are still doing the same thing that we did in all of [Unit 2](#). We are simply using a new type of data to test our hypotheses. We will see what these hypotheses look like shortly, but first, we must take a moment to address why we are doing our calculations this way.

Equity and Interpretation of Variance Components

The ANOVA table presents variability as neatly partitioned into between-group and within-group components, but this

statistical clarity can obscure real-world complexity. For example, groups that experience marginalization often show greater within-group variability due to uneven exposure to risk, opportunity, or support. Treating this variability as mere “error” can unintentionally minimize the lived consequences of inequality.

At the same time, emphasizing between-group differences without contextual interpretation risks reinforcing deficit-based narratives. The ANOVA table organizes information efficiently, but it does not—and cannot—determine how that information should be interpreted or used.

ANOVA and Type I Error

You may be wondering why we do not just use another t -test to test our hypotheses about three or more groups, the way we did in [Unit 2](#). After all, we are still just looking at group mean differences. The reason is that our t -statistic formula can only handle up to two groups, one minus the other. With only two groups, we can move our population parameters for the group means around in our null hypothesis and still get the same interpretation: the means are equal, which can also be concluded if one mean minus the other mean is equal to zero. However, if we tried adding a third mean, we would no longer be able to do this. So, in order to use t -tests to compare three or more means, we would have to run a series of individual group comparisons.

For only three groups, we would have three t -tests: Group 1 vs. Group 2, Group 1 vs. Group 3, and Group 2 vs. Group 3. This may not sound like a lot, especially with the advances in technology that have made running an analysis very fast, but it quickly scales up. With just one additional group, bringing our total to four, we would have six comparisons: Group 1 vs. Group 2, Group 1 vs. Group 3, Group 1 vs. Group 4, Group 2 vs. Group 3, Group 2 vs. Group 4, and Group 3 vs. Group 4. This makes for a logistical and computational nightmare for five or more groups.

A bigger issue, however, is our probability of committing a Type I error. Remember that a Type I error is a false positive, and the chance of committing a Type I error is equal to our significance level, α . This is true if we are only running a single analysis (such as a t -test with only two groups) on a single dataset. However, when we start running multiple analyses on the same dataset, our Type I error rate increases, raising the probability that we are capitalizing on random chance and rejecting a null hypothesis when we should not. ANOVA, by comparing all groups simultaneously with a single analysis, averts this issue and keeps our error rate at the α we set.

Hypotheses in ANOVA

So far, we have seen what ANOVA is used for, why we use it, and how we use it. Now we can turn to the formal hypotheses we will be testing. As with before, we have a null and an

alternative hypothesis to lay out. Our null hypothesis is still the idea of “no difference” in our data. Because we have multiple group means, we simply list them out as equal to each other:

H_0 : There is no difference in the group means

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

We list as many μ parameters as groups we have. In the example above, we have three groups to test, so we have three parameters in our null hypothesis. If we had more groups, say, four, we would simply add another μ to the list and give it the appropriate subscript, giving us:

H_0 : There is no difference in the group means

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

Notice that we do not say that the means are all equal to zero; we only say that they are equal to one another. It does not matter what the actual value is, so long as it holds for all groups equally.

Our alternative hypothesis for ANOVA is a little bit different. Let’s take a look at it and then dive deeper into what it means:

H_A : At least one mean is different

The first difference is obvious: there is no mathematical statement of the alternative hypothesis in ANOVA. This is due to the second difference: we are not saying which group is going to be different, only that at least one will be. Because we do not hypothesize about which mean will be different, there is no way to write it mathematically. Similarly, we do not have directional hypotheses (greater than or less than) like we did in

the previous chapter. Due to this, our alternative hypothesis is always exactly the same: at least one mean is different.

In the previous chapter, we saw that if we reject the null hypothesis, we can adopt the alternative, and this made it easy to understand what the differences looked like. In ANOVA, we will still adopt the alternative hypothesis as the best explanation of our data if we reject the null hypothesis. However, when we look at the alternative hypothesis, we can see that it does not give us much information. We will know that a difference exists somewhere, but we will not know where that difference is. Is only Group 1 different, but Groups 2 and 3 are the same? Is only Group 2 different? Are all three of them different? Based on just our alternative hypothesis, there is no way to be sure. We will come back to this issue later and see how to find out specific differences. For now, just remember that we are testing for any difference in group means, and it does not matter where that difference occurs.

Critical Perspective: What the Hypotheses Do Not Say

The null hypothesis in ANOVA asserts that group means are equal, while the alternative hypothesis states that at least one mean differs. Notably absent from both hypotheses is any explanation for *why* differences might exist. This absence is deliberate from a statistical standpoint, but it creates space for misinterpretation.

When researchers or practitioners implicitly attribute

differences to personal ability, motivation, or merit—without evidence—they move beyond what ANOVA can justify. A critical approach requires resisting the urge to fill explanatory gaps with assumptions that may reflect bias rather than data.

Now that we have our hypotheses for ANOVA, let's work through an example. We will continue to use the data from Figure 33.1, Figure 33.2, and Figure 33.3 for continuity.

Example Scores on Job-Application Tests

Our data come from three groups of 10 people each, all of whom applied for a single job opening: those with no college degree, those with a college degree that is not related to the job opening, and those with a college degree from a relevant field. We want to know if we can use this group membership to account for our observed variability and, by doing so, test if there is a difference between our three group means. We will start, as always, with our hypotheses.

Step 1: State the Hypotheses

Our hypotheses are concerned with the means of groups based on education level, so:

H_0 : There is no difference between the means of the education groups

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

H_A : At least one mean is different

Again, we phrase our null hypothesis in terms of what we are actually testing, and we use a number of population parameters equal to our number of groups. Our alternative hypothesis is always exactly the same.

Step 2: Find the Critical Values

Our test statistic for ANOVA, as we saw above, is F. Because we are using a new test statistic, we will get a new table: the F distribution table, a portion of which is shown in Table 33.2. (The complete F table can be found in [Appendix C](#).)

Table 33.2. Critical Values for F (F Table) with df Denominator (Within) Vertical

df: Denominator (Within)(vertical) df Numerator (Between) (horizontal)	1	2	3	4	5	6	7	8
1	161	200	216	225	230	234	237	239
2	18.51	19.00	19.16	19.25	19.30	19.33	19.36	19.37
3	10.13	9.55	9.28	9.12	9.01	8.94	8.88	8.84
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23

df: Denominator (Within)(vertical) df Numerator (Between) (horizontal)	1	2	3	4	5	6	7	8
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95
12	4.75	3.88	3.49	3.26	3.11	3.00	2.92	2.85
13	4.67	3.80	3.41	3.18	3.02	2.92	2.84	2.77
14	4.60	3.74	3.34	3.11	2.96	2.85	2.77	2.70
15	4.54	3.68	3.29	3.06	2.90	2.79	2.70	2.64
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59
17	4.45	3.59	3.20	2.96	2.81	2.70	2.62	2.55
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51

df: Denominator (Within)(vertical) df Numerator (Between) (horizontal)	1	2	3	4	5	6	7	8
19	4.38	3.52	3.13	2.90	2.74	2.63	2.55	2.48
20	4.35	3.49	3.10	2.87	2.71	2.60	2.52	2.45
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42
22	4.30	3.44	3.05	2.82	2.66	2.55	2.47	2.40
23	4.28	3.42	3.03	2.80	2.64	2.53	2.45	2.38
24	4.26	3.40	3.01	2.78	2.62	2.51	2.43	2.36
25	4.24	3.38	2.99	2.76	2.60	2.49	2.41	2.34
26	4.22	3.37	2.98	2.74	2.59	2.47	2.39	2.32
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.30

The F table only displays critical values for $\alpha = .05$.

This is because other significance levels are uncommon, and so it is not worth it to use up the space to present them. There are now two degrees of freedom we must use to find our critical value: the numerator and the denominator. These correspond to the numerator and denominator of our test statistic, which, if you look at the ANOVA table presented earlier (Table 33.1), are our Between and Within rows, respectively. The df_B is the “df: Numerator (Between)” because it is the degrees of freedom value used to calculate the Mean Square Between, which in turn is the numerator of our F statistic. Likewise, the df_W is the “df: Denominator (Within)” because it is the degrees of freedom value used to calculate the Mean Square Within, which is our denominator for F.

The formula for df_B is $k - 1$; remember that k is the number of groups we are assessing. In this example, $k = 3$ so our $df_B = 2$. This tells us that we will use the second column, the one labeled 2, to find our critical value. To find the proper row, we simply calculate the df_W , which was $N - k$. The original prompt told us that we have “three groups of 10 people each,” so our total sample size is 30. This makes our value for $df_W = 27$. If we follow the second column down to the row for 27, we find that our critical value is 3.35.

We use this critical value the same way as we did before: it is our criterion against which we will compare our obtained test statistic to determine statistical significance.

Step 3: Calculate the Test Statistic and Effect Size

Now that we have our hypotheses and the criteria we will use to test them, we can calculate our test statistic. To do this, we will fill in the ANOVA table, working our way from left to right and filling in each cell to get our final answer. We will assume that we are given the SS values as shown below:

Table 33.3. Job-Application Tests And

Source	SS		df
Between	8246		
Within	3020		
Total			

These may seem like random numbers, but

remember that they are based on the distances between the groups themselves and within each group. Figure 33.4 shows the plot of the data with the group means and grand mean included. If we wanted to, we could use this information, combined with our earlier information that each group has 10 people, to calculate the between-groups sum of squares by hand. However, doing so would take some time, and without the specific values of the data points, we would not be able to calculate our within-groups sum of squares, so we will trust that these values are the correct ones.

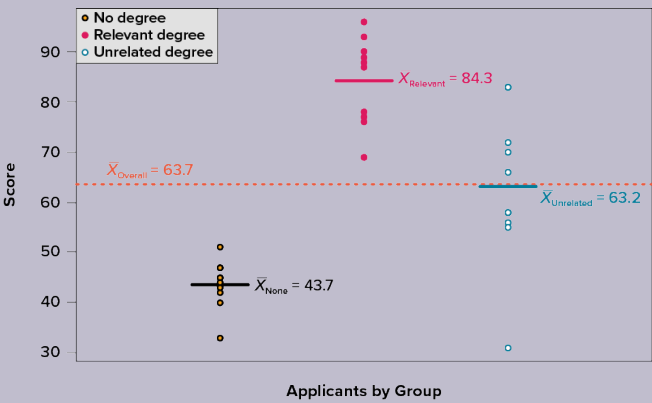


Figure 33.4. Means. [\[Image Description\]](#)

We were given the sums of squares values for our

first two rows, so we can use those to calculate the total sum of squares.

Table 33.3. Job-Application Tests An

Source	SS		c
Between	8246		
Within	3020		
Total	11266		

We also calculated our degrees of freedom earlier, so we can fill in those values. Additionally, we know that the total degrees of freedom is $N - 1$, which is 29. This value of 29 is also the sum of the other two degrees of freedom, so everything checks out.

Table 33.4 Job-Application Tests Anova C

Source	SS	df	MS	F
Between	8246	2		
Within	3020	27		
Total	11266	29		

Now we have everything we need to calculate our mean squares. Our MS values for each row are just the SS divided by the df for that row, giving us:

Table 33.5 Job-Application Tests Anova C

Source	SS	df
Between	8246	2
Within	3020	27
Total	11266	29

Remember that we do not calculate a Total Mean

Square, so we leave that cell blank. Finally, we have the information we need to calculate our test statistic. F is our MS_B divided by MS_W .

Table 33.6 Job-Application Tests And

Source	SS	df		MS
Between	8246	2		4123
Within	3020	27		111.85
Total	11266	29		

So, working our way through the table, given only two SS values and the sample size and group size from before, we calculate our test statistic to be $F_{\text{obt}} = 36.86$, which we will compare to the critical value in Step 4.

Effect Size: Variance Explained

Recall that the purpose of ANOVA is to take observed variability and see if we can explain those differences based on group membership. To that end, our effect size will be just that: the variance explained. You can think of variance explained as the

proportion or percent of the differences we are able to account for based on our groups. We know that the overall observed differences are quantified as the total sum of squares, and that our observed effect of group membership is the between-groups sum of squares. Our effect size, therefore, is the ratio of these two sums of squares. Specifically:

$$\eta^2 = \frac{SS_B}{SS_T}$$

The effect size η^2 is called “eta-squared” and represents variance explained. For our example, our values give an effect size of:

$$\eta^2 = \frac{8246}{11266} = .73$$

So, we are able to explain 73% of the variance in job-test scores based on education. This is, in fact, a huge effect size, and most of the time we will not explain nearly that much variance. Our guidelines for the size of our effects are:

Table 33.7. Job-Application Test Effect

η^2	Size
.01	Small
.09	Medium
.25	Large

So, we found that not only do we have a statistically significant result, but that our observed effect was very large! However, we still do not know specifically which groups are different from each other. It could be that they are all different, or that only those job seekers who have a relevant degree are different from the others, or that only those who have no degree are different from the others. To find out which is true, we need to do a special analysis called a post hoc test.

Step 4: Make the Decision

Our obtained test statistic was calculated to be $F_{obt} = 36.86$, and our critical value was found to be $F^* =$

3.35. Our obtained statistic is larger than our critical value, so we can reject the null hypothesis.

Reject H^0 . The results of the ANOVA indicated that there were significant differences in job skills test scores for applicants in each of the three education groups, and the effect size was large, $F(2, 27) = 36.86$, $p < .05$, $\eta^2 = .73$. Post hoc tests (see [the next section](#)) were performed to determine where the differences were.

Notice that when we report F , we include both degrees of freedom. We always report the numerator and then the denominator, separated by a comma. We must also note that, because we were only testing for any difference, we cannot yet conclude which groups are different from the others. To do so, we need to perform a post hoc test.

Critical Perspective: Explained Variance and Moral Weight

Explaining a large proportion of variance does not imply moral justification for differential treatment. Even when group membership accounts for a substantial share of outcome differences, those differences may reflect historical

disadvantage, exclusion, or unequal opportunity rather than meaningful distinctions in ability or worth.

Effect sizes quantify the strength of an association, not its legitimacy. Researchers should therefore avoid presenting large effect sizes as self-evident reasons for stratification, selection, or exclusion in applied settings.

Post Hoc Tests

A post hoc test is used only after we find a statistically significant result and need to determine where our differences truly came from. The term post hoc comes from the Latin for “after the event.” Many different post hoc tests have been developed, and most of them will give us similar answers. We will only focus here on the most commonly used ones. We will also only discuss the concepts behind each and will not worry about calculations.

Bonferroni Test

A Bonferroni test is perhaps the simplest post hoc analysis. A Bonferroni test is a series of t -tests performed on each pair of groups. As we discussed earlier, the number of groups quickly increases the number of comparisons, which inflates Type I error rates. To avoid this, a Bonferroni test divides our significance level α by the number of comparisons we are making so that when they are all run, they sum back up to our

original Type I error rate. Once we have our new significance level, we simply run independent samples t -tests to look for differences between our pairs of groups. This adjustment is sometimes called a Bonferroni Correction, and it is easy to do by hand if we want to compare obtained p -values to our new corrected α level, but it is more difficult to do when using critical values like we do for our analyses, so we will leave our discussion of it to that.

Tukey's Honestly Significant Difference

Tukey's honestly significant difference (HSD) is a popular post hoc analysis that, like Bonferroni's, makes adjustments based on the number of comparisons; however, it makes adjustments to the test statistic when running the comparisons of two groups. These comparisons give us an estimate of the difference between the groups and a confidence interval for the estimate. We use this confidence interval in the same way we use a confidence interval for a regular independent samples t -test: if it contains 0.00, the groups are not different, but if it does not contain 0.00, then the groups are different.

Below are the differences between the group means and the Tukey's HSD confidence intervals for the differences:

Table 33.8. Differences Between Group Means and Tukey’s H

Comparison	Difference
None vs. relevant	40.60
None vs. unrelated	19.50
Relevant vs. unrelated	21.10

As we can see, none of these intervals contain 0.00, so we can conclude that all three groups are different from one another.

Scheffé Test

Another common post hoc test is the Scheffé test. Like Tukey’s HSD, the Scheffé test adjusts the test statistic for how many comparisons are made, but it does so in a slightly different way. The result is a test that is “conservative,” which means that it is less likely to commit a Type I error, but this comes at the cost of less power to detect effects. We can see this by looking at the confidence intervals that the Scheffé test gives us:

Table 33.9. Differences Between Group Means and

Comparison	Difference
None vs. relevant	40.60
None vs. unrelated	19.50
Relevant vs. unrelated	21.10

As we can see, these are slightly wider than the intervals we got from Tukey’s HSD. This means that, all other things being equal, they are more likely to contain zero. In our case, however, the results are the same, and we again conclude that all three groups differ from one another.

There are many more post hoc tests than just these three, and they all approach the task in different ways, with some being more conservative and others being more powerful. In general, though, they will give highly similar answers. What is important here is to be able to interpret a post hoc analysis. If you are given post hoc analysis confidence intervals, like the ones seen above, read them the same way we read confidence intervals in [Chapter 32](#). If they contain zero, there is no difference; if they do not contain zero, there is a difference.

Other ANOVA Designs

We have only just scratched the surface of ANOVA in this

chapter. There are many other variations available for the one-way ANOVA presented here. There are also other types of ANOVAs that you are likely to encounter. The first is called a factorial ANOVA. A factorial ANOVA uses multiple grouping variables, not just one, to look for group mean differences. Just as there is no limit to the number of groups in a one-way ANOVA, there is no limit to the number of grouping variables in a factorial ANOVA, but it becomes very difficult to find and interpret significant results with many factors, so usually they are limited to two or three grouping variables with only a small number of groups in each.

Another ANOVA is called a repeated measures ANOVA. This is an extension of a related samples t -test, but in this case, we are measuring each person three or more times to look for a change. We can even combine both of these advanced ANOVAs into mixed designs to test very specific and valuable questions. These topics are far beyond the scope of this text, but you should know about their existence. Our treatment of ANOVA here is a small first step into a much larger world!

When more than one independent variable is included in a factorial design, the appropriate approach is the **factorial ANOVA**. Again, the basics of the factorial ANOVA are the same as for the one-way and repeated-measures ANOVAs. The main difference is that it produces an F ratio and p value for each main effect and for each interaction. Returning to our calorie estimation example, imagine that the health psychologist tests the effect of

participant major (psychology vs. nutrition) and food type (cookie vs. hamburger) in a factorial design. A factorial ANOVA would produce separate F ratios and p values for the main effect of major, the main effect of food type, and the interaction between major and food type. Appropriate modifications must be made depending on whether the design is between-subjects, within-subjects, or mixed.

More on Factorial ANOVAs

Factorial: a term used when there are two or more independent variables (IVs; the factors). The factors could be between-groups, within-groups, repeated measures, or a combination of between and within.

- Independent factorial design: several IVs (predictors/factors) and each has been measured using different participants (between groups).
- Related factorial design: several IVs (factors/predictors) have been measured, but the same participants have been used in all conditions (repeated measures or within-subjects).
- Mixed design: several IVs (factors/predictors) have been measured. One or more factors use different participants (between-subjects) and one or more factors use the same participants (within-subjects). Thus, there is a combination of independent (between) and related

(within or repeated) designs.

The moderator is what creates an interaction. Below are traditional representations of the statistical and conceptual figures of interaction effects. We will say that Factor B moderates the relationship between Factor A (the IV) and the DV.

In a later lesson, we will work on an ANCOVA, where we will distinguish between a moderator and a covariate. In lessons on regression models, you will likely be introduced to the notion of mediator.

Graphic representations of a moderated relationship?

Workflow for Two-Way ANOVA

The following is a proposed workflow for conducting a two-way ANOVA.

Steps of the Two-Way ANOVA Workflow

1. Enter data

- predictors should be formatted as as

factors (ordered or unordered); the dependent variable should be continuously scaled

- understanding the format of data can often provide clues as to which ANOVA/statistic to use

2. Explore data

- graph the data
- compute descriptive statistics
- evaluate distributional assumptions
- assess the homogeneity of variance assumption with Levene's test
- assess the normality assumption with the Shapiro Wilk test
- determine if there are outliers; if appropriate, delete

3. Compute the omnibus ANOVA

- depending on what you found in the data exploration phase, you may need to run a robust version of the test

4. Follow-up testing based on significant main or interaction effects

- significant interactions require test of simple main effects which could be further explored with contrasts, posthoc comparisons, and/or polynomials
- the exact methods you choose will depend upon the tests of assumptions during data exploration

5. Managing Type I error

ANOVA as a Tool, Not an Explanation

Across all its forms—one-way, factorial, repeated-measures, or mixed—ANOVA remains a tool for detecting differences in means, not a mechanism for explaining social reality. Used uncritically, ANOVA can lend statistical authority to unequal systems by presenting disparities as neutral facts. Used

critically, it can help document inequality and motivate deeper investigation into its causes.

Responsible use of ANOVA requires technical competence, contextual knowledge, and ethical reflection. Statistical significance alone is never sufficient grounds for interpretation or action.

Practice Problems

Short Answer Reflections



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=161#h5p-47>

Test Your Knowledge



An interactive H5P element has been excluded from this version of the text. You

can view it online here:

[https://smc.pressbooks.pub/
critresearchmethodpsych/?p=161#h5p-48](https://smc.pressbooks.pub/critresearchmethodpsych/?p=161#h5p-48)

License & Attribution

“Analysis of Variance” by Rebecca Anguiano is adapted from “[Analysis of Variance](#)” by Linda R. Cote Ph.D.; Rupa G. Gordon Ph.D., Chrislyn E. Randell Ph.D., Judy Schmitt, and Helena Marvin, which is licensed CC BY-NC-SA 4.0; “[Some Basic Null Hypothesis Tests](#)” by Rajiv S. Jhangiani; I-Chant A. Chiang; Carrie Cuttler; and Dana C. Leighton, which is licensed CC BY-NC-SA 4.0, and “[Factorial Anova](#)” by Lynette H. Bikos, which is licensed by CC BY-NC-SA.

“Analysis of Variance” is licensed under CC BY-NC-SA 4.0.

Image Descriptions

Figure 33.1. A scatter plot displaying test scores (y-axis, ranging from 30 to 100) for 30 individual applicants (x-axis, numbered 0 to 30). The data points are represented as black dots scattered across the plot area. Scores vary considerably, with the lowest score around 30 (applicant 19) and the highest scores around 97 (applicant 13). The distribution shows substantial variation with no apparent linear trend or pattern, suggesting scores are relatively independent of applicant number. Most scores cluster between 40 and 90, with a few notable outliers on both the lower and upper ends of the range. [[Return to Figure 33.1](#)]

Figure 33.2. A scatter plot displaying test scores (y-axis, ranging from 30 to 100) for 30 individual applicants (x-axis, numbered 0 to 30), color-coded by educational background. The legend indicates three categories: No degree (black dots), Relevant degree (magenta dots), and Unrelated degree (cyan dots). The data reveals a clear pattern related to education

level. Applicants with relevant degrees (magenta) consistently score highest, with scores ranging from approximately 69 to 97, clustering primarily in the 75-95 range. Applicants with unrelated degrees (cyan) show intermediate performance, with scores ranging from approximately 30 to 83, mostly falling between 55 and 83. Applicants with no degree (black) score lowest, with scores ranging from approximately 32 to 51, concentrated in the 40-51 range. This visualization suggests a strong positive correlation between having relevant educational credentials and test performance, with educational background appearing to be a significant predictor of score outcomes. [[Return to Figure 33.2](#)]

Figure 33.3. A scatter plot displaying test scores (y-axis, ranging from 30 to 100) organized by three educational groups (x-axis). The legend identifies three categories: No degree (black dots), Relevant degree (magenta dots), and Unrelated degree (cyan dots). The data points are vertically aligned within three distinct columns representing each education group. Applicants with relevant degrees (magenta, middle column) show the highest performance, with scores tightly clustered between approximately 69 and 97, with the majority concentrated in the 76-95 range. This group demonstrates the least variability and highest overall scores. Applicants with unrelated degrees (cyan, right column) display intermediate performance with greater variability, ranging from approximately 30 to 83. Most scores fall between 54 and 72, though there is one notable outlier at around 30. Applicants

with no degree (black, left column) show the lowest scores, ranging from approximately 32 to 51, with most scores clustered tightly between 40 and 51. This group exhibits the most consistent (but lowest) performance. The visualization clearly demonstrates a strong relationship between educational credentials and test performance, with relevant education providing the greatest advantage. [[Return to Figure 33.3](#)]

Figure 33.4. A scatter plot displaying test scores (y-axis, 30-100) organized by three educational groups (x-axis): No degree (black dots), Relevant degree (magenta dots), and Unrelated degree (cyan dots). Horizontal lines indicate mean scores for each group: relevant degree holders average 84.3 with scores clustered between 69-97, unrelated degree holders average 63.2 with scores ranging from 30-83, and applicants with no degree average 43.7 with scores between 32-51. A dashed line shows the overall mean of 63.7. The visualization demonstrates a clear performance hierarchy based on educational credentials, with a 40.6-point gap between the highest and lowest performing groups. [[Return to Figure 33.4](#)]

34.

CORRELATIONS

Makenzie O'Neil

Learning Objectives

By the end of this chapter, you will be able to

1. Create and interpret a scatterplot.
2. Calculate and interpret a Pearson's r .
3. Conduct and interpret null hypothesis tests of Pearson's r .
4. Identify the primary use of correlational analyses and important considerations of using the statistic.

Thus far, all of our analyses have focused on comparing the value of a continuous variable across different groups via mean differences. We will now turn away from means and look instead at how to assess the relationship between two continuous variables in the form of correlations. As we will see, the logic behind correlations is the same as it was behind group means, but we will now have the ability to assess an entirely new data structure.

Variability and Covariance

A common theme throughout statistics is the notion that individuals will differ on different characteristics and traits, which we call variance. In inferential statistics and hypothesis testing, our goal is to find systematic reasons for differences and rule out random chance as the cause. By doing this, we are using information on a different variable—which so far has been group membership, like in ANOVA—to explain this variance. In correlations, we will instead use a continuous variable to account for the variance.

Because we have two continuous variables, we will have two characteristics or scores on which people will vary. What we want to know is whether people vary on the scores together. That is, as one score changes, does the other score also change in a predictable or consistent way? This notion of variables differing together is called covariance (the prefix co- meaning “together”).

Let’s look at our formula for variance on a single variable:

$$s^2 = \frac{\sum (X - M)^2}{N - 1}$$

We use X to represent a person’s score on the variable at hand, and M to represent the mean of that variable. The numerator of this formula is the sum of squares, which we have seen several times for various uses. Recall that squaring a value is just multiplying that value by itself. Thus, we can write the same equation as:

$$s^2 = \frac{\sum((X - M)(X - M))}{N - 1}$$

This is the same formula and works the same way as before, where we multiply the deviation score by itself (we square it) and then sum across squared deviations.

Now, let's look at the formula for covariance. In this formula, we will still use X to represent the score on one variable, and we will now use Y to represent the score on the second variable. We will still use bars to represent averages of the scores. The formula for covariance (cov_{XY} with the subscript XY to indicate covariance across the X and Y variables) is:

$$\text{cov}_{XY} = \frac{\sum((X - M_X)(Y - M_Y))}{N - 1}$$

As we can see, this is the exact same structure as the previous formula. Now, instead of multiplying the deviation score by itself on one variable, we take the deviation scores from a single person on each variable and multiply them together. We do this for each person (exactly the same as we did for variance) and then sum them to get our numerator. The numerator in this is called the sum of products (SP).

$$SP = \sum((X - M_X)(Y - M_Y))$$

We will calculate the sum of products using the same table we used to calculate the sum of squares. In fact, the table for sum of products is simply a sum of squares table for X , plus a

sum of squares table for Y, with a final column of products, as shown below.

Table 34.1. Sum of Products Table

X	$(X - M_X)$	$(X - M_X)^2$	Y	$(Y - M_Y)$	$(Y - M_Y)^2$	$(X - M_X)(Y - M_Y)$

This table works the same way it did before (remember that the column headers tell you exactly what to do in that column). We list our raw data for the X and Y variables in the X and Y columns, respectively, then add them up so we can calculate the mean of each variable. We then take those means and subtract them from the appropriate raw score to get our deviation scores for each person on each variable, and the columns of deviation scores will both add up to zero. We will square our deviation scores for each variable to get the sums of squares for X and Y so that we can compute the variance and standard deviation of each (we will use the standard deviation in our equation below). Finally, we take the deviation score from each variable and multiply them together to get our product score. Summing this column will give us our sum of products. It is very important that you multiply the raw deviation scores from each variable, not the squared deviation scores.

Our sum of products will go into the numerator of our

formula for covariance, and then we only have to divide by $N - 1$ to get our covariance. Unlike the sum of squares, both our sum of products and our covariance can be positive, negative, or zero, and they will always match (e.g., if our sum of products is positive, our covariance will always be positive). A positive sum of products and covariance indicates that the two variables are related and move in the same direction. That is, as one variable goes up, the other will also go up, and vice versa. A negative sum of products and covariance means that the variables are related but move in opposite directions when they change, which is called an inverse relationship. In an inverse relationship, as one variable goes up, the other variable goes down. If the sum of products and covariance is zero, then that means the variables are not related. As one variable goes up or down, the other variable does not change in a consistent or predictable way.

The previous paragraph brings us to an important definition about relationships between variables. What we are looking for in a relationship is a consistent or predictable pattern. That is, the variables change together, either in the same direction or opposite directions, in the same way each time. It doesn't matter if this relationship is positive or negative, only that it is not zero. If there is no consistency in how the variables change within a person, then the relationship is zero and does not exist. We will revisit this notion of direction vs. zero relationship later on.

Visualizing Relationships

As discussed in an earlier chapter, visualizing data remains an important first step in understanding and describing our data before we move into inferential statistics. Nowhere is this more important than in correlation. Correlations are visualized by a scatter plot, where our X variable values are plotted on the x-axis, the Y variable values are plotted on the y-axis, and each point or marker in the plot represents a single person's score on X and Y. Figure 34.1 shows a scatter plot for hypothetical scores on job satisfaction (X) and worker well-being (Y). We can see from the axes that each of these variables is measured on a 10-point scale, with 10 being the highest on both variables (high satisfaction and good well-being) and 1 being the lowest (dissatisfaction and poor well-being). When we look at this plot, we can see that the variables do seem to be related. The higher scores on job satisfaction tend to also be the higher scores on well-being, and the same is true of the lower scores.

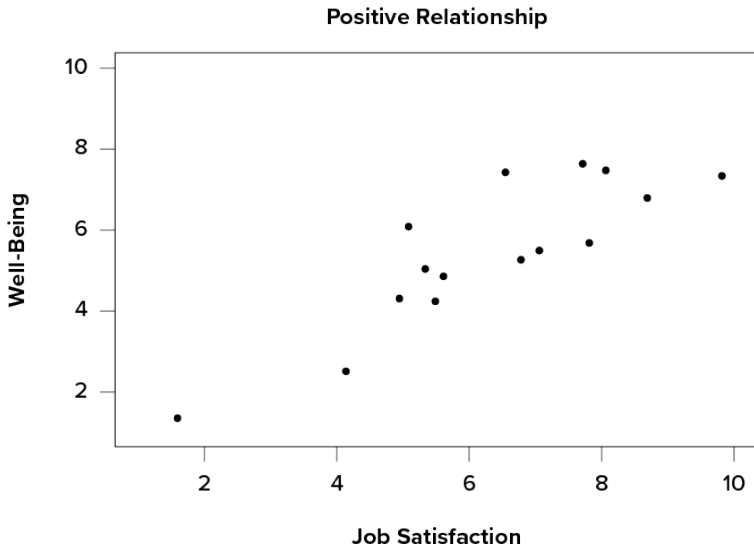


Figure 34.1. Plotting Job Satisfaction and Well-being Scores. [\[Image Description\]](#)

Figure 34.1 demonstrates a positive relationship. As scores on X increase, scores on Y also tend to increase. Although this is not a perfect relationship (if it were, the points would form a single straight line), it is nonetheless very clearly positive. This is one of the key benefits to scatter plots: they make it very easy to see the direction of the relationship. As another example, Figure 34.2 shows a negative relationship between job satisfaction (X) and burnout (Y). As we can see from this plot, higher scores on job satisfaction tend to correspond with lower scores on burnout, which is how stressed, unenergetic, and unhappy someone is at their job. As with Figure 34.1, this is not a perfect relationship, but it is still a clear one. As these

figures show, points in a positive relationship move from the bottom left of the plot to the top right, and points in a negative relationship move from the top left to the bottom right.

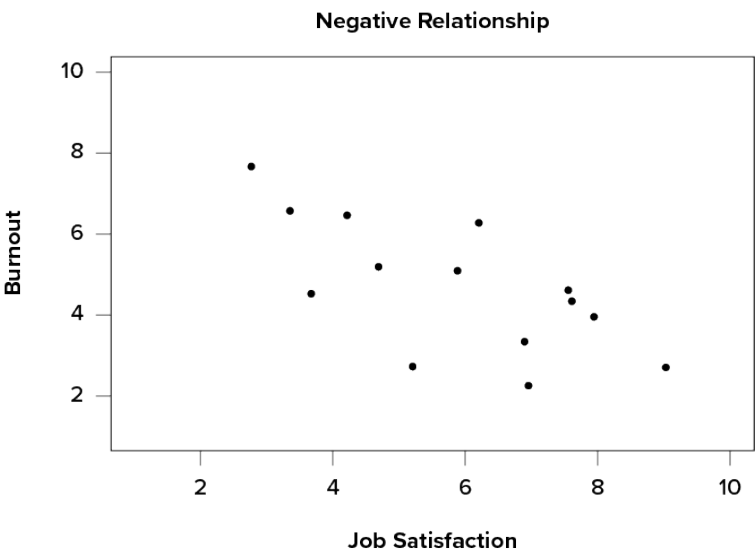


Figure 34.2. Plotting Job Satisfaction and Burnout Scores. [\[Image Description\]](#)

Scatter plots can also indicate that there is no relationship between the two variables. In these scatter plots (for example, Figure 34.3, which plots job satisfaction and job performance), there is no interpretable shape or line in the scatter plot. The points appear randomly throughout the plot. If we tried to draw a straight line through these points, it would basically be flat. The low scores on job satisfaction have roughly the same scores on job performance as do the high

scores on job satisfaction. Scores in the middle or average range of job satisfaction have some scores on job performance that are about equal to the high and low levels and some scores on job performance that are a little higher, but the overall picture is one of inconsistency.

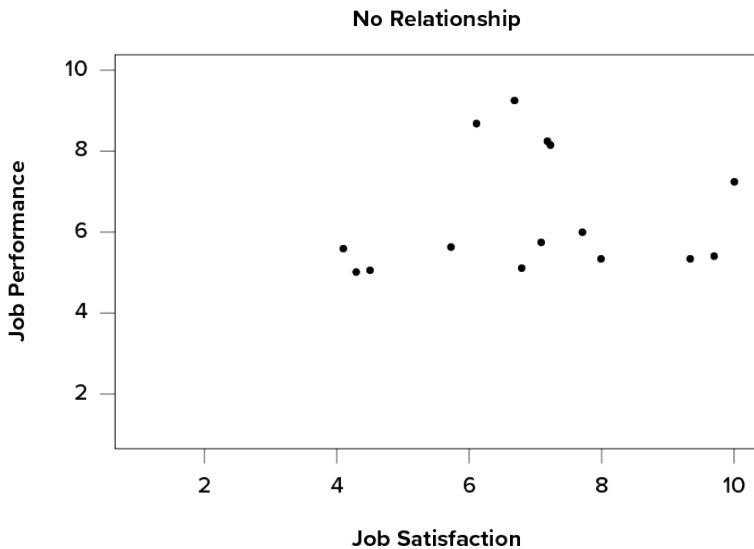


Figure 34.3. Plotting No Relationship Between Job Satisfaction and Job Performance. [\[Image Description\]](#)

As we can see, scatter plots are very useful for giving us an approximate idea of whether there is a relationship between the two variables and, if there is, whether that relationship is positive or negative. They are also the only way to determine one of the characteristics of correlations that are discussed next: form.

Three Characteristics

When we talk about correlations, there are three traits that we need to know in order to truly understand the relationship (or lack of relationship) between X and Y: **form**, **direction**, and **magnitude**. We will discuss each of them in turn.

Form

The first characteristic of relationships between variables is their form. The form of a relationship is the shape it takes in a scatter plot, and a scatter plot is the only way it is possible to assess the form of a relationship. There are three forms we look for: linear, curvilinear, or no relationship. A linear relationship is what we saw in Figure 34.1, Figure 34.2, and Figure 34.3. If we drew a line through the middle points in any of the scatter plots, we would be best suited with a straight line. The term linear comes from the word line. A linear relationship is what we will always assume when we calculate correlations. All of the correlations presented here are only valid for linear relationships. Thus, it is important to plot our data to make sure we meet this assumption.

The relationship between two variables can also be curvilinear. As the name suggests, a curvilinear relationship is one in which a line through the middle of the points in a

scatter plot will be curved rather than straight. Two examples are presented in Figure 34.4 and Figure 34.5.

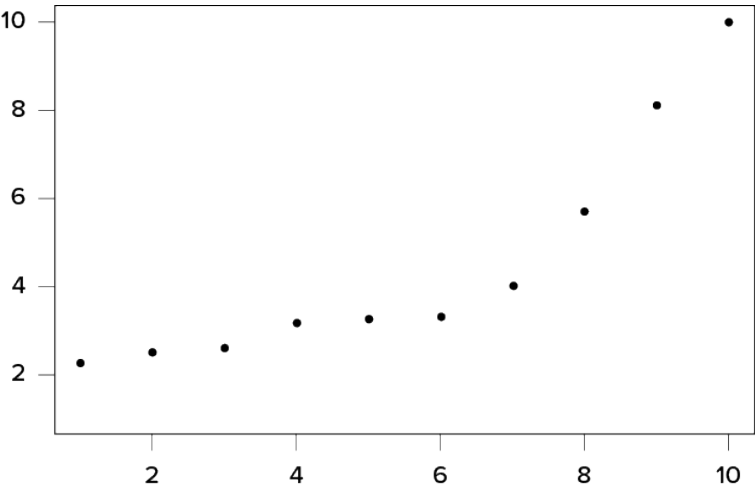


Figure 34.4. Exponentially Increasing Curvilinear Relationship. [\[Image Description\]](#)

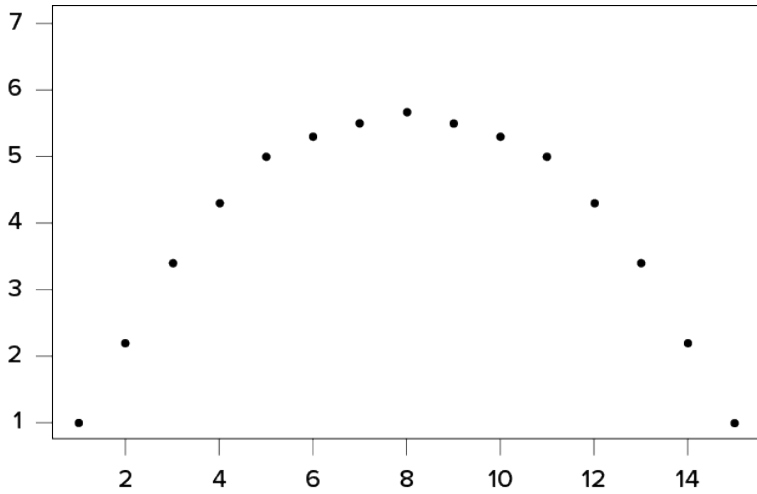


Figure 34.5. Inverted-U Curvilinear Relationship. [\[Image Description\]](#)

Curvilinear relationships can take many shapes, and the two examples above are only a small sample of the possibilities. What they have in common is that they both have a very clear pattern, but that pattern is not a straight line. If we try to draw a straight line through them, we would get a result similar to what is shown in Figure 34.6.

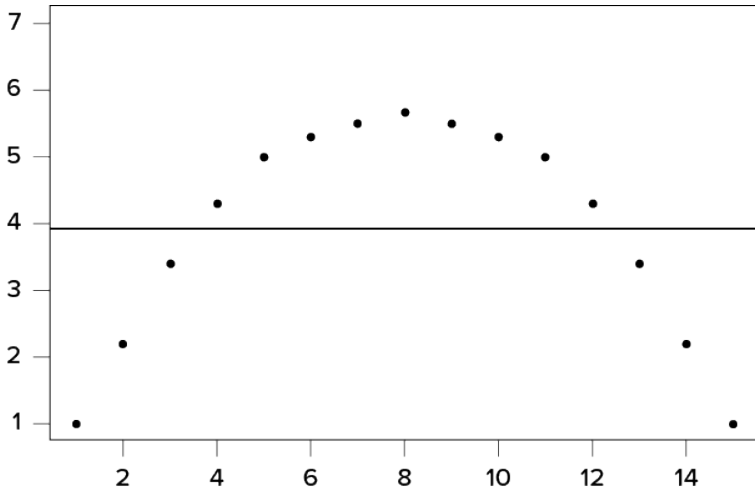


Figure 34.6. Overlaying a Straight Line on a Curvilinear Relationship. [\[Image Description\]](#)

Although that line is the closest it can be to all points at the same time, it clearly does a very poor job of representing the relationship we see. Additionally, the line itself is flat, suggesting there is no relationship between the two variables, even though the data show that there is one. This is important to keep in mind because the math behind our calculations of correlation coefficients will only ever produce a straight line—we cannot create a curved line with the techniques discussed here.

Finally, sometimes when we create a scatter plot, we end up with no interpretable relationship at all. An example of this is shown in Figure 34.7. The points in this plot show no

consistency in relationship, and a line through the middle would once again be a straight, flat line.

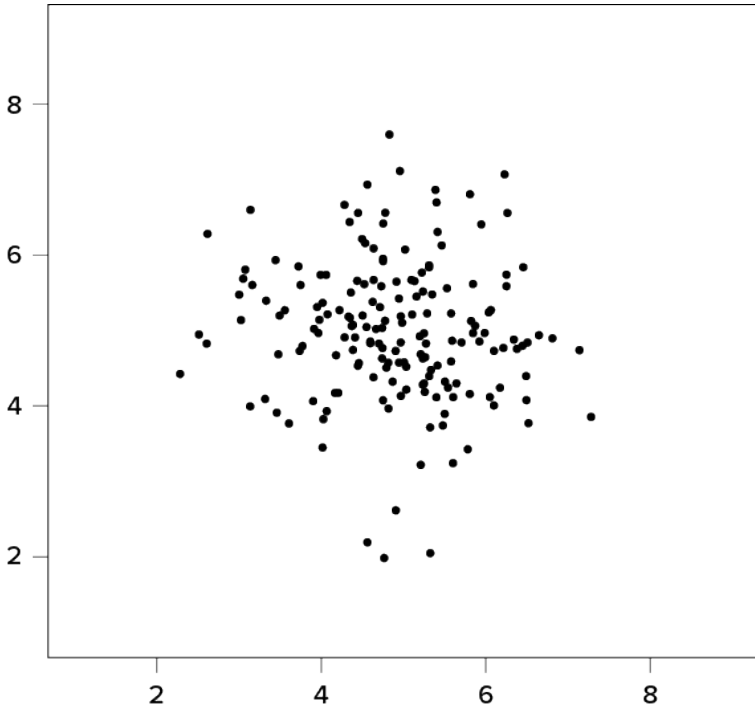


Figure 34.7. No Relationship. [\[Image Description\]](#)

Sometimes when we look at scatter plots, it is tempting to get biased by a few points that fall far away from the rest of the points and seem to imply that there may be some sort of relationship. These points are called outliers, and we will discuss them in more detail [later in the chapter](#). These can be

common, so it is important to formally test for a relationship between our variables, not just rely on visualization. This is the point of hypothesis testing with correlations, and we will go in-depth on it soon. First, however, we need to describe the other two characteristics of relationships: direction and magnitude.

Direction

The direction of the relationship between two variables tells us whether the variables change in the same way at the same time or in opposite ways at the same time. We saw this concept earlier when first discussing scatter plots, and we used the terms positive and negative. A positive relationship is one in which X and Y change in the same direction: as X goes up, Y goes up, and as X goes down, Y also goes down. A negative relationship is just the opposite: X and Y change together in opposite directions: as X goes up, Y goes down, and vice versa.

As we will see soon, when we calculate a correlation coefficient, we are quantifying the relationship demonstrated in a scatter plot. That is, we are putting a number to it. That number will be either positive, negative, or zero, and we interpret the sign of the number as our direction. If the number is positive, it is a positive relationship, and if it is negative, it is a negative relationship. If it is zero, then there is no relationship. The direction of the relationship corresponds directly to the slope of the hypothetical line we draw through

scatter plots when assessing the form of the relationship. If the line has a positive slope that moves from bottom left to top right, it is positive, and vice versa for negative. If the line is flat, that means it has no slope, and there is no relationship, which will in turn yield a zero for our correlation coefficient.

Magnitude

The number we calculate for our correlation coefficient, which we will describe in detail below, corresponds to the magnitude of the relationship between the two variables. The magnitude is how strong or how consistent the relationship between the variables is. Higher numbers mean greater magnitude, which means a stronger relationship.

Our correlation coefficients will take on any value between -1.00 and 1.00 , with 0.00 in the middle, which again represents no relationship. A correlation of -1.00 is a perfect negative relationship; as X goes up by some amount, Y goes down by the same amount, consistently. Likewise, a correlation of 1.00 indicates a perfect positive relationship; as X goes up by some amount, Y also goes up by the same amount. Finally, a correlation of 0.00 , which indicates no relationship, means that as X goes up by some amount, Y may or may not change by any amount, and it does so inconsistently.

The vast majority of correlations do not reach -1.00 or 1.00 . Instead, they fall in between, and we use rough cut-offs

for how strong the relationship is based on this number. Importantly, the sign of the number (the direction of the relationship) has no bearing on how strong the relationship is. The only thing that matters is the magnitude, or the absolute value of the correlation coefficient. A correlation of -1 is just as strong as a correlation of 1 . We generally use values of $.10$, $.30$, and $.50$ as indicating weak, moderate, and strong relationships, respectively.

The strength of a relationship, just like the form and direction, can also be inferred from a scatter plot, though this is much more difficult to do. Some examples of weak and strong relationships are shown in Figure 34.8 and Figure 34.9, respectively. Weak correlations still have an interpretable form and direction, but it is much harder to see. Strong correlations have a very clear pattern, and the points tend to form a line. The examples show two different directions, but remember that the direction does not matter for the strength, only the consistency of the relationship and the size of the number, which we will see next.

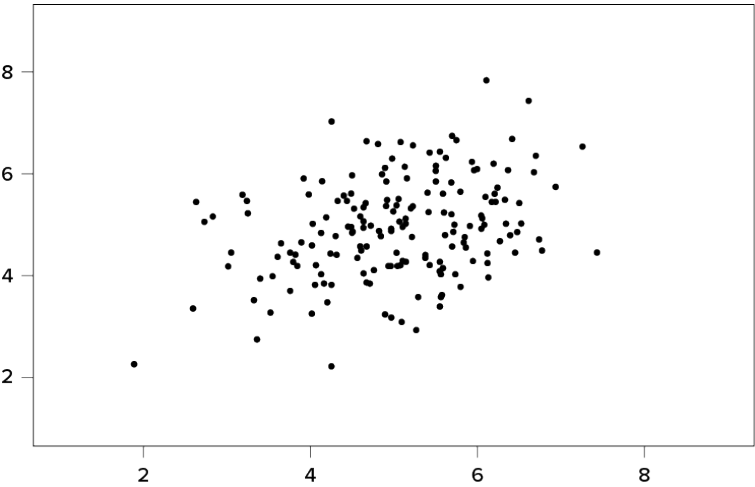


Figure 34.8. Weak Positive Correlation. [\[Image Description\]](#)

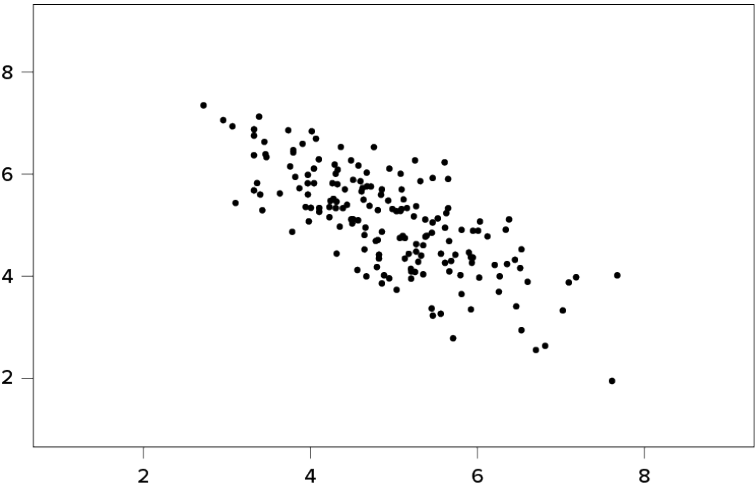


Figure 34.9. Strong Negative Correlation. [\[Image Description\]](#)

Pearson's r

There are several different types of correlation coefficients, but we will only focus on Pearson's r . Pearson's r measures the linear relationship between two continuous variables. It is the most popular correlation coefficient for assessing linear relationships, which serves as both a descriptive statistic (like M) and an inferential test statistic (like t). It is descriptive because it describes what is happening in the scatter plot; r will have both a sign ($+/-$) for the direction and a number (0 to 1 in absolute value) for the magnitude. As noted above, because it assumes a linear relationship, nothing about r will suggest what the form is—it will only tell what the direction and magnitude will be if the form is linear. (Remember: always make a scatter plot first!) The coefficient r also works as a test statistic because the magnitude of r will correspond directly to a t -value as the specific degrees of freedom, which can then be compared to a critical value. Luckily, we do not need to do this conversion by hand. Instead, we will have a table of r critical values that looks very similar to our t table, and we can compare our r directly to those.

The formula for r is very simple: it is just the covariance (defined above) divided by the standard deviations of X and Y :

$$r = \frac{\text{cov}_{XY}}{s_X s_Y} = \frac{SP}{\sqrt{(SS_X)(SS_Y)}}$$

The first formula gives a direct sense of what a correlation is: a covariance standardized onto the scale of X and Y ; the

second formula is computationally simpler and faster. Both of these equations will give the same value, and as we saw at the beginning of the chapter, all of these values are easily computed by using the sum of products table. When we do this calculation, we will find that our answer is always between -1.00 and 1.00 (if it's not, check the math again), which gives us a standard, interpretable metric, similar to what z scores did.

It was stated earlier that r is a descriptive statistic like M , and just like M , it corresponds to a population parameter. For correlations, the population parameter is the lowercase Greek letter (“rho”); be careful not to confuse with a p -value—they look quite similar. The statistic r is an estimate of ρ , just as M is an estimate of μ . Thus, we will test our observed value of r that we calculate from the data and compare it to a value of specified by our null hypothesis to see if the relationship between our variables is significant, as we will see in the following example.



Example Anxiety and Depression

Anxiety and depression are often reported to be highly linked (or “comorbid”). Our hypothesis testing procedure follows the same four-step process as before, starting with our null and alternative hypotheses. We will look for a positive relationship between our variables among a group of 10 people because that is what we would expect based on them being comorbid.

Step 1: State the Hypotheses

Our hypotheses for correlations start with a baseline assumption of no relationship, and our alternative will be directional if we expect to find a specific type of relationship.

For this example, we expect a positive relationship:

H_0 : There is no relationship between anxiety and depression

$$H_0 : \rho = 0$$

H_A : There is a positive relationship between anxiety and depression

$$H_A : \rho \neq 0$$

Remember that ρ (“rho”) is our population parameter for the correlation that we estimate with r , just like M and μ for means. Remember also that if there is no relationship between variables, the magnitude will be 0, which is where we get the null and alternative hypothesis values.

Step 2: Find the Critical Values

As a reminder, the critical value serves as a threshold to determine whether the data in one’s sample is statistically significant. The critical values for correlations can be found from the correlation table (a portion of

which appears in [Table 34.2](#)), which looks very similar to the t table (The complete correlation table can be found in [Appendix D \[New Tab\]](#)). The column of critical values is based on our significance level (α) and the directionality of our test. The row is determined by our degrees of freedom. For correlations, we have $N - 2$ degrees of freedom, rather than $N - 1$. For our example, we have 10 people, so our degrees of freedom = $10 - 2 = 8$.

Table 34.2. Critical Values for Pearson’s *r* (Corr

Degrees of freedom (df) <i>df = n – 2</i>	Critical Value		
<i>Level of Significance for One-Tailed Test</i>	.05	.025	.01
<i>Level of Significance for Two-Tailed Test</i>	.10	.05	.02
1	.988	.997	.999
2	.900	.950	.980
3	.805	.878	.934
4	.729	.811	.882
5	.669	.754	.833

Degrees of freedom (df) $df = n - 2$	Critical Value		
<i>Level of Significance for One-Tailed Test</i>	.05	.025	.01
<i>Level of Significance for Two-Tailed Test</i>	.10	.05	.02
6	.621	.707	.789
7	.582	.666	.750
8	.549	.632	.715
9	.521	.602	.685
10	.497	.576	.658
11	.476	.553	.634

Degrees of freedom (df) <i>df = n - 2</i>	Critical Value		
<i>Level of Significance for One-Tailed Test</i>	.05	.025	.01
<i>Level of Significance for Two-Tailed Test</i>	.10	.05	.02
12	.458	.532	.612
13	.441	.514	.592
14	.426	.497	.574
15	.412	.482	.558

We were not given any information about the level of significance at which we should test our hypothesis, so we will assume $\alpha = .05$ as always. From our table, we can see

that a one-tailed test (because we expect only a positive relationship) at the $\alpha = .05$ level has a critical value of $r^* = .549$. Thus, if our observed correlation is greater than .549, it will be statistically significant. This is a rather high bar (remember, the guideline for a strong relationship is $r = .50$); this is because we have so few people. Larger samples make it easier to find significant relationships.

Step 3: Calculate the Test Statistic and Effect Size

We have laid out our hypotheses and the criteria we will use to assess them, so now we can move on to our test statistic. Before we do that, we must first create a scatter plot of the data to make sure that the most likely form of our relationship is, in fact, linear. Figure 34.10 shows our data plotted out, and it looks like they are, in fact, linearly related, so Pearson's r is appropriate.

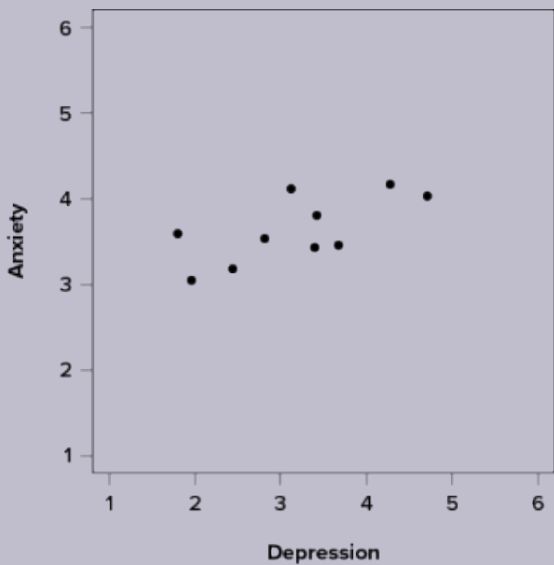


Figure 34.10. Scatter Plot of Depression and Anxiety. [\[Image Description\]](#)

The data we gather from our participants is as follows:

Table 34.3. Comorbidity of Depression and Anxiety Data

Dep.	2.81	1.96	3.43	3.40	4.71	1.80	4.27	3.68	2.44	3.13
Anx.	3.54	3.05	3.81	3.43	4.03	3.59	4.17	3.46	3.19	4.12

We will need to put these values into our sum of products table to calculate the standard deviation and covariance of our variables. We will use X for depression and Y for anxiety to keep track of our data. But be aware that this choice is arbitrary, and the math will work out the same if we decide to do the opposite. Our table is thus:

Table 34.4. Comorbidity of Depression and Anxiety
Sum of Products

X	$(X - M_X)$	$(X - M_X)^2$	Y	$(Y - M_Y)$	$(Y - M_Y)^2$	$(X - M_X)(Y - M_Y)$
2.81	-0.35	0.12	3.54	-0.10	0.01	0.04
1.96	-1.20	1.44	3.05	-0.59	0.35	0.71
3.43	0.27	0.07	3.81	0.17	0.03	0.05
3.40	0.24	0.06	3.43	-0.21	0.04	-0.05
4.71	1.55	2.40	4.03	0.39	0.15	0.60
1.80	-1.36	1.85	3.59	-0.05	0.00	0.07
4.27	1.11	1.23	4.17	0.53	0.28	0.59

X	$(X - M_X)$	$(X - M_X)^2$	Y	$(Y - M_Y)$	$(Y - M_Y)^2$	$(X - M_X)(Y - M_Y)$
3.68	0.52	0.27	3.46	-0.18	0.03	-0.09
2.44	-0.72	0.52	3.19	-0.45	0.20	0.32
3.13	-0.03	0.00	4.12	0.48	0.23	-0.01
31.63	0.03	7.97	36.39	-0.01	1.33	2.22

The bottom row is the sum of each column. We can see from this that the sum of the X observations is 31.63, which makes the mean of the X variable $M = 3.16$. The deviation scores for X sum to 0.03, which is very close to 0, given rounding error, so everything looks right so far. The next column is the squared deviations for X , so we can see that the sum of squares for X is

$SS_X = 7.97$. The same is true of the Y columns, with an average of $\mathbf{M}_Y = 3.64$, deviations that sum to zero within rounding error, and a sum of squares as $SS_Y = 1.33$. The final column is the product of our deviation scores (not of our squared deviations), which gives us a sum of products of $SP = 2.22$.

There are now three pieces of information we need to calculate before we compute our correlation coefficient: the covariance of X and Y and the standard deviation of each.

The covariance of two variables, remember, is the sum of products divided by $N - 1$. For our data:

$$\text{cov}_{XY} = \frac{SP}{N - 1} = \frac{2.22}{9} = 0.25$$

The formulas for standard deviation are the same as before. Using subscripts X and Y to denote depression and anxiety:

$$s_X = \sqrt{\frac{\sum(X - M_X)^2}{N - 1}} = \sqrt{\frac{7.97}{9}} = 0.94$$

$$s_Y = \sqrt{\frac{\sum(Y - M_Y)^2}{N - 1}} = \sqrt{\frac{1.33}{9}} = 0.38$$

Now we have all of the information we need to calculate r :

$$r = \frac{\text{cov}_{XY}}{s_X s_Y} = \frac{0.25}{(0.94)(0.38)} = .70$$

We can verify this using our other formula, which is computationally shorter:

$$r = \frac{SP}{\sqrt{(SS_X)(SS_Y)}} = \frac{2.22}{\sqrt{(7.97)(1.33)}} = .70$$

So our observed correlation between anxiety and depression is $r = .70$, which, based on sign and magnitude, is a strong, positive correlation. Now we need to compare it to our critical value to see if it is also statistically significant.

Effect Size and Pearson's r

Pearson's r is an incredibly flexible and useful statistic. Not only is it both descriptive and inferential, as we saw above, but because it is on a standardized metric (always between -1.00 and 1.00), it can also serve as its own effect size. In general, we use $r = .10$, $r = .30$, and $r = .50$ as our guidelines for small, medium, and large effects. Just like with Cohen's d , these guidelines are not absolutes, but they do serve as useful indicators in most situations. Notice as well that these are the same guidelines we used earlier to interpret the magnitude of the relationship based on the correlation coefficient.

In addition to r being its own effect size, there is an additional effect size we can calculate for our results. This effect size is r^2 , and it is exactly what it looks like—it is the

squared value of our correlation coefficient. r^2 is interpreted as the amount of variance explained in the outcome variance, and the cut scores are the same as well: .01, .09, and .25 for small, medium, and large, respectively. Notice here that these are the same cutoffs we used for regular r effect sizes, but squared ($.10^2 = .01$, $.30^2 = .09$, $.50^2 = .25$) because, again, the r^2 effect size is just the squared correlation, so its interpretation should be, and is, the same. The reason we use r^2 as an effect size is because our ability to explain variance is often important to us.

Step 4: Make a Decision

Our critical value was $r^* = .549$, and our obtained value was $r = .70$. Our obtained value was larger than our critical value, so we can reject the null hypothesis.

Reject H_0 . Based on our sample of 10 people, there is a statistically significant,

strong, positive relationship between anxiety and depression, $r(8) = .70, p < .05$.

Notice in our interpretation that, because we already know the magnitude and direction of our correlation, we can interpret that. We also report the degrees of freedom, just like with t , and we know that $p < \alpha$ because we rejected the null hypothesis. As we can see, even though we are dealing with a very different type of data, our process of hypothesis testing has remained unchanged. Unlike for our other statistics, we do not report an effect size for the correlation coefficient because the reader can easily do that for themselves by squaring r . The r^2 statistic is called the coefficient of determination and is essentially the effect size for a correlation coefficient; it tells us what percentage of the variance in the X variable is explained by the Y variable (and vice versa).

Figure 34.11 shows the output from JASP for this example.

Correlation			
Pearson's Correlations			
Variable		Depression	Anxiety
1. Depression	Pearson's r	—	
	p-value	—	
2. Anxiety	Pearson's r	0.680	—
	p-value	0.015	—
Note. All tests one-tailed, for positive correlation			

Figure 34.11. Output from JASP for the correlation described in the Anxiety and Depression example. The output provides the Pearson's r (.68; note the value provided in text using hand calculations is .70 due to rounding), and the exact p value (.015, which is less than .05). Based on our sample of 10 people, there is a statistically significant, strong, positive relationship between anxiety and depression, $r(8) = .68$, $p = .015$.
[\[Image Description\]](#)

Correlation versus Causation

We cover a great deal of material in this textbook, and many

of the principles underlying what we do in statistics can be used in your day-to-day life to help you interpret information objectively and make better decisions. We now come to what may be the most important lesson in introductory statistics: the difference between correlation and causation.

It is very, very tempting to look at variables that are correlated and assume that this means they are causally related; that is, it gives the impression that X is causing Y. However, in reality, correlations do not—and cannot—do this. Correlations do not prove causation. While this sounds like a simple idea on the surface, correlational data are often used in decision-making in large scale settings (e.g. schools, healthcare, public policy). However, no matter how logical, how obvious, or how convenient it may seem, no correlational analysis can demonstrate causality. The only way to demonstrate a causal relationship is with a properly designed and controlled experiment.

Many times, we have good reason for assessing the correlation between two variables, and often that reason will be that we suspect one causes the other. Thus, when we run our analyses and find strong, statistically significant results, it is tempting to say that we have found the causal relationship that we are looking for. The reason we cannot do this is that, without an experimental design that includes random assignment and control variables, the relationship we observe between the two variables may be caused by something else that we failed to measure—something we can only detect and

control for with an experiment. These confound variables, which we will represent with Z, can cause two variables, X and Y, to appear related when in fact they are not. They do this by being the hidden—or lurking—cause of each variable independently. That is, if Z causes X and Z causes Y, X and Y will appear to be related. However, if we control for the effect of Z (the method for doing this is beyond the scope of this text), then the relationship between X and Y will disappear.

A popular example of this effect is the correlation between ice cream sales and deaths by drowning. These variables are known to correlate very strongly over time. However, this does not prove that one causes the other. The lurking variable in this case is the weather—people enjoy swimming and eating ice cream more during hot weather as a way to cool off. As another example, consider shoe size and spelling ability in elementary school children. Although there should clearly be no causal relationship here, the variables are nonetheless consistently correlated. The confound in this case? Age. Older children spell better than younger children and are also bigger, so they have larger shoes.

When there is the possibility of confounding variables being the hidden cause of our observed correlation, we will often collect data on Z as well and control for it in our analysis. This is good practice and a wise thing for researchers to do. Thus, it would seem that it is easy to demonstrate causation with a correlation that controls for Z. However, the number of

variables that could potentially cause a correlation between X and Y is functionally limitless, so it would be impossible to control for everything. That is why we use experimental designs; by randomly assigning people to groups and manipulating variables in those groups, we can balance out individual differences in any variable that may be our cause.

It is not always possible to do an experiment, however, so there are certain situations in which we will have to be satisfied with our observed relationship and do the best we can to control for known confounds. However, in these situations, even if we do an excellent job of controlling for many extraneous (a statistical and research term for “outside”) variables, we must be careful not to use causal language. That is because, even after controls, sometimes variables are related just by chance.

Sometimes, variables will end up being related simply due to random chance, and we call these correlations spurious. Spurious just means random, so what we are seeing is random correlations because, given enough time, enough variables, and enough data, sampling error will eventually cause some variables to appear related when they are not. Sometimes, this even results in incredibly strong, but completely nonsensical, correlations. This becomes more and more of a problem as our ability to collect massive datasets and dig through them improves, so it is very important to think critically about any relationship you encounter.

Final Considerations

Correlations, although simple to calculate, can be very complex, and there are many additional issues we should consider. We will look at two of the most common issues that affect our correlations and discuss some other correlations and reporting methods you may encounter.

Range Restriction

The strength of a correlation depends on how much variability is in each of the variables X and Y. This is evident in the formula for Pearson's r , which uses both covariance (based on the sum of products, which comes from deviation scores) and the standard deviation of both variables (based on the sums of squares, which also come from deviation scores). Thus, if we reduce the amount of variability in one or both variables, our correlation will go down. Failure to capture the full range of a variability is called range restriction.

Take a look at Figure 34.12 and Figure 34.13. Figure 34.12 shows a strong relationship ($r = .67$) between two variables. An orange oval is overlaid on it to make the relationship even more distinct. Figure 34.13 shows the same data, but the bottom half of the X variable (all scores below 5) has been removed, which causes our relationship (again represented by an orange oval) to become much weaker ($r =$

.38). Thus, range restriction has truncated (made smaller) our observed correlation.

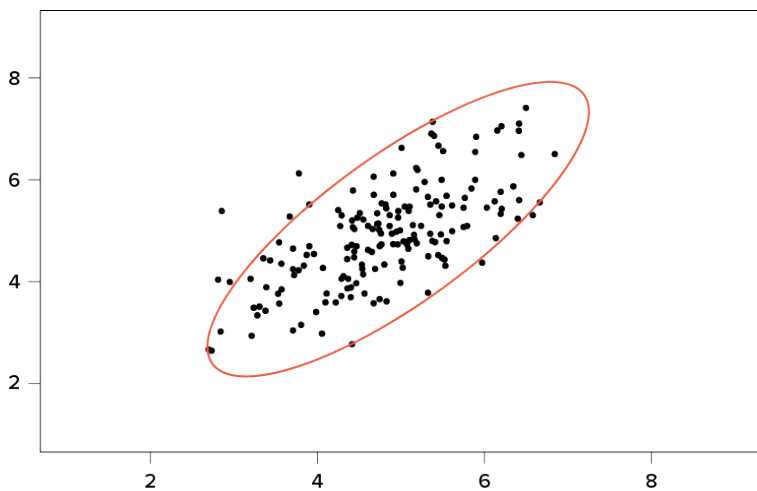


Figure 34.12. Strong Positive Correlation. [\[Image Description\]](#)

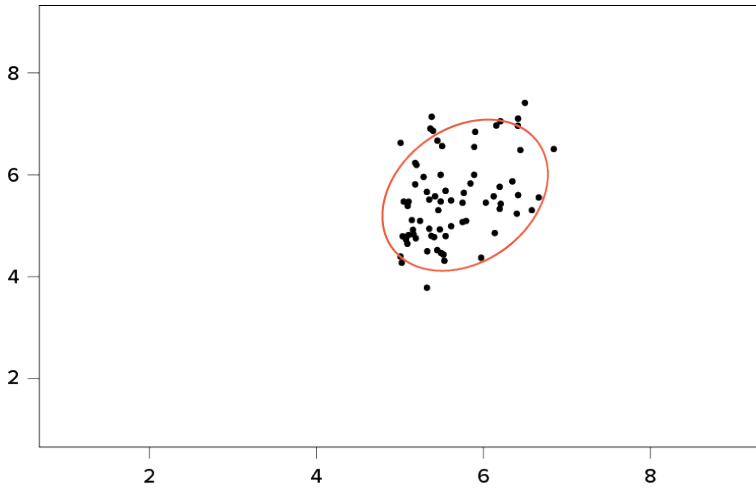


Figure 34.13. Effect of Range Restriction. [\[Image Description\]](#)

Sometimes, range restriction happens by design. For example, we rarely hire people who do poorly on job applications, so we would not have the lower range of those predictor variables. Other times, we inadvertently cause range restriction by not properly sampling our population. Although there are ways to correct for range restriction, they are complicated and require much information that may not be known, so it is best to be very careful during the data collection process to avoid it.

Outliers

Another issue that can cause the observed size of our correlation to be inappropriately large or small is the presence of outliers. An outlier is a data point that falls far away from

the rest of the observations in the dataset. Sometimes outliers are the result of incorrect data entry, poor or intentionally misleading responses, or simple random chance. Other times, however, they represent real people with meaningful values on our variables. For instance, if the only African American student in a predominately white school exhibits unusually high anxiety symptoms, this could simply indicate that this one student is highly anxious; however, it could also be signaling valuable insights into the school's culture. The distinction between meaningful and accidental outliers is a difficult one that requires close examination and the expert judgment of the researcher. Sometimes, we will remove the outlier (if we think it is an accident) or we may decide to keep it (if we find the scores to still be meaningful even though they are different).

The scatter plots in Figure 34.14 show the effects that an outlier can have on data. In the first plot, we have our raw dataset. You can see in the upper right corner that there is an outlier observation that is very far from the rest of our observations on both the X and Y variables. In the middle plot, we see the correlation computed when we include the outlier, along with a straight line representing the relationship; here, it is a positive relationship. In the third plot, we see the correlation after removing the outlier, along with a line showing the direction once again. Not only did the correlation get stronger, it completely changed direction!

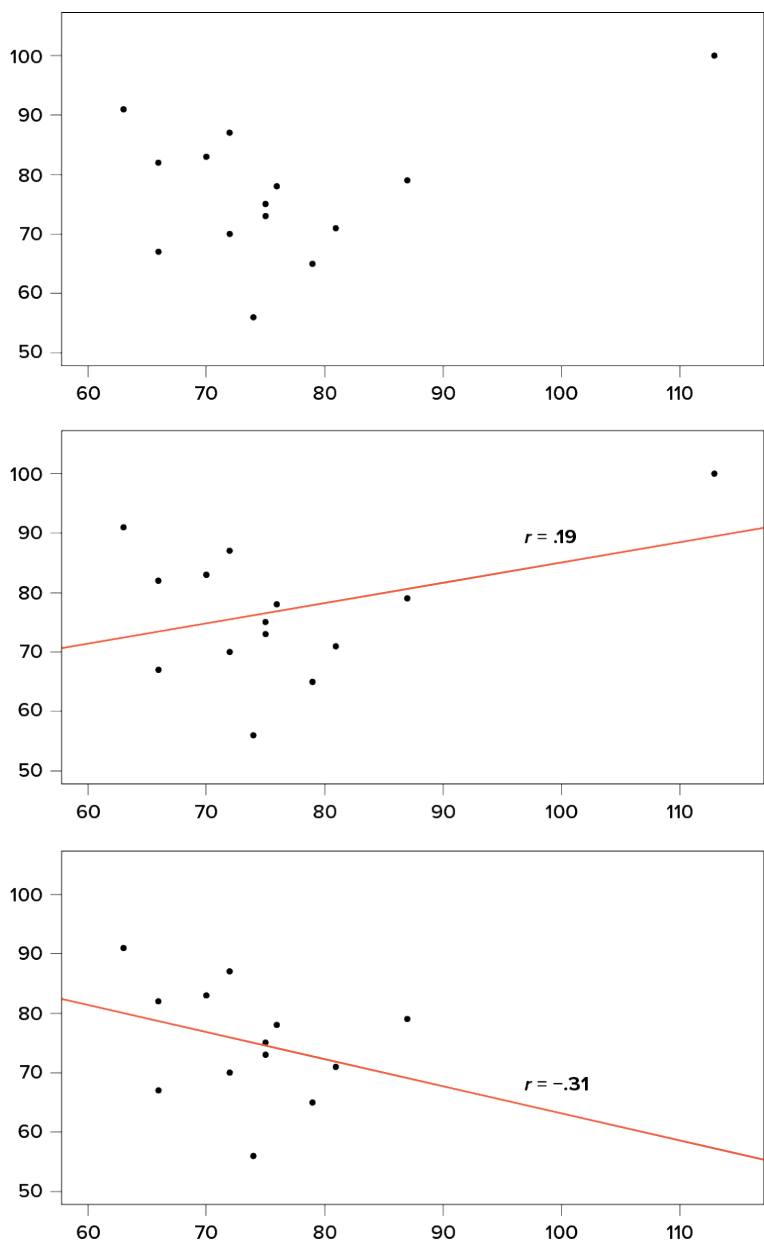


Figure 34.14. Three Scatter Plots Showing Correlations With

and Without Outliers. [\[Image Description\]](#)

In general, there are three effects that an outlier can have on a correlation: it can change the magnitude (make it stronger or weaker), it can change the significance (make a non-significant correlation significant or vice versa), and/or it can change the direction (make a positive relationship negative or vice versa). Outliers are a big issue in small datasets where a single observation can have a strong weight compared with the rest. However, as our sample sizes get very large (into the hundreds), the effects of outliers diminish because they are outweighed by the rest of the data. Nevertheless, no matter how large a dataset you have, it is always a good idea to screen for outliers, both statistically (using analyses that we do not cover here) and visually (using scatter plots).

Other Correlation Coefficients

In this chapter, we have focused on Pearson's r as our correlation coefficient because it is very common and useful. There are, however, many other correlations out there, each of which is designed for a different type of data. The most common of these is Spearman's ρ (ρ_s), which is designed to be used on ordinal data rather than continuous data. This is a useful analysis if we have ranked data or our data do not conform to the normal distribution. There are even more

correlations for ordered categories, but they are much less common and beyond the scope of this chapter.

Additionally, the principles of correlations underlie many other advanced analyses. In the next chapter, we will learn about regression, which is a formal way of running and analyzing a correlation that can be extended to more than two variables. Regression is a powerful technique that serves as the basis for even our most advanced statistical models, so what we have learned in this chapter will open the door to an entire world of possibilities in data analysis.

Correlation Matrices

Many research studies look at the relationship between more than two continuous variables. In such situations, we could simply list all of our correlations, but that would take up a lot of space and make it difficult to quickly find the relationship we are looking for. Instead, we create correlation matrices so that we can quickly and simply display our results. A matrix is like a grid that contains our values. There is one row and one column for each of our variables, and the intersections of the rows and columns for different variables contain the correlation for those two variables.

At the [beginning of the chapter](#), we saw scatter plots presenting data for correlations between job satisfaction, well-being, burnout, and job performance. We can create a

correlation matrix to quickly display the numerical values of each. Such a matrix is shown below.

Table 34.5. Correlation Matrix for Job Satisfaction, Well-being, Burnout, and Job Performance

	Satisfaction	Well-being	Burnout	Performance
Satisfaction	1.00			
Well-being	0.41	1.00		
Burnout	-0.54	-0.87	1.00	
Performance	0.08	0.21	-0.33	1.00

Notice that there are values of 1.00 where each row and column of the same variable intersect. This is because a variable correlates perfectly with itself, so the value is always exactly 1.00. Also notice that the upper cells are left blank and only the cells below the diagonal of 1.00s are filled in. This is because correlation matrices are symmetrical: they have the same values above the diagonal as below it. Filling in both sides would provide redundant information and make it a bit harder to read the matrix, so we leave the upper triangle blank.

Correlation matrices are a very condensed way of presenting many results quickly, so they appear in almost all research studies that use continuous variables. Many matrices also

include columns that show the variable means and standard deviations, as well as asterisks showing whether or not each correlation is statistically significant.

Equity Activity: School funding

A school district has a policy of providing funding to schools based on the median cost of homes in a given area. This means that areas with higher home prices tend to receive more funding from the district. A set of researchers is interested in if the amount of funding that schools receive is related to students' standardized testing scores. To test this, they take a sample of 30 schools from around the state and record how much funding each school received and the average standardized test scores for each school. The researchers find a correlation value of $r = .41$ between school funding and test scores. Test the hypothesis that there is a significant relationship between these variables at the $\alpha = 0.05$ level using the four-step hypothesis testing procedure. What conclusions might the researchers draw about how funding is allocated given the results of the hypothesis test?



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodspsysh/?p=691#h5p-60>

Practice Problems

Short Answer Reflections



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodspsysh/?p=691#h5p-62>

Test Your Knowledge



An interactive H5P element has

— been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodspsycho/?p=691#h5p-61>

License & Attribution

“Correlations” by Makenzie O’Neil is adapted from “[Correlations](#)” by Linda R. Cote Ph.D.; Rupa G. Gordon Ph.D., Chrislyn E. Randell Ph.D., Judy Schmitt, and Helena Marvin, which is licensed CC BY-NC-SA 4.0.

“Correlations” is licensed under CC BY-NC-SA 4.0.

Image Descriptions

Figure 34.1. This scatter plot illustrates a positive correlation between job satisfaction and well-being, with both axes ranging from 1 to 10. Approximately 15-17 data points are

distributed across the graph, showing a clear upward trend from the lower left (around 1.5, 1.3) to the upper right, with most points concentrated in the 7-10 range for both variables. The pattern demonstrates that as job satisfaction increases, well-being tends to increase as well, though some scatter in the data indicates individual variation. The title “Positive Relationship” reinforces this positive linear correlation between the two variables. [[Return to Figure 34.1](#)]

Figure 34.2. This scatter plot illustrates a negative correlation between job satisfaction and burnout, with both axes ranging from 1 to 10. Approximately 15 data points are distributed across the graph, showing a clear downward trend from the upper left (around 2.5 job satisfaction, 7.6 burnout) to the lower right, with points at higher job satisfaction levels corresponding to lower burnout scores. The pattern demonstrates that as job satisfaction increases, burnout tends to decrease, though some variation exists in the data. The title “Negative Relationship” emphasizes this inverse correlation between the two variables. [[Return to Figure 34.2](#)]

Figure 34.3. This scatter plot illustrates the absence of correlation between job satisfaction and job performance, with both axes ranging from 1 to 10. Approximately 15 data points are distributed across the graph in a random pattern, with no discernible upward or downward trend. The points are scattered throughout the middle and upper portions of the plot, ranging from about 4 to 10 on the job satisfaction axis and 5 to 9.5 on the job performance axis. Unlike positive or

negative correlations, this pattern shows that job satisfaction levels do not predict job performance, as high and low performance scores appear across all levels of job satisfaction. The title “No Relationship” emphasizes this lack of correlation between the two variables. [[Return to Figure 34.3](#)]

Figure 34.4. This scatter plot illustrates a strong positive correlation between two variables, with the x-axis ranging from 0 to 10 and the y-axis ranging from 0 to 10. Approximately 10 data points are plotted, forming a nearly perfect linear pattern from the lower left (around 0.5, 2.2) to the upper right (10, 10). The points show minimal scatter around an imaginary trend line, indicating a very tight relationship between the variables. This pattern demonstrates that as the x-variable increases, the y-variable increases proportionally at a consistent rate, suggesting a strong predictive relationship between the two measures. Note that this plot lacks axis labels and a title, so the specific variables being measured are not identified. [[Return to Figure 34.4](#)]

Figure 34.5. This scatter plot illustrates a curvilinear or non-linear relationship between two variables, with the x-axis ranging from 0 to 16 and the y-axis ranging from 0 to 7. Approximately 15 data points are plotted, forming an inverted U-shape or parabolic pattern. The points begin low on the left (around 1, 1), rise steadily through the lower x-values, reach their highest concentration in the middle range (around $x=7-10$, $y=5-5.5$), and then descend back down on the right side (ending near 16, 1). This pattern demonstrates that the

relationship between the variables is not linear but rather quadratic, where the y-variable increases with the x-variable up to a certain point and then decreases. Note that this plot lacks axis labels and a title, so the specific variables being measured are not identified. [[Return to Figure 34.5](#)]

Figure 34.6. This scatter plot illustrates a curvilinear or non-linear relationship between two variables, with the x-axis ranging from 0 to 16 and the y-axis ranging from 0 to 7. Approximately 15 data points are plotted, forming an inverted U-shape or parabolic pattern. A horizontal reference line at $y=4$ divides the plot into upper and lower regions. The points begin low on the left (around 1, 1), rise steadily through the lower x-values, reach their peak in the middle range (around $x=7-9$, $y=5.5$), and then descend back down on the right side (ending near 16, 1). This pattern demonstrates a quadratic relationship where the y-variable increases with the x-variable up to a midpoint and then decreases, with the reference line potentially indicating a threshold or baseline value. Note that this plot lacks axis labels and a title, so the specific variables being measured are not identified. [[Return to Figure 34.6](#)]

Figure 34.7. This scatter plot illustrates the absence of correlation between two variables, with both axes ranging from 0 to 9. Approximately 200 data points are densely plotted, forming a roughly circular or elliptical cluster centered around the middle of the graph (approximately $x=5$, $y=5$). The points are concentrated most heavily in the center, with the density decreasing toward the edges, ranging from about

2 to 7.5 on both axes. Unlike linear relationships, this pattern shows no upward or downward trend, indicating that changes in the x-variable do not predict changes in the y-variable. The circular distribution suggests the variables are independent of each other, with the data spread relatively evenly in all directions from the center point. Note that this plot lacks axis labels and a title, so the specific variables being measured are not identified. [[Return to Figure 34.7](#)]

Figure 34.8. This scatter plot illustrates a weak positive correlation between two variables, with both axes ranging from 0 to 9. Approximately 200 data points are plotted, showing a general upward trend from the lower left (around 2, 2.5) to the upper right (around 8, 6.5), though with considerable scatter around this trend. The points are most densely concentrated in the center of the plot (around $x=4-6$, $y=4-6$), with the cloud of data points appearing somewhat elongated diagonally. While there is a discernible positive relationship where higher x-values tend to correspond with higher y-values, the substantial variation in the data indicates that the x-variable is only a modest predictor of the y-variable. Note that this plot lacks axis labels and a title, so the specific variables being measured are not identified. [[Return to Figure 34.8](#)]

Figure 34.9. This scatter plot illustrates a negative correlation between two variables, with both axes ranging from 0 to 9. Approximately 200 data points are plotted, showing a clear downward trend from the upper left (around

3, 7.5) to the lower right (around 8, 2). The points form an elongated cloud pattern that slopes diagonally downward, most densely concentrated in the center region (around $x=4-6$, $y=4-6$). This pattern demonstrates that as the x -variable increases, the y -variable tends to decrease, indicating an inverse relationship between the two measures. While there is some scatter in the data, the negative trend is clearly visible, suggesting a moderate to strong negative correlation. Note that this plot lacks axis labels and a title, so the specific variables being measured are not identified. [[Return to Figure 34.9](#)]

Figure 34.10. This scatter plot illustrates a weak positive correlation between depression and anxiety, with the x -axis labeled “Depression” ranging from 1 to 6 and the y -axis labeled “Anxiety” ranging from 1 to 6. Approximately 12 data points are plotted, showing a gentle upward trend from the lower left (around 2, 3) to the upper right (around 5, 4). The points are moderately scattered, concentrated mostly in the 2-5 range on the depression axis and the 3-4.5 range on the anxiety axis. This pattern suggests that as depression levels increase, anxiety levels tend to increase as well, though the relationship is relatively weak with considerable variability. The sparse distribution of points and moderate scatter indicates that while there is some association between these variables, depression alone does not strongly predict anxiety levels. [[Return to Figure 34.10](#)]

Figure 34.11. This correlation table, titled “Correlation” with a subtitle “Pearson’s Correlations,” displays the statistical relationship between depression and anxiety variables. The

table has three columns (Variable, Depression, and Anxiety) and shows correlation statistics for two variables. For the Depression variable (row 1), Pearson's r and p -value are shown as dashes in both the Depression and Anxiety columns, as a variable's correlation with itself is not meaningful. For the Anxiety variable (row 2), the correlation with Depression shows Pearson's $r = 0.680$ and $p\text{-value} = 0.015$, while its correlation with itself is marked with dashes. A note at the bottom states "Note. All tests one-tailed, for positive correlation." This table indicates a statistically significant moderate to strong positive correlation between depression and anxiety, with the low p -value (0.015) suggesting this relationship is unlikely to be due to chance. [[Return to Figure 34.11](#)]

Figure 34.12. This scatter plot illustrates a positive correlation between two variables, with both axes ranging from 0 to 9. Approximately 250 data points are plotted, showing a clear upward trend from the lower left (around 2.5, 2.5) to the upper right (around 7.5, 7.5). A red ellipse is overlaid on the plot, encircling the data and representing the 95% confidence region or correlation ellipse. The ellipse is oriented diagonally from lower left to upper right, emphasizing the positive linear relationship between the variables. The data points are most densely concentrated in the center of the ellipse (around $x=5$, $y=5$), with the density decreasing toward the edges. This visualization demonstrates not only the positive correlation but also the spread and

distribution of the data around the trend line. Note that this plot lacks axis labels and a title, so the specific variables being measured are not identified. [[Return to Figure 34.12](#)]

Figure 34.13. This scatter plot illustrates the absence of correlation between two variables, with both axes ranging from 0 to 9. Approximately 100 data points are plotted, forming a roughly circular cluster concentrated in the center-right portion of the graph (around $x=5-7$, $y=4.5-6$). A red ellipse is overlaid on the plot, encircling the data and representing the 95% confidence region. Unlike the previous correlation plot, this ellipse is nearly circular rather than elongated, indicating no directional relationship between the variables. The data points show no upward or downward trend, with high and low y -values appearing across all x -values within the cluster. This pattern demonstrates that changes in the x -variable do not predict changes in the y -variable, suggesting the two measures are independent of each other. Note that this plot lacks axis labels and a title, so the specific variables being measured are not identified. [[Return to Figure 34.13](#)]

Figure 34.14. This figure contains three vertically stacked scatter plots displaying the same dataset to illustrate different interpretations of correlation. All three panels use identical axes, with the x -axis ranging from 60 to 120 and the y -axis ranging from 50 to 105, and contain approximately 15 identical data points scattered across the plot area. The top panel presents the raw data without any trend line, showing

points distributed without an obvious pattern. The middle panel adds a red upward-sloping trend line with the correlation coefficient labeled as “ $r = .19$,” suggesting a very weak positive relationship where y slightly increases as x increases. The bottom panel displays a red downward-sloping trend line with “ $r = -.31$,” indicating a weak negative relationship where y slightly decreases as x increases. This three-panel comparison demonstrates how the same data can be interpreted with different correlation strengths and directions, highlighting the subjective nature of fitting trend lines or the sensitivity of correlation to outliers and data interpretation. Note that the axes are unlabeled, so the specific variables are not identified.

[\[Return to Figure 34.14\]](#)

Media Attributions

- Scatter_Plot_Depression_and_Anxiety-2

Previous/next navigation

[Previous: Analysis of Variance](#)

[Next: Linear Regression](#)

Back to top

License



[Correlations](#) Copyright © by Makenzie O'Neil is licensed under a [Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License](#), except where otherwise noted.

Share This Book

[Pressbooks](#)

[Powered by Pressbooks](#)

- [Pressbooks User Guide](#)
- [|Pressbooks Directory](#)

[Pressbooks on YouTube](#)[Pressbooks on LinkedIn](#)

35.

LINEAR REGRESSION

Makenzie O'Neil

Learning Objectives

By the end of this chapter, you will be able to

1. Calculate slope, y-intercept, and standard error of the estimate.
2. Use the regression line equation to predict a value of y from a known value of x .
3. Compute a hypothesis test using regression.
4. Interpret regression findings from an ANOVA table.

In a previous chapter, we learned about ANOVA, which involves a new way of looking at how our data are structured

and the inferences we can draw from that. Then, we learned about correlations, which analyze two continuous variables at the same time to see if they systematically relate in a linear fashion. In this chapter, we will combine these two techniques in an analysis called simple linear regression, or regression for short. Regression uses the technique of variance partitioning from ANOVA to more formally assess the types of relationships looked at in correlations. Regression is the most general and most flexible analysis covered in this book, and we will only scratch the surface.

Line of Best Fit

In correlations, we referred to a linear trend in the data. That is, we assumed that there was a straight line we could draw through the middle of our scatter plot that would represent the relationship between our two variables, X and Y . Regression involves solving for the equation of that line, which is called the line of best fit.

The line of best fit can be thought of as the central tendency of our scatter plot. The term best fit means that the line is as close to all points (with each point representing both variables for a single person) in the scatter plot as possible, with a balance of scores above and below the line. This is the same idea as the mean, which has an equal weighting of scores above and below it and is the best singular descriptor of all our data points for a single variable.

We have already seen many scatter plots, so we know by now that no scatter plot has points that form a perfectly straight line. Because of this, when we put a straight line through a scatter plot, it will not touch all of the points, and it may not even touch any! This will result in some distance between the line and each of the points it is supposed to represent, just like a mean has some distance between it and all of the individual scores in the dataset.

The distances between the line of best fit and each individual data point go by two different names that mean the same thing: errors and residuals. The term **error** in regression is closely aligned with the meaning of error in statistics (think standard error or sampling error); it does not mean that we did anything wrong, it simply means that there was some discrepancy or difference between what our analysis produced and the true value we are trying to get at. The term **residual** is new to our study of statistics, and it takes on a very similar meaning in regression to what it means in everyday parlance: there is something left over. In regression, what is “left over”—that is, what makes up the residual—is an imperfection in our ability to predict values of the Y variable using our line. This definition brings us to one of the primary purposes of regression and the line of best fit: predicting scores.

Prediction

The goal of regression is the same as the goal of ANOVA: to

take what we know about one variable (X) and use it to explain our observed differences in another variable (Y). In ANOVA, we talked about—and tested for—group mean differences, but in regression, we do not have groups for our predictor variable; we have a continuous variable, like in correlation. Because of this, our vocabulary will be a little bit different, but the process, logic, and end result are all the same.

In regression, we most frequently talk about prediction, specifically predicting our outcome variable Y from our predictor variable X, and we use the line of best fit to make our predictions. Let's take a look at the equation for the line, which is quite simple:

$$\hat{Y} = a + bX$$

The terms in the equation are defined as:

$\hat{Y}$: the predicted value of Y for an individual person

a: the intercept of the line

b: the slope of the line

X: the observed value of X for an individual person

What this shows us is that we will use our known value of X for each person to predict the value of Y for that person. The predicted value, $\hat{Y}$ ("Y-hat"), is our best guess for what a person's score on the outcome is. Notice also that the form of the equation is very similar to very simple linear equations that you have likely encountered before and has only two parameter estimates: an intercept (where the line crosses the y-axis) and a slope (how steep—and the direction, positive or negative—the line is). These are parameter estimates because, like everything

else in statistics, we are interested in approximating the true value of the relationship in the population but can only ever estimate it using sample data. We will soon see that one of these parameters, the slope, is the focus of our hypothesis tests (the intercept is only there to make the math work out properly and is rarely interpretable). The formulas for these parameter estimates use very familiar values:

$$a = M_Y - bM_X$$

$$b = \frac{\text{cov}_{XY}}{s_X^2} = \frac{SP}{SS_X} = r \left(\frac{s_Y}{s_X} \right)$$

We have seen each of these before $\mathbf{M_Y}$ and $\mathbf{M_X}$ are the means of Y and X, respectively; $\mathbf{cov_{XY}}$ is the covariance of X and Y we learned about with correlations; and $\mathbf{s_X^2}$ is the variance of X. The formula for the slope is very similar to the formula for a Pearson correlation coefficient; the only difference is that we are dividing by the variance of X instead of the product of the standard deviations of X and Y. Because of this, our slope is scaled to the same scale as our X variable and is no longer constrained to be between 0 and 1 in absolute value. This formula provides a clear definition of the slope of the line of best fit, and just like with correlation, this definitional formula can be simplified into a short computational formula for easier calculations. In this case, we are simply taking the sum of products and dividing by the sum of squares for X.

Notice that there is a third formula for the slope of the line that involves the correlation between X and Y. This is because regression and correlation look for the same thing: a straight

line through the middle of the data. The only difference between a regression coefficient in simple linear regression and a Pearson correlation coefficient is the scale. So, if you lack raw data but have summary information on the correlation and standard deviations for variables, you can still compute a slope, and therefore an intercept, for a line of best fit.

It is important to point out that the Y values in the equations for a and b are our observed Y values in the dataset, not the predicted Y values ($\hat{Y}$) from our equation for the line of best fit. Thus, we will have three values for each person: the observed value of X (X), the observed value of Y (Y), and the predicted value of Y ($\hat{Y}$). You may be asking why we would try to predict Y if we have an observed value of Y , and that is a reasonable question. The answer has two explanations. First, we need to use known values of Y to calculate the parameter estimates in our equation, and we use the difference between our observed values and predicted values ($Y - \hat{Y}$) to see how accurate our equation is. And second, we often use regression to create a predictive model that we can then use to predict values of Y for other people for whom we only have information on X .

Let's look at this from an applied example. Businesses often have more applicants for a job than they have openings available, so they want to know who among the applicants is most likely to be the best employee. There are many criteria that can be used, but one is a personality test for conscientiousness, with the belief being that more

conscientious (more responsible) employees are better than less conscientious employees. A business might give their employees a personality inventory to assess conscientiousness and study existing performance data to look for a relationship. In this example, we have known values of the predictor (X , conscientiousness) and outcome (Y , job performance), so we can estimate an equation for a line of best fit and see how accurately conscientiousness predicts job performance, then use this equation to predict future job performance of applicants based only on their known values of conscientiousness from personality inventories given during the application process.

The key in assessing whether a linear regression works well is the difference between our observed and known Y values and our predicted $\hat{Y}$ values. As mentioned in passing above, we use subtraction to find the difference between them ($Y - \hat{Y}$) in the same way we use subtraction for deviation scores and sums of squares. The value ($Y - \hat{Y}$) is our residual, which, as defined above, is how close our line of best fit is to our actual values. We can visualize residuals to get a better sense of what they are by creating a scatter plot and overlaying a line of best fit on it, as shown in Figure 35.1.

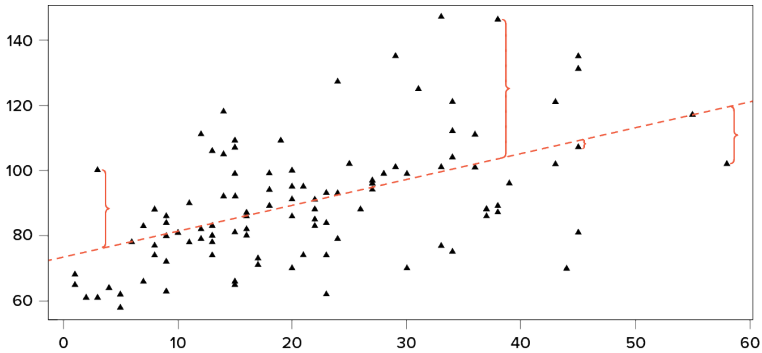


Figure 35.1. Scatter Plot With Residuals. [\[Image Description\]](#)

In Figure 35.1, the triangular dots represent observations from each person on both X and Y and the dotted line is the line of best fit estimated by the equation $\hat{Y} = a + bX$. For every person in the dataset, the line represents their predicted score. The brackets between the triangular dots and the predicted scores on the line of best fit are our residuals. (For ease of viewing, they are only drawn for four observations, but in reality there is one for every observation.) You can see that some residuals are positive and some are negative, and that some are very large and some are very small. This means that some predictions are very accurate and some are very inaccurate, and that some predictions overestimate values and some underestimate values. Across the entire dataset, the line of best fit is the one that minimizes the total (sum) value of all residuals. That is, although predictions at an individual level might be somewhat inaccurate, across our full sample and (theoretically) in future samples our total amount of error is as small as possible. We

call this property of the line of best fit the **least squares error solution**. This term means that the solution—or equation—of the line is the one that provides the smallest possible value of the squared errors (squared so that they can be summed, just like in standard deviation) relative to any other straight line we could draw through the data.

Predicting Scores and Explaining Variance

We have now seen that the purpose of regression is twofold: we want to predict scores based on our line and, as stated earlier, explain variance in our observed Y variable, just like in ANOVA. These two purposes go hand in hand, and our ability to predict scores is literally our ability to explain variance. That is, if we cannot account for the variance in Y based on X , then we have no reason to use X to predict future values of Y .

We know that the overall variance in Y is a function of each score deviating from the mean of Y (as in our calculation of variance and standard deviation). So, just like the brackets in Figure 35.1 representing residuals, given as $(Y - \hat{Y})$, we can visualize the overall variance as each score's distance from the overall mean of Y , given as $(Y - M_Y)$ our normal deviation score. This is shown in Figure 35.2.

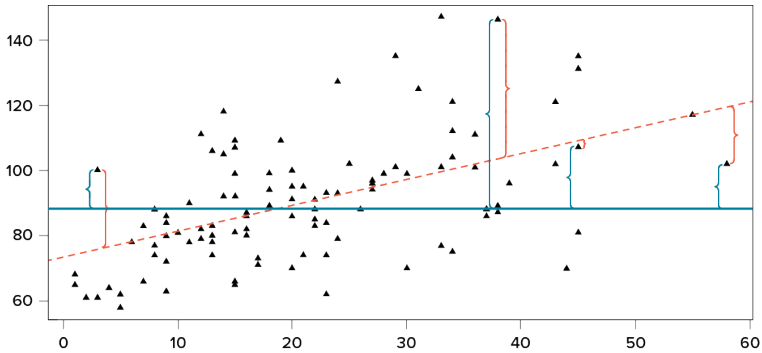


Figure 35.2. Scatter Plot with Residuals and Deviation Scores.
[\[Image Description\]](#)

In Figure 35.2, the solid line is the mean of Y , and the blue brackets are the deviation scores between our observed values of Y and the mean of Y . This represents the overall variance that we are trying to explain. Thus, the residuals and the deviation scores are the same type of idea: the distance between an observed score and a given line, either the line of best fit that gives predictions or the line representing the mean that serves as a baseline. The difference between these two values, which is the distance between the lines themselves, is our model's ability to predict scores above and beyond the baseline mean; that is, it is our model's ability to explain the variance we observe in Y based on values of X . If we have no ability to explain variance, then our line will be flat (the slope will be 0.00) and will be the same as the line representing the mean, and the distance between the lines will be 0.00 as well.

We now have three pieces of information: the distance from

the observed score to the mean, the distance from the observed score to the prediction line, and the distance from the prediction line to the mean. These are our three pieces of information needed to test our hypotheses about regression and to calculate effect sizes. They are our three sums of squares, just like in ANOVA. Our distance from the observed score to the mean is the sum of squares total, which we are trying to explain. Our distance from the observed score to the prediction line is our sum of squares error, or residual, which we are trying to minimize. Our distance from the prediction line to the mean is our sum of squares model, which is our observed effect and our ability to explain variance. Each of these will go into the ANOVA table to calculate our test statistic.

ANOVA Table

Our ANOVA table in regression follows the exact same format as it did for ANOVA (hence the name). The top row (Model) is our observed effect, the middle row is our error, and the bottom row is our total. The columns take on the same interpretations as well: from left to right, we have our sums of squares, our degrees of freedom, our mean squares, and our F statistic.

Table 35.1. ANOVA Table

Source	SS	df	MS	F
Model	$\sum(\hat{Y} - M_Y)^2$	1	$\frac{SS_M}{df_M}$	$\frac{MS_M}{MS_E}$
Error	$\sum(Y - \hat{Y})^2$	$N - 2$	$\frac{SS_E}{df_E}$	
Total	$\sum(Y - M_Y)^2$	$N - 1$		

As with ANOVA, getting the values for the SS column is a straightforward but somewhat arduous process. First, you take the raw scores of X and Y and calculate the means, variances, and covariance using the sum of products table introduced in our chapter on correlations. Next, you use the variance of X and the covariance of X and Y to calculate the slope of the line, b. (The [formula for calculating b](#) was provided earlier.) After that, you use the means and the slope to find the intercept, a, which is given alongside b. After that, you use the full prediction equation for the line of best fit to get predicted Y scores ($\hat{Y}$) for each person. Finally, you use the observed Y scores, predicted Y scores, and mean of Y (M_Y) to find the appropriate deviation scores for each person for each sum of

squares source in the table and sum them to get the sum of squares model, sum of squares error, and sum of squares total.

The other columns in the ANOVA table are all familiar. The degrees of freedom column still has $N - 1$ for our total, but now we have $N - 2$ for our error degrees of freedom and 1 for our model degrees of freedom; this is because simple linear regression only has one predictor, so our degrees of freedom for the model is always 1 and does not change. The total degrees of freedom must still be the sum of the other two, so our degrees of freedom error will always be $N - 2$ for simple linear regression. The mean square columns are still the SS column divided by the df column, and the test statistic F is still the ratio of the mean squares. Based on this, it is now explicitly clear that not only do regression and ANOVA have the same goal, but they are, in fact, the same analysis entirely. The only difference is the type of data we feed into the predictor side of the equations: continuous for regression and categorical for ANOVA.

Hypothesis Testing in Regression

Regression, like all other analyses, will test a null hypothesis in our data. In regression, we are interested in predicting Y scores and explaining variance using a line, the slope of which is what allows us to get closer to our observed scores than the mean of Y can. Thus, our hypotheses concern the slope of the line, which is estimated in the prediction equation by b (the slope of

a population, as opposed to b , which is the slope of a sample). Specifically, we want to test that the slope is not zero:

H_0 : There is no explanatory relationship between our variables

$$H_0 : \beta = 0$$

H_A : There is an explanatory relationship between our variables

$$H_A : \beta > 0$$

$$H_A : \beta < 0$$

$$H_A : \beta \neq 0$$

A non-zero slope indicates that we can explain values in Y based on X and therefore predict future values of Y based on X . Our alternative hypotheses are analogous to those in correlation: positive relationships have values above zero, negative relationships have values below zero, and two-tailed tests are possible. Just like ANOVA, we will test the significance of this relationship using the F statistic calculated in our ANOVA table compared to a critical value from the F distribution table. Let's take a look at an example and regression in action.

Example: Happiness and Well-being

Researchers are interested in explaining differences

in how happy people are based on how healthy people are. They gather data on each of these variables from 18 people and fit a linear regression model to explain the variance. We will follow the four-step hypothesis-testing procedure to see if there is a relationship between these variables that is statistically significant.

Step 1: State the Hypotheses

The null hypothesis in regression states that there is no relationship between our variables. The alternative states that there is a relationship, but because our research description did not explicitly state a direction of the relationship, we will use a non-directional hypothesis.

H_0 : There is no explanatory relationship between health and happiness

$$H_0 : \beta = 0$$

H_A : There is an explanatory relationship between health and happiness

$$H_A : \beta \neq 0$$

Step 2: Find the Critical Value

Because regression and ANOVA are the same analysis, our critical value for regression will come

from the same place: the F distribution table, which uses two types of degrees of freedom. We saw in the ANOVA table that the degrees of freedom for our numerator—the Model line—is always 1 in simple linear regression, and that the denominator degrees of freedom—from the Error line—is $N - 2$. In this instance, we have 18 people, so our degree of freedom for the denominator is 16. Going to our F table (a portion of which is shown in [Table 35.2](#)), we find that the appropriate critical value for 1 and 16 degrees of freedom is $F^* = 4.49$. (The complete F table can be found in the appendices)

Table 35.2. Critical

df: Denominator (Within)	df: Numerator (Between)					
	1	2	3	4	5	6
1	161	200	216	225	230	234
2	18.51	19.00	19.16	19.25	19.30	19.33
3	10.13	9.55	9.28	9.12	9.01	8.94
4	7.71	6.94	6.59	6.39	6.26	6.16
5	6.61	5.79	5.41	5.19	5.05	4.95
6	5.99	5.14	4.76	4.53	4.39	4.28
7	5.59	4.74	4.35	4.12	3.97	3.87
8	5.32	4.46	4.07	3.84	3.69	3.58
9	5.12	4.26	3.86	3.63	3.48	3.37

df: Denominator (Within)	df: Numerator (Between)					
	1	2	3	4	5	6
10	4.96	4.10	3.71	3.48	3.33	3.22
11	4.84	3.98	3.59	3.36	3.20	3.09
12	4.75	3.88	3.49	3.26	3.11	3.00
13	4.67	3.80	3.41	3.18	3.02	2.92
14	4.60	3.74	3.34	3.11	2.96	2.85
15	4.54	3.68	3.29	3.06	2.90	2.79
16	4.49	3.63	3.24	3.01	2.85	2.74

Step 3: Calculate the Test Statistic and Effect Size

The process of calculating the test statistic for regression first involves computing the parameter

estimates for the line of best fit. To do this, we first calculate the means, standard deviations, and sum of products for our X and Y variables, as shown below.

Table 35.3. Calculations of Means, Standard Deviations, and Sum of Products

X	(X – MX)	(X – MX)²	Y	(Y – MY)	(Y – MY)²	(X – MX)(Y – MY)
17.65	–2.13	4.53	10.36	–7.10	50.37	15.10
16.99	–2.79	7.80	16.38	–1.08	1.16	3.01
18.30	–1.48	2.18	15.23	–2.23	4.97	3.29
18.28	–1.50	2.25	14.26	–3.19	10.18	4.79
21.89	2.11	4.47	17.71	0.26	0.07	0.55
22.61	2.83	8.01	16.47	–0.98	0.97	–2.79
17.42	–2.36	5.57	16.89	–0.56	0.32	1.33
20.35	0.57	0.32	18.74	1.29	1.66	0.73
18.89	–0.89	0.79	21.96	4.50	20.26	–4.00

X	(X – MX)	(X – MX) ²	Y	(Y – MY)	(Y – MY) ²	(X – MX)(Y – MY)
18.63	-1.15	1.32	17.57	0.11	0.01	-0.13
19.67	-0.11	0.01	18.12	0.66	0.44	-0.08
18.39	-1.39	1.94	12.08	-5.37	28.87	7.48
22.48	2.71	7.32	17.11	-0.34	0.12	-0.93
23.25	3.47	12.07	21.66	4.21	17.73	14.63
19.91	0.13	0.02	17.86	0.40	0.16	0.05
18.21	-1.57	2.45	18.49	1.03	1.07	-1.62
23.65	3.87	14.99	22.13	4.67	21.82	18.08
19.45	-0.33	0.11	21.17	3.72	13.82	-1.22
356.02	0.00	76.14	314.18	0.00	173.99	58.29

From the raw data in our X and Y columns, we find that the means are $M_X = 19.78$ and $M_Y = 17.45$. The deviation scores for each variable sum to zero, so all is well there. The sums of squares for X and Y ultimately lead us to standard deviations of $s_X = 2.12$ and $s_Y = 3.20$. Finally, our sum of products is 58.29, which gives us a covariance of $cov_{XY} = 3.43$, so we know our relationship will be positive. This is all the information we need for our equations for the line of best fit.

First, we must calculate the slope of the line:

$$b = \frac{SP}{SS_X} = \frac{58.29}{76.14} = 0.77$$

This means that as X changes by 1 unit, Y will change by 0.77. In terms of our problem, as health increases by 1, happiness goes up by 0.77, which is a positive relationship. Next, we use the slope, along with the means of each variable, to compute the intercept:

$$\begin{aligned} a &= M_Y - bM_X \\ &= 17.45 - (0.77)(19.78) \\ &= 17.45 - 15.03 \\ &= 2.42 \end{aligned}$$

For this particular problem (and most regressions),

the intercept is not an important or interpretable value, so we will not read into it further. Now that we have all of our parameters estimated, we can give the full equation for our line of best fit:

$$\hat{Y} = 2.42 + 0.77X$$

We can plot this relationship in a scatter plot and overlay our line onto it, as shown in Figure 35.3.

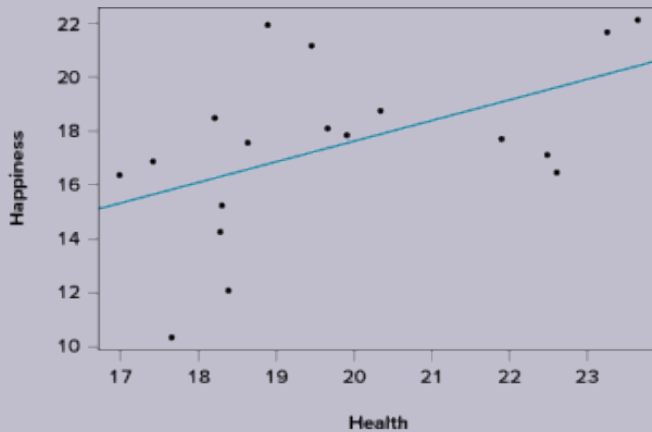


Figure 35.3. Health and Happiness Data and Line of Best Fit. [\[Image Description\]](#)

We can use the line equation to find predicted values for each observation and use them to calculate our sums of squares model and error, but this is tedious to do by hand, so we will let

the computer software do the heavy lifting in that column of our ANOVA table:

Table 35.4. Health and Happiness ANOVA, part 1

Source	SS	df	MS	F
Model	44.62			
Error	129.37			
Total				

Now that we have these, we can fill in the rest of the ANOVA table. We already found our degrees of freedom in Step 2:

Table 35.5. Health and Happiness ANOVA, part 2

Source	SS	df	MS	F
Model	44.62	1		
Error	129.37	16		
Total				

Our total line is always the sum of the other two lines, giving us:

Table 35.6. Health and Happiness ANOVA, part 3

Source	SS	df	MS	F
Model	44.62	1		
Error	129.37	16		
Total	173.99	17		

Our mean squares column is only calculated for the

model and error lines and is always our SS divided by our df, which is:

Table 35.7. Health and Happiness ANOVA, part 4

Source	SS	df	MS	F
Model	44.62	1	44.62	
Error	129.37	16	8.09	
Total	173.99	17		

Finally, our F statistic is the ratio of the mean squares:

Table 35.8. Health and Happiness ANOVA, part 5

Source	SS	df	MS	F
Model	44.62	1	44.62	5.52
Error	129.37	16	8.09	
Total	173.99	17		

This gives us an obtained F statistic of 5.52, which we will use to test our hypothesis.

Effect Size in Regression

We know that, because our statistic is significant, we should calculate an effect size. In regression, our effect size is variance explained, just like it was in ANOVA. Instead of using η^2 to represent this, we instead use R^2 as we saw in correlation—yet more evidence that all of these are the same analysis. (Note that in regression analysis, R^2 is typically capitalized, although for simple linear regression it represents the same value as r^2 we used in

correlation.) Variance explained is still the ratio of SS_M to SS_T :

$$R^2 = \frac{SS_M}{SS_T} = \frac{44.62}{173.99} = .26$$

We are explaining 26% of the variance in happiness based on health, which is a large effect size. (R^2 uses the same effect size cutoffs as η^2 .)

Step 4: Make the Decision

We now have everything we need to make our final decision. Our obtained test statistic was $F = 5.52$ and our critical value was $F^* = 4.49$. Since our obtained test statistic is greater than our critical value, we can reject the null hypothesis.

Reject H_0 . Based on our sample of 18 people, we can predict levels of happiness based on how healthy someone is, and the effect size was large, $F(1, 16) = 5.52$, $p < .05$, $R^2 = .26$.

[Figure 35.4](#) shows the output from JASP for this example.

Linear Regression

Model Summary - Happiness				
Model	R	R ²	Adjusted R ²	RMSE
H ₂	0.506	0.256	0.210	2.844

Note. Null model includes Health

ANOVA						
Model		Sum of Squares	df	Mean Square	F	p
H ₀	Regression	44.562	1	44.562	5.509	0.032
	Residual	129.431	16	8.089		
	Total	173.993	17			

Note. Null model includes Health

Coefficients						
Model		Unstandardized	Standard Error	Standardized	t	p
H ₀	(Intercept)	2.317	6.485		0.357	0.726
	Health	0.765	0.326	0.506	2.347	0.032

Figure 35.4. Output from JASP for the linear regression described in the Happiness and Well-Being example. The output provides the slope (b) of the line in the Coefficients table under Health Unstandardized (0.765), the y -intercept (a) of the line in the Coefficients table under (Intercept) Unstandardized (2.317, note slightly different from hand calculations due to rounding). The Model Summary – Happiness table provides R^2 (.256). The ANOVA table provides the result of the F test statistic. Based on our sample of 18 people, we can predict levels of happiness based on how healthy someone is, $F(1, 16) = 5.5$, $p = .032$. [\[Image Description\]](#)

Accuracy in Prediction

We found a large, statistically significant relationship between

our variables, which is what we hoped for. However, if we want to use our estimated line of best fit for future prediction, we will also want to know how precise or accurate our predicted values are. What we want to know is the average distance from our predictions to our actual observed values, or the average size of the residual ($Y - \hat{Y}$). The average size of the residual is known by a specific name: the standard error of the estimate ($s_{(Y-\hat{Y})}$), which is given by the formula

$$s_{(Y-\hat{Y})} = \sqrt{\frac{\sum(Y - \hat{Y})^2}{N - 2}}$$

This formula is almost identical to our standard deviation formula, and it follows the same logic. We square our residuals, add them up, and then divide by the degrees of freedom. Although this sounds like a long process, we already have the sum of the squared residuals in our ANOVA table! In fact, the value under the square root sign is just the SS_E divided by the df_E , which is called the **mean squared error**, or MS_E :

$$s_{(Y-\hat{Y})} = \sqrt{\frac{\sum(Y - \hat{Y})^2}{N - 2}} = \sqrt{MS_E}$$

For our example:

$$s_{(Y-\hat{Y})} = \sqrt{\frac{129.37}{16}} = \sqrt{8.09} = 2.84$$

So, on average, our predictions are just under 3 points away from our actual values. There are no specific cutoffs or guidelines for how big our standard error of the estimate can or should be; it is highly dependent on both our sample size and the scale of our original Y variable, so expert judgment should

be used. In this case, the estimate is not that far off and can be considered reasonably precise.

Multiple Regression and Other Extensions

Simple linear regression, as presented here, is only a stepping stone toward an entire field of research and application. Regression is an incredibly flexible and powerful tool, and the extensions and variations on it are far beyond the scope of this chapter. (Indeed, even entire books struggle to accommodate all possible applications of the simple principles laid out here.) The next step in regression is to study multiple regression, which uses multiple X variables as predictors for a single Y variable at the same time. The math of multiple regression is very complex, but the logic is the same: we are trying to use variables that are statistically significantly related to our outcome to explain the variance we observe in that outcome. Other forms of regression include curvilinear models that can explain curves in the data rather than the straight lines used here, as well as moderation models that change the relationship between two variables based on levels of a third. The possibilities are truly endless and offer a lifetime of discovery.

Equity Activity: School funding

In [the correlation chapter](#), we found a statistically significant correlation ($r=.41$) between school funding and standardized test scores, demonstrating that the more funding a school receives, the better the students perform on standardized tests. Using the following summary statistics, compute a line of best fit predicting test scores ($M= 78.5$, $s = 10.3$) from school funding ($M=5.2$ (i.e., 5,200 per student), $s = 1.8$), include the standard error of the estimate. How would a school that received \$3,000 per student be expected to perform on the standardized test?



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=167#h5p-49>

Practice Problems

Short Answer Reflections



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=167#h5p-50>

Test Your Knowledge



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=167#h5p-51>

License & Attribution

“Linear Regression” by Makenzie O’Neil is adapted from “[Linear Regression](#)” by Linda R. Cote Ph.D.; Rupa G. Gordon Ph.D., Chrislyn E. Randell Ph.D., Judy Schmitt, and Helena Marvin, which is licensed CC BY-NC-SA 4.0.

“Linear Regression” is licensed under CC BY-NC-SA 4.0.

Image Descriptions

Figure 35.1. This scatter plot illustrates a linear regression analysis with approximately 150 data points represented by black triangles, distributed across an x-axis ranging from 0 to 60 and a y-axis ranging from 60 to 150. A dashed red line runs diagonally upward from the lower left (around 0, 72) to the upper right (around 60, 120), representing the best-fit regression line. The data points show a moderate positive correlation with considerable scatter around the trend line.

Three vertical red lines are drawn from specific data points to the regression line, illustrating the concept of residuals (the vertical distances between observed values and predicted values). Two of these residual lines appear on the left side of the plot (around $x=2$ and $x=5$) and one on the right side (around $x=40$), highlighting how some points fall above the regression line (positive residuals) and others below it (negative residuals). This visualization demonstrates both the linear relationship between the variables and the variation in individual data points around the predicted values. Note that this plot lacks axis labels and a title, so the specific variables being measured are not identified. [[Return to Figure 35.1](#)]

Figure 35.2. This scatter plot demonstrates variance decomposition in linear regression, with approximately 150 black triangular data points plotted across an x-axis (0-60) and y-axis (60-150). A dashed red regression line shows the positive linear trend, while a solid blue horizontal line marks the mean of Y (around 88). Four sets of vertical lines illustrate how total deviation is partitioned: blue lines represent explained variation (distance from the mean to the regression line) and red lines represent unexplained variation or residuals (distance from observed points to the regression line). These demonstration lines appear at four x-values, showing points both above and below the regression line. This visualization illustrates that total variation equals explained variation plus residual variation in regression analysis. Note that axis labels

and title are not provided, so the specific variables are not identified. [[Return to Figure 35.2](#)]

Figure 35.3. This scatter plot illustrates the relationship between health and happiness, with the x-axis labeled “Health” ranging from 17 to 24 and the y-axis labeled “Happiness” ranging from 10 to 23. Approximately 20 black data points are scattered across the plot, showing considerable variation around the trend. A solid blue regression line runs diagonally upward from approximately (17, 15) to (24, 21), indicating a positive linear relationship where higher health scores are associated with higher happiness scores. The moderate scatter of points around the line suggests that while health is a predictor of happiness, there is substantial individual variation, with some people reporting high happiness despite lower health scores and others reporting lower happiness despite higher health scores. This visualization demonstrates that health explains some but not all of the variation in happiness levels. [[Return to Figure 35.3](#)]

Figure 35.4. This linear regression output titled “Linear Regression” contains three statistical tables analyzing the relationship between Health and Happiness. The Model Summary table shows model H_0 with multiple fit statistics: $R=0.506$, $R^2=0.256$, Adjusted $R^2=0.210$, and $RMSE=2.844$, with a note that “Null model includes Health.” The ANOVA table presents the model H_0 broken down by source: Regression (Sum of Squares=44.562, $df=1$, Mean Square=44.562), Residual (Sum of Squares=129.431, $df=16$,

Mean Square=8.089), and Total (Sum of Squares=173.993, $df=17$), with an F-statistic of 5.509 and p-value of 0.032, indicating statistical significance. The same note appears below this table. The Coefficients table displays unstandardized and standardized coefficients for model H_0 : the Intercept shows an unstandardized coefficient of 2.317 (Standard Error=6.485, $t=0.357$, $p=0.726$), while Health shows an unstandardized coefficient of 0.765 (Standard Error=0.326), standardized coefficient of 0.506, $t=2.347$, and $p=0.032$. These results indicate that Health is a statistically significant predictor of Happiness, explaining approximately 25.6% of the variance in happiness scores. [[Return to Figure 35.4](#)]

PART VI

WRITING &
PRESENTING
RESEARCH

36.

AMERICAN PSYCHOLOGICAL ASSOCIATION (APA) STYLE

Mireille Ukeye

Learning Objectives

By the end of this chapter, you will be able to

1. Define APA style and list several of its most important characteristics.
2. Identify three levels of APA style and give examples of each.

3. Identify multiple sources of information about APA style.
4. Identify the way those in the field of psychology present and share research, especially on equity-related issues, impacts public understanding and policy decisions.

What Is APA Style?

APA style is a set of guidelines for writing in psychology and related fields. These guidelines are set down in the **Publication Manual of the American Psychological Association** (APA, 2010)¹. The *Publication Manual* originated in 1929 as a short journal article that provided basic standards for preparing manuscripts to be submitted for publication (Bentley et al., 1929)². It was later

-
1. American Psychological Association. (2010). *Publication Manual of the American Psychological Association* (6th ed.). Washington, D.C.: American Psychological Association.
 2. Bentley, M., Peerenboom, C. A., Hodge, F. W., Passano, E. B., Warren, H. C., & Washburn, M. F. (1929). Instructions in regard to preparation of manuscript. *Psychological Bulletin*, 26, 57–63.

expanded and published as a book by the association and is now in its sixth edition. The primary purpose of APA style is to facilitate scientific communication by promoting clarity of expression and by standardizing the organization and content of research articles and book chapters. It is easier to write about research when you know what information to present, the order in which to present it, and even the style in which to present it. Likewise, it is easier to read about research when it is presented in familiar and expected ways.

APA style is best thought of as a “genre” of writing that is appropriate for presenting the results of psychological research—especially in



academic and professional contexts. It is not synonymous with “good writing” in general. You would not write a literary analysis for an English class, even if it were based on psychoanalytic concepts, in APA style. You would write it in Modern Language Association (MLA) style instead. And you would not write a newspaper article, even if it were about a new breakthrough in behavioral neuroscience, in APA style. You would write it in Associated Press (AP) style instead. At the same time, you would not write an empirical research report in MLA style, in AP style, or in the style of a romance novel, an email to a friend, or a shopping list. You would write it in APA style. Part of being a good writer in general is

adopting a style that is appropriate to the writing task at hand, and for writing about psychological research, this is APA style.

The Levels of APA Style

Because APA style consists of a large number and variety of guidelines—the *Publication Manual* is nearly 300 pages long—it can be useful to think about it in terms of three basic levels. The first is the overall organization of an article. Empirical research reports, in particular, have several distinct sections that always appear in the same order:

- **Title page.** Presents the article title and author names and affiliations.
- **Abstract.** Summarizes the research.
- **Introduction.** Describes previous research and the rationale for the current study.
- **Method.** Describes how the study was conducted.
- **Results.** Describes the results of the study.
- **Discussion.** Summarizes the study and discusses its implications.
- **References.** Lists the references cited throughout the article.



The second level of APA style can be referred to as **high-level style** (covered in “Writing Clearly and Concisely” of the *Publication Manual*),

which includes guidelines for the clear expression of ideas. There are two important themes here. One is that APA-style writing is formal rather than informal. It adopts a tone that is appropriate for communicating with professional colleagues—other researchers and practitioners—who share an interest in the topic. Beyond this shared interest, however, these colleagues are not necessarily similar to the writer or to each other. A graduate student in British Columbia might be writing an article that will be read by a young psychotherapist in Toronto and a respected professor of psychology in Tokyo. Thus, formal writing avoids slang, contractions, pop culture references, humor, and other elements that would be acceptable in talking with a friend or in writing informally.

The second theme of high-level APA style is that it is straightforward. This means that it communicates ideas as simply and clearly as possible, putting the focus on the ideas themselves and not on how they are communicated. Thus, APA-style writing minimizes literary devices such as metaphor, imagery, irony, suspense, and so on. Again, humor is kept to a minimum. Sentences are short and direct. Technical terms must be used, but they are used to improve communication,

not simply to make the writing sound more “scientific.” For example, if participants immersed their hands in a bucket of ice water, it is better just to write this than to write that they “were subjected to a pain-inducement apparatus.” At the same time, however, there is no better way to communicate that a between-subjects design was used than to use the term “between-subjects design.”

APA Style and the Values of Psychology

Robert Madigan and his colleagues have argued that APA style has a purpose that often goes unrecognized (Madigan, Johnson, & Linton, 1995)³. Specifically, it promotes psychologists’ scientific values and assumptions. From this perspective, many features of APA style that at first seem arbitrary actually make good sense. The following are several features of APA-style writing and the scientific values or assumptions they reflect.

3. Madigan, R., Johnson, S., & Linton, P. (1995). The language of psychology: APA style as epistemology. *American Psychologist*, 50, 428–436.

Table 36.1 APA-Style Writing and Reflected Values or Assumptions

APA style feature	Scientific value or assumption
There are very few direct quotations of other researchers.	The phenomena and theories of psychology are objective and do not depend on the specific words a particular researcher used to describe them.
Criticisms are directed at other researchers' work but not at them personally.	The focus of scientific research is on drawing general conclusions about the world, not on the personalities of particular researchers.
There are many references and reference citations.	Scientific research is a large-scale collaboration among many researchers.
Empirical research reports are organized with specific sections in a fixed order.	There is an ideal approach to conducting empirical research in psychology (even if this ideal is not always achieved in actual research).
Researchers tend to "hedge" their conclusions, e.g., "The results <i>suggest</i> that..."	Scientific knowledge is tentative and always subject to revision based on new empirical results.

Another important element of high-level APA style is the avoidance of language that is biased against particular groups. This is not only to avoid offending people—why would you want to offend people who are interested in your work?—but also for the sake of scientific objectivity and accuracy. For

example, the term *sexual orientation* should be used instead of *sexual preference* because people do not generally experience their orientation as a “preference,” nor is it as easily changeable as this term suggests (APA Committee on Lesbian, Gay, and Bisexual Concerns Joint Task Force on Guidelines for Psychotherapy With Lesbian, Gay, and Bisexual Clients, 2000)⁴.

The general principles for avoiding biased language are fairly simple. First, be sensitive to labels by avoiding terms that are offensive or have negative connotations. This includes avoiding terms that identify people with a disorder or other problem they happen to have. Instead, refer to the individual, what the *APA Publication Manual* refers to as putting the “person first.” For example, *people diagnosed with schizophrenia* is better than *schizophrenics*. Second, use more specific terms rather than more general ones. For example, *Chinese Americans* is better than *Asian Americans* if everyone in the group is, in fact, Chinese American. Third, avoid objectifying research participants. Instead, acknowledge their active contribution to the research. For example, “The *students completed* the questionnaire” is better than

4. American Psychological Association, Committee on Lesbian, Gay, and Bisexual Concerns Joint Task Force on Guidelines for Psychotherapy With Lesbian, Gay, and Bisexual Clients. (2000). Guidelines for psychotherapy with lesbian, gay, and bisexual clients. Retrieved from <https://www.apa.org/pi/lgbt/resources/guidelines>

“The *subjects were administered* the questionnaire.” Note that this principle also makes for clearer, more engaging writing. Table 36.2 shows several more examples that follow these general principles.

Table 36.2. Examples of Avoiding Biased Language

Instead of...	Use...
man, men	men and women, people
firemen	firefighters
homosexuals, gays, bisexuals	lesbians, gay men, bisexual men, bisexual women
minority	specific group label (e.g., African American)
neurotics	people scoring high in neuroticism
special children	children with learning disabilities

The previous edition of the *Publication Manual* strongly discouraged the use of the term *subjects* (except for nonhumans) and strongly encouraged the use of *participants* instead. The current edition, however, acknowledges that *subjects* can still be appropriate in referring to human participants in areas in which it has traditionally been used (e.g., basic memory research). But it also encourages the use of more specific terms when possible: *university students*, *children*, *respondents*, and so on.

The third level of APA style can be referred to as **low-level style**. Low-level style includes all the specific guidelines

pertaining to spelling, grammar, references and reference citations, numbers and statistics, figures and tables, and so on. There are so many low-level guidelines that even experienced professionals need to consult the *Publication Manual* from time to time. Table 36.3 contains some of the most common types of APA style errors based on an analysis of manuscripts submitted to one professional journal over a 6-year period (Onwuegbuzie, Combs, Slate, & Frels, 2010)⁵. These errors were committed by professional researchers but are probably similar to those that students commit the most too. See also Note 11.8 “Online APA Style Resources” in this section and, of course, the *Publication Manual* itself.

5. Onwuegbuzie, A. J., Combs, J. P., Slate, J. R., & Frels, R. K. (2010). Editorial: Evidence-based guidelines for avoiding the most common APA errors in journal article submissions. *Research in the Schools*, 16, ix–xxxvi

Table 36.3. Top 10 APA Style Errors

Error type	Example
1. Use of numbers	Failing to use numerals for 10 and above
2. Hyphenation	Failing to hyphenate compound adjectives that precede a noun (e.g., “role playing technique” should be “role-playing technique”)
3. Use of <i>et al.</i>	Failing to use it after a reference is cited for the first time
4. Headings	Not capitalizing headings correctly
5. Use of <i>since</i>	Using <i>since</i> to mean <i>because</i>
6. Tables and figures	Not formatting them in APA style; repeating information that is already given in the text
7. Use of commas	Failing to use a comma before <i>and</i> or <i>or</i> in a series of three or more elements
8. Use of abbreviations	Failing to spell out a term completely before introducing an abbreviation for it
9. Spacing	Not consistently double-spacing between lines
10. Use of “&” in references	Using & in the text or <i>and</i> in parentheses

Online APA Style Resources

The best source of information on APA style is

the *Publication Manual* itself. However, there are also many good websites on APA style, which do an excellent job of presenting the basics for beginning researchers. Here are a few of them.

[*APA Style*](#)

[*Purdue Online Writing Lab*](#)

[*Douglas Degelman's APA Style Essentials*](#)

[*Doc Scribe's APA Style Lite*](#)

APA-Style References and Citations

Because science is a large-scale collaboration among researchers, references to the work of other researchers are extremely important. Their importance is reflected in the extensive and detailed set of rules for formatting and using them.

References

At the end of an APA-style article or book chapter is a list that contains **references** to all the works cited in the text (and *only* the works cited in the text). The reference list begins on its

own page, with the heading “References,” centered in upper and lower case. The references themselves are then listed alphabetically according to the last names of the first-named author for each citation. (As in the rest of an APA-style manuscript, *everything* is double-spaced.) Many different kinds of works might be cited in APA-style articles and book chapters, including magazine articles, websites, government documents, and even television shows. Of course, you should consult the *Publication Manual* or Online APA Style Resources for details on how to format them. Here we will focus on formatting references for the three most common kinds of works cited in APA style: journal articles, books, and book chapters.

Journal Articles

For journal articles, the generic format for a reference is as follows:

Author, A. A., Author, B. B., & Author, C. C. (year).
Title of article. *Title of Journal*, volume(issue),
pp–pp. doi:xx.xxxxxxxxxx

Here is a concrete example:

Adair, J. G., & Vohra, N. (2003). The explosion of knowledge, references, and citations: Psychology's unique response to a crisis. *American Psychologist*, 58(1), 15–23. doi: 10.1037/0003-066X.58.1.15

There are several things to notice here. The reference includes a hanging indent. That is, the first line of the reference is not indented, but all subsequent lines are. The authors' names appear in the same order as on the article, which reflects the authors' relative contributions to the research. Only the authors' last names and initials appear, and the names are separated by commas with an ampersand (&) between the last two. This is true even when there are only two authors. Only the first word of the article title is capitalized. The only exceptions are for words that are proper nouns or adjectives (e.g., "Freudian") or if there is a subtitle, in which case the first word of the subtitle is also capitalized. In the journal title, however, all the important words are capitalized. The journal title and volume number are italicized; however, the issue number (listed within parentheses) is not. At the very end of the reference is the digital object identifier (DOI), which provides a permanent link to the location of the article on the Internet. Include this if it is available. It can generally be

found in the record for the item on an electronic database (e.g., PsycINFO) and is usually displayed on the first page of the published article.

Books

For a book, the generic format and a concrete example are as follows:

Author, A. A. (year). *Title of book*. Location:
Publisher.

Kashdan, T., & Biswas-Diener, R. (2014). *The upside of your dark side*. New York, NY: Hudson Street Press.

Book Chapters

For a chapter in an edited book, the generic format and a concrete example are as follows:

Author, A. A., Author, B. B., & Author, C. C. (year).

Title of chapter. In A. A. Editor, B. B. Editor, & C. C. Editor (Eds.), *Title of book* (pp. xxx–xxx). Location: Publisher.

Lilienfeld, S. O., & Lynn, S. J. (2003). Dissociative identity disorder: Multiple personalities, multiple controversies. In S. O. Lilienfeld, S. J. Lynn, & J. M. Lohr (Eds.), *Science and pseudoscience in clinical psychology* (pp. 109–142). New York, NY: Guilford Press.

Notice that references for books and book chapters are similar to those for journal articles, but there are several differences too. For an edited book, the names of the editors appear with their first and middle initials followed by their last names (not the other way around)—with the abbreviation “Eds.” (or “Ed.,” if there is only one) appearing in parentheses immediately after the final editor’s name. Only the first word of a book title is capitalized (with the exceptions noted for article titles), and the entire title is italicized. For a chapter in an edited book, the page numbers of the chapter appear in parentheses after the book title with the abbreviation “pp.” Finally, both formats end with the location of publication and the publisher, separated by a colon.

Reference Citations

When you refer to another researcher's idea, you must include a **reference citation** (in the text) to the work in which that idea originally appeared and a full reference to that work in the reference list. What counts as an idea that must be cited? In general, this includes phenomena discovered by other researchers, theories they have developed, hypotheses they have derived, and specific methods they have used (e.g., specific questionnaires or stimulus materials). Citations should also appear for factual information that is not common knowledge, so that other researchers can check that information for themselves. For example, in an article on the effect of cell phone usage on driving ability, the writer might cite official statistics on the number of cell phone–related accidents that occur each year. Among the ideas that do not need citations are widely shared methodological and statistical concepts (e.g., between-subjects design, *t* test) and statements that are so broad that they would be difficult for anyone to argue with (e.g., “Working memory plays a role in many daily activities.”). Be careful, though, because “common knowledge” about human behavior is often incorrect. Therefore, when in doubt, find an appropriate reference to cite or remove the questionable assertion.

When you cite a work in the text of your manuscript, there are two ways to do it. Both include only the last names of the authors and the year of publication. The first method is to use

the authors' last names in the sentence (with no first names or initials), followed immediately by the year of publication in parentheses. Here are some examples:

Burger (2008) conducted a replication of Milgram's (1963) original obedience study.

Although many people believe that women are more talkative than men, Mehl, Vazire, Ramirez-Esparza, Slatcher, and Pennebaker (2007) found essentially no difference in the number of words spoken by male and female college students.

Notice several things. First, the authors' names are treated grammatically as names of people, not as things. It is better to write "a replication of Milgram's (1963) study" than "a replication of Milgram (1963)." Second, when there are two authors, the names are not separated by commas, but when there are three or more authors, they are. Third, the word *and* (rather than an ampersand) is used to join the authors' names. Fourth, the year follows immediately after the final author's name. An additional point, which is not illustrated in these examples but is illustrated in the sample paper in Section 11.2 "Writing a Research Report in American Psychological Association (APA) Style," is that the year only needs to be

included the first time a particular work is cited in the same paragraph.

The second way to cite an article or a book chapter is parenthetically—including the authors' last names and the year of publication in parentheses following the idea that is being credited. Here are some examples:

People can be surprisingly obedient to authority figures (Burger, 2008; Milgram, 1963).

Recent evidence suggests that men and women are similarly talkative (Mehl, Vazire, Ramirez-Esparza, Slatcher, & Pennebaker, 2007).

One thing to notice about such parenthetical citations is that they are often placed at the end of the sentence, which minimizes their disruption to the flow of that sentence. In contrast to the first way of citing a work, this way always includes the year—even when the citation is given multiple times in the same paragraph. Notice also that when there are multiple citations in the same set of parentheses, they are organized alphabetically by the name of the first author and separated by semicolons.

There are no strict rules for deciding which of the two citation styles to use. Most articles and book chapters contain

a mixture of the two. In general, however, the first approach works well when you want to emphasize the person who conducted the research—for example, if you were comparing the theories of two prominent researchers. It also works well when you are describing a particular study in detail. The second approach works well when you are discussing a general idea, and especially when you want to include multiple citations for the same idea.

The third most common error in Table 36.3 has to do with the use of *et al.* This is an abbreviation for the Latin term *et alia*, which means “and others.” In APA style, if an article or a book chapter has *more than two authors but fewer than six*, you should include all their names when you first cite that work. After that, however, you should use the first author’s name followed by “et al.” If the article has only two authors, then both should be included in every citation. If an article has six or more authors, then you should only list the name of the first author followed by et al. each and every time you cite that work (even the first time). Here are some examples:

Recall that Mehl et al. (2007) found that women and men spoke about the same number of words per day on average.

There is a strong positive correlation between the number of daily hassles and the number of symptoms people experience (Kanner et al., 1981).

Notice that there is no comma between the first author's name and "et al." Notice also that there is no period after "et" but there is one after "al." This is because "et" is a complete word and "al." is an abbreviation for the word *alia*.

Equity Activity: Equity in Gifted Education

Review this [article by Grissom and Redding \(2016\)](#) on the underrepresentation of high-achieving students of color in gifted programs.



An interactive H5P element has been excluded from this version of the text. You can view it online here:

[https://smc.pressbooks.pub/
critresearchmethodpsych/?p=170#h5p-52](https://smc.pressbooks.pub/critresearchmethodpsych/?p=170#h5p-52)

Practice Problems

Test Your Knowledge



An interactive H5P element has been excluded from this version of the text. You can view it online here:

[https://smc.pressbooks.pub/
critresearchmethodpsych/?p=170#h5p-53](https://smc.pressbooks.pub/critresearchmethodpsych/?p=170#h5p-53)

License & Attribution

“APA Style” by Mireille Ukeye is adapted from “[American Psychological Association \(APA\) Style](#)” by Rajiv S. Jhangiani; I-Chant A. Chiang; Carrie Cuttler; and Dana C. Leighton is licensed CC BY 4.0.

“American Psychological Association (APA) Style” is licensed under CC BY-NC-SA 4.0.

37.

DISSEMINATION

Mireille Ukeye

Learning Objectives

By the end of this chapter, you will be able to

1. Identify and describe both traditional and nontraditional channels for sharing research findings and the audiences they reach.
2. Develop strategies for effectively disseminating research, including tailoring messages, choosing appropriate formats, and ensuring ethical and accurate communication.

After completing a research study and writing it up in an APA-

style report, the next critical step is dissemination; sharing your findings with others so that your work can inform science, practice, and the broader community. Dissemination differs from writing up research. Writing organizes and presents your study in a coherent report, while dissemination focuses on how your research reaches its audience. Effective dissemination ensures that your findings are noticed, understood, and used. Research that remains unread cannot influence theory, inform practice, or educate the public.

Dissemination can take many forms. Some methods are traditional, aimed at the academic community and peer review, while others are nontraditional, designed to reach broader or more diverse audiences. Choosing the right dissemination strategies requires careful consideration of your goals, audience, and ethical responsibilities, including maintaining participant confidentiality and accurately representing your results.



Traditional Channels of Dissemination

The most familiar methods for disseminating research are traditional channels, which include peer-reviewed journal articles, conference presentations, and research reports. Peer-reviewed journal articles are the standard for academic dissemination. They provide a permanent, citable record of your study and are evaluated for quality and credibility through a formal review process. For example, publishing an article on cognitive behavioral therapy for child anxiety in the *Journal of Clinical Psychology* ensures that other researchers can build on your work and compare results across studies.

Conference presentations and posters are another traditional channel. Presenting at professional meetings allows researchers to communicate their findings directly to other

scholars, receive immediate feedback, and engage in discussions that can spark new ideas. For instance, a student might present a poster on social media use and mental health at the annual American Psychological Association (APA) convention, highlighting key results and receiving questions that refine their interpretations.

Technical or research reports are often prepared for specific organizations, funding agencies, or institutions. These reports can provide more detailed information than typical journal articles and may be designed for audiences who are implementing programs or policies. For example, a report summarizing program evaluation data for a school district can inform administrators about effective interventions and guide decision-making.

Table 1. Traditional Dissemination Channels

Channel	Audience	Purpose	Example
Peer-reviewed journal article	Academic researchers	Provide a permanent, rigorous, and citable record	Publishing findings on CBT for child anxiety in the <i>Journal of Clinical Psychology</i>
Conference presentation/poster	Academic researchers and practitioners	Share preliminary findings, receive feedback, and network	Poster on social media use and mental health at APA convention
Technical or research report	Organizations, policymakers, program administrators	Provide detailed data and actionable recommendations	School district report evaluating reading intervention programs

Nontraditional Channels of Dissemination

Nontraditional dissemination methods are increasingly important for reaching broader audiences, including practitioners, policymakers, and the public. Blogs and online articles allow researchers to present findings in accessible language and summarize key takeaways in ways that nonexperts can understand. For example, a student could write

a blog post explaining strategies for reducing test anxiety based on their study results, providing practical tips for teachers and students.

Podcasts and video platforms provide opportunities to discuss research in engaging audio or visual formats. Researchers might host a podcast episode on the effects of sleep on memory, highlighting their study and interviewing experts or participants, while videos can provide dynamic explanations that are more likely to be shared widely. Social media platforms such as Instagram and TikTok can be used to disseminate findings through threads, infographics, or short video summaries, allowing research to reach thousands of people quickly. For instance, a thread summarizing results from a study on exercise and mood can draw attention from other researchers, journalists, or the public.

Newsletters and community outreach also play a role in dissemination, particularly for research intended to inform local programs or policies. Translating study findings into practical recommendations for schools, clinics, or community organizations ensures that the research has a tangible impact. For example, a school newsletter explaining results from a study on classroom interventions for attention can guide teachers' practices and improve student outcomes.

Table 2. Nontraditional Dissemination Channels

Channel	Audience	Purpose	Example
Blog or online article	General public, practitioners	Summarize research in an accessible language	Blog post on strategies for reducing test anxiety
Podcast or video	General public, practitioners	Engage audiences with audio-visual content	Podcast on sleep and memory research
Social media	Broad online audience	Rapid dissemination and engagement	A social media thread summarizing exercise and mood study
Newsletter or community outreach	Local organizations, schools, or policymakers	Translate findings into actionable recommendations	School newsletter on classroom interventions for attention

Strategies for Effective Dissemination

Dissemination is most effective when it is planned and intentional. Researchers should consider who their audience is, what message they want to communicate, and which format will maximize reach and comprehension. Tailoring language to the audience, highlighting the relevance and practical implications of the research, and using visuals or summaries can enhance understanding. Even when reaching nonacademic audiences, it is essential to maintain accuracy, avoid overstating results, and respect ethical considerations.

Planning dissemination early, alongside study design, ensures that your research will have maximum impact once it is completed.

In summary, dissemination is a critical component of the research process. It is not the same as writing a research report, but it is closely linked to it, as the written report often forms the basis for further sharing. By thoughtfully choosing dissemination channels and strategies, researchers can ensure that their work reaches the appropriate audiences, informs science and practice, and contributes to public understanding of psychology.

Equity Activity: Research on discrimination

Read this article: [Racial/ethnic discrimination and well-being during adolescence: A meta-analytic review](#). [New Tab].



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=172#h5p-54>

Practice Problems

Test Your Knowledge



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=172#h5p-55>

License & Attribution

“Dissemination” by Mireille Ukeye is adapted from “[Writing a Research Report in American Psychological Association \(APA\) Style](#)” by Rajiv S. Jhangiani; I-Chant A. Chiang; Carrie Cuttler; and Dana C. Leighton is licensed CC BY 4.0.

“Writing a Research Report in American Psychological Association (APA) Style” is licensed under CC BY-NC-SA 4.0.

38.

POSTERS & PRESENTATIONS

Stephanie D'Costa

Learning Objectives

By the end of this chapter, you will be able to

1. Identify the rudiments of oral and poster presentations.
2. Articulate positionalities when writing & presenting research findings.
3. List actionable steps in their research presentations to address the social issue.

Types of Conference Presentations

Conference presentations take many forms. Before submitting an abstract to a conference, be sure to consider what kind of presentation you want to make. Below, we discuss some common presentation types:

- **Traditional Paper/Oral Presentation:** This is the standard oral presentation (usually 15 minutes plus additional time at the end for questions) where one or more speakers (joint-presenters) share research results, completed works, innovative concepts, theoretical applications, methodologies, or tools.
- **Student Presentation:** These are similar to the traditional paper/oral presentations described above, but with an emphasis on students' work. By providing a separate avenue for students to share their work or labelling the presentation as "students", the pressure can be lessened. Sometimes, students have separate sessions, but other times, they are grouped with other paper presentations. If this is the case, the presentation is usually identified as student presentations in the program.
- **Poster Presentation:** This is a less formal opportunity to share your work in a visual format. We discuss this in greater depth later in the chapter.

- **Panel Presentation:** This is where multiple speakers present their perspective on a common issue, usually for 60 to 90 minutes. While many students prefer to present posters or shorter oral presentations, if a group of students has a common research interest or concern, they can apply to a conference to present on a panel. The speakers are responsible for coordinating the panel and assigning roles (such as moderator). Each speaker on a panel is usually given at least one individual question as well as an introductory and closing remark.
- **Roundtables:** are similar to panels in the sense that a group of discussants seated around a table comment on a theme. Roundtable presenters bring targeted questions to pose to participants at the table in order to learn from and with those attending. It is quite unlikely that you will present your work on a roundtable, but you can check out conference websites if you wish to learn more (see Box for a list of potential conferences).
- **Lightning Round-Tables:** These are opportunities to network by briefly summarizing your work to a small audience (usually in 15 minutes or less), followed by an interactive discussion. Discussants will then move to another table and repeat the procedure. This provides the opportunity to get more intimate connections for other participants and attendees.

In addition to the above presentations, at a conference, you

will likely see *expert lectures, keynote addresses*, and *debates*. These are presented by established academics in the field, so we will not discuss them. However, it is a great idea to go to these presentations at conferences. For the rest of the chapter, we will focus on oral presentations and posters because these are what you will most likely present at conferences. If you wish to submit an abstract for other presentation types, be sure to discuss it with your advisor, supervisor, or mentor.

Oral Presentations

A well-organized oral presentation typically has the following elements: a background, research question, outline of research and the talk (methods), findings/discussion, and conclusion. Below, we elaborate on 12 tips to help you successfully showcase these elements and execute a noteworthy presentation.

Planning

Planning is fundamental to you delivering a successful oral presentation. Almost every conference you attend will have a set of guidelines for you to adhere to (e.g., the time limit). Begin by familiarizing yourself with those guidelines. Remember that you will not be able to share your entire research project or all the interesting findings on one 18×24

poster or in a 15-minute presentation. You will need to zoom in on a *specific issue or research question* from your project.

Table 38.1 shows an example of how one might plan a 30-minute oral presentation. When planning, highlight the significant portions of each section: the introduction, positionality, literature review, methods, findings, discussion, and conclusion. Give rough outlines of how much time each section will take, and test this outline to ensure you will not be going over (example of fifteen-minute speech plan in Box 12.3.1.1). It is best to try to get your practice done at least one to two minutes less than the designated time (e.g., aim to finish a 30-minute presentation in 28 minutes). On the day, nerves, technical issues, and other factors can make you go for longer than you practiced. In general, program chairs keep very strict timing and give you frequent time updates during your presentation. Planning with built-in flexibility can help calm your nerves before and during your presentation.

Example Outline of Oral Presentation

Table 38.1. Oral Presentation Outline

Section	Time (30 minutes total)
Introduction (Hook, engaging example etc.)	2 minutes
Positionality Statement	2 minutes
Research Question & Outline of the talk	2 minutes
Literature Review (optional)	5 minutes
Methods	5 minutes
Findings	8 minutes
Discussion/Conclusion	5 minutes

The Introduction

Because oral presentations at conferences are short, you must aim to quickly entice your audience. Common ways to do this include starting with a historical anecdote or story related

to your topic, unpacking a key quote from your qualitative study, introducing a paradox in your field, asking a provoking question to your audience, and inquiring what a seemingly straightforward concept in your field really means in practice.

Positionality Statement

Make sure to briefly describe who you are and why you conducted this research. Highlight other team members and identify their positionality and connection to the research topic. Include aspects of your professional and personal identity that help you understand the research topic in nuanced ways. Make sure to keep this brief.

Research Question(s) & Order

Once you have introduced your topic, immediately state your research question(s) and use that momentum to guide your listeners through the methods and findings. If outlined on a slide, put it on the same slide or on the next. After this, aim to incorporate it in the context of your presentation. Answer how your presentation will be structured and tell your audience how this structure will address your research question.

Literature Review (optional)

This is a section that you can skip in a presentation, but if you

do decide to keep it, make it sparse. We suggest picking one or two *key authors* that inspired your study or separating the key concepts in the literature that have inspired your study. As with all types of research, outline the literature with close attention to the gap you are going to fill.

Methods (necessary, but shortened)

Give the short version of your methods. You are allowed, in an oral presentation, to just be as simple as saying a “grounded theory approach”. Suggest to your audience that they can ask you further about your methods in the question period. This is a place where you should also consider talking about the limitations of your research. However, saving it for after the conclusion to make clear to your audience that the implications of your study can be strengthened in future research is another useful strategy.

Findings (the heart, but keep it concise)

Consider shortening your findings to just two or three themes. Especially in qualitative research, going down every rabbit hole with regard to your findings will distract from the core point of your presentation: your contribution. Highlight only those findings which you think are (1) unique, (2) useful to others, (3) best answer your research question, and (4) capable of

being conveyed in your very limited timeframe. For instance, Wilson's (2021) research on Uber had five themes which he shortened to three for his fifteen-minute conference presentation. As he explained, "I chose the three themes that addressed the legislative impact of Uber's framing in order to best address one of my RQ's: "did Uber's framing in the media affect the final legislative decisions?"

Discussion

The discussion is a section that can be easily truncated into your conclusion or at the end of the findings section. It is essential, however, that you implicate the meaning of your findings for the field. What was the gap you filled? How do your findings corroborate with past research on your subject? Whenever covering any of your findings, consider how they affect the field of your audience: what does your work say about their work? Likewise, in disseminating your research for the community, this final part is essential: how do your findings affect their day-to-day lives? Will it affect a policy that governs their behavior, etc?

Conclusion, Limitations, and Implications

The conclusion is where you outline what you have said, what is missing from your study, and what can be done in the future.

Clearly summarize the key findings of your talk before talking about what is missing. Once you have summarized the findings, be humble! Talk about the limitations of your research and briefly discuss how they could be addressed in future studies. Once the ground is laid, you are now ready to resoundingly end your presentation: summarize the major themes of your research into implications – the contribution of your work. Why should everyone remember the work you do? It is entirely based upon your ability to convince them that the research is worth remembering into the future: in future work, research, and reflection. Implications often take two forms: for *future research* and for *action* outside of research. When speaking to academics, the first is more important. When discussing with the broader community, the latter will likely be higher valued.

Designing Your Slides

The most important thing to remember about designing your slides is to keep them clean, clear, and engaging. Do not include too much text on one page and ensure that the colors used are accessible. You might also consider using concept maps (Google slides has lots of pyramid animations). Whatever you do, keep slides sparse, do not pick Roman fonts, be consistent, and use bold quotes. For additional tips, see [Brian Campbell's website](#) for suggestions.

Trimming the Excess

It is unlikely that you will get the timing right on your first practice run. It is okay to allow yourself to go over (or under) to begin, but ensure you can make the necessary adjustments to each section of your presentation. If you are still extremely over the time limit (and you should aim to go a little under, so you can take it slow for your presentation), then you should cut full sections. Consider removing your literature review, compressing your discussion into your findings or conclusion, and/or taking off one of your findings sections. If you are still over the limit, consider shortening your research question or focusing on fewer research questions. If your presentation is too short, consider expanding on your findings and the discussions.

Fear of Public Speaking/Cooling Off Before You Speak

Composure and confidence will make your presentation go over smoother. Speaking with confidence—in a clear, steady voice—is essential to winning the confidence of your audience. However, sometimes you are just overrun by nerves. If that is the case, you are not alone. Fear of public speaking is extremely common, but there are things that you can do to help calm those nerves. See [Mayo Clinic's website](#) for a list of suggestions.

Student testimony: Negotiating with your nerves

A quick online search for “presentation tips” will yield an overwhelming number of suggestions, but everyone’s nervousness may come from different places. Before you dive into looking for advice, ask yourself why you are nervous. Worried about presentation content? Create a list of major points you want to get across. Worried about going overtime? Cut down on unnecessary content and time each slide. Worried that you’ll feel intimidated by the audience? Plant a friend in the audience and look at them. Regardless, it’s important to give yourself enough time to prepare and practice for the presentation.

Here is four pieces of advice that helps me get through every presentation:

1. Remember that you know something that the audience doesn’t. (No audience is all-knowing. Presenting your ideas and teaching others should be an empowering

experience.)

2. Mention some things you find interesting and are passionate about related to the content. (This could be a surprising finding in your research, or an interesting encounter during data collection.)
3. Prepare a script, rely on bullet points sentences, and avoid long paragraphs. (Bullet points help you stay on track with all the information you want to cover and give you room to improvise if needed.)
4. Create a presentation ritual. (Find something that calms you down or makes you feel confident. This can be wearing a shirt you feel confident in or drinking some warm tea before you practice and present.)

Our nervousness often comes from a prediction of how we think the audience is going to perceive us, and our brains are great at coming up with reasons why we might not deliver a satisfactory presentation. Don't be fooled – these thoughts are often inaccurate. Learn to doubt your doubts!

*Youcheng (Mark) Ding, UBC Sociology Honours
Student, 2019-2020*

Poster Presentations

All of the rules for oral presentations apply to some degree to poster presentations, but with some important qualifications. As most poster presentations do not have the same oral component (online conferences excepted), the structure of your talk is not constrained by what you can say in ten minutes. It is constrained by what you can put in an 18×24 poster. Most people will not spend more than ten minutes on your poster, and if they do, their eyes will have to be flagged by vital information early. It is therefore important that your poster is able to quickly convey the bigger picture of your argument, quickly shouting why your audience should care about it before going into the other relevant details. The following tips are aimed at that purpose.

The General Rule about Font and Spacing

Posters typically use a 16 pt font, and it is an unspoken rule

that 60% of your poster should be empty space and the rest image and text. We suggest picking two fonts: 18 pt for bolded statements and titles, and 16 pt for regular text. Also, academic conferences will often state their own rules for the poster presentation. Write down these rules and organize everything else accordingly.

Titles Are Your Vitals

The titles will give your reader a sense of the direction of your entire paper. It is therefore important that they are exceptionally clear and purposeful. While it is okay to say “introduction, RQ, Methods,” we suggest that you be even more to the point: “Washington School Districts Reading Scores, Single Case Analysis, Results: Bilingual students’ growth, etc.” Make use of subtitles (even with a slightly different colored font which thematically fits your color schema) to etch the entire purpose of your poster and grab the reader’s attention.

Put That RQ in Bold

Try to put your research question in a separate, bolded, enlarged section so your reader will be able to immediately decide if your poster is related to their interest.

Concept Maps & Diagrams

Even if you do not use the concept map, it is good to draw out your poster or oral presentation on a concept map before constructing your final script. This way, you can easily narrow down the structure of your presentation. And just as it makes things easier for you to conceptualize, so will it make things easier for your audience, so consider adding them as an outline of your paper or to explain a complicated relationship in your findings. PowerPoint and Google Slides both have solid tools for designing diagrams and concept maps.

Avoid Big Text Blurbs

Do not use wordy paragraphs on your poster. A poster must aim to balance telling with showing. Make use of diagrams, select key quotations, and one-sentence summaries of your key findings. In quantitative research, this task will be much easier, since the poster can be simply structured to highlight graphs of the key findings.

Data Visualizations

Data visualization is important on posters, especially for quantitative research. Consider the types of graphs that present your findings and are easy to understand.

Aesthetic Considerations

Aesthetic concerns apply, but balanced alongside clarity. You want to grab your audience's attention, but then justify that attention by elucidating an important point in their field. Use consistent and complementary color schemes. Try primary colors, but nothing too gaudy like neon. Look up a color wheel and research a scheme that will work for you. Consider downloading and using the templates of posters. Do not use different fonts, etc.

Engaging Your Audience

At academic conferences, it is not necessary that you talk to everyone who looks at your poster. You are more than welcome to play the sheepish store clerk who smiles and waits patiently to see if the customers in their store will buy their goods. An alternative to this, however, is to attempt to engage passersby in your work. For instance, ask them politely if they want an overview of the work or if they have also done research in the same field. Direct them to the research question and to the key findings of your research. Let them know if you think your research will have implications for their field. By touching on these questions near the heart of their disciplines, you can show them your value as a contributor.



One or more interactive elements has been excluded from this version of the text. You

can view them online here:

[https://smc.pressbooks.pub/](https://smc.pressbooks.pub/critresearchmethodspsysh/?p=178#oembed-1)

[critresearchmethodspsysh/?p=178#oembed-1](https://smc.pressbooks.pub/critresearchmethodspsysh/?p=178#oembed-1)

*Video 38.1. [How To Present an Academic Research Poster](#) by
[Syracuse University iSchool](#)*

Equity Activity: Non-traditional dissemination of research

Posters, presentations, and research papers are the traditional ways researchers disseminate research. The purpose of research dissemination is to make sure that the data gets into the hands of the communities that it is serving. There are some limitations on who can access our research findings. However, there are creative ways to share research, such as podcasts, infographics, and even TikTok videos! Here are some examples for you to consider (select the image to enlarge):

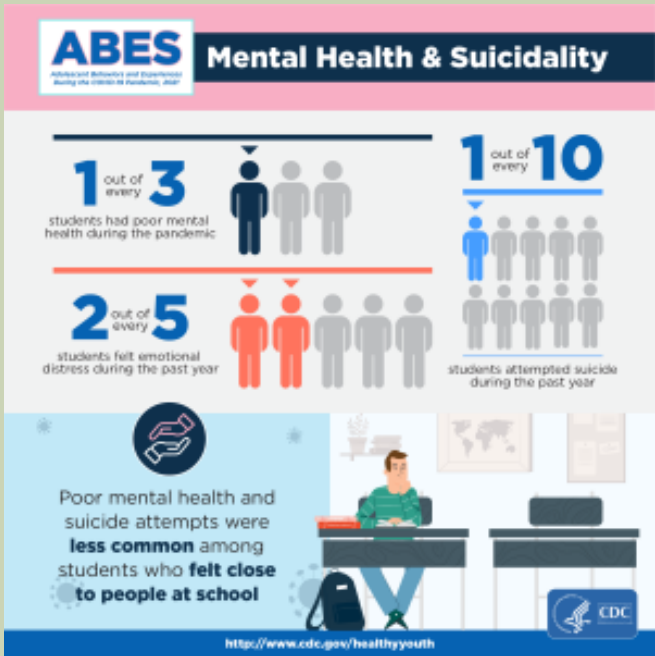


Figure 38.1. An infographic on mental health and suicidality statistics among LGBTQ+ youth. [ABES Infographic Accessible]

Speaking of Psychology podcast logo. A brain and a conversation bubble on yellow background

Figure 38.2. APA Podcast Logo

Liberation now! podcast logo with a stylized X
symbol in white within an orange and red circle,
surrounded by yellow radiating sound waves, with
"podcast" written in red

Figure 38.3. Liberation Now! Podcast Logo



Figure 38.4. Honoring Immigrant Mothers Through Storytelling.



One or more interactive elements has been excluded from this version of the

text. You can view them online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=178#oembed-2>

Video 38.2. *Indigenous Research: Beadworking as an Indigenous Research Paradigm* by *Tiffany Prete*



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://smc.pressbooks.pub/critresearchmethodpsych/?p=178#h5p-56>

License & Attribution

“Posters & Presentations” by Stephanie D’Costa

is adapted from “[Types of Conference Presentations](#)”, “[Oral Presentations](#)”, “[Poster Presentations](#)” by Oral Robinson and Alexander Wilson is licensed CC BY-NC-SA 4.0.

“Posters & Presentations” is licensed under CC BY-NC-SA 4.0.

Media Attributions

- DASH_ABES_Social_Mental-Health_Suicidality_IG

APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z TABLE)

Stephanie D'Costa; Mireille Ukeye;
Makenzie O'Neil; and Rebecca Anguiano

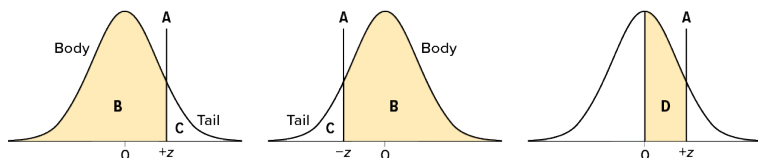


Figure A.1 Three Normal Curves.

Figure A1 Image Description

A row of three beige, bell-curve diagrams on a black background illustrates how areas are labeled on a z -distribution. Left panel: the area from the mean ($z = 0$) to a cutoff left of center is shaded and labeled B—the “body” between the mean and a negative z . Middle panel: the area from the mean to a cutoff right of center is shaded and labeled B—the “body” between the mean and a positive z . Right panel: only the extreme right tail beyond a positive z is shaded and labeled D—the tail probability to the right of z .

Table A.1. Standard Normal Distribution Table (z Table)

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
0.00	.5000	.5000	.0000
0.01	.5040	.4960	.0040
0.02	.5080	.4920	.0080
0.03	.5120	.4880	.0120
0.04	.5160	.4840	.0160
0.05	.5199	.4801	.0199
0.06	.5239	.4761	.0239
0.07	.5279	.4721	.0279
0.08	.5319	.4681	.0319
0.09	.5359	.4641	.0359
0.10	.5398	.4602	.0398
0.11	.5438	.4562	.0438

APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z
TABLE) | 899

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
0.12	.5478	.4522	.0478
0.13	.5517	.4483	.0517
0.14	.5557	.4443	.0557
0.15	.5596	.4404	.0596
0.16	.5636	.4364	.0636
0.17	.5675	.4325	.0675
0.18	.5714	.4286	.0714
0.19	.5753	.4247	.0753
0.20	.5793	.4207	.0793
0.21	.5832	.4168	.0832
0.22	.5871	.4129	.0871
0.23	.5910	.4090	.0910

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
0.24	.5948	.4052	.0948
0.25	.5987	.4013	.0987
0.26	.6026	.3974	.1026
0.27	.6064	.3936	.1064
0.28	.6103	.3897	.1103
0.29	.6141	.3859	.1141
0.30	.6179	.3821	.1179
0.31	.6217	.3783	.1217
0.32	.6255	.3745	.1255
0.33	.6293	.3707	.1293
0.34	.6331	.3669	.1331
0.35	.6368	.3632	.1368

APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z
TABLE) | 901

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
0.36	.6406	.3594	.1406
0.37	.6443	.3557	.1443
0.38	.6480	.3520	.1480
0.39	.6517	.3483	.1517
0.40	.6554	.3446	.1554
0.41	.6591	.3409	.1591
0.42	.6628	.3372	.1628
0.43	.6664	.3336	.1664
0.44	.6700	.3300	.1700
0.45	.6736	.3264	.1736
0.46	.6772	.3228	.1772
0.47	.6808	.3192	.1808

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
0.48	.6844	.3156	.1844
0.49	.6879	.3121	.1879
0.50	.6915	.3085	.1915
0.51	.6950	.3050	.1950
0.52	.6985	.3015	.1985
0.53	.7019	.2981	.2019
0.54	.7054	.2946	.2054
0.55	.7088	.2912	.2088
0.56	.7123	.2877	.2123
0.57	.7157	.2843	.2157
0.58	.7190	.2810	.2190
0.59	.7224	.2776	.2224

APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z
TABLE) | 903

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
0.60	.7257	.2743	.2257
0.61	.7291	.2709	.2291
0.62	.7324	.2676	.2324
0.63	.7357	.2643	.2357
0.64	.7389	.2611	.2389
0.65	.7422	.2578	.2422
0.66	.7454	.2546	.2454
0.67	.7486	.2514	.2486
0.68	.7517	.2483	.2517
0.69	.7549	.2451	.2549
0.70	.7580	.2420	.2580
0.71	.7611	.2389	.2611

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
0.72	.7642	.2358	.2642
0.73	.7673	.2327	.2673
0.74	.7704	.2296	.2704
0.75	.7734	.2266	.2734
0.76	.7764	.2236	.2764
0.77	.7794	.2206	.2794
0.78	.7823	.2177	.2823
0.79	.7852	.2148	.2852
0.80	.7881	.2119	.2881
0.81	.7910	.2090	.2910
0.82	.7939	.2061	.2939
0.83	.7967	.2033	.2967

APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z
TABLE) | 905

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
0.84	.7995	.2005	.2995
0.85	.8023	.1977	.3023
0.86	.8051	.1949	.3051
0.87	.8078	.1922	.3078
0.88	.8106	.1894	.3106
0.89	.8133	.1867	.3133
0.90	.8159	.1841	.3159
0.91	.8186	.1814	.3186
0.92	.8212	.1788	.3212
0.93	.8238	.1762	.3238
0.94	.8264	.1736	.3264
0.95	.8289	.1711	.3289

906 | APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z TABLE)

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
0.96	.8315	.1685	.3315
0.97	.8340	.1660	.3340
0.98	.8365	.1635	.3365
0.99	.8389	.1611	.3389
1.00	.8413	.1587	.3413
1.01	.8438	.1562	.3438
1.02	.8461	.1539	.3461
1.03	.8485	.1515	.3485
1.04	.8508	.1492	.3508
1.05	.8531	.1469	.3531
1.06	.8554	.1446	.3554
1.07	.8577	.1423	.3577

APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z
TABLE) | 907

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
1.08	.8599	.1401	.3599
1.09	.8621	.1379	.3621
1.10	.8643	.1357	.3643
1.11	.8665	.1335	.3665
1.12	.8686	.1314	.3686
1.13	.8708	.1292	.3708
1.14	.8729	.1271	.3729
1.15	.8749	.1251	.3749
1.16	.8770	.1230	.3770
1.17	.8790	.1210	.3790
1.18	.8810	.1190	.3810
1.19	.8830	.1170	.3830

908 | APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z TABLE)

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
1.20	.8849	.1151	.3849
1.21	.8869	.1131	.3869
1.22	.8888	.1112	.3888
1.23	.8907	.1093	.3907
1.24	.8925	.1075	.3925
1.25	.8944	.1056	.3944
1.26	.8962	.1038	.3962
1.27	.8980	.1020	.3980
1.28	.8997	.1003	.3997
1.29	.9015	.0985	.4015
1.30	.9032	.0968	.4032
1.31	.9049	.0951	.4049

APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z
TABLE) | 909

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
1.32	.9066	.0934	.4066
1.33	.9082	.0918	.4082
1.34	.9099	.0901	.4099
1.35	.9115	.0885	.4115
1.36	.9131	.0869	.4131
1.37	.9147	.0853	.4147
1.38	.9162	.0838	.4162
1.39	.9177	.0823	.4177
1.40	.9192	.0808	.4192
1.41	.9207	.0793	.4207
1.42	.9222	.0778	.4222
1.43	.9236	.0764	.4236

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
1.44	.9251	.0749	.4251
1.45	.9265	.0735	.4265
1.46	.9279	.0721	.4279
1.47	.9292	.0708	.4292
1.48	.9306	.0694	.4306
1.49	.9319	.0681	.4319
1.50	.9332	.0668	.4332
1.51	.9345	.0655	.4345
1.52	.9357	.0643	.4357
1.53	.9370	.0630	.4370
1.54	.9382	.0618	.4382
1.55	.9394	.0606	.4394

APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z
TABLE) | 911

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
1.56	.9406	.0594	.4406
1.57	.9418	.0582	.4418
1.58	.9429	.0571	.4429
1.59	.9441	.0559	.4441
1.60	.9452	.0548	.4452
1.61	.9463	.0537	.4463
1.62	.9474	.0526	.4474
1.63	.9484	.0516	.4484
1.64	.9495	.0505	.4495
1.65	.9505	.0495	.4505
1.66	.9515	.0485	.4515
1.67	.9525	.0475	.4525

912 | APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z TABLE)

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
1.68	.9535	.0465	.4535
1.69	.9545	.0455	.4545
1.70	.9554	.0446	.4554
1.71	.9564	.0436	.4564
1.72	.9573	.0427	.4573
1.73	.9582	.0418	.4582
1.74	.9591	.0409	.4591
1.75	.9599	.0401	.4599
1.76	.9608	.0392	.4608
1.77	.9616	.0384	.4616
1.78	.9625	.0375	.4625
1.79	.9633	.0367	.4633

APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z
TABLE) | 913

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
1.80	.9641	.0359	.4641
1.81	.9649	.0351	.4649
1.82	.9656	.0344	.4656
1.83	.9664	.0336	.4664
1.84	.9671	.0329	.4671
1.85	.9678	.0322	.4678
1.86	.9686	.0314	.4686
1.87	.9693	.0307	.4693
1.88	.9699	.0301	.4699
1.89	.9706	.0294	.4706
1.90	.9713	.0287	.4713
1.91	.9719	.0281	.4719

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
1.92	.9726	.0274	.4726
1.93	.9732	.0268	.4732
1.94	.9738	.0262	.4738
1.95	.9744	.0256	.4744
1.96	.9750	.0250	.4750
1.97	.9756	.0244	.4756
1.98	.9761	.0239	.4761
1.99	.9767	.0233	.4767
2.00	.9772	.0228	.4772
2.01	.9778	.0222	.4778
2.02	.9783	.0217	.4783
2.03	.9788	.0212	.4788

APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z
TABLE) | 915

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
2.04	.9793	.0207	.4793
2.05	.9798	.0202	.4798
2.06	.9803	.0197	.4803
2.07	.9808	.0192	.4808
2.08	.9812	.0188	.4812
2.09	.9817	.0183	.4817
2.10	.9821	.0179	.4821
2.11	.9826	.0174	.4826
2.12	.9830	.0170	.4830
2.13	.9834	.0166	.4834
2.14	.9838	.0162	.4838
2.15	.9842	.0158	.4842

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
2.16	.9846	.0154	.4846
2.17	.9850	.0150	.4850
2.18	.9854	.0146	.4854
2.19	.9857	.0143	.4857
2.20	.9861	.0139	.4861
2.21	.9864	.0136	.4864
2.22	.9868	.0132	.4868
2.23	.9871	.0129	.4871
2.24	.9875	.0125	.4875
2.25	.9878	.0122	.4878
2.26	.9881	.0119	.4881
2.27	.9884	.0116	.4884

APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z
TABLE) | 917

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
2.28	.9887	.0113	.4887
2.29	.9890	.0110	.4890
2.30	.9893	.0107	.4893
2.31	.9896	.0104	.4896
2.32	.9898	.0102	.4898
2.33	.9901	.0099	.4901
2.34	.9904	.0096	.4904
2.35	.9906	.0094	.4906
2.36	.9909	.0091	.4909
2.37	.9911	.0089	.4911
2.38	.9913	.0087	.4913
2.39	.9916	.0084	.4916

918 | APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z TABLE)

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
2.40	.9918	.0082	.4918
2.41	.9920	.0080	.4920
2.42	.9922	.0078	.4922
2.43	.9925	.0075	.4925
2.44	.9927	.0073	.4927
2.45	.9929	.0071	.4929
2.46	.9931	.0069	.4931
2.47	.9932	.0068	.4932
2.48	.9934	.0066	.4934
2.49	.9936	.0064	.4936
2.50	.9938	.0062	.4938
2.51	.9940	.0060	.4940

APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z
TABLE) | 919

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
2.52	.9941	.0059	.4941
2.53	.9943	.0057	.4943
2.54	.9945	.0055	.4945
2.55	.9946	.0054	.4946
2.56	.9948	.0052	.4948
2.57	.9949	.0051	.4949
2.58	.9951	.0049	.4951
2.59	.9952	.0048	.4952
2.60	.9953	.0047	.4953
2.61	.9955	.0045	.4955
2.62	.9956	.0044	.4956
2.63	.9957	.0043	.4957

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
2.64	.9959	.0041	.4959
2.65	.9960	.0040	.4960
2.66	.9961	.0039	.4961
2.67	.9962	.0038	.4962
2.68	.9963	.0037	.4963
2.69	.9964	.0036	.4964
2.70	.9965	.0035	.4965
2.71	.9966	.0034	.4966
2.72	.9967	.0033	.4967
2.73	.9968	.0032	.4968
2.74	.9969	.0031	.4969
2.75	.9970	.0030	.4970

APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z
TABLE) | 921

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
2.76	.9971	.0029	.4971
2.77	.9972	.0028	.4972
2.78	.9973	.0027	.4973
2.79	.9974	.0026	.4974
2.80	.9974	.0026	.4974
2.81	.9975	.0025	.4975
2.82	.9976	.0024	.4976
2.83	.9977	.0023	.4977
2.84	.9977	.0023	.4977
2.85	.9978	.0022	.4978
2.86	.9979	.0021	.4979
2.87	.9979	.0021	.4979

922 | APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z TABLE)

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
2.88	.9980	.0020	.4980
2.89	.9981	.0019	.4981
2.90	.9981	.0019	.4981
2.91	.9982	.0018	.4982
2.92	.9982	.0018	.4982
2.93	.9983	.0017	.4983
2.94	.9984	.0016	.4984
2.95	.9984	.0016	.4984
2.96	.9985	.0015	.4985
2.97	.9985	.0015	.4985
2.98	.9986	.0014	.4986
2.99	.9986	.0014	.4986

APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z
TABLE) | 923

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
3.00	.9987	.0013	.4987
3.01	.9987	.0013	.4987
3.02	.9987	.0013	.4987
3.03	.9988	.0012	.4988
3.04	.9988	.0012	.4988
3.05	.9989	.0011	.4989
3.06	.9989	.0011	.4989
3.07	.9989	.0011	.4989
3.08	.9990	.0010	.4990
3.09	.9990	.0010	.4990
3.10	.9990	.0010	.4990
3.11	.9991	.0009	.4991

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
3.12	.9991	.0009	.4991
3.13	.9991	.0009	.4991
3.14	.9992	.0008	.4992
3.15	.9992	.0008	.4992
3.16	.9992	.0008	.4992
3.17	.9992	.0008	.4992
3.18	.9993	.0007	.4993
3.19	.9993	.0007	.4993
3.20	.9993	.0007	.4993
3.21	.9993	.0007	.4993
3.22	.9994	.0006	.4994
3.23	.9994	.0006	.4994

APPENDIX A: STANDARD NORMAL DISTRIBUTION TABLE (Z
TABLE) | 925

(A) z	(B) Proportion in Body	(C) Proportion in Tail	(D) Proportion between Mean and z
3.24	.9994	.0006	.4994
3.30	.9995	.0005	.4995
3.40	.9997	.0003	.4997
3.50	.9998	.0002	.4998
3.60	.9998	.0002	.4998
3.70	.9999	.0001	.4999
3.80	.99993	.00007	.49993
3.90	.99995	.00005	.49995
4.00	.99997	.00003	.49997

License & Attribution

“Appendix A: Standard Normal Distribution Table (z Table)” is pulled from “[Appendix A: Standard Normal Distribution Table \(z Table\)](#)” by Linda R. Cote Ph.D.; Rupa G. Gordon Ph.D., Chrislyn E. Randell Ph.D., Judy Schmitt, and Helena Marvin, which is licensed CC BY-NC-SA 4.0.

APPENDIX B: T DISTRIBUTION TABLE (T TABLE)

Stephanie D'Costa; Mireille Ukeye;
Makenzie O'Neil; and Rebecca Anguiano

Table B.1. *t* Distribution Table (*t* Table)

Degrees of Freedom (df)	Critical Values of T-Distribution							
<i>Proportion (a) in One tail</i>	.25	.20	.15	.10	.05	.025	.01	.005
<i>Proportion (a) in Two tails combined</i>	.50	.40	.30	.20	.10	.05	.02	.01
1	1.000	1.376	1.963	3.078	6.314	12.706	31.821	63.657
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925
3	0.765	1.078	1.250	1.638	2.353	3.182	4.541	5.841
4	0.741	1.041	1.190	1.533	2.132	2.776	3.747	4.604
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032
6	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707
7	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499
8	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355

Degrees of Freedom (df)	Critical Values of T-Distribution							
<i>Proportion (a) in One tail</i>	.25	.20	.15	.10	.05	.025	.01	.005
<i>Proportion (a) in Two tails combined</i>	.50	.40	.30	.20	.10	.05	.02	.01
9	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250
10	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169
11	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106
12	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055
13	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012
14	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977
15	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947
16	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921

Degrees of Freedom (df)	Critical Values of T-Distribution							
<i>Proportion (a) in One tail</i>	.25	.20	.15	.10	.05	.025	.01	.005
<i>Proportion (a) in Two tails combined</i>	.50	.40	.30	.20	.10	.05	.02	.01
17	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898
18	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878
19	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861
20	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845
21	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831
22	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819
23	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807
24	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797

Degrees of Freedom (df)	Critical Values of T-Distribution							
<i>Proportion (a) in One tail</i>	.25	.20	.15	.10	.05	.025	.01	.005
<i>Proportion (a) in Two tails combined</i>	.50	.40	.30	.20	.10	.05	.02	.01
25	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787
26	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779
27	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771
28	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763
29	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756
30	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750
40	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704
60	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660

Degrees of Freedom (df)	Critical Values of T-Distribution							
<i>Proportion (a) in One tail</i>	.25	.20	.15	.10	.05	.025	.01	.005
<i>Proportion (a) in Two tails combined</i>	.50	.40	.30	.20	.10	.05	.02	.01
120	0.677	0.845	1.041	1.289	1.658	1.980	2.358	2.617
∞	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576

License & Attribution

“Appendix B: t Distribution Table (t Table)” is pulled from “[Appendix B: t Distribution Table \(t Table\)](#)” by Linda R. Cote Ph.D.; Rupa G. Gordon Ph.D., Chrislyn E. Randell Ph.D., Judy Schmitt, and Helena Marvin, which is licensed CC BY-NC-SA 4.0.

APPENDIX C: CRITICAL VALUES FOR F (F TABLE)

Stephanie D'Costa; Mireille Ukeye;
Makenzie O'Neil; and Rebecca Anguiano

Note: The critical values in this table are for $p = .05$.

Table C.1. Critical Values for F

df: Denominator (Within)	df: Numerator (Between)							
	1	2	3	4	5	6	7	8
1	161	200	216	225	230	234	237	239
2	18.51	19.00	19.16	19.25	19.30	19.33	19.36	19.37
3	10.13	9.55	9.28	9.12	9.01	8.94	8.88	8.84
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95

df: Denominator (Within)	df: Numerator (Between)							
	1	2	3	4	5	6	7	8
12	4.75	3.88	3.49	3.26	3.11	3.00	2.92	2.85
13	4.67	3.80	3.41	3.18	3.02	2.92	2.84	2.77
14	4.60	3.74	3.34	3.11	2.96	2.85	2.77	2.70
15	4.54	3.68	3.29	3.06	2.90	2.79	2.70	2.64
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59
17	4.45	3.59	3.20	2.96	2.81	2.70	2.62	2.55
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51
19	4.38	3.52	3.13	2.90	2.74	2.63	2.55	2.48
20	4.35	3.49	3.10	2.87	2.71	2.60	2.52	2.45
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42
22	4.30	3.44	3.05	2.82	2.66	2.55	2.47	2.40

df: Denominator (Within)	df: Numerator (Between)							
	1	2	3	4	5	6	7	8
23	4.28	3.42	3.03	2.80	2.64	2.53	2.45	2.38
24	4.26	3.40	3.01	2.78	2.62	2.51	2.43	2.36
25	4.24	3.38	2.99	2.76	2.60	2.49	2.41	2.34
26	4.22	3.37	2.98	2.74	2.59	2.47	2.39	2.32
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.30
28	4.20	3.34	2.95	2.71	2.56	2.44	2.36	2.29
29	4.18	3.33	2.93	2.70	2.54	2.43	2.35	2.28
30	4.17	3.32	2.92	2.69	2.53	2.42	2.34	2.27
32	4.15	3.30	2.90	2.67	2.51	2.40	2.32	2.25
34	4.13	3.28	2.88	2.65	2.49	2.38	2.30	2.23
36	4.11	3.26	2.86	2.63	2.48	2.36	2.28	2.21

df: Denominator (Within)	df: Numerator (Between)							
	1	2	3	4	5	6	7	8
38	4.10	3.25	2.85	2.62	2.46	2.35	2.26	2.19
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18
42	4.07	3.22	2.83	2.59	2.44	2.32	2.24	2.17
44	4.06	3.21	2.82	2.58	2.43	2.31	2.23	2.16
46	4.05	3.20	2.81	2.57	2.42	2.30	2.22	2.14
48	4.04	3.19	2.80	2.56	2.41	2.30	2.21	2.14
50	4.03	3.18	2.79	2.56	2.40	2.29	2.20	2.13
55	4.02	3.17	2.78	2.54	2.38	2.27	2.18	2.11
60	4.00	3.15	2.76	2.52	2.37	2.25	2.17	2.10
65	3.99	3.14	2.75	2.51	2.36	2.24	2.15	2.08
70	3.98	3.13	2.74	2.50	2.35	2.23	2.14	2.07

df: Denominator (Within)	df: Numerator (Between)							
	1	2	3	4	5	6	7	8
80	3.96	3.11	2.72	2.48	2.33	2.21	2.12	2.05
100	3.94	3.09	2.70	2.46	2.30	2.19	2.10	2.03
125	3.92	3.07	2.68	2.44	2.29	2.17	2.08	2.01
150	3.91	3.06	2.67	2.43	2.27	2.16	2.07	2.00
200	3.89	3.04	2.65	2.41	2.26	2.14	2.05	1.98
400	3.86	3.02	2.62	2.39	2.23	2.12	2.03	1.96
1,000	3.85	3.00	2.61	2.38	2.22	2.10	2.02	1.95
∞	3.84	2.99	2.60	2.37	2.21	2.09	2.01	1.94

License & Attribution

“Appendix C: Critical Values for F (F Table)” is pulled from “[Appendix C: Critical Values for F \(F Table\)](#)” by Linda R. Cote Ph.D.; Rupa G. Gordon Ph.D., Chrislyn E. Randell Ph.D., Judy Schmitt, and Helena Marvin, which is licensed CC BY-NC-SA 4.0.

APPENDIX D: CRITICAL VALUES FOR PEARSON'S R (CORRELATION TABLE)

Stephanie D'Costa; Mireille Ukeye;
Makenzie O'Neil; and Rebecca Anguiano

Table D.1. Critical Values for Pearson's r (Correlation Table)

Degrees of freedom (df) <i>df = n - 2</i>	Critical Value			
<i>Level of Significance for One-Tailed Test</i>	.05	.025	.01	.005
<i>Level of Significance for Two-Tailed Test</i>	.10	.05	.02	.01
	.10	.05	.02	.01
1	.988	.997	.9995	.9999
2	.900	.950	.980	.990
3	.805	.878	.934	.959
4	.729	.811	.882	.917
5	.669	.754	.833	.875
6	.621	.707	.789	.834
7	.582	.666	.750	.798

Degrees of freedom (df) <i>df = n - 2</i>	Critical Value			
<i>Level of Significance for One-Tailed Test</i>	.05	.025	.01	.005
<i>Level of Significance for Two-Tailed Test</i>	.10	.05	.02	.01
	.10	.05	.02	.01
8	.549	.632	.715	.765
9	.521	.602	.685	.735
10	.497	.576	.658	.708
11	.476	.553	.634	.684
12	.458	.532	.612	.661
13	.441	.514	.592	.641
14	.426	.497	.574	.623

Degrees of freedom (df) <i>df = n - 2</i>	Critical Value			
<i>Level of Significance for One-Tailed Test</i>	.05	.025	.01	.005
<i>Level of Significance for Two-Tailed Test</i>	.10	.05	.02	.01
	.10	.05	.02	.01
15	.412	.482	.558	.606
16	.400	.468	.543	.590
17	.389	.456	.529	.575
18	.378	.444	.516	.561
19	.369	.433	.503	.549
20	.360	.423	.492	.537
21	.352	.413	.482	.526

Degrees of freedom (df) <i>df = n - 2</i>	Critical Value			
<i>Level of Significance for One-Tailed Test</i>	.05	.025	.01	.005
<i>Level of Significance for Two-Tailed Test</i>	.10	.05	.02	.01
	.10	.05	.02	.01
22	.344	.404	.472	.515
23	.337	.396	.462	.505
24	.330	.388	.453	.496
25	.323	.381	.445	.487
26	.317	.374	.437	.479
27	.311	.367	.430	.471
28	.306	.361	.423	.463

Degrees of freedom (df) <i>df = n - 2</i>	Critical Value			
<i>Level of Significance for One-Tailed Test</i>	.05	.025	.01	.005
<i>Level of Significance for Two-Tailed Test</i>	.10	.05	.02	.01
	.10	.05	.02	.01
29	.301	.355	.416	.456
30	.296	.349	.409	.449
35	.275	.325	.381	.418
40	.257	.304	.358	.393
45	.243	.288	.338	.372
50	.231	.273	.322	.354
60	.211	.250	.295	.325

Degrees of freedom (df) <i>df = n - 2</i>	Critical Value			
<i>Level of Significance for One-Tailed Test</i>	.05	.025	.01	.005
<i>Level of Significance for Two-Tailed Test</i>	.10	.05	.02	.01
	.10	.05	.02	.01
70	.195	.232	.274	.302
80	.183	.217	.257	.283
90	.173	.205	.242	.267
100	.164	.195	.230	.254

License & Attribution

“Appendix D: Critical Values for Pearson’s r (Correlation Table)” is pulled from “[Appendix D: Critical Values for Pearson’s \$r\$ \(Correlation Table\)](#)” by Linda R. Cote Ph.D.; Rupa G. Gordon Ph.D., Chrislyn E. Randell Ph.D., Judy Schmitt, and Helena Marvin, which is licensed CC BY-NC-SA 4.0.

APPENDIX E: CHI-SQUARE TABLE

Stephanie D'Costa; Mireille Ukeye;
Makenzie O'Neil; and Rebecca Anguiano

Table E.1. Critical Values for Chi-Square (X^2 Table)

Degrees of Freedom (df)	Proportion in Critical Region				
	.1	.05	.02	.01	.005
1	2.706	3.841	5.024	6.635	7.879
2	4.605	5.991	7.378	9.210	10.597
3	6.251	7.815	9.348	11.345	12.838
4	7.779	9.488	11.143	13.277	14.860
5	9.236	11.070	12.833	15.086	16.750
6	10.645	12.592	14.449	16.812	18.548
7	12.017	14.067	16.013	18.475	20.278
8	13.362	15.507	17.535	20.090	21.955
9	14.684	16.919	19.023	21.666	23.589
10	15.987	18.307	20.483	23.209	25.188

Degrees of Freedom (df)	Proportion in Critical Region				
	.1	.05	.02	.01	.005
11	17.275	19.675	21.920	24.725	26.757
12	18.549	21.026	23.337	26.217	28.300
13	19.812	22.362	24.736	27.688	29.819
14	21.064	23.685	26.119	29.141	31.319
15	22.307	24.996	27.488	30.578	32.801
16	23.542	26.296	28.845	32.000	34.267
17	24.769	27.587	30.191	33.409	35.718
18	25.989	28.869	31.526	34.805	37.156
19	27.204	30.144	32.852	36.191	38.582
20	28.412	31.410	34.170	37.566	39.997
21	29.615	32.671	35.479	38.932	41.401

Degrees of Freedom (df)	Proportion in Critical Region				
	.1	.05	.02	.01	.005
22	30.813	33.924	36.781	40.289	42.796
23	32.007	35.172	38.076	41.638	44.181
24	33.196	36.415	39.364	42.980	45.559
25	34.382	37.652	40.646	44.314	46.928
26	35.563	38.885	41.923	45.642	48.290
27	36.741	40.113	43.195	46.963	49.645
28	37.916	41.337	44.461	48.278	50.993
29	39.087	42.557	45.722	49.588	52.336
30	40.256	43.773	46.979	50.892	53.672
40	51.805	55.758	59.342	63.691	66.766
50	63.167	67.505	71.420	76.154	79.490

Degrees of Freedom (df)	Proportion in Critical Region				
	.1	.05	.02	.01	.005
60	74.397	79.082	83.298	88.379	91.952
70	85.527	90.531	95.023	100.425	104.215
80	96.578	101.879	106.629	112.329	116.321
90	107.565	113.145	118.136	124.116	128.299
100	118.498	124.342	129.561	135.807	140.169

License & Attribution

“Appendix E: Chi-Square Table” is pulled from “[Appendix E: Chi-Square Table](#)” by Linda R. Cote Ph.D.; Rupa G. Gordon Ph.D., Chrislyn E. Randell Ph.D., Judy Schmitt, and Helena Marvin, which is licensed CC BY-NC-SA 4.0.